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# Current Advances in AGRICULTURAL SCIENCES

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## Sustainable management of crop residues in India

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### ABSTRACT

Indian agriculture produces about 650 million tonnes of crop residues annually. The residues are used for animal feed, soil mulch and manure, thatching for rural homes and fuel for domestic and industrial use and thus are of tremendous value to farmers. However, approximately 90-140 Mt of the residues are burned on-farm primarily to clear the field from straw and stubble of the preceding crop for sowing of the succeeding crop. Recently, the problem of on-farm burning of residues has intensified due to unavailability of labour, high cost of removing the residues and use of combines without straw spreading mechanism. The problem is severe in the mechanized rice-wheat system of the northwest India. Burning of crop residues leads to release of soot particles and smoke causing human health problems; emission of greenhouse gases such as carbon dioxide, methane and nitrous oxide causing global warming; loss of plant nutrients such as N, P, K and S; and adverse impacts on soil properties. The paper discusses the amounts of crop residues available in the country and extent of on-farm burning, and identifies the competing uses of crop residues, and the research needs. The residues can be gainfully utilized for livestock feed, composting, power generation, production of biofuel, mushroom cultivation and extraction of bioactive compounds using secondary agriculture technologies. Conservation agriculture (CA), can be effectively practiced if need-based region-specific, crop residue management plans are developed taking into consideration generation, demand, quality, feasibility and economics of residue management.

**Key words:** Air pollution, Biomass burning, Crop residue, On-farm burning, Residue management options

Crop residues include any biomass left in the field (straws, stubbles and other vegetative parts of crops) after grains and other economic components have been harvested. Processing of produce through milling also produces substantial amount of residues. The disposal of such huge amount of residues is a major concern. India being an agriculture-dependent country generates a large quantity of agricultural wastes and produces about 650 million tonnes (Mt) of crop residues annually (Pathak *et al.*, 2012). With the growing population there is a great need for increasing the agricultural production and thus, more agricultural biomass will be generated. Crop residue is a valuable renewable resource and an important component in ecosystem stability of world's agricultural land. The residues are used mainly as animal feed, thatching for rural homes, residential cooking fuel and industrial fuel. However, it is not properly managed and utilized. A huge quantity of crop residue is burnt in the open fields which have significant impacts on regional air quality and human health (Arbex *et al.*, 2004; Agarwal *et al.*, 2010).

Combine harvesting technologies, which have become common in rice-wheat system in India, leave behind huge quantities of straw in the field. Harvesting of rice wheat crops with combine harvesters is popular in Indo-Gangatic plains of India specially Punjab, Haryana and western Uttar Pradesh. According to an estimate 75-80% of rice in Punjab is machine harvested which leaves behind enormous amount of crop

residue (Badrinath *et al.*, 2006). The combines cut the cereal crop at a certain height above the ground, thereby creating two distinct straw components after harvesting: (i) the standing stubble or anchored crop residues; and (ii) the windrows of straw or loose crop residues-big uneven heaped lines of straw. It is particularly the latter that are a nuisance for establishing the subsequent crop. In combine harvested rice wheat fields both the anchored and loose rice straw is generally left in the field and burned *in situ* as a means of clearing land rapidly and inexpensively thereby allowing tillage practices to proceed unimpeded by residual crop material. In case of manual harvesting most of the residue is removed from the fields and utilized as feed for livestock, fuel and thatching and the remaining residue is burnt in the fields. Crop residue burning has a strong regional and crop specific variation with considerable spatial and temporal heterogeneity. Emissions from burning are not consistent over the calendar year, the majority of emissions occur during the months of mid April-May and October to November. Burning of agricultural residue is associated with emission of greenhouse gases such as CO<sub>2</sub>, N<sub>2</sub>O, CH<sub>4</sub>, emission of air pollutants such as CO, NH<sub>3</sub>, NO<sub>x</sub>, SO<sub>2</sub>, NMHC, volatile organic compounds (VOCs) and semi-volatile organic compounds (SVOCs) and particulates (Andreae *et al.*, 1996). These pollutants emitted due to burning can be transported over long distances and across political boundaries and result in decrease in crop productivity, elevated oxidant

levels, acid deposition and visibility impairment. It also cause nutrient and resource loss and adversely affects soil properties. Need of the hour is to develop an appropriate policy for promoting multiple uses of crop residues and prevent their on-farm burning. This review paper aims to (i) quantify the amount of crop residues generated in the country and the extent of their on-farm burning, (ii) assess the environmental impacts of on-farm burning of crop residues, (iii) identification of different uses of crop residues and the research needs.

### CROP RESIDUE GENERATION IN INDIA

There were large uncertainties as well as variability in crop residues generation and their use depending on the cropping intensity, productivity and crops grown in different states of India. According to estimates of Jain *et al.* (2014) 620 Mt crop residues were generated in the year 2009 in India. Sahai *et al.* (2007) have estimated 253 Mt of crop residue generation in the year 2010. Their estimates were extremely low as compared to other researchers. Ministry of New and Renewable Energy (MNRE), Govt. of India have estimated that approximately 500 Mt of crop residue is generated every year (Table 1). Residue generated by different crops was grouped in five categories for the ease of representation, based on the type of crop, like cereals, pulses, oilseeds, fibers and others. Crops included in each category are given in Table 2. Residue

Table 1. Generation and burning of crop residues in various states of India

| States            | Residue generation <sup>#</sup><br>Mt yr <sup>-1</sup> | Residue surplus <sup>**</sup> |
|-------------------|--------------------------------------------------------|-------------------------------|
| Andhra Pradesh    | 43.89                                                  | 6.96                          |
| Arunachal Pradesh | 0.4                                                    | 0.07                          |
| Assam             | 11.43                                                  | 2.34                          |
| Bihar             | 25.29                                                  | 5.08                          |
| Chhattisgarh      | 11.25                                                  | 2.12                          |
| Goa               | 0.57                                                   | 0.14                          |
| Gujarat           | 28.73                                                  | 8.9                           |
| Haryana           | 27.83                                                  | 11.22                         |
| Himachal Pradesh  | 2.85                                                   | 1.03                          |
| Jammu and Kashmir | 1.59                                                   | 0.28                          |
| Jharkhand         | 3.61                                                   | 0.89                          |
| Karnataka         | 33.94                                                  | 8.98                          |
| Kerala            | 9.74                                                   | 5.07                          |
| Madhya Pradesh    | 33.18                                                  | 10.22                         |
| Maharashtra       | 46.45                                                  | 14.67                         |
| Manipur           | 0.9                                                    | 0.11                          |
| Meghalaya         | 0.51                                                   | 0.09                          |
| Mizoram           | 0.06                                                   | 0.01                          |
| Nagaland          | 0.49                                                   | 0.09                          |
| Orissa            | 20.07                                                  | 3.68                          |
| Punjab            | 50.75                                                  | 24.83                         |
| Rajasthan         | 29.32                                                  | 8.52                          |
| Sikkim            | 0.15                                                   | 0.02                          |
| Tamil Nadu        | 19.93                                                  | 7.05                          |
| Tripura           | 0.04                                                   | 0.02                          |
| Uttarakhand       | 2.86                                                   | 0.63                          |
| Uttar Pradesh     | 59.97                                                  | 13.53                         |
| West Bengal       | 35.93                                                  | 4.29                          |
| India             | 501.76                                                 | 140.84                        |

<sup>#</sup> MNRE, Average Data for the year 2002 to 2004

<sup>\*\*</sup> surplus crop residue is subjected to burning

Table 2. Crops considered under different crop types

| Crop types | Major crops                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------|
| Cereals    | Rice, wheat, maize, jowar, bajra, ragi and small millets                                                |
| Pulses     | Arhar, urad, moong, horse gram, gram, cow gram                                                          |
| Fibers     | Sunhemp, mesta, jute, coconut, cotton                                                                   |
| Oilseed    | Til, sunflower, safflower, mustard, niger, linseed, soybean, groundnut, castor                          |
| Others     | Sugarcane, turmeric, chilies, coffee, banana, potato, onion, arecanut, rubber, tea and other vegetables |

generation was highest in Uttar Pradesh (60 Mt), followed by Punjab (51 Mt) and Maharashtra (46 Mt) (Table 1).

Among different crop types, cereal crops generated 352 Mt residue followed by fibre crops (66 Mt), oilseed crops (29 Mt), pulses (13 Mt) and sugarcane (12 Mt). The cereal crops (rice, wheat, maize, millets) contributed 70% while rice crop alone contributed 48% and wheat ranked second with 32% of cereal crop residues (Fig. 1 and Fig. 2). Fibre crops contributed

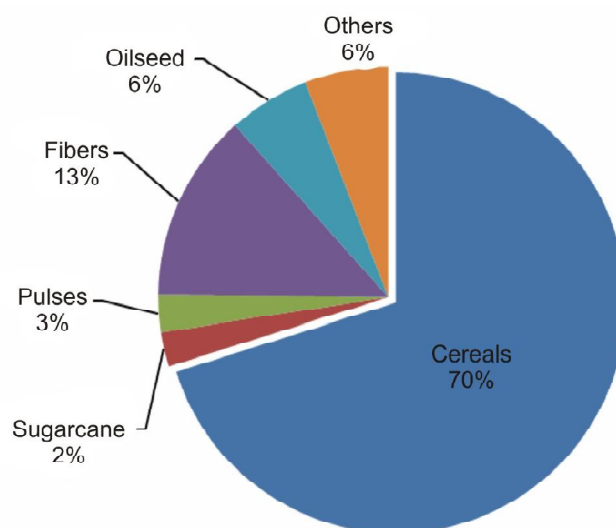


Fig. 1. Contribution of various crops for crop residue generation in India (Calculated from MNRE report 2009)

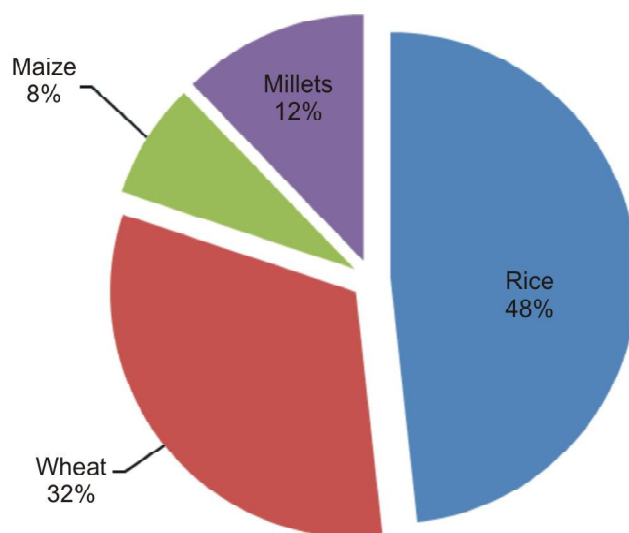


Fig. 2. Contribution of different cereal crops for crop residue generation in India (Calculated from MNRE report 2009)

13% of residues generated from all crops. Among fibre crops, cotton crop generated maximum (53 Mt) with 11% of crop residues. Coconut ranked second among fibre crops with 12 Mt of residue generation. Other oilseed crops generated less than 2 Mt of residue annually. Sugarcane residues comprising tops and leaves generated 12 Mt i.e. 2% of crop residues in India. Generation of cereal crop residues was highest in Uttar Pradesh (54.5 Mt), followed by Punjab (50 Mt) and West Bengal (35.8 Mt). Maharashtra contributed maximum to the generation of crop residues of pulse crops (3 Mt) while residues from fibre crops were dominant in Andhra Pradesh (14 Mt). Gujarat and Rajasthan generated about 6 Mt each of residues from oilseed crops (Table 3).

Table 3. Crop-wise generation of crop residues in different states of India

| States            | Cereals             | Pulses | Fibres | Oilseeds | Others |
|-------------------|---------------------|--------|--------|----------|--------|
|                   | Mt yr <sup>-1</sup> |        |        |          |        |
| Andhra Pradesh    | 24.413              | 0.910  | 13.949 | 4.293    | 0.328  |
| Arunachal Pradesh | 0.324               | 0.000  | 0.000  | 0.046    | 0.030  |
| Assam             | 9.440               | 0.018  | 0.398  | 0.348    | 1.230  |
| Bihar             | 22.599              | 0.903  | 0.020  | 0.532    | 1.238  |
| Chhattisgarh      | 10.580              | 0.373  | 0.000  | 0.247    | 0.052  |
| Goa               | 0.365               | 0.000  | 0.193  | 0.000    | 0.012  |
| Gujarat           | 10.463              | 0.730  | 7.111  | 5.898    | 4.524  |
| Haryana           | 23.630              | 0.043  | 2.656  | 1.461    | 0.043  |
| Himachal Pradesh  | 2.843               | 0.000  | 0.000  | 0.000    | 0.008  |
| Jammu and Kashmir | 1.590               | 0.000  | 0.000  | 0.000    | 0.001  |
| Jharkhand         | 3.146               | 0.108  | 0.000  | 0.060    | 0.300  |
| Karnataka         | 20.689              | 0.566  | 6.583  | 3.372    | 2.728  |
| Kerala            | 1.504               | 0.000  | 5.131  | 0.004    | 3.102  |
| Madhya Pradesh    | 19.435              | 2.167  | 4.199  | 0.967    | 6.415  |
| Maharashtra       | 22.738              | 2.910  | 13.381 | 2.466    | 4.956  |
| Manipur           | 0.893               | 0.000  | 0.000  | 0.000    | 0.004  |
| Meghalaya         | 0.328               | 0.000  | 0.050  | 0.010    | 0.123  |
| Mizoram           | 0.038               | 0.000  | 0.000  | 0.011    | 0.012  |
| Nagaland          | 0.486               | 0.000  | 0.000  | 0.000    | 0.006  |
| Orissa            | 18.087              | 0.720  | 0.252  | 0.873    | 0.138  |
| Punjab            | 44.156              | 0.037  | 5.895  | 0.192    | 0.471  |
| Rajasthan         | 18.161              | 2.262  | 2.234  | 5.593    | 1.072  |
| Sikkim            | 0.141               | 0.000  | 0.000  | 0.006    | 0.002  |
| Tamil Nadu        | 6.812               | 0.205  | 2.445  | 2.145    | 8.325  |
| Tripura           | 0.000               | 0.000  | 0.040  | 0.000    | 0.001  |
| Uttarakhand       | 2.517               | 0.000  | 0.000  | 0.024    | 0.319  |
| Uttar Pradesh     | 53.297              | 1.083  | 0.000  | 0.158    | 5.436  |
| West Bengal       | 33.280              | 0.052  | 1.830  | 0.046    | 0.726  |
| India             | 351.769             | 13.085 | 66.366 | 25.628   | 29.463 |

(Calculated from MNRE report 2009)

## CROP RESIDUE UTILIZATION

Residues have numerous competing uses such as animal feed, fodder, fuel, roof thatching, packaging and composting. In India residues have been traditionally used as a household and industrial fuel (combustion with coal, wood, etc). The competitive uses for various residues are different in different states depending on their availability and requirements. Residues of cereal crops are mainly used as cattle feed. The farmer use residue themselves as fodder, fuel, roof thatching, etc. or sell it to other landless households or intermediaries which in turn sells it to the industrial units. The remaining residues are just left unused or burnt in the fields by the farmers.

The residues of most of the cereal crops except rice and 50% of pulses are used for fodder. Rice straw is not used as fodder in northern India as it has a low feed value because the nutrients present in rice straw are not readily available to the livestock due to its high silica and lingo-cellulosic content. The rice straw is fed to animals only in case of acute scarcity of fodder availability. In states like Punjab and Haryana, where rice straw is not utilized as cattle feed, large quantity is subjected to burning in the fields. Rice straw and husk is used as domestic fuel or in boilers for parboiling rice in states like West Bengal, heating water and straw is also used as separators in packaging. Rice husk was generally not subjected to open burning prior to the introduction of combine harvesters. However combine harvesters leave husk also in the field which is later on subjected to burning. Straw and cobs are the two types of residue generated from maize crop. Maize straw is mainly used as fodder or domestic fuel. Maize cobs are very hard, and therefore are not consumed in significant quantities as fodder (Meshram, 2002). It is rather used as fuel in homes or thrown away, which ends up in open burning. Sugarcane tops in most of the areas is either used for feeding of dairy animals or burnt in field for ratoon crop, whereas in some states like UP it is directly burnt in the fields (Yadav and Solomon, 2006). Cotton, chilli, pulses and oilseeds residues are mainly used as fuel for household needs. Cotton stalk are used for fencing, thatching and wall construction and the remaining stalk is left in fields and burnt on-farm. A few small paper mills in the cotton belt utilizes cotton crops seasonal residue (harvested mostly in January-March and October-December), which is otherwise used as domestic fuel. Residues of groundnut are burnt as fuel in brick kilns and lime kilns. Coconut shell, stalks of rapeseed and mustard, pigeon pea and jute and mesta, and sun flower are used as domestic fuel. Coconut generates about 3 Mt of husk annually. Almost 25-30% husk is utilized for making coir (Smith *et al.*, 2009) and nearly, a million ton burnt as fuel.

There are several technologies in India for utilizing crop residues. However, there are several constraints which limit the large scale adoption of these technologies. One major problem of utilizing this large amount of straw has been the high cost and labour requirement for collection and transportation. Some countries have developed strategies for successful management of crop residues to avoid on-farm burning. In countries like China, USA, Philippines and Indonesia some amount of crop residues are burnt on-farm. In China approximately 700 Mt crop residues are generated annually, out of this 31% of crop residues are left in the field, 31% is used for animal feed, 19% for bio-energy production and 15% as fertilizer (Jiang *et al.*, 2012). In USA burning has been regulated in some of the states. For example, in California farmers require a permit for burning, and it has to be carried out only on 'burn-days' decided by local Air Districts in consultation with the California Air Resource Board depending on the weather conditions (Pettus, 2006). Before burning the crop residues are required to be shredded in to small size and piled. In some countries crop residues are used as a source of energy (e.g. Indonesia, Nepal, Thailand, Malaysia, Philippines,

Indonesia and Nigeria), making compost (Philippines, Israel, China), as animal feed (Lebanon, Pakistan, Syria, Iraq, Israel, Tanzania, China and countries in Africa), and for mushroom cultivation (Vietnam) (Cooper and Liang, 2007; Machl, 2009).

### UNUTILIZED CROP RESIDUE

The unutilized or surplus residue is the amount of crop residue estimated by subtracting the amount of crop residue used for various purposes (thatching, fodder, dung cakes and household fuel) from the total crop residues generated. The surplus crop residues are typically burned in the field by farmers. According to the estimates of MNRE (2009) total crop residue surplus in India was 141 Mt yr<sup>-1</sup> where cereals and fiber crops contributed 58% and 23% fraction respectively (Fig. 3). Rest 19% was from sugar cane, pulses oilseeds and other crops. Out of 82 Mt yr<sup>-1</sup> surplus crop residues from the cereal crops, 44 Mt yr<sup>-1</sup> was from rice followed by 24.5 Mt yr<sup>-1</sup> of wheat crop which was mostly subjected to burning in open fields. In case of fiber crops (33 Mt of surplus residue) approximately 80% was cotton crop residue.

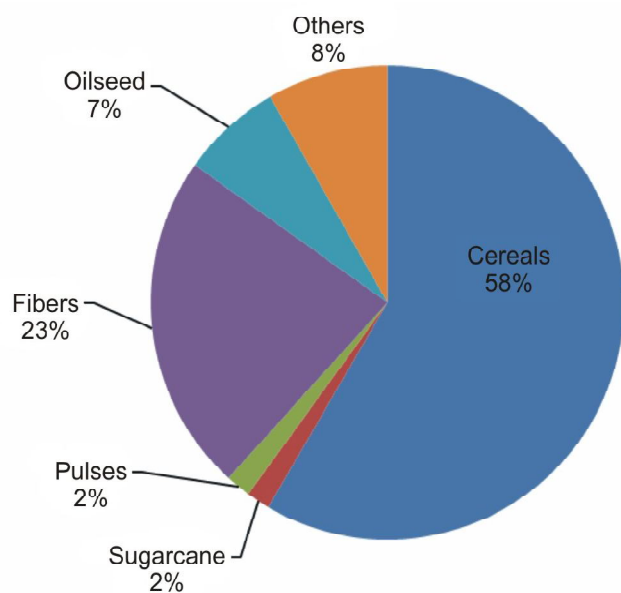


Fig. 3. Contribution of various crops types for unutilized crop residue in India (Calculated from MNRE report 2009).

The state of Punjab ranked first in surplus crop residue generation (24.7 Mt), followed by Maharashtra (14.4 Mt) and Uttar Pradesh (12.4 Mt). Large amount of surplus crop residue in state of Maharashtra was because of the contribution of fibre crop residues. Although crop residue generation from fibre crops was dominant in Andhra Pradesh (14 Mt) but the surplus was highest in Maharashtra (6.7 Mt), followed by Punjab (5.3 Mt) and Gujarat (5 Mt) due to major fraction coming from cotton biomass. Punjab and Uttar Pradesh generated around 19 and 12 Mt of surplus crop residue from cereal crops (Fig. 4).

### CROP RESIDUE BURNING

On a global basis, burning of agricultural waste is the

second major source, representing nearly 2020 Tg (approx 25% of total biomass burned) (Andreae *et al.*, 1996; Chang and Song, 2010). Street *et al.* (2003) reported out of, 730 Tg of biomass burned in a typical year, from both anthropogenic and natural causes in Asia with 18% contribution from India's. Large uncertainties exist in the estimates of utilization as well as on-farm burning of crop residues depending upon the crops considered, residue to grain ratio and fraction of residues subjected to burning. According to various estimates approximately 32-127 Mt of the crop residues are burnt on-farm (Pathak *et al.*, 2006; Pathak *et al.*, 2010; Sahai *et al.*, 2011). The substantial difference may be attributed to the use of various conversion factors, the inclusion or deletion of different crops.

Increased mechanization, declining number of livestock, long period required for composting and no economically viable alternate use of residues are some of the reasons for residues being burnt in field. The number of combine harvester in the country, particularly in the IGP has increased from nearly 2000 in 1986 up to 10000 in 2010 (Gupta *et al.*, 2003). North western part (Punjab, Haryana and western Uttar Pradesh) of the IGP has about 75% of the cropped area under combine harvesting (Sidhu and Beri, 2008). Combine harvesters are used extensively in central and eastern Uttar Pradesh, Uttarakhand, Bihar, Rajasthan, Madhya Pradesh and southern states as well for harvesting rice and wheat. The major reasons for increase in use of combine are labour shortage, high wage during harvesting season, ease of harvesting and thrashing and uncertainty of weather (Verma and Jayakumar, 2012). With combine harvesting, however, about 80% of the residues are left in the field as loose straw that finally ends up being burnt. It is estimated that about 15-22 Mt rice straw is burned every year in Punjab alone (Sidhu and Beri, 2008). Other reasons for intentional burning include clearing of fields, fertility enhancement, and pest and pasture management. The time gap between rice harvesting and wheat sowing in northwest India is around 15-20 days. In this short duration farmers prefer burning the rice stalk in the field instead of harvesting it for fodder. Burning provides a fast way of controlling weeds, insects and diseases, both by eliminating them directly or by altering their natural habitat. It is also perceived to boost soil fertility, although burning actually has a differential impact on soil fertility. It increases the short-term availability of some nutrients (e.g. P and K) and reduces soil acidity, but leads to a loss of other nutrients (e.g. N and S) and organic matter (Verma and Jayakumar, 2012).

### IMPACTS OF FIELD BURNING OF CROP RESIDUES

#### *Emission of gaseous and aerosol species*

Burning of agricultural residues, represent a significant source of GHGs, chemically and radiatively important trace gases and aerosols such as CH<sub>4</sub>, CO<sub>2</sub>, N<sub>2</sub>O, CO, SO<sub>x</sub>, NO<sub>x</sub>, particulate and other hydrocarbons to the atmosphere affecting the atmospheric composition (Graedel and Crutzen, 1993; Jain *et al.*, 2014). This change in composition of the atmosphere

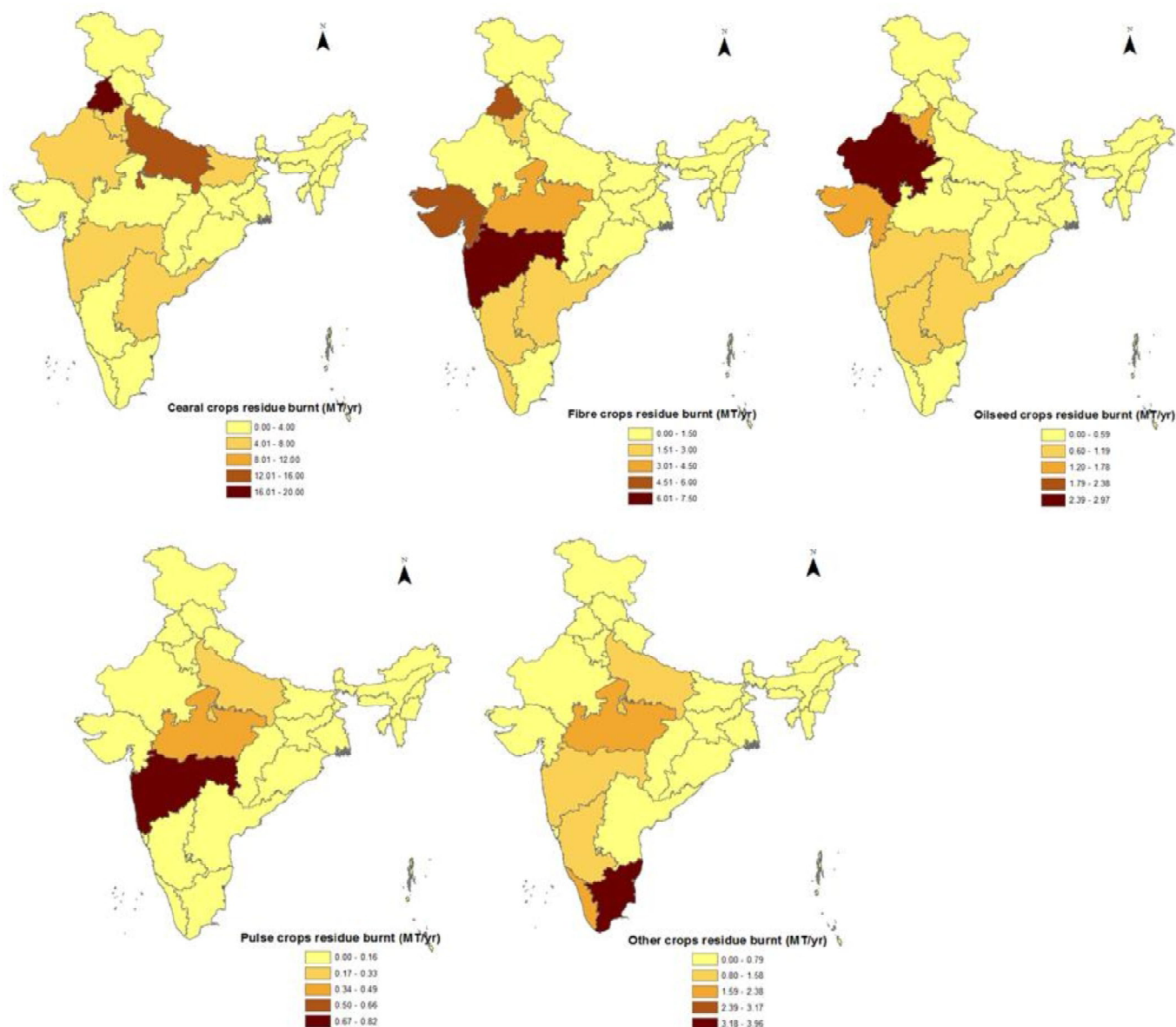


Fig. 4. State-wise distribution of unutilized crop residue in India (calculated from MNRE)

may have a direct or indirect effect on the radiation balance (Cattani *et al.*, 2005). Burning of crop residues also emits large amount of particulates, volatile organic compounds (VOCs) and semi-volatile organic compounds (SVOCs) including polycyclic aromatic hydrocarbons (PAHs) and polychlorinated biphenyls (PCBs) (Koppmann *et al.*, 2005). These gases may lead to a regional increase in the levels of aerosols, acid deposition, increase in tropospheric ozone and depletion of the stratospheric ozone layer. They may subsequently undergo trans-boundary migration depending upon the wind speed/direction, reactions with oxidants like OH<sup>•</sup> leading to physico-chemical transformation and wash out by precipitation. Several authors have estimated the emissions of major pollutants from burning crop residues. For example, 70, 7 and 0.66% of C present in rice straw is emitted as CO<sub>2</sub>, CO and CH<sub>4</sub>, respectively, while 20 and 2.1% of N in straw is emitted as NO<sub>x</sub> and N<sub>2</sub>O, respectively, and 17% as S in straw is emitted as SO<sub>x</sub> upon

burning (Samra *et al.*, 2003). According to Jenkins and Bhatnagar (1991) one ton straw on burning emits approximately 3 kg particulate matter, 60 kg CO, 1460 kg CO<sub>2</sub>, 199 kg ash and 2 kg SO<sub>2</sub>. According to the estimates of Yevich and Logan (2003) burning of crop residues in India resulted in the emissions of 91, 4.1, 0.6, 0.1 and 1.2 Tg yr<sup>-1</sup> of CO<sub>2</sub>, CO, CH<sub>4</sub>, NO<sub>x</sub> and total particulate matter for the year 1985. Emissions from open biomass burning have been estimated by various researchers (Venkataraman *et al.*, 2006; Chang and Song, 2010) over tropical Asia/India. Sahai *et al.* (2011) have estimated that burning of 63 Mt of crop residue emitted 4.86 Mt of CO<sub>2</sub> equivalents of GHGs 3.4 Mt of CO and 0.14 Mt of NO<sub>x</sub>. In the present study it was estimated that burning of crop residues emitted 380 Gg of CH<sub>4</sub>, 9.86 Gg of N<sub>2</sub>O, 56.34 Gg of SO<sub>x</sub> and 352 Gg of NO<sub>x</sub>.

#### Loss of residues nutrient

Burning of crop residue results in loss of entire amount

of C, along with approximately 80% N, 25% of P, 20% of K and 50% of S and present in straw, thereby leading to atmospheric pollution (Raison, 1979; Ponnamperna, 1984; Lefroy, 1994). If these crop residues are incorporated or retained, the soil will be enriched with organic carbon and N. Burning of paddy straw causes a loss of about 79.38 kg ha<sup>-1</sup> of nitrogen, 108.86 kg ha<sup>-1</sup> of potassium and 183.71 kg ha<sup>-1</sup> of phosphorus (Gupta *et al.*, 2003). According to Sahai *et al.* (2011) 26.1 Mt of C, 0.35 Mt of N is emitted per year due to burning of crop residues. In the present study it was estimated that burning of crop residues resulted in the loss of 0.44 Mt of N, 0.02 Mt of P and 0.38 Mt of K from cereal crop residues (82 Mt), 0.04 Mt of N, 0.003 Mt of P and 0.014 Mt of K from pulse crop residues and 0.008 Mt of N, 0.001 Mt of P, 0.003 Mt of K from sugarcane residues.

### **Impact of burning on soil properties**

Heat from burning residues elevates soil temperature kills the beneficial bacterial and fungal populations. However, the death is temporary as the microbes regenerate after few days. Repeated burning in the field, however, permanently diminishes the microbial population. However burning of residue also kills the pests and disease causing pathogens (Abrol *et al.*, 2005). Burning immediately increases the exchangeable NH<sub>4</sub><sup>+</sup>-N and bicarbonate extractable P content but there is no build up of nutrients in the profile. Long-term burning reduces total N and C and potentially mineralized N in the 0-15 cm soil layer (Sidhu and Beri, 1989). Burning also degrades the soil structure (Hubbart *et al.*, 2004).

## **ALTERNATIVE MANAGEMENT OF CROP RESIDUES**

There are several options which can be practiced to manage crop residues in productive manners such as composting, generation of energy, biofuel production, mushroom cultivation, recycling in the field through biochar, etc.

### **Energy from crop residues**

Biomass can be efficiently utilized as a source of energy and is of interest worldwide because of its environmental advantages. In recent years, usage of crop residue for energy production has been proposed as a substitute for fossil fuels. It also offers an immediate solution for the reduction of the CO<sub>2</sub> content in the atmosphere. Besides this, it is renewable and can be used without damaging the environment. In comparison with the other renewable energy resources such as solar and wind energy, biomass is a storable resource, inexpensive, energy efficient and eco-friendly. According to the estimates of MNRE the total power production potential of surplus agro-residues is 18729 MWe<sup>2</sup>. However, crop residues have low bulk-density and low energy yield per unit weight basis (Biomass atlas). The transportation of large volumes of crop residues required for efficient energy generation represents a major cost enhancing factor irrespective of the available bio-energy technology. Kalpataru Power Transmission Limited (KPTL), Rajasthan is successfully generating energy from crop residues

in Ganganagar and Tonk districts of Rajasthan. The plant is utilizing 80,000 tonnes of biomass of mustard crop, annually to generate 1.5 lakh kW energy per day (Gupta, personal communication). However, the plant also produces a large amount of ash which has to be managed in a profitable and environment friendly manner.

### **Ethanol from crop residues**

The conversion of ligno-cellulosic biomass into alcohol is of immense importance and is a researchable issue as ethanol can be either blended with gasoline as a fuel extender and octane-enhancing agent, or used as a neat fuel in internal combustion engines. The theoretical estimates of ethanol production from different feedstock (corn grain, rice straw, wheat straw, bagasse and saw dust) varies from 382-471 l tonne<sup>-1</sup> of dry matter (Demirbas and Sahin, 2009).

### **Composting of residues**

The residues can be composted by using it as animal bedding and then heaping in dung pit. Each kg of straw absorbs about 2-3 kg of urine from the animal shed (Sidhu and Beri, 2008). It can also be composted by alternative methods such as vermi composting or batch composting methods on the farm itself. The residues of rice from one hectare give about 3.2 tonnes of manure as rich in nutrients as farmyard manure (FYM) (Sidhu and Beri, 2008). Indian Agricultural Research Institute (IARI), New Delhi, has successfully developed a biomass-compost unit for making of good quality compost. This mechanized unit efficiently uses waste biomass and crop residues generated in the IARI farm. The decomposition process, which is hastened by a consortium of microorganisms, takes 75-90 days (Singh, personal communication).

### **Biomethanation**

Biomass such as rice straw can be converted to biogas, a mixture of carbon dioxide and methane and used as fuel. The process yields good quality of gas 55-60% of methane. It is reported that methane yield of 214 m<sup>3</sup> t<sup>-1</sup> of rice straw can be obtained by biomethanation process (Aggarwal *et al.*, 2007). The solid biogas spent slurry can be used as manure or can be converted into good quality compost within a short period. This process promises a method to utilize crop residues in a non destructive way to extract high quality fuel gas and produce compost to be recycled in soil.

### **Gasification of biomass**

Gasification is a thermo-chemical process in which gas is formed due to partial combustion of any organic material. The process breaks down residue completely to yield energy rich gaseous products after initial pyrolysis. The products of pyrolysis depend upon the temperature, type of input material, treatment process, etc (Lei, 2010). The main problem in biomass gasification for power generation is the cleaning of gas to remove the impurities. In some states gasifiers with more than 1MW capacity has been installed for generation of producer gas which is fed to the engines coupled to the alternators for

electricity generation. One ton of biomass can be used for generation of 300 kWh of electricity (Singh and Gu, 2010). As per another estimate energy potential of rice (41 Mt yr<sup>-1</sup>), maize (6.2 Mt yr<sup>-1</sup>), sugarcane (240 Mt yr<sup>-1</sup>) and others residues (163.5 Mt yr<sup>-1</sup>; cotton, and coconut shell and fronds) is 4700 MW, 700 MW, 8900 MW and 28000 MW, respectively ([www.icac.org](http://www.icac.org)).

### **Bio-oil production**

Bio-oil can be produced from crop residues by the process of fast pyrolysis or flash pyrolysis. It takes place in less than two seconds with temperatures between 300 and 550°C in the absence of oxygen. Fast pyrolysis of crop residue requires the temperature of biomass to be raised to 400-500°C within few seconds. This results in a remarkable change in the nature of the thermal disintegration process. Approximately 70-75% of dry weight of biomass is converted into condensable vapours, which on cooling within a couple of seconds, yields a dark brown viscous liquid known as bio-oil (TERI). The calorific value of bio-oil varies between 16-20 MJ kg<sup>-1</sup> (Demirbas, 2008).

### **Biochar production**

Biochar is high carbon material produced from the slow pyrolysis or vacuum pyrolysis (heating in the absence of oxygen) of crop residue. It has got advantages in terms of its efficiency as an energy source, its use as a fertilizer when mixed with soil, its ability to stabilize as well as reduce emissions of harmful gases in the atmosphere. It can potentially play a major role in the long-term storage of carbon. Biochar increases the fertility, water retention capability of the soil as well as increasing the rate of mineral delivery to roots of the plants (Biomass network report, Verheijen *et al.*, 2009). With the current available technologies, it is not economically viable and cannot be popularized among the farmers. There is a need to develop low cost pyrolysis kiln for the generation of biochar to utilize surplus crop residues, which are otherwise burnt on-farm. However, if all the products and by-products such as heat energy, H<sub>2</sub> and bio-oil are captured and used in the biochar generation process, it would become economically-viable.

### **Recycling of crop residues in fields**

The resource conserving technologies (RCTs) involving no- or minimum-tillage, direct seeding, bed planting, use of leaf colour chart for N management, and crop diversification are being advocated as alternatives to the conventional rice-based system for improving productivity and sustainability. The RCTs with innovations in residue management avoids straw burning, improve soil organic C, are input efficient and have potential to reduce GHG emissions (Sohi *et al.*, 2010; Pathak *et al.*, 2010). Recycling of crop residues is an integral part of conservation agriculture (CA), as alternative to the conventional production system for improving productivity and sustainability. Recent estimates revealed that CA based resource conserving technologies (RCTs) that include laser assisted precision land levelling, zero/reduced tillage, direct drilling into the residues, direct seeded rice, unpuddled mechanical transplanted rice,

raised bed planting and diversification/intensification are being practiced over nearly 3.9 Mha of South Asia (Derpsch and Friedrich, 2010). Incorporation of crop residues in the soil or retention on surface has several positive impacts on physical, chemical and biological properties of soil. It increases hydraulic conductivity, cation exchange capacity (CEC), water infiltration and moisture retention capacity of the soil, improves aggregate stability and soil structure, reduces bulk density, surface sediment and water runoff, surface crust formation, water evaporation from the top few inches of soil and prevents leaching of nutrients. It also increases the microbial biomass and enhances activities of enzymes such as dehydrogenase and alkaline phosphatase. Mulching with crop residues increases the minimum soil temperature in winter by reducing upward heat flux from soil and decreases soil temperature during summer due to shading effect. The crop residues play an important role in amelioration of soil acidity by releasing bases such as hydroxyls during the decomposition of crop residues with higher C:N, and soil alkalinity through application of residues from lower C:N crops such as legumes, oilseeds and pulses (Pathak *et al.*, 2012). The crop residues also help in carbon sequestration in the soil. But higher residue levels in conservation agriculture can pose problem of different disease occurrence, insect or weed attacks, difficulties in tillage practices and proper placement of seed, fertilizer and pesticide with more residues on the surface. This may require increased use of herbicides and application of specific nutrients with specialized equipment for proper fertilizer placement, contributing to higher costs in adopting CA.

No-till can also complicate manure application, thereby, leading to nutrient stratification within soil profile from repeated surface applications without any mechanical intervention for residue incorporation. Although a lot of improvement has been done in the zero-till seed-cum-fertilizer drill machinery, but there is still a lot of scope for further improvement to give farmers a hassle free technology.

## **RESEARCH NEEDS FOR CROP RESIDUE MANAGEMENT**

Management of crop residues with conservation agriculture is vital for long-term sustainability of Indian agriculture. Crop residues should not be burnt and some amount of residues should be used for CA for improving soil health and reducing environment pollution. Several technologies are available; they require improvement for adoption by resource poor, low skilled farmers. Some of the areas where research activities could be taken up are : (1) Inventorization of amount of crop residues generated by different crops in different regions of the country, (2) Identification of the major uses of crop residues and comparative assessment of their competing uses, (3) Assessing the quality of crop residues and their suitability for various purposes, (4) Quantifying the permissible amount of residues of different crops which can be incorporated/retained in the soil depending on cropping systems, soil type, and climate without creating operational problems for the next crop, (5) Enhancing decomposition rate

of residues for *in-situ* incorporation, (6) Analysis of benefit to cost ratio and socio-economic impacts of residue retention or incorporation with conservation agriculture vis-à-vis residue burning, (7) Developing complete package of practices of CA for prominent cropping system in each agro-ecological region, (8) Complete life cycle analysis of residue retention and CA vis-à-vis residue burning other uses of agricultural residues.

The residues are of great economic value as fodder, fuel and raw material for various industries. However, a large amount of residue is burnt in fields leading to emission of air pollutants. Problems with the crop residues are different in different region of India and are associated with the socio-economic needs of the farmers. Therefore, policy needs to be formulated for each region separately as the policy in one region of India may not work in region or state. There is also a need to create awareness among the farming communities about the adverse impacts of burning and importance of crop residues incorporation for maintaining soil health and sustainable agricultural productivity.

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## Productivity, economic viability and energy efficiency of intercropping winter maize (*Zea mays*) and rajmash bean (*Phaseolus vulgaris*) in potato (*Solanum tuberosum*) with border ridge technique

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### ABSTRACT

A field experiment was carried out for two consecutive winter (*rabi*) seasons of 2006-07 and 2007-08 at Kanpur to assess the production potential, economic viability and energy efficiency of potato (*Solanum tuberosum* L.) intercropped with winter maize (*Zea mays* L.) and rajmash bean (*Phaseolus vulgaris* L.) for central plain zone of Uttar Pradesh. Values of land equivalent ratio (LER) and area-time equivalent ratio (ATER) with all the intercropping systems were greater indicating advantage in yield, land-use efficiency and monetary return unit<sup>-1</sup> time and space over the respective monocultures. Potato (45 x 20 cm) + winter maize (cob purpose) (3:2) was proved to be the most efficient, productive and remunerative cropping system as it gave the highest mean potato equivalent yield (34.55 t ha<sup>-1</sup>) and also accounted for highest values for LER (1.67), ATER (1.65), production efficiency (317 kg ha<sup>-1</sup> day<sup>-1</sup>), monetary advantage index (41584), and monetary efficiency (₹ 574 ha<sup>-1</sup> day<sup>-1</sup>) compared to other intercropping systems. Potato (45 x 20 cm) + winter maize (cob purpose) registered higher energy output (335.82 x 10<sup>3</sup> MJ ha<sup>-1</sup>) over other cropping systems. The energy produced by this system was 19.5, 40.4 and 48.3% higher than potato (60 x 15 cm) + winter maize (cob purpose), potato (45 x 20 cm) + winter maize (grain purpose) and potato (60 x 15 cm) + winter maize (grain purpose), respectively. Higher net return of ₹ 62.6 x 10<sup>3</sup> ha<sup>-1</sup>, benefit: cost ratio of 2.52 and energy output efficiency of 3081 MJ ha<sup>-1</sup> day<sup>-1</sup> were obtained with potato (45 x 20 cm) + winter maize (cob purpose), followed by potato (60 x 15 cm) + winter maize (cob purpose) intercropping system with corresponding values of ₹ 58.32 x 10<sup>3</sup> ha<sup>-1</sup>, 2.46 and 2577 MJ ha<sup>-1</sup> day<sup>-1</sup>, respectively.

**Key words:** Border ridge technique, Economic efficiency, Energy efficiency, Maize, *Phaseolus vulgaris*, Potato, Rajmash bean, *Solanum tuberosum*, Intercropping, Monetary advantage, *Zea mays*

In recent days there is mounting interest in diversified agricultural production systems to obtain improved crop protection, increased productivity and profitability offered by many intercropping systems. This may be due to some of the established and speculated advantages for intercropping systems such as higher yields, greater land-use efficiency and improvement in soil fertility status through the addition of atmospheric N by fixation and excretion from the component legume (Willey, 1979; Ofori and Stern, 1987). As a consequence, an increase in the total biological productivity per area unit of land and in sustainability in production occurs. The efficiency and the advantage of an intercropping system are fundamentally dependent of the complementarities between the component crops. On the other hand, when differences in plant architecture of the component crops help a better utilization of the available resources, or when biochemical differences exist among crops in their response to environmental resources, spatial or physiological complementarity occurs (Liebman, 2002).

Besides, intercropping also acts as insurance for resource poor farmers if one crop fails, they get some yield of another crop. Cultivation of potato (*Solanum tuberosum* L.) as sole crop is unstable and it is sometimes uneconomical due to its

sensitiveness to adverse weather conditions and market forces (Rana and Saran, 1998). Maize (*Zea mays* L.) is a choice which has got good market value in different forms particularly in spring season. Similarly, rajmash bean (*Phaseolus vulgaris* L.) is a potential crop for intercropping with potato as it is sown at the same time as potato but due to its fast initial growth compared to potato as well as early maturity leads to lesser competition for resources. Singh *et al.* (2008) reported that wheat and maize are suitable component crops in potato intercropping for West Central and North-Eastern plains.

There is very meager information available on intercropping studies in potato under central plain zone of Uttar Pradesh. Hence, the present work had the aim to estimate the effect of the association between the component crops in potato-winter maize/ rajmash bean intercropping systems, as well as to determine the best system for environmental resource management in regard to productivity and bio-economic indicators.

### MATERIALS AND METHODS

A field experiment was carried out during the winter (*rabi*) seasons of 2006-07 and 2007-08 at Students' Instructional Farm

in C.S. Azad University of Agriculture and Technology, Kanpur, India. The soil was clay loam alluvial type, low in organic carbon, and available nitrogen, medium in available phosphorus and available potassium with slightly alkaline in soil reaction. The experiment comprising 12 treatments was conducted in randomized complete block design replicated thrice, had 6 sole crops and 6 combinations of potato intercropping with other 2 crops with 2 plant geometries in 3:2 row proportions (Table 4). The crop varieties 'Chipsona 2' of potato, 'Sharadmani' of winter maize and 'VL 63' of rajmash bean were sown on 25 October in 2006 and 1 November in 2007. Potato, maize (grain), maize (cob) and rajmash bean were harvested on 13 February, 13 February, 14 March and 2 March 2007 and 15 February, 13 February, 17 March and 5 March 2008, respectively. Potato was sown at 2 plant geometries i.e. 45 cm x 20 cm and 60 cm x 15 cm apart through regular and border ridge technique (BRT) – a novel agro-technique in which 3 rows of main crop are sown and every 4<sup>th</sup> row is kept vacant/unsown which can be utilized for intercropping purpose. Sole maize and rajmash bean were sown at a spacing of 60 cm and 45 cm in rows, respectively. The plant-to-plant distance in maize and rajmash bean was 20 and 10 cm, respectively. In intercropping treatments, 2 rows of each maize and rajmash bean were accommodated in 4<sup>th</sup> vacant row of potato in replacement series.

Recommended package of practices was followed to raise the healthy crop. Potato was fertilized with 150 kg N, 100 kg P<sub>2</sub>O<sub>5</sub> and 100 kg K<sub>2</sub>O ha<sup>-1</sup>, while 120 kg N, 80 kg P<sub>2</sub>O<sub>5</sub> and 60 kg K<sub>2</sub>O ha<sup>-1</sup> were applied to winter maize and rajmash bean in both sole and intercrops. In intercropping, the crops received the fertilizers on the basis of proportionate area under each crop. Full recommended doses of P and K along with 50% N to potato and rajmash bean and one-third N to maize were applied as basal to all sole as well as intercropping systems. Remaining two-third N to winter maize was side banded in 2 equal splits at knee high and tasseling stages. In potato, rest 50% N was applied at the time of earthing, while in rajmash bean, remaining 50% N was given in 2 equal splits as top dressing. Two thinning in winter maize were done at 20 and 35 days after sowing (DAS) to the required plant density. Potato equivalent yield (PEY) and economics were calculated on the basis of prevailing market prices of component crops for the main produce/ by-products and inputs. Production efficiency in term of kg ha<sup>-1</sup> day<sup>-1</sup> was worked out by total economic yield in terms of PEY divided by total duration of the crops in a system. Economic/ monetary efficiency in term of ₹ ha<sup>-1</sup> day<sup>-1</sup> was worked out by dividing the total net monetary returns by total duration of the crops. The energy output of different cropping systems was calculated on the basis of economic yield as given by Devasenapathy *et al.* (2009) and expressed as total energy (x 10<sup>3</sup> MJ ha<sup>-1</sup>). Energy intensiveness was expressed as energy output of intercropping system in MJ ₹<sup>-1</sup> invested. Energy intensiveness and energy output efficiency were calculated using the following formula:

$$\text{Energy intensiveness (MJ ₹}^{-1}\text{)} = \text{Energy output (MJ ha}^{-1}\text{)} / \text{Net return (₹ ha}^{-1}\text{)}$$

$$\text{Energy output efficiency (MJ ha}^{-1}\text{ day}^{-1}\text{)} = \text{Energy output (MJ ha}^{-1}\text{)} / \text{Duration of the system (days)}$$

The biological efficiency of intercropping is determined by comparing the productivity of a given area of intercropping with that of sole crops. The competition functions proposed to describe the competitive relationships in intercropping were calculated on mean basis as suggested by Willey (1979).

The land equivalent ratio (LER) was used as the first criterion for mixed stand advantage for all three component crops used in this experiment by using the formula suggested by Mead and Willey (1980). In particular, LER verifies the effectiveness of intercropping for using the resources of the environment compared to sole cropping. When LER is greater than 1, the intercropping favours the growth and yield of the species. In contrast, when LER is lower than 1, the intercropping negatively affects the growth and yield of plants grown in mixtures (Willey, 1979). The LER values were calculated as:

$$\text{LER} = (\text{LER}_{\text{potato}} + \text{LER}_{\text{intercrop}})$$

$$\text{where, } \text{LER}_{\text{potato}} = (\text{Y}_{\text{pi}} / \text{Y}_{\text{p}}), \text{ and } \text{LER}_{\text{intercrop}} = \text{Y}_{\text{ip}} / \text{Y}_{\text{i}}$$

Where, Y<sub>p</sub> and Y<sub>i</sub> are the yields of potato and intercrops (maize/ rajmash bean) as sole crops, respectively, and Y<sub>pi</sub> and Y<sub>ip</sub> are the yields of potato and maize/ rajmash bean as intercrops, respectively.

The second coefficient was the relative crowding coefficient (K) which is a measure of the relative dominance of one species over the other in a mixture. It was suggested by de Wit (1960) and later developed by Hall (1974). The K is calculated by the following expressions:

$$\text{K} = (\text{K}_{\text{potato}} \times \text{K}_{\text{intercrop}})$$

Where,

$$\text{K}_{\text{potato}} = \text{Y}_{\text{pi}} \times \text{Z}_{\text{ip}} / [(\text{Y}_{\text{p}} - \text{Y}_{\text{pi}}) \times \text{Z}_{\text{pi}}], \text{ and}$$

$$\text{K}_{\text{intercrop}} = \text{Y}_{\text{ip}} \times \text{Z}_{\text{pi}} / [(\text{Y}_{\text{i}} - \text{Y}_{\text{ip}}) \times \text{Z}_{\text{ip}}]$$

Where, Z<sub>pi</sub> and Z<sub>ip</sub> were the proportions of potato and intercrops (maize/ rajmash bean) in the mixture, respectively. When the value of K is greater than 1, there is a yield advantage; when K is equal to 1, there is no yield advantage; and, when it is less than 1, there is a disadvantage (Willey, 1979).

The third index was aggressivity (A) which is often used to determine the competitive relationship between 2 crops used in the mixed cropping. Aggressivity represents a simple measure of how much the relative yield increase in 'a' crop is greater than that of 'b' crop in an intercropping system. This was proposed by McGilchrist (1965). The aggressivity was formulated as follows:

$$\text{A}_{\text{potato}} = (\text{Y}_{\text{pi}} / \text{Y}_{\text{p}} \times \text{Z}_{\text{pi}}) - (\text{Y}_{\text{ip}} / \text{Y}_{\text{i}} \times \text{Z}_{\text{ip}}), \text{ and}$$

$$\text{A}_{\text{intercrop}} = (\text{Y}_{\text{ip}} / \text{Y}_{\text{i}} \times \text{Z}_{\text{ip}}) - (\text{Y}_{\text{pi}} / \text{Y}_{\text{p}} \times \text{Z}_{\text{pi}})$$

For potato example; if A<sub>potato</sub> = 0, both crops are equally competitive, if A<sub>potato</sub> is positive, then the potato species is dominant, if A<sub>potato</sub> is negative, then the potato is weak (Willey, 1979).

Also, competitive ratio (CR) is another way to assess competition between different species. This was obtained through the formulae suggested by Willey and Rao (1980). The CR gives more desirable competitive ability for the crops and is also advantageous as an index over K. The CR represents simply the ratio of individual LERs of the 2 component crops and takes into account the proportion of the crops in which they are initially sown (Willey, 1979). Then, the CR index was calculated using the following formula:

$$CR_{\text{potato}} = (LER_{\text{potato}} / LER_{\text{intercrop}}) (Z_{\text{ip}} / Z_{\text{pi}}), \text{ and}$$

$$CR_{\text{intercrop}} = (LER_{\text{intercrop}} / LER_{\text{potato}}) (Z_{\text{pi}} / Z_{\text{ip}})$$

Since land equivalent ratio does not take into account the time for which land is occupied by the component crops of an intercropping system, area-time equivalency ratio (ATER) was also determined (Heibsch and McCollum, 1987).

$$ATER = L_p t_p + L_i t_i / T$$

Where,

$L_p L_i$  = LER of potato and intercrops (maize/rajmash bean) in intercropping, respectively

$t_p t_i$  = The duration (days) of potato and intercrops

$T$  = The duration (days) of the whole intercropping system

Finally, the monetary advantage index (MAI) was calculated since none of the above competition indices provides any information on the economic advantage of the intercropping system (Willey, 1979). The calculation of MAI was as follows:

$MAI = (\text{value of combined intercrops}) (LER - 1) / LER$ ; the higher the MAI value, the more profitable cropping system.

## RESULTS AND DISCUSSION

### Effect of intercropping

**Potato:** The growth parameter (dry weight of haulms  $\text{plant}^{-1}$ ), yield attributes (tubers  $\text{plant}^{-1}$  and dry weight of tubers  $\text{plant}^{-1}$ ) and tuber yield of potato showed significant variation due to intercropping (Table 1). Higher values of dry weight of haulms

$\text{plant}^{-1}$ , tubers  $\text{plant}^{-1}$  and dry weight of tubers  $\text{plant}^{-1}$  were noticed under intercropping treatments and sole cropping of potato through border ridge technique-BRT (3:0) than under sole cropping of potato by regular method. It might be due to more availability of space and plant nutrients as intercrops also fertilized with additional dose of recommended fertilizers. Besides, chemicals secreted by plant roots of maize and/or rajmash bean might be beneficial for potato growth. The beneficial association of maize and potato has also been reported by Tripathi *et al.* (2010). Among the intercropping treatments, potato (45 x 20 cm) + maize (cob purpose) recorded higher values of tubers  $\text{plant}^{-1}$ , followed by potato (45 x 20 cm) + maize (grain purpose). However, and dry weight of tubers  $\text{plant}^{-1}$  was recorded maximum under potato (60 x 15 cm) + maize (grain purpose). The reason may be explained that better performance of these characters under replacement series in intercropping might be due to more space available for development of potato plants as there was best competition with other component crops because of time space in growth behaviour of potato and intercrops. Comparatively poor yield attributes in sole potato by regular method might be owing to lesser availability of nutrients as fertilizer were given only potato crop as well as lesser space provided to potato plants.

The tuber yield of potato was significantly higher under regular planting of potato ( $T_1$  and  $T_2$ ) than under sole potato through BRT and intercropping treatments (Table 1) The higher yield obtained under regular planting of potato are attributed to higher plant population of potato. On the other hand, intercropping as well as sole potato through BRT reduced the tuber yield of potato compared with regular planting of potato due to lesser area sown with potato.

**Intercrops:** All the yield attributing characters of winter maize and rajmash bean were higher in intercropped stands with potato over their sole cropping (Table 2 and 3). Among intercropping systems, appreciable increase in yield attributes of winter maize and rajmash bean were observed when maize/rajmash bean intercropped with potato at 60 x 15 cm spacing than maize/rajmash bean intercropped with potato at 45 x 20 cm owing to lesser completion for plant nutrients, light, space and moisture, and took more advantage of solar radiation. The seed yield of maize and rajmash bean under intercropping system

Table 1. Effect of intercropping system on growth parameters, yield attributes and tuber yield of potato (mean data of two years)

| Cropping system                                                          | Dry weight of haulms<br>$\text{plant}^{-1}$ at harvest (g) | Tubers $\text{plant}^{-1}$ | Dry weight of tubers<br>$\text{plant}^{-1}$ | Tuber yield<br>(t $\text{ha}^{-1}$ ) |
|--------------------------------------------------------------------------|------------------------------------------------------------|----------------------------|---------------------------------------------|--------------------------------------|
| $T_1$ : Sole potato regular (45 x 20 cm)                                 | 19.65                                                      | 8.77                       | 63.46                                       | 24.11                                |
| $T_2$ : Sole potato regular (60 x 15 cm)                                 | 21.57                                                      | 8.51                       | 66.08                                       | 24.48                                |
| $T_3$ : Sole potato through BRT (45 x 20 cm) (3:0)                       | 21.79                                                      | 9.29                       | 68.69                                       | 21.96                                |
| $T_4$ : Sole potato through BRT (60 x 15 cm) (3:0)                       | 22.42                                                      | 9.10                       | 75.81                                       | 22.36                                |
| $T_7$ : Potato through BRT (45 x 20 cm) + maize for grain purpose (3:2)  | 21.46                                                      | 9.34                       | 69.33                                       | 21.88                                |
| $T_8$ : Potato through BRT (60 x 15 cm) + maize for grain purpose (3:2)  | 22.14                                                      | 8.89                       | 75.97                                       | 22.25                                |
| $T_9$ : Potato through BRT (45 x 20 cm) + maize for cob purpose (3:2)    | 22.18                                                      | 9.40                       | 72.12                                       | 21.74                                |
| $T_{10}$ : Potato through BRT (60 x 15 cm) + maize for cob purpose (3:2) | 22.69                                                      | 9.06                       | 75.06                                       | 22.19                                |
| $T_{11}$ : Potato through BRT (45 x 20 cm) + rajmash bean (3:2)          | 21.09                                                      | 9.16                       | 68.58                                       | 21.54                                |
| $T_{12}$ : Potato through BRT (60 x 15 cm) + rajmash bean (3:2)          | 22.02                                                      | 8.99                       | 74.13                                       | 22.21                                |
| SE(d)±                                                                   | 0.49                                                       | 0.05                       | 1.26                                        | 0.43                                 |
| CD (P=0.05)                                                              | 1.04                                                       | 0.10                       | 2.67                                        | 0.91                                 |

BRT = Border ridge technique (3:0)

Table 2. Effect of intercropping system on growth parameters, yield attributes and yield of winter maize (mean data of two years)

| Cropping system                                                                  | Cobs plant <sup>-1</sup> | Grain rows cob <sup>-1</sup> | Grains cob <sup>-1</sup> | Grain weight cob <sup>-1</sup> | 100-grain weight (g) | Stover/ green fodder yield (t ha <sup>-1</sup> ) | Grain/ equivalent yield (t ha <sup>-1</sup> ) |
|----------------------------------------------------------------------------------|--------------------------|------------------------------|--------------------------|--------------------------------|----------------------|--------------------------------------------------|-----------------------------------------------|
| T <sub>5</sub> : Sole winter maize                                               | 1.42                     | 14.87                        | 418.34                   | 63.35                          | 22.33                | 9.57                                             | 5.53                                          |
| T <sub>7</sub> : Potato through BRT (45 x 20 cm) + maize for grain purpose (3:2) | 1.46                     | 15.22                        | 421.71                   | 68.62                          | 23.37                | 6.16                                             | 3.37                                          |
| T <sub>8</sub> : Potato through BRT (60 x 15 cm) + maize for grain purpose (3:2) | 1.53                     | 16.15                        | 447.33                   | 74.29                          | 24.56                | 5.66                                             | 3.02                                          |
| T <sub>9</sub> : Potato through BRT (45 x 20 cm) + maize for cob purpose (3:2)   | 1.45                     | 15.19                        | 426.63                   |                                |                      | 11.23                                            | 3.77                                          |
| T <sub>10</sub> : Potato through BRT (60 x 15 cm) + maize for cob purpose (3:2)  | 1.52                     | 15.70                        | 448.89                   |                                |                      | 8.58                                             | 3.17                                          |
| SE(d)±                                                                           | 0.05                     | 0.29                         | 9.87                     | 1.91                           | 0.66                 | 0.41                                             | 0.28                                          |
| CD (P=0.05)                                                                      | 0.10                     | 0.64                         | 21.35                    | 4.44                           | 1.51                 | 0.97                                             | 0.63                                          |

Table 3. Effect of intercropping system on growth parameters, yield attributes and yield of rajmash bean (mean data of two years)

| Cropping system                                                        | Dry weigh plant <sup>-1</sup> | Pods plant <sup>-1</sup> | Pod weight plant <sup>-1</sup> | Grains pod <sup>-1</sup> | Grains plant <sup>-1</sup> | Grain weight plant <sup>-1</sup> | Grain yield (t ha <sup>-1</sup> ) |
|------------------------------------------------------------------------|-------------------------------|--------------------------|--------------------------------|--------------------------|----------------------------|----------------------------------|-----------------------------------|
| T <sub>6</sub> : Sole rajmash bean                                     | 23.17                         | 9.80                     | 11.24                          | 2.29                     | 22.06                      | 9.80                             | 1.48                              |
| T <sub>11</sub> : Potato through BRT (45 x 20 cm) + rajmash bean (3:2) | 25.15                         | 10.32                    | 11.81                          | 2.46                     | 24.35                      | 10.50                            | 0.93                              |
| T <sub>12</sub> : Potato through BRT (60 x 15 cm) + rajmash bean (3:2) | 26.00                         | 10.43                    | 11.92                          | 2.51                     | 24.80                      | 10.61                            | 0.65                              |
| SE(d)±                                                                 | 1.74                          | 0.59                     | 0.55                           | 0.19                     | 0.52                       | 0.17                             | 0.08                              |
| CD (P=0.05)                                                            | NS                            | NS                       | NS                             | NS                       | 1.46                       | 0.45                             | 0.22                              |

decreased significantly compared to their sole stands might be due to lesser number of plants unit<sup>-1</sup> area under intercropping. Among intercropping systems, seed yield of winter maize and rajmash bean were numerically higher when these component crops intercropped with potato at 45 x 20 cm spacing. Uddin *et al.* (2009) and Tripathi *et al.* (2010) too reported such beneficial effects.

### Total productivity and production efficiency

All the intercropping systems showed superiority over sole cropping of potato, winter maize or rajmash bean, as evident by potato equivalent yield (PEY) (Table 4). The additional yield advantage due to intercropping and also the higher economic

value of intercrop may be accredited to higher total productivity under intercropping systems, irrespective of planting geometries compared to sole cropping. The PEY was recorded with intercropping pattern of potato + maize (cob purpose) under both the geometries i.e. 45 x 20 cm and 60 x 15 cm (T<sub>9</sub> and T<sub>10</sub>), followed by potato + maize (grain purpose) (T<sub>7</sub> and T<sub>8</sub>). Higher PEY under intercropping systems could be attributed to balance competition and complementary effect of component crops for better utilization of available resources. Secondly, maize and rajmash bean as intercrops provided least completion to potato crop and also made available some extra nutrients thus enhanced grain yield of both the intercrops. Similar beneficial effects of potato and maize intercropping in relation

Table 4. Effect of intercropping on potato equivalent yield (PEY) and economics of potato-based cropping systems (mean data of two years)

| Cropping system                                                                  | PEY (t ha <sup>-1</sup> ) | Cost of cultivation (x 10 <sup>3</sup> ₹ ha <sup>-1</sup> ) | Gross return (x 10 <sup>3</sup> ₹ ha <sup>-1</sup> ) | Net return (x 10 <sup>3</sup> ₹ ha <sup>-1</sup> ) | B:C ratio |
|----------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------|------------------------------------------------------|----------------------------------------------------|-----------|
| T <sub>1</sub> : Sole potato regular (45 x 20 cm)                                | 24.11                     | 42.85                                                       | 72.33                                                | 29.48                                              | 1.69      |
| T <sub>2</sub> : Sole potato regular (60 x 15 cm)                                | 24.48                     | 42.85                                                       | 73.44                                                | 30.59                                              | 1.71      |
| T <sub>3</sub> : Sole potato through BRT (45 x 20 cm) (3:0)                      | 21.96                     | 34.55                                                       | 65.88                                                | 31.33                                              | 1.91      |
| T <sub>4</sub> : Sole potato through BRT (60 x 15 cm) (3:0)                      | 22.36                     | 34.55                                                       | 67.08                                                | 32.53                                              | 1.94      |
| T <sub>5</sub> : Sole winter maize                                               | 17.07                     | 31.55                                                       | 51.21                                                | 19.66                                              | 1.62      |
| T <sub>6</sub> : Sole rajmash bean                                               | 12.33                     | 19.50                                                       | 36.99                                                | 17.49                                              | 1.90      |
| T <sub>7</sub> : Potato through BRT (45 x 20 cm) + maize for grain purpose (3:2) | 32.35                     | 49.95                                                       | 97.05                                                | 47.10                                              | 1.94      |
| T <sub>8</sub> : Potato through BRT (60 x 15 cm) + maize for grain purpose (3:2) | 31.65                     | 47.15                                                       | 94.95                                                | 47.80                                              | 2.01      |
| T <sub>9</sub> : Potato through BRT (45 x 20 cm) + maize for cob purpose (3:2)   | 34.55                     | 41.05                                                       | 103.65                                               | 62.60                                              | 2.52      |
| T <sub>10</sub> : Potato through BRT (60 x 15 cm) + maize for cob purpose (3:2)  | 32.79                     | 40.05                                                       | 98.37                                                | 58.32                                              | 2.46      |
| T <sub>11</sub> : Potato through BRT (45 x 20 cm) + rajmash bean (3:2)           | 29.29                     | 44.15                                                       | 87.87                                                | 43.72                                              | 1.99      |
| T <sub>12</sub> : Potato through BRT (60 x 15 cm) + rajmash bean (3:2)           | 27.63                     | 42.20                                                       | 82.89                                                | 40.69                                              | 1.96      |
| SE(d)±                                                                           | 0.81                      |                                                             | 1.15                                                 | 0.79                                               | 0.09      |
| CD (P=0.05)                                                                      | 1.75                      |                                                             | 2.43                                                 | 1.64                                               | 0.18      |

BRT = Border ridge technique (3:0); B: C ratio = Benefit: cost ratio

Table 5. Effect of intercropping on biological efficiency of potato-based cropping systems (mean data of two years)

| Cropping system                                                                  | LER    |           |       | ATER | CR     |           | Aggressivity (A) |           | RCC                      |                             |                                              |
|----------------------------------------------------------------------------------|--------|-----------|-------|------|--------|-----------|------------------|-----------|--------------------------|-----------------------------|----------------------------------------------|
|                                                                                  | Potato | Intercrop | Total |      | Potato | Intercrop | Potato           | Intercrop | Potato (K <sub>p</sub> ) | Intercrop (K <sub>i</sub> ) | Product (K = K <sub>p</sub> K <sub>i</sub> ) |
| T <sub>7</sub> : Potato through BRT (45 x 20 cm) + maize for grain purpose (3:2) | 1.00   | 0.61      | 1.61  | 1.39 | 1.09   | 0.92      | 1.77             | -1.77     | 182.33                   | 2.34                        | 426.70                                       |
| T <sub>8</sub> : Potato through BRT (60 x 15 cm) + maize for grain purpose (3:2) | 1.00   | 0.55      | 1.55  | 1.33 | 1.20   | 0.83      | 1.89             | -1.89     | 134.85                   | 1.80                        | 243.37                                       |
| T <sub>9</sub> : Potato through BRT (45 x 20 cm) + maize for cob purpose (3:2)   | 0.99   | 0.68      | 1.67  | 1.65 | 0.97   | 1.03      | 1.61             | -1.61     | 65.88                    | 3.21                        | 211.68                                       |
| T <sub>10</sub> : Potato through BRT (60 x 15 cm) + maize for cob purpose (3:2)  | 0.99   | 0.57      | 1.56  | 1.54 | 1.16   | 0.86      | 1.83             | -1.83     | 87.02                    | 2.01                        | 175.33                                       |
| T <sub>11</sub> : Potato through BRT (45 x 20 cm) + rajmash bean (3:2)           | 0.98   | 0.62      | 1.60  | 1.46 | 1.05   | 0.95      | 1.68             | -1.68     | 34.19                    | 2.54                        | 86.72                                        |
| T <sub>12</sub> : Potato through BRT (60 x 15 cm) + rajmash bean (3:2)           | 0.99   | 0.44      | 1.43  | 1.29 | 1.50   | 0.67      | 2.11             | -2.11     | 98.71                    | 1.17                        | 115.95                                       |

LER = Land equivalent ratio; Land equivalent coefficient; ATER = Area-time equivalency ratio; CR = Competitive ratio; RCC = Relative crowding coefficient

to higher system productivity and profitability have also been reported by Uddin *et al.* (2009) and Tripathi *et al.* (2010). The lowest PEY was registered under intercropping of potato + rajmash bean with both plant geometries.

Intercropping systems were more efficient than their respective sole crops and production efficiency ranged from 2 to 49%, 85.4 to 157.7% and 123.7 to 138.1 over sole cropping of potato, maize and rajmash bean, respectively (Table 6). However, among the intercropping systems, potato + maize (cob purpose) under both the planting geometries i.e. 45 x 20 cm and 60 x 15 cm (T<sub>9</sub> and T<sub>10</sub>) seemed to be more productive recorded 317 and 301 kg ha<sup>-1</sup> day<sup>-1</sup> production efficiency, respectively, followed by potato + maize (grain purpose) (T<sub>7</sub> and T<sub>8</sub>) with 233 and 228 kg ha<sup>-1</sup> day<sup>-1</sup> production efficiency, respectively.

### Biological efficiency

Land equivalent ratio (LER) of all the intercropping systems was greater than one, indicating higher total productivity of the system and yield advantage due to intercropping (Table 6). The LER was ranged from 1.43-1.67 in various potato-based intercropping systems, indicating greater

biological efficiency of crops grown in association and it was probably due to greater temporal and spatial complementarity effect and thereby giving corresponding yield advantages. The higher value of LER (67%) was recorded in potato (45 x 20 cm) + maize (cob purpose) intercropping system, followed by potato (45 x 20 cm) + maize (grain purpose) intercropping system (61%). However, potato planted (60 x 15 cm) with maize/ rajmash bean recorded the lowest values of LER. In all planting patterns, positive A<sub>potato</sub> values showed that potato was the dominant species. The main crop potato appeared more competitive than companion crops i.e. winter maize/ rajmash bean by giving positive values of aggressivity (Table 6). Intercropped potato had higher competitive ratios (CRs) in both planting geometries and in all planting patterns, except potato (45 x 20 cm) + maize (cob purpose) intercropping system (Table 6). The relative crowding co-efficient indicated that it was advantageous and biologically sustainable to grow maize/ rajmash bean as intercrops with potato under irrigated conditions, which was further established by the product of relative crowding co-efficient K (Table 6), may be due to mutual co-operation. The intercropped potato had higher K<sub>potato</sub> values than the intercropped maize and rajmash bean in the potato + maize/

Table 6. Production efficiency, energy efficiency and economics efficiency of different of potato-based cropping systems (mean data of 2 years)

| Cropping system                                                                  | Production efficiency (kg ha <sup>-1</sup> day <sup>-1</sup> ) | Energy efficiency                                     |                                            |                                                                   | Monetary advantages index | Monetary efficiency (₹ ha <sup>-1</sup> day <sup>-1</sup> ) |
|----------------------------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------|---------------------------|-------------------------------------------------------------|
|                                                                                  |                                                                | Total energy (x 10 <sup>3</sup> MJ ha <sup>-1</sup> ) | Energy intensiveness (MJ ₹ <sup>-1</sup> ) | Energy output efficiency (MJ ha <sup>-1</sup> day <sup>-1</sup> ) |                           |                                                             |
| T <sub>1</sub> : Sole potato regular (45 x 20 cm)                                | 221                                                            | 86.80                                                 | 2.94                                       | 796                                                               |                           | 270                                                         |
| T <sub>2</sub> : Sole potato regular (60 x 15 cm)                                | 224                                                            | 88.13                                                 | 2.88                                       | 808                                                               |                           | 281                                                         |
| T <sub>3</sub> : Sole potato through BRT (45 x 20 cm) (3:0)                      | 201                                                            | 79.06                                                 | 2.52                                       | 725                                                               |                           | 287                                                         |
| T <sub>4</sub> : Sole potato through BRT (60 x 15 cm) (3:0)                      | 205                                                            | 80.50                                                 | 2.47                                       | 738                                                               |                           | 298                                                         |
| T <sub>5</sub> : Sole winter maize                                               | 123                                                            | 253.55                                                | 11.98                                      | 1824                                                              |                           | 141                                                         |
| T <sub>6</sub> : Sole rajmash bean                                               | 97                                                             | 21.76                                                 | 1.24                                       | 171                                                               |                           | 138                                                         |
| T <sub>7</sub> : Potato through BRT (45 x 20 cm) + maize for grain purpose (3:2) | 233                                                            | 239.19                                                | 5.08                                       | 1721                                                              | 36770                     | 339                                                         |
| T <sub>8</sub> : Potato through BRT (60 x 15 cm) + maize for grain purpose (3:2) | 228                                                            | 226.37                                                | 4.74                                       | 1629                                                              | 33692                     | 344                                                         |
| T <sub>9</sub> : Potato through BRT (45 x 20 cm) + maize for cob purpose (3:2)   | 317                                                            | 335.82                                                | 5.36                                       | 3081                                                              | 41584                     | 574                                                         |
| T <sub>10</sub> : Potato through BRT (60 x 15 cm) + maize for cob purpose (3:2)  | 301                                                            | 280.92                                                | 4.82                                       | 2577                                                              | 35312                     | 535                                                         |
| T <sub>11</sub> : Potato through BRT (45 x 20 cm) + rajmash bean (3:2)           | 231                                                            | 91.21                                                 | 2.11                                       | 718                                                               | 32951                     | 341                                                         |
| T <sub>12</sub> : Potato through BRT (60 x 15 cm) + rajmash bean (3:2)           | 217                                                            | 89.52                                                 | 2.20                                       | 705                                                               | 24925                     | 320                                                         |

rajmash bean intercropping systems (Table 6). The total K was much higher in potato (45 x 20 cm) + maize (grain purpose) system than other intercropping associations. When the values of area time equivalent ratio (ATER) was considered to assess the yield advantage due to intercropping, it was found that all the intercropping systems showed ATER values greater than unity indicating better land-utilization efficiency under these systems.

### Energy efficiency

Potato (45 x 20 cm) + maize (cob purpose) intercropping system registered the maximum total energy ( $335.82 \times 10^3$  MJ ha<sup>-1</sup>), followed by potato (60 x 15 cm) + maize (cob purpose) intercropping system ( $280.92 \times 10^3$  MJ ha<sup>-1</sup>) (Table 6). This was due to relatively more green fodder yield of maize under these treatments which resulted in higher energy output. The energy produced by these intercropping systems was 325 and 249% higher over sole potato planted through BRT ( $79.06 \times 10^3$  MJ ha<sup>-1</sup> and  $80.50 \times 10^3$  MJ ha<sup>-1</sup>, T<sub>3</sub> and T<sub>4</sub>, respectively). However, potato + rajmash bean under both the planting patterns was comparable in respect of energy output with sole stand of potato (all planting patterns) but higher over sole rajmash bean. The energy intensiveness of different cropping systems revealed that the highest value was noted with sole winter maize (11.98 MJ ₹<sup>-1</sup>). Among the intercropping systems, minimum values pertaining to energy intensiveness was recorded in potato + rajmash bean intercropping systems (2.11 and 2.20 MJ ₹<sup>-1</sup>, T<sub>11</sub> and T<sub>12</sub>, respectively). Lower energy intensiveness results in higher productive and resource efficient cropping system.

### Economic efficiency

All the intercropping systems showed their superiority in terms of economic viability and sustainability over monoculture cropping of all component crops. Potato + winter maize (cob purpose) intercropping system raised with 45 x 20 cm and 60 x 15 cm planting pattern fetched higher gross returns (₹ 103.65 x 10<sup>3</sup> ha<sup>-1</sup> and ₹ 98.37 x 10<sup>3</sup> ha<sup>-1</sup>), net returns (₹ 62.60 x 10<sup>3</sup> ha<sup>-1</sup> and ₹ 58.32 x 10<sup>3</sup> ha<sup>-1</sup>) and B:C ratio (2.52 and 2.46), respectively, followed by potato + winter maize (grain purpose) intercropping systems than all sole stands (Table 4). This might be due to difference in potato yield and additional advantage of maize grown for cob purpose, which resulted in higher net returns under these intercropping systems. This finding is in close conformity with those of Tripathi *et al.* (2010). The minimum net return and B:C ratio was recorded in potato + rajmash bean intercropping system, perhaps due to relatively lower potato equivalent yield.

The monetary advantage evaluated over sole crops indicated a definite gain from the intercropping systems. The values of monetary advantage index (MAI) was higher in potato - maize intercropping than the potato- rajmash bean intercropping and the highest MAI was observed for potato (45 x 20 cm) + maize (cob purpose) intercropping (41584), followed by potato (45 x 20 cm) + maize (grain purpose) (36770) due to higher land-equivalent ratio and value of combined produce in intercrops (Table 6). These observations are in line

with the finding of Tripathi *et al.* (2010). Compared to planting pattern in potato, narrow row spacing (45 x 20 cm) yielded better MAI values than did wider spacing (60 x 15 cm). Potato (60 x 15 cm) + rajmash bean intercropping system fetched the minimum MAI which may be attributed to the lower value of combined intercrop yield. The maximum monetary efficiency (₹ 574 ha<sup>-1</sup> day<sup>-1</sup>) was obtained from potato (45 x 20 cm) + maize (cob purpose) and minimum from potato (60 x 15 cm) + rajmash bean intercropping system (₹ 320 ha<sup>-1</sup> day<sup>-1</sup>). Compared to potato + maize intercropping, cob purpose maize had better monetary efficiency values than did grain purpose maize (Table 6).

Thus, it can be concluded that intercropping of potato + winter maize (cob purpose) in replacement series with 3:2 row ratio can be suggested as a productive and remunerative and biologically efficient intercropping system under irrigated conditions of central plain zone of Uttar Pradesh.

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## Impact of climate change on wheat (*Triticum aestivum*) productivity in late sown condition at Allahabad, Uttar Pradesh

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### ABSTRACT

Increasing evidences over past few decades indicate that significant changes in climate are taking place worldwide due to anthropogenic activities. The increase in temperature was observed at annual, post monsoon and winter season time scales over India. The rising trends of annual maximum and minimum temperatures were noticed at the rate of  $0.01^{\circ}\text{C year}^{-1}$  and  $0.04^{\circ}\text{C year}^{-1}$ , respectively over Allahabad during the last 40 years. It was observed that rate of rising was more in case of minimum temperature as compared to maximum temperature at post monsoon and winter season time scales. A decreasing trend of annual rainfall was noticed at the rate of  $2.8 \text{ mm year}^{-1}$  and rainfall during post monsoon and winter season was decreased at the almost equal rate of  $0.4 \text{ mm year}^{-1}$ . It was observed that annual maximum and minimum relative humidity increased at the rate of  $0.06\% \text{ year}^{-1}$  and  $0.2\% \text{ year}^{-1}$ , respectively. Variability in the climatic elements like temperature influenced growing conditions and productivity of thermo-sensitive crop like wheat (*Triticum aestivum* L.). The simulation study assessed that yield of wheat was reduced by 10.4 to 16.8% with the increase of 2 to  $3^{\circ}\text{C}$  temperature. Thus, adaptation strategies have to be identified to nullify the negative impacts of climate change on crop productivity.

**Key words:** Climate change, Rainfall, Relative humidity, Temperature, Wheat

The scientific assessment report of the Intergovernmental Panel on Climate Change (IPCC, 2013) showed that between 1880 and 2012, globally averaged temperature increased by  $0.85^{\circ}\text{C}$ . Atmospheric levels of the main greenhouse gases (carbon dioxide, methane, nitrous oxide) have all risen since the start of the industrial era and climate change over the next few decades will be largely governed by greenhouse gases in the atmosphere. Annual mean surface temperature over Indian sub-continent is likely to range between  $3.5$  and  $5.5^{\circ}\text{C}$  by 2080 (Lal, 2001). An analysis of the mean annual surface air temperature over India indicates a significant warming of about  $0.3$ - $0.6^{\circ}\text{C}$  since the 1860 (Pant, 2003). The warming trends in mean annual temperature were observed over the India with higher rate of temperature increase during winter and post-monsoon seasons compared to that of annual (Rupa *et al.*, 2002). Besides that, the amount and distribution of rainfall is becoming more and more erratic over India in last 50 years (Goswami, 2006). The increasing temperature and carbon dioxide and uncertainties in rainfall coupled with global climate change may have serious consequences on crop production and future food security. In north India, wheat is one of the most important food grain crops. It is, therefore, important to quantify the consequences of climatic change and variability on wheat crop sustainability contributing to our food security. The productivity of wheat shows wide variability due to its

wide adaptability in different climatic regions. Temperature has direct correlation with grain filling period and physiological maturity. Under late sown conditions, wheat encounters higher temperatures during milking to dough stages of crop growth and detrimental to crop yield. Again, rising trend of temperature in reproductive stage of crop growth is of major concern due to sensitivity of this stage to higher temperatures (Saikia *et al.*, 2009). When the temperature rises, the number of days reduces to reach normal maturity date (Satyanarayana *et al.*, 2009). In north India, a simulation study showed that an increase in temperature by  $2^{\circ}\text{C}$  above normal has reduced potential yield of wheat in many places (Aggarwal and Sinha, 1993). The effects of changes in temperature, precipitation and  $\text{CO}_2$  concentrations on crop productivity have been studied extensively (Pathak and Wassmann, 2009) using crop simulation models.

The crop modelling is a potential tool to quantify the impact of climate, weather conditions, soil environment, crop management and genotype as well as their interactions on crop growth, yield, resource use efficiency and environmental impact (Boote *et al.*, 1996). The Decision Support System for Agrotechnology Transfer (DSSAT) models of CERES-wheat have been used most extensively in predicting the effect of various climate change scenarios on crop yields. Keeping into account the expected climatic changes, the effects on productivity of wheat crop under Vindhyan agro climatic zone of Uttar Pradesh was studied by using CERES-wheat simulation model.

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## MATERIALS AND METHODS

The long term climatic variability study was made by trend analysis of past 40 years (1971-2010) climatological data of maximum and minimum temperatures, rainfall, maximum and minimum relative humidity for Allahabad district which is situated in Vindhyan agro climatic zone of Uttar Pradesh.

The climate change and its impact on crop productivity were studied with CERES- wheat 4.5 model. Calibration of model was based on the past weather and crop yield data for the period 2002-03 to 2007-08 and validation is based on 2008-09 and 2009-10 years weather as well as crop yield data of Allahabad. In calibration, genotypic coefficients for the cultivar 'Malviya234' were determined by adjusting the coefficients until close matches were achieved between the simulated and observed yield. The genetic coefficients of late (December) sown wheat variety Malviya 234 are given in the Table 1.

The climate change study was based on the assumption that crop was sown under irrigated condition and the crop may undergo moisture stress condition without irrigations. Simulation also considered nitrogen application with an assumption that there was no loss of crop yield from insect-pest. On the basis of observed climatic variability trends over the region and earlier assessments, the impact of synthetic picture of climate change in relation to crop yield was studied. The synthetic scenario of climate change was created by increasing maximum and minimum temperatures from the normal.

## RESULTS AND DISCUSSION

### Temperature variability trends

The trend analysis of the mean maximum and minimum temperatures was done for the period of 1971-2010 and trend lines and regression equations were obtained for Allahabad against annual, post monsoon and winter season time scales which are presented in the Fig. 1. The rising trends of annual maximum and minimum temperatures were noticed at the rate of  $0.01^{\circ}\text{C year}^{-1}$  and  $0.04^{\circ}\text{C year}^{-1}$ , respectively. The study showed that rate of rising is more in case of minimum temperature as compared to maximum temperature. The long term trend analysis of mean maximum and minimum temperatures at seasonal scales revealed that maximum and minimum temperatures were increased at the rate of  $0.02^{\circ}\text{C year}^{-1}$  and  $0.06^{\circ}\text{C year}^{-1}$ , respectively during post monsoon season which coincides with field preparation, sowing and early vegetative growth phase of wheat. Winter season which coincides with further vegetative phase as well as reproductive phase of crop, increasing trend of maximum and minimum temperatures were noticed with a rate of  $0.01^{\circ}\text{C year}^{-1}$  and  $0.05^{\circ}\text{C year}^{-1}$ , respectively. Again, the rate of rising was higher in case of minimum temperature as compared to maximum temperature during post monsoon and winter season. In India, rising temperature trends at annual scale and post-monsoon season and winter season scales were also reported by Rupa *et al.* (2002).

### Rainfall variability trends

The trend analysis of the rainfall was done for the period

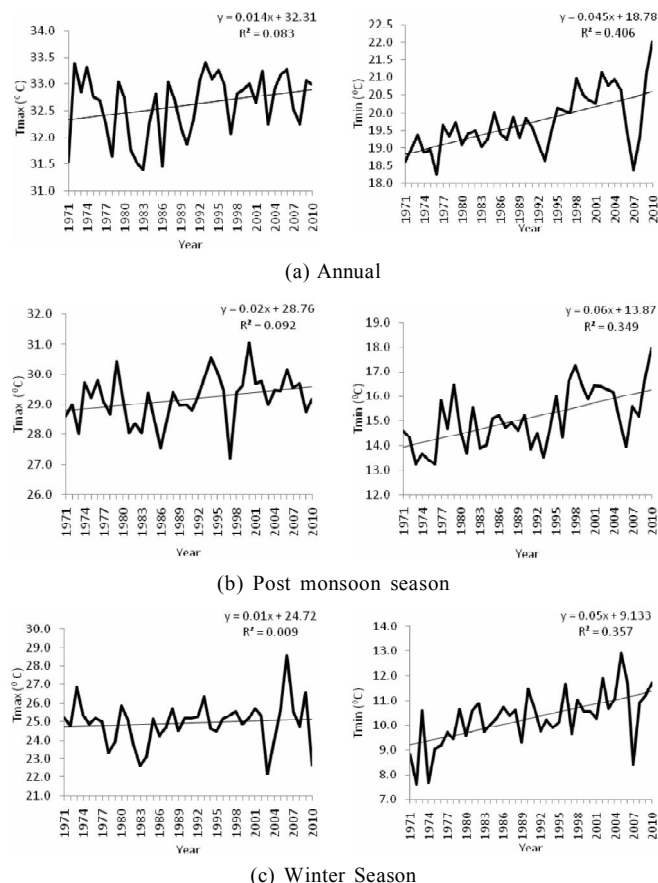


Fig. 1. Trends of (a) annual, (b) post monsoon season, and (c) winter season maximum (Tmax) and minimum (Tmin) temperatures at Allahabad for the period 1971-2010

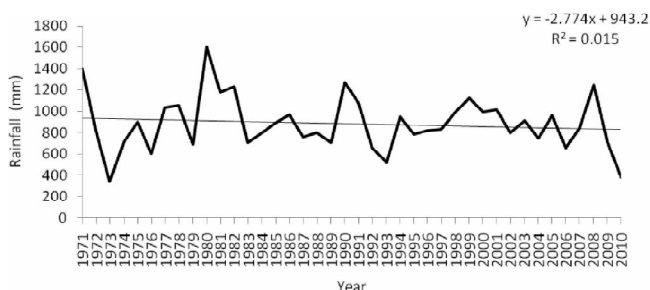
of 1971-2010 and trend lines and regression equations were obtained for Allahabad against annual, post monsoon and winter season time scales which are presented in the Fig. 2. Overall, decreasing trends of rainfall were observed at annual and seasonal time scales. The trend analysis revealed that annual rainfall was decreased at the rate of  $2.8 \text{ mm year}^{-1}$ . Likewise, rainfall during post monsoon and winter seasons was also decreased at the almost equal rate of  $0.4 \text{ mm year}^{-1}$ .

### Relative humidity variability trends

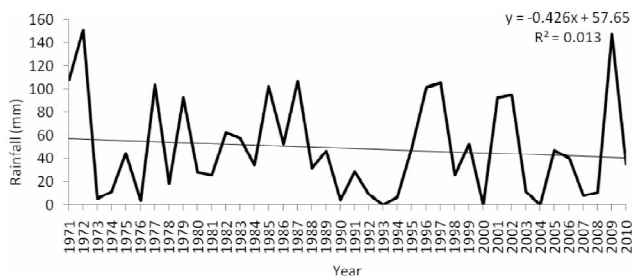
Atmospheric moisture is measured in terms of relative humidity. The time-trend analysis of the mean morning and evening relative humidity was done for the period of 1971-2010 and trend lines and regression equations were obtained for Allahabad against annual, post monsoon and winter season time scales which are presented in the Fig. 3. In general, increasing trends of morning and evening relative humidity were observed for annual and seasonal time scales. It was observed that annual morning relative humidity was increased at the rate of  $0.06\% \text{ year}^{-1}$ . The analysis revealed that rate of rising of morning relative humidity was more in winter season (around  $0.07\% \text{ year}^{-1}$ ) than post monsoon season ( $0.02\% \text{ year}^{-1}$ ). On the other hand, the annual evening relative humidity was increased at the rate of  $0.2\% \text{ year}^{-1}$ .

### Simulations of climate change impact on crop yield

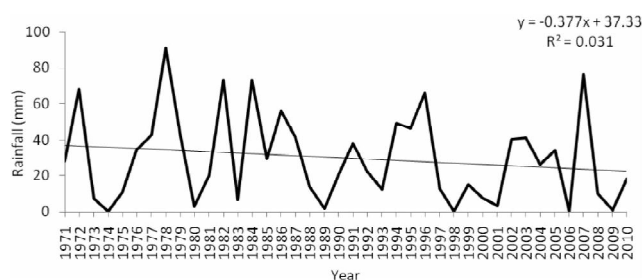
The grain yield of wheat were simulated with increase in



(a) Annual



(b) Post monsoon season



(c) Winter Season

Fig. 2. Trends of (a) annual, (b) post monsoon season, and (c) winter season rainfall at Allahabad for the period 1971-2010

Table 1. Genetic coefficients of wheat variety Malviya 234

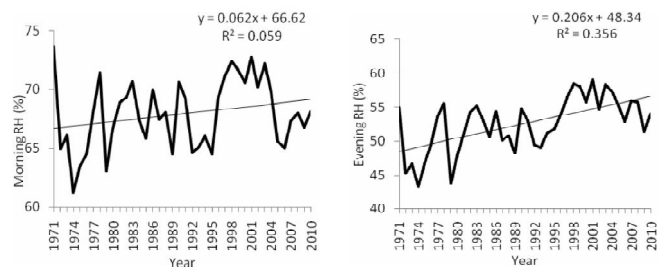
| Genetic coefficients | Description                                                                                                                    | Value |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------|-------|
| PIV                  | Days at optimum vernalizing temperature required to complete vernalization.                                                    | 20    |
| PID                  | Percentage reduction in development rate in a photoperiod 10 hour shorter than the threshold relative to that at the threshold | 80    |
| P5                   | Grain filling (excluding lag) phase duration (°C.d)                                                                            | 650   |
| G1                   | Kernel number per unit canopy weight at anthesis (#/g)                                                                         | 19    |
| G2                   | Standard kernel size under optimum conditions (mg)                                                                             | 35    |
| G3                   | Standard, non-stressed dry weight (total, including grain) of a single tiller at maturity (g)                                  | 1.5   |

maximum and minimum temperatures from normal while holding the other climatic parameters constant and presented in Table 2. It was observed that simulated yield of wheat was considerably reduced approximately by 10 to 17% from normal

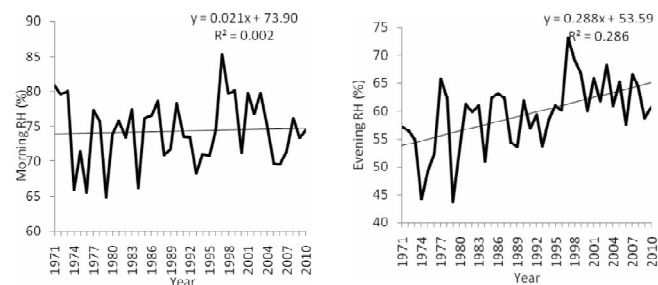
Table 2. Effect of increasing maximum and minimum temperatures (°C) from normal on productivity of late sown wheat

| Yield parameter                    | Increase in temperature (°C) |                 |                |                |                 |                 |                 |
|------------------------------------|------------------------------|-----------------|----------------|----------------|-----------------|-----------------|-----------------|
|                                    | Normal                       | +0.5            | +1.0           | +1.5           | +2.0            | +2.5            | +3.0            |
| Grain yield (kg ha <sup>-1</sup> ) | 2969                         | 2896<br>(-2.5)* | 2856<br>(-3.8) | 2754<br>(-7.2) | 2661<br>(-10.4) | 2546<br>(-14.2) | 2469<br>(-16.8) |

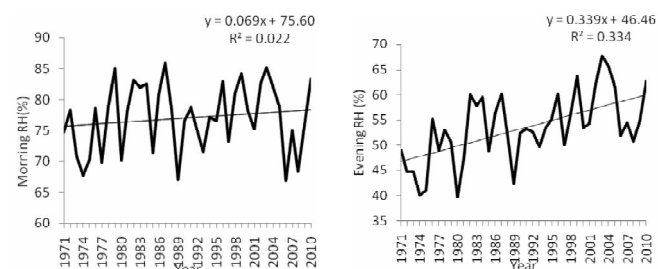
\*Values in parentheses are per cent deviation of yield from normal



(a) Annual



(b) Post monsoon season



(c) Winter season

Fig. 3. Trends of (a) annual, (b) post monsoon season, and (c) winter season morning and evening relative humidity (RH) at Allahabad for the period 1971-2010

with the increase of 2 to 3°C temperature. The reduction in yield was due to forced maturity (Satyanarayana *et al.*, 2009).

The study supported unfavourable impacts of future anticipated climate change on productivity of wheat under irrigated condition. The crop which is sown in late conditions, already suffers from unfavourable higher temperature conditions during reproductive stage. Additionally, increasing temperature due to climate change may further lead to critical conditions. North India is vulnerable to climate change which may bring yield reduction of cereal crop like wheat. The potential adaptation strategies have to be identified to combat the negative impacts of climate change on crop productivity. Introduction of agroforestry system could be an effective measure in creating better microclimate and to reduce excess heat load in crop. In climate change studies, limitations are to be considered in the crop simulation modelling like crop damage

by insect-pest that could not be simulated. Nevertheless, such simulation studies could help us to understand the likely impact of future climate change on crop productivity.

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## Nutrient status in sugarcane growing soils of Haryana

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### ABSTRACT

A soil survey was carried out in different sugarcane growing agro-climatic and edaphic regions of Haryana. Soil samples were collected and analyzed for physico-chemical properties. Soil pH, EC and organic carbon content ranged from 7.2-8.9, 0.09-1.10 dS m<sup>-1</sup> and 0.25-0.97%, respectively across the regions. Available nitrogen, phosphorus (P<sub>2</sub>O<sub>5</sub>) and potassium (K<sub>2</sub>O) contents widely varied from 203.8-319.9, 19.6-96.6 and 121.4-582.4 kg ha<sup>-1</sup>, respectively. On the basis of nutrient index values, sugarcane growing soils of Haryana were categorized as deficient in available nitrogen, medium in P<sub>2</sub>O<sub>5</sub> and high in K<sub>2</sub>O. DTPA-extractable Zn, Cu, Fe and Mn, were above the critical limits. A significant and positive correlation was found between organic carbon and available nitrogen (r=0.958\*), P<sub>2</sub>O<sub>5</sub> (r=0.597\*), K<sub>2</sub>O (r=0.317\*), and DTPA-extractable Zn (r=0.272\*), Cu (r=0.272\*), Fe (r=0.398\*), and Mn (r=0.429\*).

**Key words:** Available nutrients, Macro and micro nutrients, Nutrient index, Organic carbon

Sugarcane cultivation in India with efficient management of inputs and resources proves to be a vehicle of rural prosperity and ensures livelihood for millions. The crop is widely cultivated both under tropical and sub-tropical regions of the country. The sub-tropical states namely Uttar Pradesh, Punjab, Haryana, Uttarakhand and Bihar register lower yields (60-70 t ha<sup>-1</sup>) as compared to that obtained in tropical states (90-110 t ha<sup>-1</sup>). Soil health and nutrient management along with climatic factors play major role for sugarcane yield as the crop remains in the field for 12-18 months and on an average removes about 205 kg N, 55 kg P<sub>2</sub>O<sub>5</sub>, 275 kg K<sub>2</sub>O, 30 kg S, 3.5 kg Fe, 1.2 kg Mn, 0.6 kg Zn and 0.2 kg Cu from the soil for a cane yield of 100 t ha<sup>-1</sup> (Yadav and Dey, 1997). In subtropical region, sugarcane is regularly cultivated with input intensive and exhaustive crop rotation of rice-wheat which often aggravates imbalanced mining of nutrients leading to deterioration in soil fertility and productivity. Under such a scenario, determination of soil health parameters is necessary for nutrient management of crops. Since, Haryana is an important sugarcane growing state under sub-tropical region envisage one lakh hectare area under the crop and 71 t ha<sup>-1</sup> productivity, therefore, analysis of soil samples from various sugar mill command areas was done to assess the soil fertility status that would facilitate balanced scheduling as well as management of fertilizer nutrients.

### MATERIALS AND METHODS

An extensive soil survey was carried out in four representative sugar mills command area, viz. Saraswati Sugar Mills Ltd., Karnal Co-Operative Sugar Mills Ltd., Jind Co-Operative Sugar Mills Ltd., and Palwal Co-Operative Sugar Mills Ltd. of Yamuna Nagar, Karnal, Jind and Palwal districts of Haryana, respectively. Total eighty soil samples, ten each from surface (0-15 cm) and sub-surface (15-30 cm) soil were collected

from each sugar mill command area. Soil samples were analyzed by using standard procedures as described for organic carbon (Walkley and Black, 1934), available nitrogen (Subbiah and Asija, 1956), available phosphorus (Olsen *et al.*, 1954), available potash (Jackson, 1973) and DTPA extractable Fe, Mn, Zn and Cu (Lindsay and Narvell, 1978) were determined by atomic absorption spectrophotometer. Nutrients index was enumerated as described by (Motsara *et al.*, 1982). On the basis of nutrient index (NI), fertilizer recommendation in respect of different nutrients was categorized as under:

| Range of NI | Fertility status |
|-------------|------------------|
| <1.67       | Low              |
| 1.67-2.33   | Medium           |
| >2.33       | High             |

### RESULTS AND DISCUSSION

#### Soil properties of surface soil

The pH of surface soil varied from 7.20-8.90 (Table 1) and 80% samples ranged between 7.0 to 8.5 and 12% samples had pH higher than 8.50. Soil pH more than 8.50 was mainly found in sugar mills of Yamuna Nagar, Karnal and Jind districts. Alkaline reaction of soil may be associated with higher water table due to the capillary rise of salt and its evaporation. The EC varied from 0.09-1.35 dS m<sup>-1</sup> with an average value of 0.18 dS m<sup>-1</sup> in Yamuna Nagar, 0.27 dS m<sup>-1</sup> in Karnal, 0.56 dS m<sup>-1</sup> in Jind and 0.76 dS m<sup>-1</sup> in Palwal. The majority of soil samples (about 82.50%) had normal EC. The low EC may be ascribed to leaching of salts to lower horizons because most of the soils of sugar mill command areas are light in texture and well percolated.

The soil organic carbon content varied from 0.25-0.97%

with an average value of 0.62% (Table 1). The 32.5, 45 and 22.5% of studied area falls in category of low, medium and high of organic carbon, respectively. Medium and low status of organic carbon was due to mono culture of sugarcane in exhaustive cropping systems involving rice and wheat which is being practiced widely. Sugarcane being a long duration crop removes high quantity of nutrients, resulting in medium and lower status of organic carbon content. On the other hand, higher content of organic carbon was recorded in Palwal district sugar mill command area. These variations in organic carbon may be attributed to the difference in soil properties, crop management practices and recycling of farm biomass in Palwal district.

Available nitrogen ranged between 203.8-319.9 kg ha<sup>-1</sup> with an average value of 257.9 kg ha<sup>-1</sup> (Table 1). Most of the tested surface soil samples were found deficient in nitrogen. Palwal soils were found comparatively rich in available nitrogen in comparison to Yamuna Nagar, Karnal and Jind districts of the State. Low nitrogen was observed due to the fact that N is lost through various mechanisms like removal by crops ammonia volatilization, nitrification, succeeding de-nitrification, leaching and runoff (De Datta and Buresh, 1989), which may be responsible for low content of available nitrogen in these soils. Sugarcane being exhaustive crop removes about 2.05 kg N per tonne of crop yield during a year which conspicuously contributes towards N depletion in the soil.

The available phosphorus (P<sub>2</sub>O<sub>5</sub>) of soils varied from 19.9-96.6 kg ha<sup>-1</sup> and averaged to 46.3 kg ha<sup>-1</sup>. Nearly 7.50% samples were deficient, 70% in medium and 22.5% in high. High phosphorus content in soil was observed possibly due to higher application of phosphatic fertilizers to different crops under input intensive cropping system of the state.

The available K<sub>2</sub>O content in soil ranged from 121.4-582.4 kg ha<sup>-1</sup> with a mean value of 274 kg ha<sup>-1</sup>. Most of the soil samples

were medium in available potassium status whereas, Jind and Palwal districts were found higher in potassium status. Medium to high potassium status in these soils may be attributed to the prevalence of potassium rich minerals. Bhanu and Sindhu (1991) also observed that the soils of Punjab are medium to high in available potassium.

### *Micronutrients*

The DTPA- extractable Zn, Fe, Cu and Mn contents in surface soil in sugar mill command area of Haryana ranged from 0.8-5.28, 13.04-73.48, 0.80-3.12 and 17.12-32.72 mg kg<sup>-1</sup> with mean value of 2.38, 1.89 and 24.34 mg kg<sup>-1</sup>, respectively (Table 1). Zinc was recorded highest 2.4 mg kg<sup>-1</sup> at Yamuna Nagar followed by 2.16, 1.76 and 1.68 mg kg<sup>-1</sup> at Karnal, Jind and Palwal districts. Soils of Yamuna Nagar had highest average content of Fe (57.78 mg kg<sup>-1</sup>) as compared to Karnal (44.44 mg kg<sup>-1</sup>), Jind (26.79 mg kg<sup>-1</sup>) and Palwal (23 mg kg<sup>-1</sup>) districts. Similarly, soils of Palwal had comparatively high content of 25.26 mg kg<sup>-1</sup> Mn as compared to Yamuna Nagar (25.18 mg kg<sup>-1</sup>), Karnal (23.54 mg kg<sup>-1</sup>) and Jind (23.37 mg kg<sup>-1</sup>). Considering 0.6 mg kg<sup>-1</sup> for Zn, 0.24 mg kg<sup>-1</sup> for Cu, 4.5 mg kg<sup>-1</sup> for Fe and 3 mg kg<sup>-1</sup> for Mn as critical limits none soil samples were found below the critical limits.

### *Soil properties of sub-surface soil*

Sub-surface soils' pH varied from 7.4-9 with mean value of 8.5 (Table 2). About 52.50% soil samples, pH was in between 7 to 8.5 and remaining 47.5% sample had 8.5 pH. Electrical conductivity of 80% soil samples was normal and 20% was in injurious range. All samples of sub-surface soil were found low in organic carbon that varied from 0.11-0.48% with mean value of 0.27%. All samples were found deficient in available nitrogen. Available phosphorus varied from 10.7 to 27.6 kg ha<sup>-1</sup> with mean value of 16.1 kg ha<sup>-1</sup>. About 12.5% soil samples were found in medium category and remaining 87.5% samples

Table 1. Range and mean value of various soil characteristics in surface soil (0-15 cm) under different sugarcane growing zones in Haryana

| Characteristics                                      | Zones        |             |             |             |             |
|------------------------------------------------------|--------------|-------------|-------------|-------------|-------------|
|                                                      | Yamuna Nagar | Karnal      | Jind        | Palwal      | All soils   |
| pH                                                   | 7.5-8.5      | 7.6-8.5     | 8.3-8.9     | 7.2-8.9     | 7.2-8.9     |
| mean                                                 | 8.0          | 8.2         | 8.5         | 8.3         | 8.2         |
| EC (dS m <sup>-1</sup> )                             | 0.09-0.27    | 0.14-0.43   | 0.34-1.35   | 0.29-1.11   | 0.09-1.11   |
| mean                                                 | 0.18         | 0.27        | 0.56        | 0.76        | 0.44        |
| Organic carbon (%)                                   | 0.26-0.82    | 0.38-0.92   | 0.25-0.83   | 0.67-0.97   | 0.25-0.97   |
| mean                                                 | 0.54         | 0.60        | 0.51        | 0.84        | 0.62        |
| N (kg ha <sup>-1</sup> )                             | 203.8-282.2  | 226.0-276.0 | 203.8-272.8 | 266.6-319.9 | 203.8-319.9 |
| mean                                                 | 248.4        | 255.0       | 240.2       | 288.2       | 257.9       |
| P <sub>2</sub> O <sub>5</sub> (kg ha <sup>-1</sup> ) | 20.7-96.6    | 19.9-97.3   | 20.7-80.5   | 23.8-72.8   | 19.9-96.6   |
| mean                                                 | 50.5         | 52.4        | 41.3        | 41.1        | 46.3        |
| K <sub>2</sub> O (kg ha <sup>-1</sup> )              | 127.5-251.9  | 121.4-449.9 | 203.8-582.4 | 185.3-663.4 | 121.4-582.4 |
| mean                                                 | 163.2        | 266.4       | 343.6       | 324         | 274.9       |
| Zn (mg kg <sup>-1</sup> )                            | 1.04-5.28    | 1.56-2.96   | 0.80-4.24   | 0.88-3.56   | 0.80-5.28   |
| mean                                                 | 2.40         | 2.16        | 1.76        | 1.68        | 2           |
| Cu (mg kg <sup>-1</sup> )                            | 0.84-3.12    | 1.84-3.12   | 1.20-2.64   | 0.80-2.36   | 0.80-3.12   |
| mean                                                 | 1.65         | 2.43        | 1.78        | 1.71        | 1.89        |
| Fe (mg kg <sup>-1</sup> )                            | 40.12-73.48  | 33.04-66.76 | 16.76-41.28 | 13.04-32.48 | 13.04-73.48 |
| mean                                                 | 57.78        | 44.44       | 26.79       | 23          | 38          |
| Mn (mg kg <sup>-1</sup> )                            | 20.40-31.68  | 19.56-30.28 | 17.84-35.52 | 17.12-32.72 | 17.12-35.52 |
| mean                                                 | 25.18        | 23.54       | 23.37       | 25.26       | 24.34       |

were deficient in available  $P_2O_5$ . Available  $K_2O$  in soil widely varied from 112.5-487.3 kg ha<sup>-1</sup> with mean value of 237.6 kg ha<sup>-1</sup>. About 10% soil samples were found deficient, 60% moderate and 30% adequate in available  $K_2O$ . DTPA extractable Zn of sub surface soil ranged from 0.56 to 3.56 mg kg<sup>-1</sup> with mean value of 1.03 mg kg<sup>-1</sup>. About 5% soil sample were found deficient in available Zn. Available Fe, Cu and Mn in sub surface soil of ranged from 7.56-57.6, 0.72-3.16, 15.8-29.6 mg kg<sup>-1</sup> with mean value of 25.07, 1.42, and 21.05, respectively. Although, Fe, Cu, and Mn, values were comparatively low in sub-surface soil in comparison to the surface soil but all values were found above their critical limits. Sub-surface soils of sugar mills command area of Haryana showed lower values of electrical conductivity, organic carbon, available N,  $P_2O_5$ ,  $K_2O$ , Zn, Fe, Cu and Mn as compared to the surface soil; whereas, soil pH showed higher values in subsurface soil. Higher values of EC in surface soil might be due to accumulation of soluble salts on surface soil through capillary action whereas, lower content of organic carbon, available N,  $P_2O_5$ ,  $K_2O$ , Zn, Fe, Cu and Mn in sub-surface soils may be because of the mostly crop residue, organic manure and fertilizers are applied on surface of the soil.

### Relationship between soil properties and available nutrients

The available nitrogen was significantly and positively correlated with organic carbon ( $r=0.958$ ), available phosphorus ( $r=0.597$ ), available potash ( $r=0.317$ ) DTPA extractable Fe (0.398), Mn ( $r=0.429$ ), Zinc ( $r=0.272$ ) and Copper ( $r=0.272$ ) (Table 2). Available potash was significantly and positively correlated with organic carbon ( $r=0.317$ ). Sharma *et al.* (2008) also observed significant and positive correlation of available nitrogen and potash with organic carbon in soils of Uttar Pradesh. Fe was significant and negatively correlated with pH ( $r=0.474$ ) but had significant positive relationship with organic carbon ( $r=0.398$ ). Like-wise, Mn showed significant and negative correlation with

pH ( $r=0.125$ ) and significant positive correlation with organic carbon ( $r=0.429$ ). Available Zn and Cu were significant and negatively correlated with pH ( $r=0.519$ ). This is possibly due to precipitation of these cations at higher pH. These finding were in agreement with Chattopadhyay *et al.* (1996). Zn and Cu had significant and positive correlation with soil organic carbon ( $r=0.272$  and  $0.272$ ). Similar trend was also observed by Singh *et al.* (2001) in sugarcane soils of central Uttar Pradesh.

### Nutrient index

On the basis of nutrient index values, N values varied from 1-1.4 with respect to N status in different sugar mill command areas of Haryana (Table 3). Generally, nitrogen content was low in all the sugar mill command areas of the state. The low available nitrogen may be attributed to the low soil organic matter content owing to high biomass production by the crop of sugarcane and restoration mechanism of soil organic matter is fairly checked. Prasunarani *et al.* (1992) also reported that the low nitrogen status in the surface soils could be attributed to low organic carbon in Andhra Pradesh soils. The available phosphorus was medium in Yamuna Nagar and Karnal whereas, Palwal and Jind sugar mill command areas were low in nutrient index values for  $P_2O_5$ . The overall nutrient index value of each Yamuna Nagar, Karnal and Palwal districts sugar mill command areas was 1.9; whereas, Jind district sugar mill

Table 3. Correlation between soil properties and available nutrients

| Nutrients | pH      | EC      | Organic carbon |
|-----------|---------|---------|----------------|
| N         | -0.369* | 0.207   | 0.958*         |
| P         | -0.416* | -0.012  | 0.597*         |
| K         | 0.269*  | 0.367*  | 0.317*         |
| Fe        | -0.474* | -0.050  | 0.398*         |
| Mn        | -0.125  | -0.027  | 0.429*         |
| Zn        | -0.519* | -0.467* | 0.272*         |
| Cu        | -0.519  | -0.467* | 0.272*         |

\*Indicates significance level at  $P = 0.05$

Table 2. Range and mean value of various soil characteristics in sub-surface soil (15-30) under different sugarcane growing zones in Haryana

| Characteristics                 | Zones        |             |             |             |             |
|---------------------------------|--------------|-------------|-------------|-------------|-------------|
|                                 | Yamuna Nagar | Karnal      | Jind        | Palwal      | All soils   |
| pH                              | 8.0-8.8      | 8.2-8.7     | 8.5-9.0     | 7.4-8.9     | 7.4-9.0     |
| mean                            | 8.3          | 8.4         | 8.7         | 8.4         | 8.5         |
| EC (dS m <sup>-1</sup> )        | 0.08-0.24    | 0.17-0.26   | 0.31-0.98   | 0.20-0.76   | 0.08-0.98   |
| mean                            | 0.16         | 0.20        | 0.54        | 0.52        | 0.36        |
| Organic carbon (%)              | 0.11-0.41    | 0.20-0.45   | 0.11-0.27   | 0.24-0.48   | 0.11-0.48   |
| mean                            | 0.27         | 0.28        | 0.17        | 0.38        | 0.27        |
| N (kg ha <sup>-1</sup> )        | 125.4-197.6  | 147.4-207.0 | 141.1-181.9 | 188.2-210.1 | 125.4-210.1 |
| mean                            | 178.5        | 184.1       | 161.8       | 199.1       | 180.9       |
| $P_2O_5$ (kg ha <sup>-1</sup> ) | 10.7-27.6    | 10.7-20.7   | 11.5-24.5   | 14.6-23.8   | 10.2-27.6   |
| mean                            | 16.9         | 14.5        | 15.6        | 17.2        | 16.1        |
| $K_2O$ (kg ha <sup>-1</sup> )   | 112.2-171.3  | 112.2-394.9 | 190.1-487.3 | 161.0-607.7 | 112.5-487.3 |
| mean                            | 143.2        | 234.8       | 308.2       | 264.1       | 237.6       |
| Zn (mg kg <sup>-1</sup> )       | 0.72-3.56    | 0.56-2.16   | 0.72-1.88   | 0.64-1.20   | 0.56-3.56   |
| mean                            | 1.31         | 0.97        | 0.97        | 0.85        | 1.03        |
| Cu (mg kg <sup>-1</sup> )       | 0.72-1.68    | 1.44-3.16   | 0.80-1.64   | 0.72-1.96   | 0.72-3.16   |
| mean                            | 1.06         | 1.93        | 1.23        | 1.47        | 1.42        |
| Fe (mg kg <sup>-1</sup> )       | 29.72-57.60  | 19.44-40.20 | 7.56-16.64  | 12.48-36.16 | 7.56-57.60  |
| mean                            | 39.21        | 28.14       | 12.46       | 20.45       | 25.07       |
| Mn (mg kg <sup>-1</sup> )       | 18.08-26.04  | 15.80-29.60 | 16.08-28.04 | 16.40-21.80 | 15.80-29.60 |
| mean                            | 21.36        | 21.81       | 21.81       | 19.22       | 21.05       |

Table 4. Nutrient index values of different zones of Haryana soils

| Sugar mill command area | Nutrient index /nutrient status |     |      |        |      |        |
|-------------------------|---------------------------------|-----|------|--------|------|--------|
|                         | N                               |     | P    |        | K    |        |
| Yamuna Nagar            | 1.1                             | Low | 2.2  | Medium | 2.0  | Medium |
| Karnal                  | 1.0                             | Low | 2.2  | Medium | 2.3  | Medium |
| Jind                    | 1.0                             | Low | 1.0  | Low    | 2.6  | High   |
| Palwal                  | 1.4                             | Low | 2.2  | Medium | 2.5  | High   |
| All Haryana             | 1.13                            | Low | 1.90 | Medium | 2.35 | High   |

command area, the nutrient index value was 1. The medium nutrient index of phosphorus could be attributed to the application of high analysis phosphatic fertilizer to crops which resulted in build-up of phosphorus and low nutrient index could be attributed to fixation of phosphorus by clay minerals as calcium phosphate compound due to calcareous soils of the sugar mill command area of the state. The nutrient index value of potash content of surface soil samples were 2 in Yamuna Nagar, 2.3 in Karnal, 2.6 in Jind and 2.5 in Palwal sugar mill command areas which indicates that soils of Yamuna Nagar and Karnal sugar mill command areas were medium in available potassium, whereas Jind and Palwal sugar mills command area soils were adequate in available potash, which may be attributed to the prevalence of potassium rich minerals like *illite* and *feldspar*.

On the basis of results obtained it can be concluded that organic carbon status of soils of Haryana showed declining trend because of the adoption of intensive cropping systems. However, continuous and adequate use of phosphatic fertilizers in these soils resulted in moderate availability of phosphorus. The potassium showed its high availability. Micronutrients were found above critical limits throughout the state. The varying nutrient status in the present study from one sampling site to another suggest that use of balanced fertilizers should be based on soil test values for application of macro and micro nutrients in sugarcane crop for enhancing sustainable sugarcane production.

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# Prospects of fruits and vegetables processing in Rajasthan

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## ABSTRACT

Rajasthan is rich in raw material for agro-processing industries. Production of fruits and vegetables in Rajasthan was 676 and 800 million tonnes, respectively in 2009-10. In the decade 1990-2000, the area under fruits grew at the rate of 2.21%, production at 4.37% and productivity at 1.98%. However, during the decade 2001-2010, the growth rate of fruits jumped to 2.97, 5.49 and 2.32% per annum for area, production and productivity, respectively. The similar trend was observed in case of vegetables too. In rural and urban area during 2004-05, the income elasticity of demand for fruits was 1.65 and 1.24%, respectively indicating elastic demand in both areas. It shows that fruit consumption increased by 1.65% in rural areas and 1.24% in urban regions due to one per cent increase in income. In the case of vegetables, the income elasticity of demand was 0.51% in rural areas and 0.57% in urban areas indicating relatively inelastic demand in both areas. The income elasticity of demand for fruits has increased over the years and reached 1.79% in rural and 1.68% in urban areas during 2009-10 (66<sup>th</sup> round). In respect of vegetables, there has not been much increase in income elasticity (0.62% in rural and 0.68% in urban regions) during this period.

**Key words:** Agro-processing industries, Fruit, Vegetables

Rajasthan has geographical area of 3.42 lakhs sq.km. It has attained the status of being the largest state of India. The state represents 10.4% land surface area with 6% population of India. Almost 66% population is dependent on agriculture for their livelihood. The diverse agro-ecological conditions prevailing in the state is amenable for growing fruits, vegetables, spices, flowers, root and tuber crops, medicinal and aromatic crops. Out of the net cultivated area of about 165 lakh hectares in Rajasthan, horticultural crops are grown on an area of about 10 lakh hectares with an annual production of about 14 lakh million tonnes. Rajasthan has favorable climate for production of quality seed spices, ber, mandarin, kinnow, pomegranate, aonla, *khari* onion and pea. Marketing, processing, production, farm supplies, research, extension, government policies and programs are important areas for agri-business. The focus in this paper is on status and growth in processing of fruits and vegetables. Due to diverse agro-climatic conditions, Rajasthan is rich in raw material for agro-processing industries. Due to lack of post-harvest handling and processing facilities, nearly 30% of fruits and vegetables are lost or damaged. Fruits and vegetables are highly perishable in nature, and processing assumes paramount importance for exploiting the economic potential by creating time, form and space utilities. The demand for processed fruit and vegetable products is bound to rise with increased real income and improvement in the standard of living, modern life style, urbanization and contribution of women to household income. Keeping in view the above facts, this paper explores the consumption pattern of fruits and vegetables in Rajasthan using National Sample Survey Organization (NSSO) data and potential of growth for the fruit

and vegetable processing industry.

## MATERIALS AND METHODS

Data on area and production of fruits and vegetables, number of processing units, installed capacity, capacity utilization and related parameters were collected from published sources i.e. Annual Reports of the Ministry of Food Processing Industry, Government of India; Directorate of Economics and Statistics, Rajasthan, Economic Survey of India, etc. To study the consumption pattern of fruits and vegetables, data on monthly per capita expenditure was collected from NSSO. To determine changes in demand for fruit and vegetable products, the income elasticity of demand was calculated. The following formula was used to work out income elasticity:

$$Y = aX^b U_i$$

$$\text{Log } Y = \text{Log } a + b \text{ Log } X + \text{Log } U_i$$

Where,

Y = consumption expenditure on fruit /vegetable products (in rupees)

X = total consumption expenditure on all food and nonfood products (in rupees)

b = regression coefficient (income elasticity)

To determine growth rates (ACGR), the following exponential function was used:

$$Y = ab^t U_t$$

$$\text{Log } Y = \text{Log } a + t \text{ Log } b + \text{Log } U_t$$

Where,

$Y$  = Area/Production of fruits and vegetables,

$a$  = Constant,

$b$  = Regression coefficient,

$t$  = Time

Compound growth rate =  $(\text{Anti log } b - 1) \times 100$

## RESULTS AND DISCUSSION

### Potential of horticultural crops

Rajasthan with its huge geographical area and diverse agro-climatic conditions favors growing of large number of horticultural crops like fruits vegetable, spices, flowers and medicinal and aromatic plants. The state is one of the biggest producers of coriander, cumin, fenugreek, isabgol and mehndi in the country. The state also produces variety of other horticultural crops like oranges, kinnow, lime, aonla, chillies, garlic, ajowain, suwa, onion, tomato, pea, cucurbitaceous vegetable and medicinal and aromatic crops like sonamukhi and ashwagandha providing surplus produce for processing and export. The climatic conditions of Rajasthan allow growing various types of seed spices. Rajasthan is having prominent position in production of seed spices in the country. The contribution of state to National production of horticultural crops is 66.51% of coriander, 33% of cumin, 82% of fenugreek, 14% of garlic, 6% of fennel, 100% psyllium husk (isabgol), 100% of mehndi, 100% of ajwain, 7% of mandarin, producing export quality Kinnow and is one of the largest producers of aonla. Rajasthan offers excellent horticultural development potential inspite of several biophysical as well as developmental constraints. The endeavors over the past decade for planned and systematic development of horticulture in the state have now started yielding results. This is a beginning and the huge untapped potentials are yet to be utilized for the betterment of state. The varied agro climatic conditions of the state favor growing of a large number of crops. This diversity in climatic conditions creates scope to develop belts of horticultural crops in the state like mandarin in warm humid areas of Jhalawar;

kinnow in dry and cool climate of Ganganagar and Hanumangarh; pomegranate in arid irrigated parts of state; ber in western parts; aonla in central semi-arid parts; papaya in central parts; mango in southern humid parts; cumin in Barmer, Jalore, Pali, Jodhpur, Nagaur; coriander in Kota, Baran, Jhalawar, Bundi, Chittorgarh; fennel in Sirohi, Tonk; garlic in Jodhpur, Chittorgarh, Baran, Jhalawar, Kota; isabgol in Barmer, Jalore; and mehandi in Pali. The area under horticultural crops increased from 7.14 lakh hectares in 2004-05 to 9.21 lakh hectares in 2008-09 and production increased from 13.85 lakh tonnes to 19.78 lakh tonnes during the same period (Table 1).

### Growth rates

Compound growth rates in area, production and productivity of fruits and vegetables in Rajasthan state have been presented in Table 2. The data revealed that area under fruits grew at the rate of 2.21%, production at 4.37% and productivity at 1.98% during 1990-2000. However during the decade 2001-2010, the ACGRs of fruits jumped to 2.97, 5.49 and 2.32% per annum for area, production and productivity, respectively. The similar trend was observed in case of vegetables too.

### State-level patterns of food consumption

The changes in the monthly per capita expenditure on high value food products (milk and milk products, meat, eggs and fish, and fruits and vegetables) over two time periods (2001-02 and 2007-08) in rural and urban areas in state was worked out and the same are presented in Table 3. The share of expenditure on fruits and vegetables has increased in the state between 2001-02 and 2007-08 in rural and urban areas. It was 8.5% in rural areas during 2001-02 and it increased to 11.8% during 2007-08. In case of urban areas, it was 13.2% in 2001-02 and it increased to 15% in 2007-08. The per cent share of expenditure on cereals has gone down and for non-food items, it has gone up.

### Income elasticity

The income elasticity of demand for fruits and vegetables has been computed for the rural and urban areas for 2004-05 (61<sup>st</sup> round) and 2009-10 (66<sup>th</sup> round) based on data collected

Table 1. Area and production of different horticultural crops in Rajasthan (area in lakh hectares and production in lakh tonnes)

| Crop group                   | 2004-05 |            | 2005-06 |            | 2006-07 |            | 2007-08 |            | 2008-09 |            |
|------------------------------|---------|------------|---------|------------|---------|------------|---------|------------|---------|------------|
|                              | Area    | Production | Area    | Production | Area    | Production | Area    | Production | Area    | Production |
| Fruits                       | 0.23    | 2.56       | 0.25    | 4.18       | 0.27    | 4.02       | 0.28    | 5.62       | 0.3     | 5.95       |
| Vegetable                    | 1.23    | 6.23       | 1.23    | 7.41       | 1.24    | 7.88       | 1.43    | 8.53       | 1.25    | 7.37       |
| Spices                       | 4.16    | 4.24       | 3.46    | 3.02       | 3.8     | 3.56       | 5.67    | 5.29       | 5.37    | 5.36       |
| Flower                       | 0.03    | 0.03       | 0.03    | 0.02       | 0.03    | 0.02       | 0.03    | 0.04       | 0.03    | 0.04       |
| Medicinal and Aromatic plant | 1.49    | 0.79       | 1.51    | 0.7        | 2.16    | 2.16       | 1.98    | 0.94       | 2.26    | 1.06       |
| Total                        | 7.14    | 13.85      | 6.48    | 15.33      | 7.5     | 17.64      | 9.39    | 20.42      | 9.21    | 19.78      |

Source: NHB (2009)

Table 2. ACGRs of fruit and vegetables in Rajasthan over the period (% per annum)

| Year      | Area  |           | Production |           | Productivity |           |
|-----------|-------|-----------|------------|-----------|--------------|-----------|
|           | Fruit | Vegetable | Fruit      | Vegetable | Fruit        | Vegetable |
| 1990-2000 | 2.21  | 4.01      | 4.37       | 5.38      | 1.98         | 0.87      |
| 2001-2010 | 2.97  | 4.47      | 5.49       | 6.00      | 2.32         | 1.29      |

Table 3. Per cent share of expenditure on food and high value commodities in Rajasthan

| Year         | Cereals | Pulses | Milk and milk products | Meat, fish and eggs | Fruits& vegetables | Total food | Non-food |
|--------------|---------|--------|------------------------|---------------------|--------------------|------------|----------|
| <i>Rural</i> |         |        |                        |                     |                    |            |          |
| 2001-02      | 28.9    | 4.5    | 35.1                   | 1.2                 | 8.5                | 62.3       | 37.7     |
| 2007-08      | 27.0    | 3.9    | 30.4                   | 1.2                 | 11.8               | 53.9       | 46.1     |
| <i>Urban</i> |         |        |                        |                     |                    |            |          |
| 2001-02      | 22.4    | 4.6    | 29.9                   | 2.2                 | 13.2               | 56.7       | 43.3     |
| 2007-08      | 23.2    | 0.2    | 29.4                   | 2.0                 | 15.0               | 42.4       | 57.6     |

Source: Computed from NSSO Report Household Consumer Expenditure in India, 2007-08

Table 4. Income elasticity of fruits and vegetables

| Year         | Fruits | Vegetables |
|--------------|--------|------------|
| <i>Rural</i> |        |            |
| 2004-05      | 1.65   | 0.51       |
| 2009-10      | 1.79   | 0.62       |
| <i>Urban</i> |        |            |
| 2004-05      | 1.24   | 0.57       |
| 2009-10      | 1.68   | 0.68       |

from NSSO rounds (Table 4). In rural and urban area during 2004-05, the income elasticity of demand for fruits was 1.65 and 1.24%, respectively indicating elastic demand in both the areas. It shows that fruits consumption increased by 1.65% in rural areas and by 1.24% in urban regions due to one per cent increase in income. In case of vegetables, the income elasticity of demand was 0.51% in rural areas and 0.57% in urban areas indicating relatively inelastic demand in both the areas during 2004-05. The income elasticity of demand for fruits has increased over the years and reached 1.79% in rural and 1.68% in urban areas during 2009-10 (66<sup>th</sup> round). As regards vegetables, there has not been much increase in income elasticity (0.62% in rural and 0.68% in urban regions) during this period.

### ***Status of fruit and vegetable processing industries***

There are not too many large-scale processing industries in the state. Most of the industries are small scale. In Rajasthan, kinnow and mandarin have market linkages and have strong demand. Aonla has strong potential for processing in high value added products as well as industrial applications in pharmaceutical industry. Ber is peculiar to Rajasthan and caters to the demand of other states as well. The number of state-wise processing industries in India is presented in Table 5 and data reveal that highest numbers of processing industries are in Andhra Pradesh, followed by Tamil Nadu and Uttar Pradesh. Rajasthan has only 515 processing industries.

Table 6 reveals growth in the processed fruits and vegetables sector in Rajasthan. It shows that production of processed fruits and vegetables has increased from 1.23 lakh tonnes in 2001-03 to 3.45 lakh tonnes in 2007-09. However, production of processed fruits and vegetables is much less than the installed capacity of fruit and vegetable processing

Table 6. Growth in the processed fruits and vegetables sector in Rajasthan

| Year    | Production of processed fruit and vegetable products (lakh tonnes) | Installed capacity of fruit and vegetables processing (lakh tonnes) | Capacity utilization of fruit and vegetables processing (%) |
|---------|--------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------|
| 2001-03 | 1.23                                                               | 3.78                                                                | 32.53                                                       |
| 2004-06 | 2.10                                                               | 4.78                                                                | 43.93                                                       |
| 2007-09 | 3.45                                                               | 6.43                                                                | 53.65                                                       |

Source: GOI (2010)

Table 5. State-wise processing industry in India

| Sl.No. | State            | Number of units |
|--------|------------------|-----------------|
| 1      | Andhra Pradesh   | 10183           |
| 2      | Assam            | 734             |
| 3      | Bihar            | 433             |
| 4      | Chandigarh       | 36              |
| 5      | Daman & Diu      | 5               |
| 6      | Delhi            | 125             |
| 7      | Pondicherry      | 42              |
| 8      | Goa              | 34              |
| 9      | Gujarat          | 1270            |
| 10     | Haryana          | 600             |
| 11     | Himachal Pradesh | 46              |
| 12     | Jammu & Kashmir  | 69              |
| 13     | Karnataka        | 1221            |
| 14     | Kerala           | 1110            |
| 15     | Madhya Pradesh   | 1302            |
| 16     | Maharashtra      | 2420            |
| 17     | Manipur          | 9               |
| 18     | Meghalaya        | 3               |
| 19     | Nagaland         | 5               |
| 20     | Orissa           | 425             |
| 21     | Punjab           | 1196            |
| 22     | Rajasthan        | 515             |
| 23     | Tamil Nadu       | 3792            |
| 24     | Tripura          | 22              |
| 25     | Uttar Pradesh    | 2652            |
| 26     | West Bengal      | 1089            |
| 27     | Others           | 9               |
| Total  |                  | 29407           |

Source: GOI (2010)

units reflecting low capacity utilization. Fruits and vegetables processing industry capacity has increased from 3.78 lakh tonnes in 2001-03 to 6.43 lakh tonnes in 2007-09. As regards capacity utilization, this was only 32.53% in 2001-03. Capacity utilization slowly increased and reached 53.65% in 2007-09.

### ***Constraints faced by the Fruit and Vegetable processing industries***

It is estimated that 30-40% of the total harvest goes waste and only 2% of total production is processed in Rajasthan. Constraints inhibiting the growth of the fruit and vegetable processing industries include:

1. Fruits and vegetables are highly perishable and require cold storage facilities and refrigerated transportation system. The lack of these facilities causes huge annual wastage.
2. Production is concentrated mainly in unorganized and tiny sectors where an economy of scale is not possible. Attainment of international acceptable quality products in these small units is difficult.
3. The high cost of raw material, machinery and packaging material; poor technology in processing, packaging and distribution; and inadequate and expensive transportation facilities, also inhibit the growth of this industry.

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## Processing effect on saponins of rajmash beans (*Phaseolus vulgaris*)

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### ABSTRACT

Sapogenol A and B are responsible for various health benefits such as cholesterol lowering effect, protection against colon cancer and protective effect on liver injury. Effect of processing such as soaking, cooking and pressure cooking on sapogenols, the aglycone of saponins of seed of five genotypes of rajmash beans (*Phaseolus vulgaris* L.), viz. HUR 15, Amber, Uday, Utkarsh and EC 406072 was studied. During soaking, sapogenol A and B reduced to an extent of 6.2% and 9.1%, respectively. Cooking caused a reduction of 26.8% in sapogenol A content and a further loss of 9.2% was observed during pressure cooking of pre-soaked grain, whereas sapogenol B reduced by 90.1% during cooking of pre-soaked seed. Complete loss of sapogenol B was observed during pressure cooking of grain. Cooking and pressure cooking led to a loss of 64.1% and 73.7%, respectively in total sapogenol content of grain of rajmash beans.

**Key words:** Cooking, Frenchbeans, Pressure cooking, Sapogenols, Saponins, Soaking

Common beans (*Phaseolus vulgaris* L.) are widely grown and consumed in various regions of the world and are a rich source of protein, carbohydrates, dietary fibre, minerals and vitamins (Rehman *et al.*, 2001). They are beneficial for health and have low glycemic index (Foster-Powell and Brand-Miller, 1995). India is the largest producer of beans in the world with a production of 4.87 million metric tonnes (FAOSTAT, 2010). *Phaseolus vulgaris* is also a rich source of saponin. Saponins are amphiphilic compounds present in a wide variety of plants and herbs. Structurally, saponins in food exist as glycosides, with a hydrophobic triterpenoid or steroid (sapogenin) group linked to water-soluble sugar residues (Oakenfull, 1981). The amount and type of sugar residues vary between saponin species, the most common being glucose, glucuronic acid, arabinose, rhamnose, xylose, and fucose attached at either the C-3 position (monodesmoside saponins) or on both the C-3 and C-22 position (bidesmoside saponins) (Lasztity *et al.*, 1998). The major saponins present in *Phaseolous vulgaris* were identified as soyasaponin I, V, and phaseoleamide (Curl *et al.*, 1988; Kinjo *et al.*, 1998). Saponins are surfactants, and were initially thought to be harmful due to their strong haemolytic activity *in vitro*. Gestetner *et al.* (1968) observed that neither saponins nor sapogenins could be detected in blood after feeding soyflour diet to mice, rats and chicks. Saponins were the major form present in the small intestine and sapogenins were primarily detected in the caecum and colon after hydrolysis by microflora. The saponins found in dry beans are the same triterpenoid type of saponins found in soy beans.

Saponins have been shown to have anti-carcinogenic and anti-mutagenic properties in a variety of *in vitro*

approaches. The saponins used in these studies were from soybeans. Since dry bean saponins are similar to soy saponins, it is expected that dry bean saponins would also produce similar results. Soyasaponins reduced the growth of HCT-15 and HT-29 colon carcinoma cells and also significantly decreased TPA associated protein kinase C activation. Because sapogenins are the major form of saponins present in the colon, Gurfinkel and Rao (2003) looked at the effect of the chemical structure of soyasaponins on anti-carcinogenic activity. Soyasaponins (I, II, III) were found to be ineffective up to 50 ppm in inhibiting cell growth, whereas soyasapogenols A and B (aglycones) effectively suppressed growth in a dose-dependent manner (6-50 ppm). The microflora produces soyasapogenol A and soyasapogenol B in a ratio of 1:3 (Gurfinkel and Rao, 2002). Only one study with saponins on carcinogenesis has been conducted *in vivo*. Koratkar and Rao (1997) found that incorporation of soyasaponins into the diet of mice (3%) reduced the incidence of mice with ACF (Aberrant Crypt Foci) and significantly decreased the number of ACF in colon. The saponins and sapogenols of Indian rajmash beans have not yet been studied. Some information has recently been generated on sapogenols, the aglycone of saponins of chickpea, lentil and pigeonpea (Vasishtha and Srivastava, 2011; Srivastava and Vasishtha, 2012; Vasishtha and Srivastava, 2013). The sapogenols A and B in unprocessed seed of rajmash beans were reported earlier (Vasishtha and Srivastava, 2012). The aim of the present investigation was to study the effect of common domestic processings such as soaking, cooking and pressure cooking on sapogenols of rajmash beans. Since rajmash beans are widely consumed in northern and central India as well as other parts of the world, the information generated hereby will be of much scientific importance from health point of view.

## MATERIALS AND METHODS

Variegated genotypes of rajmash beans, viz. PDR 14 or Uday (maroon), IIPR 96-4 or Amber (maroon), IPR 98-5 or Utkarsh (maroon), HUR 15 (white) and EC 406072 (black) were selected for this study. One lot of these genotypes was dried at 70°C and powdered to a uniform particle size in a seed grinder Perten model 3303, and the remaining seeds were used for processing techniques such as soaking, cooking and pressure cooking. For soaking treatment, seeds were soaked in water in seed:water ratio of 1:7 for 10-12 hrs and analysed after crushing in a mortar and pestle. Moisture content was worked out in soaked seed and necessary correction was made while calculating the data so as to report results on dry weight basis. Cooking of soaked seed was done as per standard cooking practice i.e. the grain was boiled in water till it became soft and suitable for eating. The pressure cooking of soaked seed was done in a pressure cooker for 10-15 minutes till the seeds became soft and edible. The cooked and pressure cooked seeds were crushed in a mortar and pestle and analysed for sapogenols A and B as per method described by Vasishtha and Srivastava (2011). Each sample was analysed in triplicate and results were calculated on dry weight basis. The statistical analysis was carried out for sapogenol A, B and total using SPSS version 13. A multiple comparison of the treatment means was performed by Duncan's new multiple range test and results are reported in Table 1. The mean and standard deviation of means were also calculated for all the genotypes under all treatments (Table 2).

## RESULTS AND DISCUSSION

Soaking caused a reduction of 6.2% in sapogenol A (Table 1). The sapogenol A of seed reduced to 233.1 mg 100 g<sup>-1</sup> on soaking and different genotypes had sapogenol A in the range of 178.6 to 292.3 mg 100 g<sup>-1</sup> (Table 2). Cooking caused further reduction in sapogenol A and a significant loss of 26.8% was observed during soaking and cooking of grain. When the seed was subjected to soaking followed by pressure cooking, a significant loss of 36% in sapogenol A was recorded. HUR 15 and IIPR 96-4 (Amber) contained relatively higher sapogenol A in cooked as well as pressure cooked seed, whereas PDR 14 (Uday) and IPR 98-5 (Utkarsh) had the least sapogenol A in cooked as well as pressure cooked grain.

The sapogenol B reduced significantly by 9.1% on soaking of grain. Shi *et al.* (2009) have also reported a significant reduction in sapogenol B on soaking of beans. The saponin B are more soluble in water due to their sugar chain structure

Table 1. Effect of soaking and cooking on sapogenol of frenchbean

| Processing techniques | Sapogenol A              | Sapogenol B            | Total Sapogenol         |
|-----------------------|--------------------------|------------------------|-------------------------|
| Raw grain             | 248.4±45.7 <sup>a</sup>  | 355.7±4.9 <sup>a</sup> | 604.1±49.4 <sup>a</sup> |
| Soaked                | 233.1±53.7 <sup>ab</sup> | 323.5±6.2 <sup>b</sup> | 556.6±51.3 <sup>a</sup> |
| Cooked                | 181.9±41.8 <sup>bc</sup> | 35.1±14.1 <sup>c</sup> | 217.0±44.4 <sup>b</sup> |
| Pressure cooked       | 158.9±36.9 <sup>c</sup>  | 0.0±0.0 <sup>d</sup>   | 158.9±36.9 <sup>b</sup> |
| CD (P=0.05)           | 77.8                     | 12.1                   | 68.7                    |

Values are Mean ± standard deviation

All means bearing different superscripts in columns are significantly different on application of Duncan's new multiple range test (P<0.05)

composition. Therefore, water leaching resulted in a loss of saponins B (Gurfinkel and Rao, 2003; Shi *et al.*, 2004) in soaked beans during the soaking process. Saponins are soluble in water. The hydrated bean seeds allow more water to penetrate more deeply into their matrix to release saponins from the bean tissue matrix by simple leaching (Shi *et al.*, 2009). Sapogenol B decreased remarkably during cooking of soaked grain and a loss of 90.1% in sapogenol B content was observed in rajmash beans (Table 1). The average sapogenol B in cooked grain of frenchbean was 35.1 mg 100 g<sup>-1</sup>. The range of sapogenol B in different genotypes in cooked grain was 14.6 to 46.7 mg 100 g<sup>-1</sup> (Table 2). The pressure cooking had however caused complete loss of sapogenol B in the grains of rajmashbeans. Shi *et al.* (2009) have reported a loss of 70% in sapogenol B during ordinary cooking of pre-soaked navy beans and 100% loss of sapogenol B during autoclaving or pressure cooking of pre-soaked navy beans. Muzquiz *et al.* (1999) have reported soyasapogenol in the range 0.44 to 2.05 g kg<sup>-1</sup> in frenchbean varieties. Ireland and Dziedzic (1987) have reported 0.149% soyasapogenol B in haricot beans and 0.102% in red kidney beans on a defatted dry weight basis.

Average total sapogenol in soaked seed of frenchbean was 556.6 mg 100 g<sup>-1</sup> (Table 1). The range of total sapogenol in soaked grain was 509.2 to 618.8 mg 100 g<sup>-1</sup>. A reduction of 7.9% in total sapogenol was observed on soaking, but the reduction was not significant (Table 1). Average total sapogenol in cooked grain was 217 mg 100 g<sup>-1</sup> and the range of total sapogenol in different genotypes in cooked grain was 169.9 to 274.1 mg 100 g<sup>-1</sup>. A reduction of 64.1% was observed in total sapogenol on cooking of rajmashbeans. Similar findings (66.3 and 61.9% destruction of total sapogenols) were reported in seeds of

Table 2. Effect of different processing on sapogenols (A, B and Total) of frenchbean

| Genotypes                                     | Raw seed  | Soaked    | Cooked    | Pressure cooked |
|-----------------------------------------------|-----------|-----------|-----------|-----------------|
| <i>Sapogenol A (mg 100g<sup>-1</sup>)</i>     |           |           |           |                 |
| HUR 15                                        | 294.9±0.4 | 292.3±0.3 | 227.8±0.3 | 201.6±0.2       |
| IIPR 96-4 (Amber)                             | 299.0±1.0 | 289.6±0.2 | 226.9±0.9 | 195.8±0.2       |
| EC 406072                                     | 206.2±0.2 | 201.6±0.2 | 156.8±0.2 | 142.3±0.3       |
| PDR 14 (Uday)                                 | 234.1±0.2 | 203.2±0.2 | 155.3±0.3 | 126.3±0.3       |
| IPR 98-5 (Utkarsh)                            | 208.0±2.0 | 178.6±0.2 | 142.7±0.7 | 128.6±0.2       |
| Mean                                          | 248.4     | 233.1     | 181.9     | 158.9           |
| CD (P=0.05)                                   | 1.9       | 0.4       | 1.0       | 0.4             |
| <i>Sapogenol B (mg 100g<sup>-1</sup>)</i>     |           |           |           |                 |
| HUR 15                                        | 362.3±0.3 | 326.5±0.5 | 46.3±0.30 | 0               |
| IIPR 96-4 (Amber)                             | 356.2±0.2 | 315.7±0.2 | 26.3±0.30 | 0               |
| EC 406072                                     | 348.9±0.4 | 326.4±0.4 | 41.6±0.20 | 0               |
| PDR 14 (Uday)                                 | 357.0±1.0 | 318.4±0.4 | 14.6±0.40 | 0               |
| IPR 98-5 (Utkarsh)                            | 353.9±0.3 | 330.6±0.6 | 46.7±0.20 | 0               |
| Mean                                          | 355.7     | 323.5     | 35.1      | 0               |
| CD (P=0.05)                                   | 0.9       | 0.8       | 0.5       |                 |
| <i>Total Sapogenol (mg 100g<sup>-1</sup>)</i> |           |           |           |                 |
| HUR 15                                        | 657.2±0.4 | 618.8±0.5 | 274.1±0.1 | 201.6±0.2       |
| IIPR 96-4 (Amber)                             | 655.2±0.2 | 605.3±0.3 | 253.2±0.2 | 195.8±0.2       |
| C 406072                                      | 555.1±0.2 | 528.0±2.0 | 198.4±0.4 | 142.3±0.3       |
| PDR 14 (Uday)                                 | 591.1±0.2 | 521.6±0.2 | 169.9±0.4 | 126.3±0.3       |
| IPR 98-5 (Utkarsh)                            | 561.9±0.1 | 509.2±0.2 | 189.4±0.2 | 128.6±0.2       |
| Mean                                          | 604.1     | 556.6     | 217.0     | 158.9           |
| CD (P=0.05)                                   | 0.4       | 1.7       | 0.5       | 0.4             |

Mean values of three determinations ± standard deviation

chickpeas and lentils during cooking in our earlier communication (Srivastava and Vasishtha, 2013; Vasishtha and Srivastava, 2013), whereas complete degradation of sapogenol B was observed in chickpeas and lentils during cooking. Total sapogenol pressure cooking of grain reduced to 158.9 mg 100 g<sup>-1</sup>, which was 73.7% lower than unprocessed grain. The total sapogenol in pressure cooked grain was found in the range of 126.3 mg 100 g<sup>-1</sup> to 201.6 mg 100 g<sup>-1</sup>. HUR 15 and IPR 96-4 (Amber) contained higher total sapogenol in cooked as well as pressure cooked grain, whereas PDR 14 (Uday) and IPR 98-5 (Utkarsh) contained least total sapogenols. Genotypes with high total sapogenols in cooked or pressure cooked grain may be better from health point of view because of health advantages of sapogenols.

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## Antioxidant activity of leaf extracts of *Murraya koenigii*

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### ABSTRACT

Antibacterial and antioxidant activity of crude leaf extracts of *Murraya koenigii* (curry leaf) in methanol, ethanol and water was evaluated *in vitro* using DPPH (1, 1-Diphenyl-2-picryl hydrazyl) assay. Free radical scavenging activity of *Murraya koenigii* leaf crude extracts was higher in methanol. The extracts had lower antioxidant activity as compared to ascorbic acid. Methanol and ethanol extracts of *Murraya koenigii* exhibited the highest radical scavenging potential with an  $IC_{50}$  value of  $140 \mu g\ ml^{-1}$  and  $155 \mu g\ ml^{-1}$ , respectively. The methanol, ethanol and aqueous extracts of *Murraya koenigii* also had antioxidant activity. *Murraya koenigii* crude extracts had activity in order of ethanol > methanol > aqueous.

**Key words:** Antioxidant activity, DPPH (1, 1-Diphenyl-2-picryl hydrazyl) assay,  $IC_{50}$  value, Reducing power assay

Medicinal plants possess therapeutic properties or exert beneficial pharmacological effects on animal body. The medicinal value of these plants lies in some chemically active substances that produce a definite physiological action on the human body. The most important of these bioactive constituents of plants are alkaloids, tannins, flavonoids and phenolic compounds. Antioxidants are vital substances which possess the ability to protect the body from damage caused by free radical induced oxidative stress. A variety of free radical scavenging antioxidants exists within the body which many of them are derived from dietary sources like fruits, vegetables and teas (Souria and Hassan, 2007). It has been established that oxidative stress is among the major causative factors in the induction of many chronic and degenerative diseases including atherosclerosis, ischemic heart disease, ageing, diabetes mellitus, cancer, immunosuppressant, neurodegenerative diseases and others. The most effective way to eliminate free radicals which cause the oxidative stress is with the help of antioxidants (Shyura *et al.*, 2005). Plant derived drugs remain important resource especially in developing countries, to combat serious disease. Approximately 62–80% of the world's population still relies on traditional medicines for the treatment of common illness. In fact, plants produce a diverse range of bioactive molecules making them a rich source of different types of medicines. Higher plants, as sources of medicinal compounds, have continued to play a dominant role in the maintenance of human health since ancient times (Farombi, 2003). Over 50% of all modern clinical drugs are of natural origin. Natural products play important role in drug development in the pharmaceutical industry. Currently, there is a growing interest toward natural antioxidants and natural antimicrobials of herbal resources. Epidemiological and *in vitro* studies on medicinal plants strongly supported that plant

constituents with antioxidant activities are capable of exerting protective effects against oxidative stress in biological systems.

Curry leaf (*Murraya koenigii*) is a good source of vitamin A, calcium and folic acid. Its richness in vitamin A and antioxidants may explain its use in preventing early development of cataract. Being a fairly good source of folic acid, the leaves can also help in absorption of iron. Other proposed benefits include boost in circulation and anti-inflammation. It is also antidiabetic, antioxidant, antimicrobial, hepatoprotective, hypocholesterolemic, and delays premature graying. With its anti-inflammatory benefits, it is used in treating bruises and skin eruptions. Fresh leaves on steam distillation under pressure yield a volatile oil. Besides the oil, the leaves contain a residual glycoside named as koenigin. Their mineral and vitamin contents are calcium, phosphorus, iron, nicotinic acid and vitamin C. Currently, there is a growing interest toward natural antioxidant herbal resources. Epidemiological and *in vitro* studies on medicinal plants strongly supported that plant constituents with antioxidant are capable of exerting protective effects against oxidative stress in biological systems.

### MATERIALS AND METHODS

**Collection of plant material:** The leaves of *Murraya koenigii* were collected from the Horticulture Department, Faculty of Agriculture, Sam Higginbottom Institute of Agriculture, Technology and Sciences, Naini, Allahabad.

**Sample preparation:** The fresh leaves were harvested and properly washed in tap water and then rinsed in sterile distilled water. The leaf was dried in the hot air oven at  $40^{\circ}C$  for 3 days. The dried leaves were pulverized using sterile laboratory mortar and pestle to obtain it in the powdered form. These were stored in airtight glass containers protected from sunlight until required

for analysis.

**Preparation of extract:** The extract was prepared by the method of Sonia *et al.*, 2000. The dried leaf powder (10 g) of *Murraya koenigii* was weighed and transferred into conical flasks for extraction purpose. 100 ml of methanol, ethanol and water was added to each of the flask, shaken and left for 24 hours undisturbed. Each extract was filtered using whatman No.1 filter paper. The solvents in the extract were removed under reduced pressure at 40°C using rotary evaporator or kept in water bath at 67°C (methanol), 78°C (ethanol) and 100°C (aqueous). The standard extracts were obtained and sealed with aluminum foils and stored in the refrigerator at 4°C until required for antibacterial activity.

**Solvent used for dissolving extract:** Dimethyl sulfoxide (DMSO), a colourless hygroscopic liquid with BP 189°C was used for dissolving various extracts for testing antimicrobial efficacy. DMSO is miscible with water and alcohol and is a very good solvent for experimental purposes (Beyer and Walter, 1997).

**Reducing power Assay:** The reducing power of the test sample was determined by taking different concentration of the leaf extract (200, 400, 600, 800 µg ml<sup>-1</sup>) in 1 ml methanol. They were mixed with 2.5 ml of phosphate buffer and 2.5 ml of potassium ferric cyanide in test tubes. The mixtures were incubated for 20 minutes at 50°C. At the end of the incubation, 2.5 ml of trichloroacetic acid was added to the mixtures followed by centrifuging at 500 rpm for 10 min. The upper layer (2.5 ml) was mixed in 2.5 ml distilled water and 0.5 ml of ferric chloride and the absorbance was measured at 700 nm. The reducing power tests were run in triplicates. Increase in absorbance of the reaction mixture indicated the reducing power of samples (Yen and Duh, 1993).

**DPPH method:** Different dilutions of the extract (200, 400, 600, 800 µg ml<sup>-1</sup>) were prepared. DPPH solution was also prepared by dissolving 6.0 mg of DPPH in 100 ml methanol. Then, 1 ml of extract from each dilution was added into the test tube containing 2 ml of DPPH solution. Control was prepared by adding 1 ml of methanol to 2 ml of DPPH solution. Ascorbic acid was used as standard. The mixture was shaken vigorously and left to stand in the dark for 30 minutes. The absorbance of the resulting solution was measured spectrophotometrically at 517 nm. The scavenging activity of each extract on DPPH radical was calculated using the following equation:

Scavenging activity = [Absorbance of control - Absorbance of sample / Absorbance of control] x 100

## RESULTS AND DISCUSSION

### Reducing power activity

The reducing power of the methanol, ethanol and aqueous extracts of *Murraya koenigii* was determined and it was found significant but less as compared to the ascorbic acid. It was observed that as the concentration of extracts increases (200, 400, 600 and 800 µg ml<sup>-1</sup>), the absorbance of samples increased gradually.

In 200 µg ml<sup>-1</sup> of ethanol extract of *Murraya koenigii*, highest reducing power was observed i.e. 0.244 as compared to aqueous extract 0.188 and methanol extract 0.184. In case of methanol extract of sample in 400 µg ml<sup>-1</sup>, highest reducing power was 0.315 than ethanol extract (0.295) and aqueous extract (0.234). Ethanol extract of test sample of 600 µg ml<sup>-1</sup> showed highest reducing power of 0.437 as compared to methanol extract (0.379) and aqueous extract (0.372). Methanol extract (800 µg ml<sup>-1</sup>) of *Murraya koenigii* showed highest reducing power (0.530) than ethanol extract (0.526) and aqueous extract (0.411). The data showed that all the samples increased their reducing ability when the concentration of extracts was increased. There was slight difference in reducing ability of all the three extracts. Hence leaf extracts of *Murraya koenigii* had good antioxidant activity. All the extracts of leaves showed lower reducing power as compared to ascorbic acid (0.792) (Table 1). Similar results were reported by Savitha and Rathnavijaya (2011) at 100 µg ml<sup>-1</sup> concentration of extract in which the highest reducing power was observed in aqueous extract of the *Andrographis paniculata* i.e. 0.530 as compared to 0.850 of Standard ascorbic acid. According to Faujan *et al.* (2009), *Murraya koenigii* showed the highest extraction yields (1.65%) and total phenolic content (3.6%). Reducing power ability of *Murraya koenigii* extracts was attributed to these compounds.

The ability to reduce Fe<sup>3+</sup> attributed from hydrogen donation from phenolic compounds was related to presence of reductant agent (Shimada *et al.*, 1992). The number and position of hydroxyl group of phenolic compounds rule their antioxidant activity that are responsible for the radical scavenging effect mainly due to redox properties (Rice *et al.*, 1995). According to Gorden (1990), antioxidant action of reductant was based on the breaking of the free radical chain by donating a hydrogen atom. The antioxidants in *Murraya koenigii* extracts affects the reduction of the Fe<sup>3+</sup> ferricyanide complex to the ferrous form. Therefore, the Fe<sup>2+</sup> can be monitored by measuring the formation of Perl's Prussian blue at 700 nm (Plate 1). It was observed that,

Table 1. Reducing power assay of *Murraya koenigii* extracts (optical density) at 700 nm

| Concentration (µg ml <sup>-1</sup> ) | Methanol    | Ethanol     | Aqueous     | Ascorbic acid |
|--------------------------------------|-------------|-------------|-------------|---------------|
| 200                                  | 0.184±0.053 | 0.244±0.041 | 0.188±0.008 | 0.665±0.044   |
| 400                                  | 0.315±0.058 | 0.295±0.008 | 0.234±0.021 | 0.721±0.051   |
| 600                                  | 0.379±0.049 | 0.437±0.025 | 0.372±0.022 | 0.753±0.023   |
| 800                                  | 0.530±0.007 | 0.526±0.004 | 0.411±0.014 | 0.792±0.037   |
| SE±                                  | 0.025       |             |             |               |
| CD (P=0.05)                          | 0.048       |             |             |               |

Values were expressed as mean ± SD (n=3)

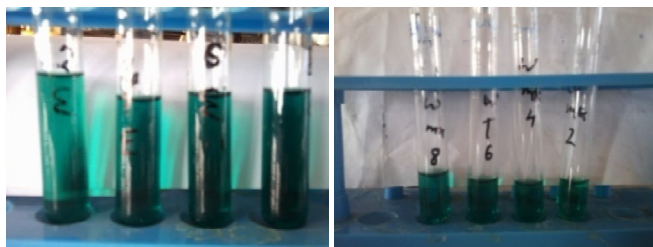


Plate 1. Antioxidant activity of *Murraya koenigii* by reducing power assay

as the concentration of extracts increases the absorbance of samples increased gradually. There was slight difference in reducing ability of all the extracts and showed lower reducing power as compared to ascorbic acid (0.792). The leaf extracts at 800  $\mu\text{g ml}^{-1}$  concentration showed highest activity in methanol (0.530), followed by ethanol (0.526) and aqueous (0.411) extract. The effect of extracts and concentrations on antioxidant activity was found to be statistically significant.

### DPPH free radical scavenging activity

Extracts of *Murraya koenigii* possess antioxidant properties. It inhibits formation of oxygen derived free radicals such as superoxide, hydroxyl radicals, lipid peroxidation and nitric oxide (Philip, 2010). Decolouration due to reaction of antioxidant in samples with the stable free DPPH radical was measured on spectrophotometer. Reduction of DPPH radicals can be observed by the decrease in absorbance at 517 nm. It was observed that as the concentration of extracts increases (200, 400, 600 and 800  $\mu\text{g ml}^{-1}$ ), the per cent of free radical scavenging activity increased in all the methanol, ethanol and aqueous extracts (Table 2). It has been determined that the antioxidant effect of plant products were mainly due to radical scavenging activity of phenolic compounds such as flavonoids, polyphenols, tannins, and phenolic terpenes (Rahman and Moon, 2007).

On comparison of individual doses of the three extracts of *Murraya koenigii*, 200  $\mu\text{g ml}^{-1}$  of methanol extract exhibited free radical scavenging potential of 71.68%, which was higher than ethanol extract (61.88%) and aqueous extract (50.96%). Likewise in 400  $\mu\text{g ml}^{-1}$ , methanol extract showed the highest free radical scavenging activity (76.93%) as compared to ethanol extract (79%) and aqueous extract (58.42%). Methanol extract of 600  $\mu\text{g ml}^{-1}$  had higher free radical scavenging activity (83.28%) than ethanol extract (81.35%) and aqueous extract (65.74%), whereas 800  $\mu\text{g ml}^{-1}$  of methanol extract had highest

free radical scavenging activity (85.08%) than ethanol extract (83.97%) and aqueous extract (75.13%). Further, the results revealed that methanol and ethanol extracts had slight difference in percent of free radical scavenging activity, whereas aqueous extract exhibited lowest free radical scavenging activity than methanol and ethanol extracts. Ascorbic acid is well known potent antioxidant. It was observed that all the three extracts had lower antioxidant activity as compared to the ascorbic acid (90%). The effect of extracts and concentrations on antioxidant activity was found to be statistically significant ( $F_{\text{cal}} 5.91 > F_{\text{tab}} 3.86$  due to extracts). The free radical scavenging activity  $\text{IC}_{50}$  Value of *Murraya koenigii* for methanol extracts was 130  $\mu\text{g ml}^{-1}$ , ethanol extracts 155  $\mu\text{g ml}^{-1}$  and aqueous extracts 195  $\mu\text{g ml}^{-1}$  respectively.  $\text{IC}_{50}$  value of ascorbic acid was found to be 70  $\mu\text{g ml}^{-1}$ .  $\text{IC}_{50}$  value was a concentration of antioxidant required for 50% scavenging of DPPH radicals (Maisuthisakul *et al.*, 2007). Smaller value corresponds to a higher antioxidant activity of the plant extract.

The dried leaf samples of *Murraya koenigii* were extracted using three solvents, viz. methanol, ethanol and aqueous. It was observed that as the concentration of extracts increases (200, 400, 600 and 800  $\mu\text{g ml}^{-1}$ ), the absorbance of samples also increased gradually. Antioxidant assay was determined by DPPH free radical scavenging activity. It was observed that 800  $\mu\text{g ml}^{-1}$  of all the extracts (methanol, ethanol, aqueous) of dried leaf of *Murraya koenigii* possessed antioxidant activity. It was found that free radical scavenging activity of leaf extract was higher in methanol (85.08%) than ethanol (83.97%) and aqueous (75.13%) extract. It showed lower DPPH free radical scavenging activity as compared to ascorbic acid (94.61%). The methanol and ethanol extract of *Murraya koenigii* exhibited the highest radical scavenging potential with an  $\text{IC}_{50}$  value of 140  $\mu\text{g ml}^{-1}$  and 155  $\mu\text{g ml}^{-1}$ , respectively. Antioxidant assay is determined by reducing power assay. There was slight difference in reducing ability of all the extracts. Hence, methanol, ethanol and aqueous extracts of *Murraya koenigii* had antioxidant activity. All samples of extracts at 800  $\mu\text{g ml}^{-1}$  concentration of crude extracts had activity in order of ethanol (0.530) > methanol (0.526) > aqueous (0.411).

It is concluded that *Murraya koenigii* leaves possess potent antioxidant activities. It inhibits production of free radicals. The herbal medicinal plants have been attracting more interest due to their potential to produce a diverse range of chemicals and biologically active compounds. The potential

Table 2. DPPH free radical scavenging assay (%) of leaf extracts of *Murraya koenigii*

| Concentration<br>( $\mu\text{g ml}^{-1}$ ) | Methanol          |       | Ethanol           |       | Aqueous           |       | Ascorbic acid     |       |
|--------------------------------------------|-------------------|-------|-------------------|-------|-------------------|-------|-------------------|-------|
|                                            | OD 517 nm         | %     | OD 517 nm         | %     | OD 517 nm         | %     | OD 517 nm         | %     |
| 200                                        | 0.205 $\pm$ 0.005 | 71.68 | 0.172 $\pm$ 0.007 | 61.88 | 0.355 $\pm$ 0.015 | 50.96 | 0.051 $\pm$ 0.010 | 92.95 |
| 400                                        | 0.167 $\pm$ 0.007 | 76.93 | 0.152 $\pm$ 0.008 | 79.00 | 0.301 $\pm$ 0.017 | 58.42 | 0.046 $\pm$ 0.001 | 93.64 |
| 600                                        | 0.121 $\pm$ 0.005 | 83.28 | 0.135 $\pm$ 0.014 | 81.35 | 0.248 $\pm$ 0.017 | 65.74 | 0.042 $\pm$ 0.006 | 94.19 |
| 800                                        | 0.108 $\pm$ 0.006 | 85.08 | 0.116 $\pm$ 0.012 | 83.97 | 0.180 $\pm$ 0.006 | 75.13 | 0.039 $\pm$ 0.004 | 94.61 |
| SE $\pm$                                   | 0.053             |       |                   |       |                   |       |                   |       |
| CD (P=0.05)                                | 0.097             |       |                   |       |                   |       |                   |       |

Values were expressed as mean  $\pm$  SD (n=3)

use of these plants is required to be explored in order to develop an alternate therapy for the treatment of infections caused by microorganisms.

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# Empowerment of farm women with pulses production technologies: An empirical framework

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## ABSTRACT

The study was conducted in six villages of Kanpur Dehat and Fatehpur districts of Uttar Pradesh during 2008-09 to design a framework for empowering farm women with improved pulse production technologies. A total of 180 farm women were randomly sampled for the study. Structured interview schedule with selected participatory rural appraisal (PRA) tools were utilized for drawing data from the respondents. For devising the framework, activity role analysis, the perceived technological needs and their time utilization pattern and constraints were studied. Participation of women farmers was highest in the areas like post-harvest operations like cleaning of grains (93%), value addition (89%), processing (85%), winnowing (67%) and seed cleaning (60%). Storage of the pulses was a joint activity in majority of the households (44%). Majority of women farmers expressed need for improved storage techniques (73.53%), drudgery reducing post-harvest handling techniques (73.34%) and improved varieties (54%). Limited availability of quality seeds (82%) and lack of information on suitable improved varieties (67%) were expressed as the major problems. Women farmers were found mid January - mid March months of the year and timings from 11AM to 3 PM during the day as most suitable for planning interventions related to their technological empowerment. Based on the findings, an empirical framework comprising of suitable technological interventions, extension methodology and time schedule for technological empowerment of farm women of the state was designed.

**Key words:** Technological empowerment, Women farmers, Pulses cultivation

In India, rural women significantly contribute to the farm economy (Santra and Kundu, 2001) by working as independent producers (de Haen *et al.*, 2003); agricultural partners sharing the work and responsibilities and also as agricultural laborers (Bati and Singh, 1987). But regardless of these variations, there is hardly any activity in agricultural production, except ploughing in which women are not actively involved (Shiva, 1999). In some of the farm operations like processing and storage, women predominate so strongly that men workers are numerically become insignificant (Aggarwal, 2003). More objectively, in the Indian Himalayas, a pair of bullocks works 1064 hours, a man 1212 hours and a woman 3485 hours in a year on one hectare farm, a figure that illustrates women's significant contribution to agricultural production (Shiva, 1992).

The agriculture technology dissemination is generally carried out with the assumption that the technologies are gender neutral albeit the fact remains that technological needs of men and women farmers vary according to their involvement in agricultural tasks, their education, experience, skill level, ergonomical characteristics which are often overlooked (Doss, 2002; Doss, 2010). It is because of these reasons that many of the promising agriculture technologies developed at the research and disseminated through development programmes fail to result the intended impact among the farming community (Jackson, 2005). Relevance of the new technology to the needs

and situation of the target group either the farmer or farm women holds the key for success of any technology or programme in agriculture. While suggesting the action for accomplishing the objective of empowering women, Rivera (1990) recommended programme development based on specific situational realities and diagnosed needs of women in agriculture and programs based on needs assessment data disaggregated by gender. United Nation (2000) also resolved in its millennium declaration that "To promote gender equality and empowerment of women as effective ways to combat poverty, hunger and disease to stimulate development that is truly sustainable". Further, it was also recommended that extension services need to form linkages with rural women's groups for collaborative agricultural development efforts (Kes and Swaminathan, 2006).

Uttar Pradesh is an important pulse growing state contributing about 14% to the national pulse basket (Anonymous, 2012). However, the area under pulse crops is progressively shrinking in the state due to low productivity and high production instability resulting in lower economic returns as compared to the cereal crops (Reddy *et al.*, 2012). Ensuring the higher rate of adoption of the improved pulses production technologies among the pulse growers may result in marked rise in the pulse availability in India. And hence, developing the capacities of farmers and farm women for application of improved pulse production technologies and thus making them technologically empowered becomes a subject of attention.

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Empowerment of farm women with appropriate need oriented technological options could be one of the strategies to gear the pulse production and productivity in India. Initiating large-scale extension programmes with special emphasis on need based interventions in the pulse growing pockets of different states of India may be instrumental for realizing high adoption of pulse production technologies resulting in higher availability of pulses in the country. The study was, therefore, conducted to design and propose the empirical framework for technological empowerment of women pulse farmers based on the analysis of their involvement, their technological needs as well as the constraints perceived by them. The paper also aims at highlighting the seasonal and daily work load of farm women as important indicators for identifying the most appropriate time for implementing the intervention without affecting their socio-household commitments.

## MATERIALS AND METHODS

The present study was the part of the project funded by Young Scientist Scheme of Department of Science and Technology (DST), New Delhi, India. The project was implemented in Kanpur and Fatehpur districts of Uttar Pradesh, India during 2008-09. These districts were purposively selected because of their importance in pulses production in the state both in terms of area and production. A total of 30 farm women were randomly sampled from each of the randomly selected six villages under two blocks namely Akbarpur and Malwan from Kanpur Dehat and Fatehpur districts, respectively for the present investigation. Thus, a total of 180 women farmers constituted the sample for the study.

The main research variable i.e. contribution of women farmers to pulses cultivation was measured for all the sub activities of pulses production and post-harvest handling on a four point closed-ended options, viz. activity performed solely by female, performed solely by male, performed jointly by male and female, performed by others including children, hired labour, relatives etc. Similarly, the other research variables like need perception and perceived constraints were ascertained from the viewpoints of female farmers through open-ended questions. The seasonal work load analysis and daily routine analysis were also done for the sampled women farmers using PRA tools. A pretested semi-structured interview schedule

containing all the research variables was utilized to elicit data from the sampled women farmers through personal interview method supported with selected PRA tools. Collected data were analyzed using descriptive statistics like mean, frequency and percentages. The research variables like gender differentiated activity analysis helped to identify the appropriate technological interventions points. Similarly, seasonal work load and daily work load analysis guided in identification of intervention time. Based on the findings, an empirical framework for technological empowerment of farm women with regard to improved pulse production technology was emanated.

## RESULTS AND DISCUSSION

### *Identification of appropriate intervention points*

#### **Gender differentiated activity analysis**

Activity analysis is commonly accepted term used for studying the role of male and female in intra and inter-household dynamics within a farming system. It is carried out with an aim to increase effectiveness of the development activities by formulating the participatory strategies through integration of both men and women. The activity analysis for pulse production and post-harvest operations was carried out to identify the operations where farm women played a major role.

*i. Pulses production activities:* From the data presented in the Table 1 it could be observed that with regard to pulse production activities, the frequency of sole participation of women was observed highest in cleaning of seed before sowing (60.6%) followed by cleaning of the fields and weeding wherein the frequency of participation was 45%. Harvesting was observed to be performed solely by farm women in about 40.56% of the sampled households. In contrast, sole participation of male farmers was highest in pulse production activities like field preparation (72%), construction and repair of irrigation channels (50%) and fertilizer application (48.9%).

It could be inferred from the above findings that male farmers performed those activities which involved higher manual force like field preparation, construction and repair of irrigation channels, fertilizer application and spray of plant protection chemicals while women farmers were doing activities

Table 1. Frequency distribution of respondents according to their participation in pulses production activities (N=180)

| Sl. No. | Activities                                    | Participation of male farmers |          | Participation of women farmers |          | Joint participation |          |
|---------|-----------------------------------------------|-------------------------------|----------|--------------------------------|----------|---------------------|----------|
|         |                                               | Frequency                     | Per cent | Frequency                      | Per cent | Frequency           | Per cent |
| 1       | Cleaning of field                             | 18                            | 10.00    | 81                             | 45.00    | 41                  | 22.78    |
| 2       | Field preparation                             | 130                           | 72.22    |                                |          |                     |          |
| 3       | Manure application                            | 60                            | 33.33    | 13                             | 07.22    | 51                  | 28.33    |
| 4       | Fertilizer application                        | 88                            | 48.89    | 13                             | 07.22    | 58                  | 32.22    |
| 5       | Seed cleaning                                 |                               |          | 109                            | 60.56    | 37                  | 20.56    |
| 6       | Seed treatment                                |                               |          | 18                             | 10.00    | 09                  | 05.00    |
| 7       | Sowing of seed                                | 50                            | 27.78    | 67                             | 37.22    | 45                  | 25.00    |
| 8       | Construction and repair of irrigation channel | 90                            | 50.00    | 16                             | 08.89    | 61                  | 33.89    |
| 9       | Weeding                                       |                               |          | 81                             | 45.00    | 60                  | 33.33    |
| 10      | Spraying insecticide                          | 81                            | 45.00    |                                |          |                     |          |
| 11      | Harvesting                                    | 21                            | 11.67    | 73                             | 40.56    | 53                  | 29.44    |

requiring finer motors skills like weeding, seed cleaning etc.

The finding are similar to those reported by Singh *et al.* (2004) who also observed that women farmers' involvement was higher in the activities like cleaning of the fields, manure application, weeding and transportation of harvested produce.

*ii. Post-harvest related activities:* A perusal of the data presented in the Table 2 revealed that the extent of participation of women farmers was higher (41-93%) in all the activities except for marketing, wherein the participation of male farmers was more (64%). Activities like cleaning and sorting of grains, value addition and processing were performed solely by women in about 93, 89 and 85% of the households, respectively.

Winnowing of the threshed produce was done by women farmers solely in about 67% of the sampled households and jointly in about 20% of the households. Activities like drying of harvested produce and bundling and transportation of produce from fields to house were performed exclusively in 65 and 62% of the households, respectively. While the same activities were performed jointly in about 23 and 31% of the households, respectively. Marketing of the produce was solely done by male farmers in majority of the households and only in 19% households women farmers were involved in marketing activities, but it was restricted to the female headed households only. Storage of the pulses was carried out jointly by farmers and farm women in majority of the households (44.4%).

From the data presented above, it could be concluded that post-harvest operations of pulses were primarily done by farm women, except for marketing. Storage of the produce after harvesting was however carried out jointly. Similar results were also observed by Aggarwal (2003) who found that in the farm activities like processing and storage, women predominate more frequently. In similar line, Singh *et al.* (2004) had also reported that activities like drying, cleaning of grains, processing activities were solely performed by women, while in activities like winnowing, grading and storage, their participation was to a higher extent.

Clear cut gender differentiated participation was observed among the sampled households in both the districts of Uttar Pradesh. The male farmers were solely involved to a greater extent in field preparation, manure and fertilizer application, construction and repair of irrigation channel and marketing of the produce. On the other hand, farm women were

the major role performer in activities related to seed cleaning, sorting, treatment and sowing activities. They were also key players in activities relating to harvesting and post-harvest handling of pulse crops. In majority of the sampled households they were found to be performing the activities right from harvesting to transportation, threshing, cleaning, drying, bagging, storage and finally to processing and value addition. The findings are in tandem with the reports of Nair (1989) who also reported that women were responsible for a wide range of agricultural operations like, seed treatment, sowing, weeding, harvesting, drying and storage. The gender differentiated role in pulse production has direct implications on technological needs with regard to improved pulse production and post-harvest handling.

### ***Perceived technological needs of women farmers related to improved pulse production***

“Need” refers to the positive driving force that impels a person towards certain objectives or conditions. Perceived needs of women farmers for improved agricultural technologies are important function of trial and adoption. Thus for technological empowerment of women farmers, it is imperative that the technologies identified and disseminated ought to be needs based. In the present study the perceived needs with respect to improved pulses production technology and improve post-harvest management technology for pulses were ascertained on a three point need urgency continuum i.e. urgently needed, needed and not needed and the respective score of 2, 1 and 0 was assigned. Need index for individual need item was calculated and based on it, the prioritization was done for the overall intensity of need related to a particular aspect.

*i. Need perception related to improved pulse production technologies:* With regard to improved pulse production technologies, farm women expressed highest degree of need for improved varieties (54%), followed by recommended seed rate (53%) and recommended measures for disease management (52.6%). They were found to have least perceived need for technologies related to weed management and irrigation stages, wherein they were found to have 31.2 and 23.2% need index, respectively (Table 3). It could be observed from the results that technological needs of women farmers mainly concentrated around seed component, like improved varieties, recommended

Table 2. Frequency distribution of respondents according to their participation in post-harvest activities related to pulse crops (N=180)

| Sl. No. | Activities                             | Participation of male farmers |          | Participation of women farmers |          | Joint participation |          |
|---------|----------------------------------------|-------------------------------|----------|--------------------------------|----------|---------------------|----------|
|         |                                        | Frequency                     | Per cent | Frequency                      | Per cent | Frequency           | Per cent |
| 1       | Bundling and transportation of produce | 26                            | 14.44    | 75                             | 41.67    | 57                  | 31.67    |
| 2       | Threshing                              | 09                            | 05.00    | 77                             | 42.78    | 61                  | 33.89    |
| 3       | Winnowing                              | -                             | -        | 121                            | 67.22    | 36                  | 20.00    |
| 4       | Drying                                 | 13                            | 7.22     | 96                             | 53.33    | 42                  | 23.33    |
| 5       | Bagging of produce                     | 34                            | 18.87    | 69                             | 38.33    | 57                  | 31.67    |
| 6       | Cleaning/sorting of grain              |                               |          | 168                            | 93.33    | 03                  | 1.67     |
| 7       | Marketing                              | 116                           | 64.44    | 26                             | 14.44    | 19                  | 10.56    |
| 8       | Storage                                | 40                            | 22.22    | 51                             | 28.33    | 80                  | 44.44    |
| 9       | Processing                             |                               |          | 153                            | 85.00    |                     |          |
| 10      | Value addition                         |                               |          | 161                            | 89.44    |                     |          |

spacing and seed production techniques. With regard to weed management farm women expressed lowest need despite their highest involvement. This could be because of their lesser awareness about the importance of management practices. Their higher involvement in sowing activity could be attributed for their higher need with regard to appropriate spacing and for recommended seed rate.

*ii. Need perception related to improved post-harvest handling technologies:* Data presented in Table 3 indicate that, among the various need items related to post harvest handling technologies of pulses, need index was highest with regard to improved storage techniques (73.5%) closely followed the need for recommended post-harvest handling practices (73.3%). Women farmers felt lowest need for drudgery reduction technologies (43.5%). It could be inferred from the above results that extent of perceived technological need of women farmers related to improved post-harvest technologies was higher in comparison to their technological needs related to pulse production. The technological needs of women farmers were thus in accordance to their participation which was also higher in post-harvest operations.

### ***Constraints perceived by women farmers related to pulse cultivation***

Analysis of constraints of a production situation provides a reflection of the existing socio-economic, bio-physical and situational factors that impede the production process and its progress. For the present study constraints with regard to pulse production and post-harvest handling were analyzed (Table 4).

*i. Constraints related to pulse production:* A total of 10 constraints related to pulse production activities were identified by farm women, who were then requested to accord the rank to these constraints according to their severity. Limited availability of quality seeds coupled with lack of information on improved varieties were the major constraint as felt by about 82 and 67% of the sampled farm women (Table 4). Problem of insect pests, blue bulls and lack of information on disease identification and management were also the major constraints as perceived by 62 to 65% of the farm women, respectively; while uncertainty in timely availability of fertilizers (45%) and spurious plant protection chemicals (37%) were perceived as least serious constraints.

Table 3. Need Index and ranking of various technological items related to improved pulse production technologies as perceived by women farmers (N=180)

| Sl. No.    | Need items                                               | Need index | Rank |
|------------|----------------------------------------------------------|------------|------|
| <i>I)</i>  | <i>Related to improved pulse production technologies</i> |            |      |
| 1.         | Improved varieties                                       | 54.02      | I    |
| 2          | Recommended fertilizer dose                              | 38.53      | VII  |
| 3          | Recommended seed rate                                    | 53.23      | II   |
| 4          | Appropriate spacing for optimizing yield                 | 41.06      | V    |
| 5          | Seed production techniques                               | 51.90      | IV   |
| 6          | Weed management                                          | 33.15      | VIII |
| 7          | Critical stages of irrigation                            | 23.24      | IX   |
| 8          | Recommended measures for disease management              | 52.64      | III  |
| 9          | Recommended measures for insect pest management          | 40.37      | VI   |
| <i>II)</i> | <i>Related to improved postharvest technologies</i>      |            |      |
| 1.         | Drudgery reduction in post harvest operations            | 43.50      | III  |
| 2          | Recommended post harvest handling technologies           | 73.34      | II   |
| 3          | Improved storage techniques                              | 73.53      | I    |

Table 4. Frequency distribution of respondents on the basis of constraints perceived by them with respect to improved pulse production technology (N=180)

| Sl. No.    | Constraints                                                  | Frequency | Percentage | Rank |
|------------|--------------------------------------------------------------|-----------|------------|------|
| <i>I)</i>  | <i>With respect to improved pulse production technology</i>  |           |            |      |
| 1          | Limited availability of quality seeds                        | 148       | 82.22      | I    |
| 2          | Problem of insect pest                                       | 117       | 65.00      | III  |
| 3          | Limited land and preference for cereals                      | 103       | 57.22      | VII  |
| 4          | Lack of information on disease identification and management | 112       | 62.22      | V    |
| 5          | Problem of blue bulls                                        | 114       | 63.33      | IV   |
| 6          | Lack of information on improved varieties                    | 121       | 67.22      | II   |
| 7          | Adverse climatic conditions                                  | 97        | 53.89      | VIII |
| 8          | Limited irrigation facilities                                | 105       | 58.33      | VI   |
| 9          | Uncertainty in timely availability of fertilizers            | 81        | 45.00      | IX   |
| 10         | Spurious plant protection chemicals                          | 67        | 37.22      | X    |
| <i>II)</i> | <i>Related to post harvest handling of pulses</i>            |           |            |      |
| 1          | Pest damage during storage                                   | 142       | 78.89      | I    |
| 2          | Heavy labour demand for threshing of pulse crops             | 116       | 64.44      | II   |
| 3          | Dominance of middleman in marketing                          | 89        | 49.44      | III  |

The major constraints perceived were linked to seed of pulse crops either in terms of the availability or the information on improved varieties. Constraints relating to fertilizer and plant protection chemicals was least perceived reflecting on the low use of these inputs in the pulse crops in the project area. The findings are in accordance with the perception of high technological need with regard to seeds in terms of improved varieties, recommended seed rate and seed production techniques by the sampled farm.

*ii. Constraints related to post harvest handling of pulse crops:* Pest damage during storage and heavy labour requirement for threshing of harvested pulse crops were the constraints as expressed by about 79 and 64% of the sampled women farmers. Dominance of the middleman in marketing was the another constraints but less frequently (49%) felt by them (Table 4). The above results could be attributed to higher involvement of women in storage and other post-harvest handling operations and also justify their higher technological need index related to improved storage techniques. Women perceived lesser constraint in marketing, which was a male dominated activity in the study area. It could be concluded that majority of the sampled farm women perceived constraints relating to limited availability of seeds, lack of information on improved varieties and pest damage during storage and threshing of pulses.

### Identification of appropriate time of intervention

#### Seasonal workload analysis

To assess the comparative workload of women in different seasons or months of the year, the seasonal workload analysis was carried out with different groups of women farmers. It helped in identifying the most appropriate time for scheduling the technological intervention when farm women may have lesser workload. For this analysis, farm women were asked to pictorially present the seasonal workload on the ground using small heaps of colour each depicting a unit of workload, against different months (Table 5).

It could be seen that during mid September to mid October (locally termed as *Ashwin* month) and mid December to mid January (Locally called as '*Pus*' ) farm women were most busy

with households and farm related work followed by of mid March to mid April (*Chaitra*) month. Farm women also depicted that they were least busy during mid January to mid March months (locally called as *Maghand* *Falgun* months), respectively. In view of the seasonal work load analysis, it could be suggested that the capacity enhancement activities needs to be scheduled in the least busy months which are usually mid January to mid March. This may ensure the higher participation of women farmers and better impact of the planned technological intervention.

#### Daily routine analysis

Daily routine analysis was carried out to have an understanding of the pattern of work load on the farm women in a entire day so as to identify appropriate time for scheduling technological intervention so that it may be convenient for them to attend the activities.

Selected groups of farm women were asked to assess their workload in different time intervals in the day i.e., 5-8 AM, 8-11 AM, 11AM-3 PM, 3-6 PM and 6-9 PM. The time intervals were mentioned on sheets of paper and placed on ground at equal distance and women were asked to draw a circle with colored powder, the size of which reflect on the workload at that particular time interval (Table 6). It could be further observed that farm women had maximum workload during 5-8 AM in morning followed by 8-11 AM. During 11:00 AM to 3:00 PM, women had minimum workload. Thus the time interval from 11:00 AM to 3:00 PM was observed to be most appropriate for scheduling technological interventions for farm women of the study area.

#### Framework for technological empowerment of farm women related to improved pulse cultivation technology

Based on the obtained results, the empirical framework for technological empowerment of farm women related to improved pulse cultivation technology is proposed (Fig. 1).

#### Appropriate technological options

With regard to identified technological interventions for

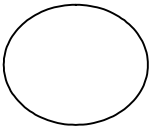
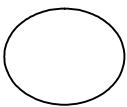
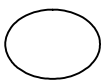
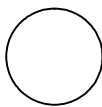
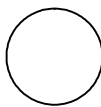
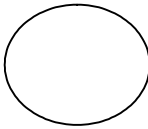
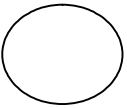
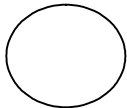
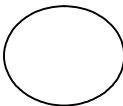
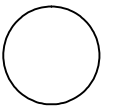
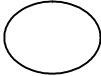
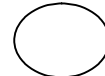
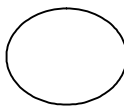
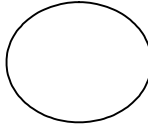
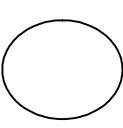
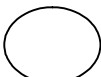
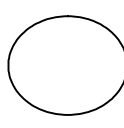
Table 5. Women farmers' workload in different months of the year

| Sl. No. | Month                                       | Kanpur Dehat | Fatehpur |
|---------|---------------------------------------------|--------------|----------|
| 1       | <i>Chaitra</i> (Mid March – mid April)      | *            | *****    |
| 2       | <i>Baisakh</i> ( Mid April- Mid May)        | *            | *****    |
| 3       | <i>Jaistha</i> ( Mid May- Mid June)         | **           | ****     |
| 4       | <i>Asadh</i> ( Mid June- Mid July)          | ***          | *****    |
| 5       | <i>Srawan</i> ( Mid July- Mid August)       | ***          | *****    |
| 6       | <i>Bhadho</i> (Mid August – Mid September)  | **           | *****    |
| 7       | <i>Ahwin</i> ( Mid September-Mid October)   | ***          | *****    |
| 8       | <i>Kartik</i> ( Mid October – Mid November) | *            | *****    |
| 9       | <i>Agahan</i> (Mid November –Mid December)  | **           | *****    |
| 10      | <i>Pus</i> ( Mid December - Mid January)    | ****         | *****    |
| 11      | <i>Magh</i> (Mid January- Mid February)     | ***          | ***      |
| 12      | <i>Falgun</i> (Mid February - Mid March)    | **           | **       |

Number of asterisks indicates the quantum of workload in increasing order

**Village analysts:** Smt.Savitri, Ram Dulari, Kaushalya, Rajkumari, Rampati of Fatehpur District and Smt. Krishna Devi, Heera Devi, Chand Devi, Kamla, Prema, NeerajYadav of Kanpur Dehat District, Uttar Pradesh, India

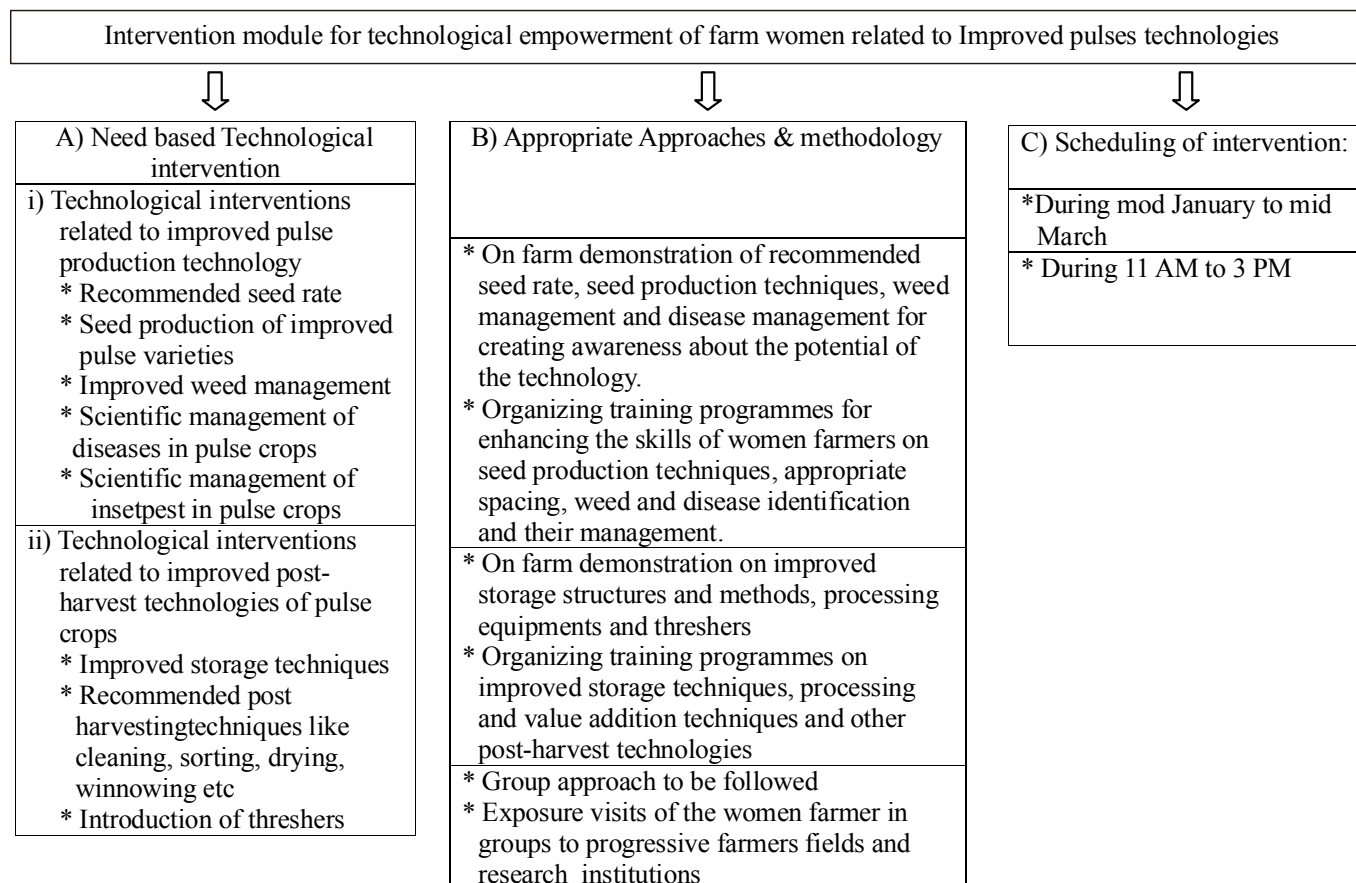
Table 6. Daily workload analysis of sample women group

| Time interval  | 5-8 morning                                                                       | 8-11 morning                                                                      | 11-3 day time                                                                     | 3-6 evening                                                                         | 6-9 night                                                                           |
|----------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1 women        |  |  |  |  |  |
| 2 woman        |  |  |  |  |  |
| 3 woman        |  |  |  |  |  |
| 4 woman farmer |  |  |  |  |  |

Size of rings depicts the work load on farm women

**Village analysts:** Smt. Lakshmi, Chandra Kali, Maya Devi, Jagdei, Ramshri, Phoolmati, Kusum, Parvati of Fatehpur District, Uttar Pradesh, India

Fig.1. Empirical framework for technological empowerment of farm women related to improved pulses technologies



farm women i.e. recommended seed rate, seed production of improved varieties, improved weed management and scientific management of diseases and insect pest in pulse crops were identified as the need based pulses production interventions for their transfer to women farmers of the study area. With regard to improved post-harvest technological interventions, improved storage techniques and recommended post-harvest operations of pulses were identified for inclusion. Further the skill enhancement related to improved processing for improving the nutritive value of pulses was also identified as an important intervention point. In addition, technological interventions for reducing the drudgery involved in threshing was included in the framework in terms of demonstration of drudgery reducing farm implements like threshers in the villages.

### ***Appropriate extension approaches***

With respect to appropriate extension approaches and methods, On-farm demonstrations supported with skill oriented training programmes related to the identified interventions are suggested methodology for technological empowerment of farm women of the selected districts. Moreover, groups approach for ensuring better participation of women in the programmes need to be considered for mitigating the social taboos which often prohibit women from interacting with the men from outside their family. Arranging exposure-visits for farm women from project villages to the pulse based research institutes to enhance their awareness on the improved pulses production and post-harvest handling is the another key activity proposed in the intervention strategy. Also, exposure visits of farm women to the nearby progressive farmers would also provide an opportunity to them to gain hands-on experiences with the technology in the real field conditions so as to consolidate their confidence.

### ***Intervention scheduling***

Scheduling of the technological interventions is another critical item for technological empowerment of women farmers and it has been suggested during the periods when women have comparatively less workload and when it is convenient for them to attend. This would ensure a greater participation of women and may result in better impact of the interventions.

Findings of the study helped to infer that women farmers are major contributor in pulse production and post-harvest related activities in the study area. They are also experiencing several constraints on the above two areas of pulses production. Therefore, their perceived needs were identified. And, keeping in mind their seasonal and daily workload, need based and locations specific intervention model was designed and recommended that ultimately may help the women farmers to enhance pulses productivity as well as availability in the state of Uttar Pradesh

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## Association of empowerment level and socio-economic condition of women in Harahua block of Varanasi, Uttar Pradesh

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### ABSTRACT

The study was conducted to assess the relationship between socio-economic indicators and women empowerment in Harahua block of district Varanasi, India during 2012. One hundred twenty five rural women were sampled for the study. The relevant variables were selected based upon the available literature and the works done in this field prior to the present investigation. Age, caste, marital status, family size, family type, land holding capacity, family income, housing pattern and occupation were used as socio-economic indicators. A structured interview schedule was used for collection of the data through personal interview method. Data was analyzed using frequency, percentage, mean, standard deviation, and chi-square test. The socio-economic profile of rural women indicated that majority of them were middle aged (51.2%), married (56%) and belonged to other backward (OBC) caste (60%). Majority belonged to small and nuclear family (63.2%) having marginal land holding (51.2%); most of the respondents belong to medium size family (59.2%) i.e. 5 to 10 members under medium income (₹ 21900 to ₹ 34700) category (44%) having Pucca house (61.6%). Among nine selected socio-economic indicators i.e., age, marital status, caste, family type, land holding, family size, income of the family, house and occupation, only four variables have relationship with women empowerment.

**Key words:** Correlation, Empowerment, Interview, Socio-economic indicators

The socio-economic status plays an important role in human development. It includes both social and economical characteristics and experiences of the individual and the related realities that help to mould one's personality, attitudes and lifestyle. The socio-economic and psychological characteristics of rural people indicate that most of the women were middle aged with low literacy level, low family income, nuclear family and most of them belonged to scheduled castes (SCs), scheduled tribes (STs) and backward caste categories with less social participation and less mass media use as highlighted in study of Singh (2001). Majority of the extension programme beneficiaries belonged to the age group of 30-40 and it was also indicated that the women participation in economic activity declined as the age advanced (Victoria and Someswar, 1998).

Empowerment is a process by which an individual gains greater control over his/her life. Empowerment indicates control over material assets, intellectual resources and ideology. It involves 'power to, power with and power within'. Some define empowerment as a process of awareness, of capacity building leading to greater participation, effective decision-making power

and control leading to transformative action. This involves ability to get what one wants and to influence others on our concerns. The term empowerment as means (a) to gain power (b) to develop power; to take or seize power; (c) to facilitate or enable power and (d) to give or grant or permit power. With reference to women, it refers to increase in the spiritual, political, social or economical strength of Women. It often involves the empowered developing confidence in their capacities (Staples, 1990). The empowerment is altering relations of power which constrain women's options and autonomy and adversely affect health and well-being (Sen, 1993). There are three domains of empowerment (adopted by the Millennium Project Task Force on Education and Gender Equality): the capabilities domain, which evaluates knowledge and health factors through indicators of education, health, and nutrition; the access to resources and opportunities domain, which primarily refers to access to political decision making and economic assets; and the security domain, which considers violence and conflict matters (Grown *et al.*, 2008).

### MATERIALS AND METHODS

The study was conducted in 5 villages of Harahua block, viz. Madwa, Lamhi, Banvaripur, Harballampur and Baniyapur of Varanasi district in Uttar Pradesh. Harahua block was purposively selected for the study in 2012. In Harahua block 5 villages were randomly selected through random sampling

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technique. From each of these villages twenty five respondents were selected by random sampling procedure, thus a total of 125 respondents were sampled. The female head or wife of the male head was chosen for interview, this was done purposively for accurate collection of information. The data was analyzed by using appropriate statistical techniques/ tools.

There were six variables selected as the indicators of empowerment i.e. education, land ownership, assets ownership, savings, income/ year and family control on which response was taken as per the scoring pattern mentioned below in the Table 1.

Table 1. Economic indicators and their scoring

| Variables           | Category                             | Score |
|---------------------|--------------------------------------|-------|
| 1. Education        | Illiterate                           | 0     |
|                     | Functionally literate/Primary school | 1     |
|                     | Middle school                        | 2     |
|                     | High school/ intermediate            | 3     |
|                     | Graduate or above                    | 4     |
| 2. Land ownership   | No ownership                         | 0     |
|                     | Combine ownership                    | 1     |
|                     | Individual ownership                 | 2     |
| 3. Assets ownership | No ownership                         | 0     |
|                     | Combine ownership                    | 1     |
|                     | Individual ownership                 | 2     |
| 4. Savings          | No saving                            | 0     |
|                     | Less than 50%                        | 1     |
|                     | More than 50%                        | 2     |
| 5. Income/ year     | <₹ 21900                             | 1     |
|                     | ₹22000-34600                         | 2     |
|                     | >₹ 34700                             | 3     |
| 6. Family control   | No control                           | 0     |
|                     | Partial control                      | 1     |
|                     | Full control                         | 2     |

Based on the score obtained from the above six variables extent of empowerment was calculated by computing the empowerment index (expressed in percentage) as follows;

$$\text{Empowerment index} = (\text{obtained score} / \text{maximum obtainable score}) \times 100$$

Based on the score obtained by each respondent, they were grouped into three categories using mean and standard deviation as measures of check.

### Categories score

Low : Below (Mean  $\pm$  SD)

Medium : Between (Mean  $\pm$  SD)

High : Above (Mean  $\pm$  SD)

## RESULTS AND DISCUSSION

### Association of empowerment with socio-economic profile of rural women

Nine relevant variables were selected as indicators of socio-economic profile i.e. age, caste, marital status, family size, family type, land holding capacity, family income, housing

pattern and occupation. Among these variables age and income are categorized on the basis of mean and standard deviation. The detail information of socio-economic profile of the respondents is as follows:

Table 2. Association between empowerment level and age of rural women

| Category | Age    |       |        |       |        |       | Total |
|----------|--------|-------|--------|-------|--------|-------|-------|
|          | Young  |       | Middle |       | Old    |       |       |
|          | Number | %     | Number | %     | Number | %     |       |
| Low      | 8      | 24.24 | 17     | 18.76 | 10     | 25    | 35    |
| Medium   | 16     | 48.48 | 35     | 54.68 | 11     | 39.28 | 62    |
| High     | 9      | 27.28 | 12     | 26.56 | 7      | 35.72 | 28    |
| Total    | 33     | 100   | 64     | 100   | 28     | 100   | 125   |

Table 2 indicated that majority of young age respondents i.e. 48.48% belong to medium empowerment level. It is same for middle and old age i.e. 54.68 and 39.28%, respectively.

Table 3. Association between empowerment level and marital status of rural women

| Category | Marital Status |       |         |       |                   |       | Total |
|----------|----------------|-------|---------|-------|-------------------|-------|-------|
|          | Unmarried      |       | Married |       | Widow / separated |       |       |
|          | Number         | %     | Number  | %     | Number            | %     |       |
| Low      | 11             | 30.56 | 18      | 29.03 | 6                 | 22.22 | 35    |
| Medium   | 12             | 33.33 | 35      | 56.46 | 15                | 55.56 | 62    |
| High     | 13             | 36.11 | 9       | 14.51 | 6                 | 22.22 | 28    |
| Total    | 36             | 100   | 62      | 100   | 27                | 100   | 125   |

The Table 3 shows that majority of unmarried women i.e. 36.11% belong to high empowerment; however, majority of married and widow or separated women belongs to medium empowerment i.e. 56.46 and 55.56%, respectively.

Table 4. Association between empowerment level and caste of rural women

| Category | Caste   |       |        |       |         |       | Total |
|----------|---------|-------|--------|-------|---------|-------|-------|
|          | General |       | OBC    |       | ST / SC |       |       |
|          | Number  | %     | Number | %     | Number  | %     |       |
| Low      | 11      | 37.94 | 18     | 24    | 6       | 28.58 | 35    |
| Medium   | 10      | 34.48 | 44     | 58.66 | 8       | 38.09 | 62    |
| High     | 8       | 27.58 | 13     | 17.34 | 7       | 33.33 | 28    |
| Total    | 29      | 100   | 75     | 100   | 21      | 100   | 125   |

Table 4 indicated that majority of general caste respondent i.e. 37.94% belongs to low empowerment while, majority of OBC and ST/SC caste belong to medium empowerment i.e. 58.66 and 38.09%, respectively.

Table 5. Association between empowerment level and family type of rural women

| Category | Family type |       |         |       | Total |
|----------|-------------|-------|---------|-------|-------|
|          | Joint       |       | Nuclear |       |       |
|          | Number      | %     | Number  | %     |       |
| Low      | 12          | 21.43 | 23      | 33.33 | 35    |
| Medium   | 26          | 46.43 | 36      | 52.18 | 62    |
| High     | 18          | 32.14 | 10      | 14.49 | 28    |
| Total    | 56          | 100   | 69      | 100   | 125   |

It is clear from the Table 5 that in joint family nearly half (46.43%) of the respondents belonged to medium empowerment category while, in nuclear family it was 52.18% of the total nuclear respondents.

Table 6. Association between empowerment level and land holding of rural women

| Category | Land holding |       |        |       |          |       | Total |
|----------|--------------|-------|--------|-------|----------|-------|-------|
|          | Medium       |       | Small  |       | Marginal |       |       |
|          | Number       | %     | Number | %     | Number   | %     |       |
| Low      | 7            | 24.13 | 6      | 18.75 | 22       | 34.37 | 35    |
| Medium   | 10           | 34.48 | 18     | 56.25 | 34       | 53.13 | 62    |
| High     | 12           | 41.39 | 8      | 25    | 8        | 12.50 | 28    |
| Total    | 29           | 100   | 32     | 100   | 64       | 100   | 125   |

The above data was categorized on the basis of the land holding capacity of respondent's family i.e. medium, small and marginal land holding. It could be observed in Table 6 that, among marginal land holding farmer more than half of the respondents (53.13%) belonged to medium empowerment. In small land holding category it is 56.26% of respondents, whereas in medium size land holding category 34.48% of the respondents had medium empowerment.

Table 7. Association between empowerment and family size of rural women

| Category | Family size |     |        |       |        |       | Total |
|----------|-------------|-----|--------|-------|--------|-------|-------|
|          | Large       |     | Medium |       | Small  |       |       |
|          | Number      | %   | Number | %     | Number | %     |       |
| Low      | 7           | 35  | 17     | 22.97 | 11     | 35.48 | 35    |
| Medium   | 7           | 35  | 42     | 56.76 | 13     | 41.94 | 62    |
| High     | 6           | 30  | 15     | 20.27 | 7      | 22.58 | 28    |
| Total    | 20          | 100 | 74     | 100   | 31     | 100   | 125   |

The above data was tabulated according to family size i.e. large, medium and small. It is clear from Table 7 that majority of respondents from large, medium and small family size belongs to medium empowerment i.e. 35, 56.76 and 41.94%, respectively.

Table 8. Association between empowerment level and income of rural women

| Category | Income |       |        |       |        |       | Total |
|----------|--------|-------|--------|-------|--------|-------|-------|
|          | High   |       | Medium |       | Low    |       |       |
|          | Number | %     | Number | %     | Number | %     |       |
| Low      | 9      | 29.05 | 11     | 20.00 | 15     | 38.46 | 35    |
| Medium   | 10     | 32.25 | 38     | 69.10 | 14     | 35.90 | 62    |
| High     | 12     | 38.70 | 6      | 10.90 | 10     | 25.64 | 28    |
| Total    | 31     | 100   | 55     | 100   | 39     | 100   | 125   |

Among low income family half (35.90%) of the respondents belonged to medium empowerment category. Among medium income family more than half (69.10%) of the respondents belonged to medium empowerment category, while among high income family it was 32.25% of the total medium income family (Table 8).

Table 9. Association between empowerment level and house of rural women

| Category                 | Frequency | Percentage |
|--------------------------|-----------|------------|
| Pucca                    | 77        | 61.60      |
| Pucca and Kaccha (mixed) | 48        | 38.40      |

The above data was categorized according to house i.e. pucca and mixed house. It is clear that from Table 9 that majority of the respondents were living in pucca houses i.e. 61.60% and the remaining were living in mixed house.

It is clear from the Table 10 that among agricultural labourers 52.83% of respondents had medium empowerment

Table 10. Association between empowerment level and occupation of rural women

| Category | Occupation          |       |                              |       | Total |
|----------|---------------------|-------|------------------------------|-------|-------|
|          | Agricultural labour |       | House wife/ non-agricultures |       |       |
|          | Number              | %     | Number                       | %     |       |
| Low      | 8                   | 15.10 | 27                           | 37.50 | 35    |
| Medium   | 28                  | 52.83 | 34                           | 47.22 | 62    |
| High     | 17                  | 32.07 | 11                           | 15.28 | 28    |
| Total    | 53                  | 100   | 72                           | 100   | 125   |

whereas, among house wife/non agricultural labors nearly half (47.22%) of respondents had medium empowerment.

### ***Relationship between empowerment and socio-economic profile of rural women***

The association between empowerment and profile of the respondents gives the relation between the selected variables which shows significance of the study related to rural women. The relevant data was tabulated based on high, medium and low empowerment categories and then the chi-square test was applied to find out their association and was analyzed and discussed as follows;

Table 11. Relationship of socio-economic indicators with empowerment

| S.No | Variables      | Degree of freedom (df) | Chi-square value |            |
|------|----------------|------------------------|------------------|------------|
|      |                |                        | Tabulated        | Calculated |
| 1.   | Age            | 4                      | 9.448            | 2.26 NS    |
| 2    | Marital status | 4                      | 9.448            | 7.73 NS    |
| 3    | caste          | 4                      | 9.448            | 6.88 NS    |
| 4    | Land holdings  | 4                      | 9.448            | 11.36**    |
| 5    | Family size    | 4                      | 9.448            | 4.24 NS    |
| 6    | Income         | 4                      | 9.448            | 17.43**    |
| 7    | Family type    | 2                      | 5.991            | 6.05**     |
| 8    | Housing        | 2                      | 5.991            | 1.05 NS    |
| 9    | Occupation     | 2                      | 5.991            | 9.49**     |

\*\*Significant at P = 0.05

From the above Table 11, the Socio-economic indicators show varied relationship with empowerment. The chi-square test was applied to find out whether there is any association between socio-economic indicators and empowerment of the respondents. At the 4 df, age, marital status, caste, land holding, family size and income were taken, in which only land holding and income were found to be significant, whereas at 2 df, family type, housing and occupation were taken, in which family type and occupation were found to be significant at 0.05 level of significance.

Degree of freedom for age, marital status, caste, land holding, family size and income were taken as 4 as it has 3\*3 (3 rows x 3 columns) and family type and occupation were taken as 2 as it has 3\*2 (3 rows x 2 columns).

Among nine selected socio-economic indicators only four variables have significant correlation with women empowerment i.e. land holding, income, family type and occupation, it means they show a noteworthy development in enhancing the activities of rural women. Among these, three indicators are more or less concerned with economical terms i.e. land holding, income, and occupation, remaining one variable is related to social terms. Thus we can say the women

empowerment is largely affected by economical indicators. Even though a lot of programmes have been come into force, still there is a need to bring new approaches which improve the living standards of the rural women which directly influence the living conditions of the family.

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## SHORT COMMUNICATION

# Effect of sowing time on yield, resource use efficiency, soil fertility status and economics of sorghum-based intercropping systems

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## ABSTRACT

A field experiment was conducted during *kharif* 2011 at Dharwad (Karnataka) to study the effect of dry and normal sowing on yield, resource use efficiency, soil fertility status and economics of sorghum based intercropping systems under rainfed condition. Higher sorghum equivalent yield (SEY) was recorded in sorghum + soybean (2:2) (7.49 t ha<sup>-1</sup>), followed by sorghum with blackgram (6.24 t ha<sup>-1</sup>). Significantly lower light transmission ratio (LTR) was recorded when sorghum was intercropped with soybean (28.98% at 60 DAS) which indicated higher light interception as compared with other intercrops. The increase in land equivalent ratio (LER) was in the range of 34 to 43% (1.34 to 1.43) with different treatment combinations. Intercropping of soybean with sorghum in 2:2 row proportion resulted in significantly higher area time equivalent ratio (ATER) (1.27) indicating higher per day productivity from the system. Soil nitrogen, phosphorus and potassium status significantly improved when sorghum was intercropped with legumes as compared to sole sorghum. Dry sowing of sorghum intercropped with soybean recorded significantly higher gross return (₹ 104.8 x 10<sup>3</sup> ha<sup>-1</sup>), net return (₹ 81.6 x 10<sup>3</sup> ha<sup>-1</sup>) and B:C ratio (3.52).

**Key words:** Dry sowing, Normal sowing, Resource use efficiency, Soil fertility, Sorghum-based intercropping systems

Sorghum is unique among the major cereals located primarily in the semi-arid tropics, which is drought tolerant and has wider adaptability. Sorghum is the staple food crop of the world's poor and the most food-insecure population. In the recent decades climate induced natural disasters like droughts, delayed monsoons, long dry spells, temperature extremities, erratic and uneven distribution of rainfall have become a major problem in enhancing agriculture production in India. In rainfed agriculture, it is not feasible to sow the crop in time if the onset of monsoon is delayed. If the land is prepared in time, taking advantage of summer rains, sowing can be taken up with the earliest monsoon rains. The other option is the concept of dry seeding based on probability of rainfall. Dry seeding is done in the anticipation of rains. It aids in establishment of crop early and gives advantage of efficient use of early rains during cropping season. It can be practiced in deep black soils, since the first drop of monsoon rain was used for crop germination, growing period was extended as against taking up of wet sowing which can be done only with considerable loss of moisture which shortens growing period of crops (Ramakichenin *et al.*, 2002). Keeping the above points in view, the experiment was conducted to evaluate sorghum-based intercropping systems in respect of yield, resource use efficiency, soil fertility and economics under rainfed condition.

The present investigation was undertaken at the Main

Agricultural Research Station, University of Agricultural Sciences, Dharwad, India during *kharif* 2011. The experiment was laid out in factorial randomized complete block design with three replications consisted of sowing time viz., dry sowing and normal sowing in the main plot and the cropping systems, viz. T<sub>1</sub>: sorghum + greengram (2:2), T<sub>2</sub>: sorghum + soybean (2:2), T<sub>3</sub>: sorghum + blackgram (2:2), T<sub>4</sub>: sole sorghum, T<sub>5</sub>: sole greengram, T<sub>6</sub>: sole soybean and T<sub>7</sub>: sole blackgram in the sub-plots. The soil of the experimental field was neutral in pH (7.4) with medium nitrogen (228.5 kg ha<sup>-1</sup>), high phosphorus (32.43 kg P<sub>2</sub>O<sub>5</sub> ha<sup>-1</sup>) and potassium (406.3 kg K<sub>2</sub>O ha<sup>-1</sup>). Seeds were sown on 30 May and 14 June 2011 in dry and normal sowing conditions, respectively. The sorghum 'CSH-1', greengram 'China mung', soybean 'JS-335' and blackgram 'DU-1' were sown using 60:40:40, 12.5:25:00, 40:80:25, 12.5:25:00 kg N, P<sub>2</sub>O<sub>5</sub> and K<sub>2</sub>O ha<sup>-1</sup>, respectively. Nitrogen, phosphorus and potassium were applied through urea, diammonium phosphate and muriate of potash, respectively. Greengram, blackgram, soybean and sorghum were harvested on 2<sup>nd</sup> August, 18<sup>th</sup> August, 1<sup>st</sup> September and 28<sup>th</sup> September, respectively in dry sown condition. Whereas, in normal sown condition crops were harvested on 20<sup>th</sup> August, 11<sup>th</sup> September, 22<sup>nd</sup> September and 11<sup>th</sup> October, respectively. The soil samples were chemically analyzed before sowing and after harvesting from each plot and analyzed for available nitrogen, phosphorus and potassium status of the soil.

The sorghum equivalent yield (SEY) was higher in intercropping of legumes with sorghum as compared to sole crops (Table 1). Dry sowing of crops recorded significantly higher SEY (5.09 t ha<sup>-1</sup>) as compared to normal sowing (4.1 t ha<sup>-1</sup>), which might be due to increased yield of crops (5.19 t ha<sup>-1</sup>) under dry sown condition (Table 1). Significantly higher SEY was recorded in sorghum with soybean in 2:2 row proportion (7.49 t ha<sup>-1</sup>), followed by sorghum with blackgram (6.24 t ha<sup>-1</sup>). The higher SEY in sorghum with soybean was due to higher grain yield obtained with sorghum (4.84 t ha<sup>-1</sup>) and soybean (1.20 t ha<sup>-1</sup>). While in sorghum and blackgram system the higher SEY was due to higher market price of blackgram as compared to greengram. The extent of increase in SEY was 28 and 16% when soybean and blackgram were intercropped with sorghum respectively as compared to sole sorghum (Table 2). These results are in conformity with the findings of Kumar *et al.* (2011).

Higher light transmission ratio (LTR) with normal sowing (33%) as compared to dry sowing (32.3%) at 60 DAS indicates higher light interception in dry sown crops, which resulted in higher yield of crops (Table 1). Significantly lower LTR was recorded when sorghum was intercropped with soybean (29% at 60 DAS). The increase in leaf canopy of intercropping increased the photosynthetic area per unit area per unit time which in turn helped in higher yield of sorghum and legumes. Similar results were reported by Kumar *et al.* (2011). Lower transmission of light in intercropped sorghum with soybean in 2:2 row proportion indicated that higher interception of light as compared to sole crops.

In the present study intercropping of legumes with sorghum increased the land equivalent ratio (LER) by 34 to 43% (LER ranged from 1.34 to 1.43) with different treatment

combinations (Table 1). Further when soybean was intercropped with sorghum, the LER was increased to an extent of 43% (LER of 1.43). In this treatment combination, the yield of sorghum (4.84 t ha<sup>-1</sup>) component was 92.2% as compared to sole crop of sorghum (5.25 t ha<sup>-1</sup>), showing no much difference in the in land utilization when sorghum was grown in combination with soybean. The reduction in LER was to an extent of 7.8% only as compared to sole sorghum. Lower LER (1.34) was observed in blackgram intercropped with sorghum due to intercrop competition. In this investigation the increased yield of intercropping of sorghum and soybean was due to better complementary use of resources such as light, nutrients and moisture as compared other intercropping systems. Rao and Willey (1980) indicated that distinct differences in maturity period of component crops usually resulted in quite large yield advantage. This type of combination clearly allowed for better use of resources over time.

Similarly, area time equivalent ratio (ATER) which considers the duration of the individual crops and the system is the better indicator of land use efficiency than LER. Sowing time did not show any significant difference between dry and normal sown conditions. Whereas intercropping of soybean with sorghum in 2:2 row proportion resulted in significantly higher ATER (1.27) indicating higher per day productivity from the system followed by sorghum with greengram (1.22) and blackgram (1.17) which were on par with each other. This was possible due to greater temporal and spatial complementarity. The higher value of LER than ATER for the same treatment indicated that ATER considers duration of crops and system is better index than LER that does not consider crop duration. These results are in agreement with Rathod *et al.* (2011).

The increase in yield is also attributed to better use of

Table 1. Effect of sowing time and cropping systems on resource use efficiency, sorghum equivalent yield (SEY), available nutrient status and economics

| Treatment                                  | Yield (t ha <sup>-1</sup> ) |         | SEY<br>(t ha <sup>-1</sup> ) | LTR at<br>60 DAS | LER  | ATER | Available nutrients (kg ha <sup>-1</sup> ) |                               |                  | Economics                                                |                                                        |              |
|--------------------------------------------|-----------------------------|---------|------------------------------|------------------|------|------|--------------------------------------------|-------------------------------|------------------|----------------------------------------------------------|--------------------------------------------------------|--------------|
|                                            | Sorghum                     | Legumes |                              |                  |      |      | N                                          | P <sub>2</sub> O <sub>5</sub> | K <sub>2</sub> O | Gross returns<br>(x 10 <sup>3</sup> ₹ ha <sup>-1</sup> ) | Net returns<br>(x 10 <sup>3</sup> ₹ ha <sup>-1</sup> ) | B:C<br>ratio |
| <i>Sowing time (S)</i>                     |                             |         |                              |                  |      |      |                                            |                               |                  |                                                          |                                                        |              |
| S <sub>1</sub> - Dry sowing                | 5.19                        | 1.27    | 5.09                         | 32.29            | 1.19 | 1.10 | 240.01                                     | 32.43                         | 383.83           | 73.6                                                     | 51.8                                                   | 2.42         |
| S <sub>2</sub> - Normal sowing             | 4.31                        | 1.03    | 4.10                         | 33.03            | 1.17 | 1.08 | 239.81                                     | 32.36                         | 383.29           | 58.6                                                     | 38.8                                                   | 1.80         |
| SEm±                                       | 1.51                        | 0.62    | 1.55                         | 0.35             | 0.03 | 0.02 | 0.54                                       | 0.23                          | 1.53             | 2.6                                                      | 2.1                                                    | 0.10         |
| CD (P=0.05)                                | 4.59                        | 1.81    | 4.52                         | NS               | NS   | NS   | NS                                         | NS                            | NS               | 7.5                                                      | 6.1                                                    | 0.28         |
| <i>Cropping system (T)</i>                 |                             |         |                              |                  |      |      |                                            |                               |                  |                                                          |                                                        |              |
| T <sub>1</sub> :Sorghum + greengram (2:2)  | 4.59                        | 0.71    | 5.34                         | 30.90            | 1.38 | 1.22 | 231.93                                     | 31.31                         | 383.03           | 97.8                                                     | 65.1                                                   | 2.89         |
| T <sub>2</sub> :Sorghum + soybean (2:2)    | 4.84                        | 1.20    | 7.49                         | 28.98            | 1.43 | 1.27 | 245.06                                     | 32.51                         | 385.05           | 104.8                                                    | 81.6                                                   | 3.52         |
| T <sub>3</sub> : Sorghum + blackgram (2:2) | 4.32                        | 0.39    | 6.24                         | 30.61            | 1.34 | 1.17 | 233.53                                     | 30.64                         | 382.26           | 78.9                                                     | 56.7                                                   | 2.55         |
| T <sub>4</sub> :Sole sorghum               | 5.25                        |         | 5.25                         | 40.71            | 1.00 | 1.00 | 216.40                                     | 28.49                         | 378.69           | 86.9                                                     | 66.0                                                   | 3.15         |
| T <sub>5</sub> :Sole greengram             |                             | 1.45    | 1.74                         | 32.84            | 1.00 | 1.00 | 248.70                                     | 34.09                         | 385.25           | 27.8                                                     | 15.2                                                   | 1.21         |
| T <sub>6</sub> :Sole soybean               |                             | 2.36    | 4.93                         | 31.60            | 1.00 | 1.00 | 252.21                                     | 35.45                         | 386.21           | 51.2                                                     | 31.2                                                   | 1.65         |
| T <sub>7</sub> :Sole blackgram             |                             | 0.79    | 1.16                         | 32.98            | 1.00 | 1.00 | 251.52                                     | 34.29                         | 384.50           | 18.4                                                     | 6.3                                                    | 0.52         |
| SEm±                                       | 1.51                        | 1.07    | 2.91                         | 0.66             | 0.05 | 0.05 | 1.01                                       | 0.40                          | 2.86             | 4.8                                                      | 3.9                                                    | 0.18         |
| CD (P=0.05)                                | 4.59                        | 3.14    | 8.45                         | 1.91             | 0.14 | 0.13 | 4.17                                       | 1.65                          | NS               | 14.0                                                     | 11.4                                                   | 0.53         |
| <i>Interaction</i>                         |                             |         |                              |                  |      |      |                                            |                               |                  |                                                          |                                                        |              |
| SEm±                                       | 3.03                        | 1.52    | 4.11                         | 0.93             | 0.07 | 0.06 | 1.43                                       | 0.57                          | 4.05             | 6799                                                     | 5568                                                   | 0.26         |
| CD (P=0.05)                                | NS                          | 4.44    | NS                           | NS               | NS   | NS   | NS                                         | NS                            | NS               | NS                                                       | NS                                                     | NS           |

**Initial nutrient content:** N = 228.50 kg ha<sup>-1</sup>; P<sub>2</sub>O<sub>5</sub> = 32.43 kg ha<sup>-1</sup>; K<sub>2</sub>O = 406.30 kg ha<sup>-1</sup>; DAS = Days after sowing; NS = Non significant; LTR = Light transmission ratio; LER = Land equivalent ratio; ATER = Area time equivalent ratio; *Market price:* Sorghum = ₹ 1500 q<sup>-1</sup>; Greengram = ₹ 1800 q<sup>-1</sup>; Soybean = ₹ 2000 q<sup>-1</sup>; Blackgram = ₹ 2200 q<sup>-1</sup>

available nutrients by the component crops. Available nitrogen in soil was more in sole legume treatments as compared to sole sorghum. Sole soybean recorded significantly higher available nitrogen ( $252.2 \text{ kg ha}^{-1}$ ) in soil than the rest of the treatments. The higher number of nodules  $\text{plant}^{-1}$  could be attributed to addition of nitrogen to soil through symbiotic nitrogen fixation and by defoliations which in turn enhanced the activity of micro-organisms in the soil. Similar results were reported by Kumar *et al.* (2011). Phosphorus availability in soil was significantly higher under sole soybean ( $35.45 \text{ kg ha}^{-1}$ ) and it was on par with sole blackgram ( $34.29 \text{ kg ha}^{-1}$ ) and sole greengram ( $34.09 \text{ kg ha}^{-1}$ ). Significantly lower phosphorus availability was noticed in sole sorghum ( $28.45 \text{ kg ha}^{-1}$ ) treatment. This could be attributed to the fact that sorghum crop is non-leguminous and it consumes more phosphorus for its root development and protein synthesis in plant. However, the higher potassium availability was recorded in sole soybean treatment ( $386.21 \text{ kg ha}^{-1}$ ), followed by sole greengram ( $385.25 \text{ kg ha}^{-1}$ ). Dahmardeh *et al.* (2010) found that intercropping increased the amount of nitrogen (N), phosphorous (P) and potassium (K) contents as compared to sole maize.

The impact of sowing time and resource utilization is depicted in terms of economic benefits. Dry sowing recorded significantly higher gross return ( $\text{₹ } 73.6 \times 10^3 \text{ ha}^{-1}$ ), net return ( $\text{₹ } 51.8 \times 10^3 \text{ ha}^{-1}$ ) and benefit: cost ratio (2.42) as compared to normal sowing ( $\text{₹ } 58.60 \times 10^3 \text{ ha}^{-1}$ ,  $\text{₹ } 38.8 \times 10^3 \text{ ha}^{-1}$  and 1.80 of gross return, net return and B:C ratio, respectively). Sorghum with soybean recorded significantly higher gross return ( $\text{₹ } 104.8 \times 10^3 \text{ ha}^{-1}$ ), net return ( $\text{₹ } 81.6 \times 10^3 \text{ ha}^{-1}$ ) and B:C ratio (3.52) as compared to sorghum with greengram intercropping system (Table 1). Similar results were reported by Reddy *et al.* (2010). The interaction effect due to sowing time and cropping system

was insignificant.

On the basis of present investigation, it can be concluded that sorghum can be grown as intercrop with soybean under dry sown conditions for higher profitability and effective use of scarce resources without affecting the nutrient status of soil.

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## SHORT COMMUNICATION

# Spikelet sterility in hybrid rice (*Oryza sativa*) as influenced by sources and levels of nutrients

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## ABSTRACT

An experiment was conducted for two years during *kharif* 2009 and 2010 at College Farm, College of Agriculture, Rajendranagar, Hyderabad, India to study the effect of organic sources and fertilizer levels on spikelet sterility in hybrid rice (*Oryza sativa* L.). The experiment was laid out in split plot design with three replications. The treatments included organic manures (control – no organic manuring, subabul incorporation @ 5 t ha<sup>-1</sup>, rice straw incorporation @ 2.5 t ha<sup>-1</sup>) as main plot treatments and fertilizer levels comprising of N:K<sub>2</sub>O kg ha<sup>-1</sup> (150:75, 175:50, 175:25, 200:50, 200:25, 225:0) as sub plot treatments. Among the organic sources, incorporation of subabul @ 5 t ha<sup>-1</sup> recorded the highest number of total grains panicle<sup>-1</sup>, filled grains panicle<sup>-1</sup>, lowest number of unfilled grains panicle<sup>-1</sup> and spikelet sterility in both the years. Among the fertilizer levels tested, 200:50 N:K<sub>2</sub>O kg ha<sup>-1</sup> recorded the highest number of total grains panicle<sup>-1</sup>, filled grains panicle<sup>-1</sup>, while 150:75 N:K<sub>2</sub>O kg ha<sup>-1</sup> recorded the lowest spikelet sterility. The highest spikelet sterility was recorded with the application of 225:0 N:K<sub>2</sub>O kg ha<sup>-1</sup> in both the years. Interaction effect was found significant on total grains panicle<sup>-1</sup> and filled grains panicle<sup>-1</sup>. Subabul incorporation @ 5 t ha<sup>-1</sup> + 200:50 N:K<sub>2</sub>O kg ha<sup>-1</sup> recorded the highest number of total and filled grains panicle<sup>-1</sup> and was comparable to subabul incorporation @ 5 t ha<sup>-1</sup> + 200:25 N:K<sub>2</sub>O kg ha<sup>-1</sup>.

**Key words:** Fertilizer levels, Filled grains, Hybrid rice, Rice straw, Spikelet sterility, Subabul, Unfilled grains

Rice occupies a pivotal position in the food security system of India contributing to 41.5% of total food grain production. In order to meet our future demand for rice, enhancing its production is the only way which could be attained only through the exploitation of hybrid vigour. Hybrid rice has a yield advantage of about 15-20% higher than that of the best commercial high yielding varieties (Yuan and Virmani, 1988). Hybrid rice accumulates more dry matter, which results in more spikelets per panicle (sink) and increased sink size often results in a decrease in grain filling percentage (Peng *et al.*, 1996). Hence, for obtaining higher yield in hybrid rice, sink is not the limiting factor and fertilizer management should focus on spikelet filling. Keeping in view, an attempt was made to study the effect of organic sources and fertilizer levels on spikelet sterility of hybrid rice.

An experiment was conducted for two years during *kharif* 2009 and 2010 at College Farm, College of Agriculture, Rajendranagar, Hyderabad. The soil of the experimental site was sandy clay loam in texture, low in available nitrogen (242 kg ha<sup>-1</sup>), medium in available phosphorus (39.4 kg ha<sup>-1</sup>) and high in available potassium (368 kg ha<sup>-1</sup>). The experiment was laid out in split plot design with three replications. The

treatments consisted of organic sources (control – no organic manuring, subabul incorporation @ 5 t ha<sup>-1</sup>, rice straw incorporation @ 2.5 t ha<sup>-1</sup>) as main plots, and fertilizer levels comprising of N:K<sub>2</sub>O kg ha<sup>-1</sup> (150:75, 175:50, 175:25, 200:50, 200:25, 225:0) as sub-plots. A common dose of 75 kg P<sub>2</sub>O<sub>5</sub> ha<sup>-1</sup> was applied to all the plots. Measured quantities of subabul twigs and rice straw were incorporated in the respective treatment plot twelve days before transplanting. The entire dose of P<sub>2</sub>O<sub>5</sub> and half dose of K<sub>2</sub>O were applied basally while N was applied in three equal splits i.e. at transplanting, maximum tillering and at panicle initiation stage. The remaining K<sub>2</sub>O was applied at flowering stage of the crop. The hybrid used was KRH-2. Twenty five and twenty one days old seedlings were transplanted during 2009 and 2010 respectively. Data on total grains panicle<sup>-1</sup>, filled grains panicle<sup>-1</sup>, unfilled grains panicle<sup>-1</sup>, spikelet sterility and 1000 grain weight was subjected to statistical analysis by applying analysis of variance for split plot design and significance was tested by F-test (Snedecor and Cochran, 1967).

In both the years, subabul incorporation @ 5 t ha<sup>-1</sup> (M<sub>2</sub>) recorded significantly higher number of total and filled grains panicle<sup>-1</sup> compared to rice straw incorporation @ 2.5 t ha<sup>-1</sup> (M<sub>3</sub>)

and control ( $M_1$ ) (Table 1). Adequate availability of nutrients might be the reason for higher number of total and filled grains panicle<sup>-1</sup> in  $M_2$ . The results are in line with the findings of Neelima (2005). Among the fertilizer levels tested, 200:50 (N:K<sub>2</sub>O kg ha<sup>-1</sup>) recorded the highest number of total and filled grains panicle<sup>-1</sup>. Balanced fertilization of nitrogen and potassium might have helped in registering higher number of total and filled grains panicle<sup>-1</sup> in  $F_4$  (200:50 N:K<sub>2</sub>O kg ha<sup>-1</sup>). Mondal *et al.* (1982) also emphasized on the importance of balanced fertilization of N and K in rice. Interaction effect between organic sources and fertilizer levels was found significant on number of total grains panicle<sup>-1</sup> and filled grains panicle<sup>-1</sup> (Table 2 and 3). Fertilizer levels  $F_2$  and  $F_3$  (nitrogen constant at 175 kg ha<sup>-1</sup> and potassium at 50 and 25 kg ha<sup>-1</sup>, respectively) gave on par results under manurial treatments  $M_2$  and  $M_3$ . Similarly, fertilizer levels  $F_4$  and  $F_5$  (nitrogen constant at 200 kg ha<sup>-1</sup> and potassium at 50 and 25 kg ha<sup>-1</sup>, respectively) gave on par results indicating that application of organic sources along with high levels of nitrogen

can help in reducing the application of potassium by 25 kg ha<sup>-1</sup>. Green leaf manuring with subabul @ 5 t ha<sup>-1</sup> coupled with 200:50 N:K<sub>2</sub>O kg ha<sup>-1</sup> ( $M_2F_4$ ) recorded maximum number of filled grains panicle<sup>-1</sup> in both the years and remained on par with  $M_2F_5$ .

The lowest number of unfilled grains panicle<sup>-1</sup> was recorded with subabul incorporation @ 5 t ha<sup>-1</sup> ( $M_2$ ) which was significantly superior over rice straw incorporation @ 2.5 t ha<sup>-1</sup> ( $M_3$ ) and no organic manuring ( $M_1$ ) (Table 1). Better filling of grains and reduced spikelet sterility resulted in less number of unfilled grains panicle<sup>-1</sup>. The beneficial effect of green manuring in rice to realize less number of unfilled grains was also reported by Neelima (2005). Among the fertilizer levels, application of 150:75 N:K<sub>2</sub>O kg ha<sup>-1</sup> recorded the lowest number of unfilled grains panicle<sup>-1</sup>. The reduced number of unfilled grains under high level of K application was ascribed due to increasing photosynthetic activity as K stimulates some vital-biochemical

Table 1. Total, filled, unfilled grains panicle<sup>-1</sup> and spikelet sterility (%) in hybrid rice as influenced by organic sources and fertilizer levels

| Treatment                                                      | Total grains panicle <sup>-1</sup> |      | Filled grains panicle <sup>-1</sup> |      | Unfilled grains panicle <sup>-1</sup> |      | Spikelet sterility (%) |       |
|----------------------------------------------------------------|------------------------------------|------|-------------------------------------|------|---------------------------------------|------|------------------------|-------|
|                                                                | 2009                               | 2010 | 2009                                | 2010 | 2009                                  | 2010 | 2009                   | 2010  |
| <i>Organic sources</i>                                         |                                    |      |                                     |      |                                       |      |                        |       |
| $M_1$ : No organic manuring (control)                          | 150                                | 152  | 116                                 | 123  | 34.0                                  | 29.0 | 22.66                  | 19.08 |
| $M_2$ : Subabul incorporation @ 5 t ha <sup>-1</sup>           | 165                                | 164  | 136                                 | 136  | 29.0                                  | 28.0 | 17.58                  | 17.07 |
| $M_3$ : Rice straw incorporation @ 2.5 t ha <sup>-1</sup>      | 155                                | 159  | 125                                 | 130  | 30.0                                  | 29.0 | 19.35                  | 18.24 |
| SEm $\pm$                                                      | 2                                  | 1    | 3                                   | 2    | 0.7                                   | 0.07 | 0.01                   | 0.03  |
| CD (P=0.05)                                                    | 4                                  | 3    | 7                                   | 5    | 1.9                                   | 0.22 | 0.02                   | 0.06  |
| <i>Fertilizer levels (N:K<sub>2</sub>O kg ha<sup>-1</sup>)</i> |                                    |      |                                     |      |                                       |      |                        |       |
| $F_1$ : 150:75                                                 | 143                                | 147  | 117                                 | 123  | 26.0                                  | 24.0 | 18.18                  | 16.33 |
| $F_2$ : 175:50                                                 | 156                                | 157  | 127                                 | 130  | 29.0                                  | 27.0 | 18.59                  | 17.20 |
| $F_3$ : 175:25                                                 | 148                                | 150  | 119                                 | 123  | 29.0                                  | 27.0 | 19.59                  | 18.00 |
| $F_4$ : 200:50                                                 | 168                                | 168  | 135                                 | 138  | 33.0                                  | 30.0 | 19.64                  | 17.86 |
| $F_5$ : 200:25                                                 | 164                                | 166  | 130                                 | 134  | 34.0                                  | 32.0 | 20.73                  | 19.28 |
| $F_6$ : 225:0                                                  | 160                                | 162  | 125                                 | 130  | 35.0                                  | 32.0 | 21.88                  | 19.75 |
| SEm $\pm$                                                      | 1                                  | 1    | 1                                   | 1    | 1.0                                   | 0.6  | 0.04                   | 0.04  |
| CD (P=0.05)                                                    | 2                                  | 2    | 3                                   | 2    | 2.2                                   | 1.2  | 0.08                   | 0.09  |
| <i>M x F</i>                                                   |                                    |      |                                     |      |                                       |      |                        |       |
| SEm $\pm$                                                      | 2                                  | 2    | 3                                   | 3    | 1.8                                   | 0.2  | 0.08                   | 0.09  |
| CD (P=0.05)                                                    | 5                                  | 4    | 8                                   | 6    | NS                                    | NS   | NS                     | NS    |

Table 2. Interaction effect of organic sources and fertilizer levels on total number of grains panicle<sup>-1</sup> and number of filled grains panicle<sup>-1</sup> in hybrid rice

| Organic sources                                                    | Fertilizer levels (N:K <sub>2</sub> O kg ha <sup>-1</sup> ) |                |                |                |                |                |                |                |                |                |                |                |
|--------------------------------------------------------------------|-------------------------------------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
|                                                                    | F <sub>1</sub>                                              | F <sub>2</sub> | F <sub>3</sub> | F <sub>4</sub> | F <sub>5</sub> | F <sub>6</sub> | F <sub>1</sub> | F <sub>2</sub> | F <sub>3</sub> | F <sub>4</sub> | F <sub>5</sub> | F <sub>6</sub> |
|                                                                    | 150:75                                                      | 175:50         | 175:25         | 200:50         | 200:25         | 225:0          | 150:75         | 175:50         | 175:25         | 200:50         | 200:25         | 225:0          |
|                                                                    | Total number of grains panicle <sup>-1</sup>                |                |                |                |                |                |                |                |                |                |                |                |
|                                                                    | 2009                                                        |                |                |                |                |                | 2010           |                |                |                |                |                |
| M <sub>1</sub> : No manuring (control)                             | 137                                                         | 155            | 132            | 162            | 157            | 154            | 143            | 157            | 137            | 162            | 159            | 155            |
| M <sub>2</sub> :Subabul incorporation @ 5 t ha <sup>-1</sup>       | 150                                                         | 162            | 161            | 174            | 172            | 170            | 150            | 160            | 159            | 174            | 173            | 168            |
| M <sub>3</sub> : Rice straw incorporation @ 2.5 t ha <sup>-1</sup> | 143                                                         | 150            | 150            | 168            | 164            | 157            | 148            | 154            | 153            | 169            | 165            | 163            |
|                                                                    | SEm±                                                        |                |                | CD (P=0.05)    |                |                | SEm±           |                |                | CD (P=0.05)    |                |                |
| F at same level of M                                               | 2                                                           |                |                | 4              |                |                | 2              |                |                | 4              |                |                |
| M at same or different level of F                                  | 2                                                           |                |                | 5              |                |                | 2              |                |                | 6              |                |                |
|                                                                    | Number of filled grains panicle <sup>-1</sup>               |                |                |                |                |                |                |                |                |                |                |                |
|                                                                    | 2009                                                        |                |                |                |                |                | 2010           |                |                |                |                |                |
| M <sub>1</sub> : No manuring (control)                             | 110                                                         | 124            | 104            | 126            | 119            | 115            | 118            | 128            | 111            | 131            | 125            | 122            |
| M <sub>2</sub> :Subabul incorporation @ 5 t ha <sup>-1</sup>       | 125                                                         | 134            | 133            | 143            | 141            | 137            | 126            | 133            | 132            | 144            | 142            | 136            |
| M <sub>3</sub> : Rice straw incorporation @ 2.5 t ha <sup>-1</sup> | 117                                                         | 122            | 121            | 135            | 131            | 124            | 124            | 128            | 126            | 138            | 134            | 130            |
|                                                                    | SEm±                                                        |                |                | CD (P=0.05)    |                |                | SEm±           |                |                | CD (P=0.05)    |                |                |
| F at same level of M                                               | 2                                                           |                |                | 5              |                |                | 2              |                |                | 5              |                |                |
| M at same or different level of F                                  | 3                                                           |                |                | 8              |                |                | 3              |                |                | 6              |                |                |

process like oxidative phosphorylation and photophosphorylation. Similar findings regarding the role of potassium were reported by Raju *et al.* (1999) and Madhavilatha (2001). Interaction effects were found non-significant.

The highest spikelet sterility was recorded with control ( $M_1$ ) treatment and lowest in the subabul green leaf manuring treatment ( $M_2$ ) (Table 2). The sufficient supply of carbohydrates might have resulted in better filling of spikelets which in turn reduced the sterility percentage of spikelets in  $M_2$ . Rice straw incorporation @  $2.5 \text{ t ha}^{-1}$  ( $M_3$ ) also recorded lower values of spikelet sterility when compared to control during both the years. Similar results were reported by Balaji Naik (2002). Among the sub plot treatments, application of  $150:75 \text{ N:K}_2\text{O kg ha}^{-1}$  and  $225:0 \text{ N:K}_2\text{O kg ha}^{-1}$  recorded the lowest and the highest spikelet sterility respectively. Spikelet sterility increased with increasing levels of nitrogen. The highest spikelet sterility in  $225:0 \text{ N:K}_2\text{O kg ha}^{-1}$  might be due to the deficiency of potassium. K deficient plants show poor germination of pollen in the floret which lead to high spikelet sterility (Von Uexhull, 1978 and Madhavilatha, 2001).

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## SHORT COMMUNICATION

# Yield and nutrient uptake of winter maize (*Zea mays*) with vegetable intercropping

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## ABSTRACT

A field experiment was carried out during winter season of 2010-2011 at Varanasi, Uttar Pradesh to assess the effect of winter maize (*Zea mays* L.) intercropped with vegetables such as radish (*Raphanus sativus* L.), spinach (*Spinacia oleracea* L.) and carrot (*Daucus carota*) on yield and nutrient uptake. Grain and straw yield, harvest index and shelling percentage of maize with all the intercropping systems were lower than the sole cropping of maize. Protein content (8.59%) and protein yield (593.41 kg ha<sup>-1</sup>) were highest in sole cropping of maize (normal planting) than intercropped maize, but organic carbon (0.30%) and organic matter (0.53%) were recorded highest in intercropping treatments. Values of land equivalent ratio with all the intercropping systems were greater indicating advantage in yield, land-use efficiency and monetary return unit<sup>-1</sup> time and space over the respective monocultures. Maize (paired)+ carrot was proved to be the most efficient, productive and remunerative cropping system as it gave the highest maize equivalent yield (28.25 t ha<sup>-1</sup>) and also accounted for higher values for net returns (₹184.8 x 10<sup>3</sup> ha<sup>-1</sup>) and B:C ratio (3.86) compared to other intercropping systems. Intercropping increased total N, P and K uptake by grain and stover of maize. Among intercropping systems, highest total NPK uptake were recorded in the maize (paired) + spinach intercropping system, followed by maize + carrot and minimum in maize + radish.

**Key words:** Carrot, Maize, Nutrient uptake, Radish, Spinach, Yield attributes, Yield

Substantial yield advantages can be achieved by intercropping compared to sole cropping which facilitates better use of resources like nutrients, and space when grown together rather than separately (Willey, 1979). Introduction of high yielding and thermo-insensitive hybrids of maize (*Zea mays* L.) has made its cultivation well adapted in winter season. Inter-row space in maize during the initial slow growth period provides ample scope to cultivate the compatible crop in between 2 rows of maize and increase the productivity per unit area and time. Radish (*Raphanus sativus* L.) and spinach (*Spinacia oleracea* L.) had been reported as the most profitable intercrops with winter maize in the Northern plains of the country (Singh and Kumar, 2002). Fertilizer requirement of intercropping system may also vary from sole cropping owing to inclusion of crop of dissimilar nature. The role of fertilization in winter maize and maize based cropping systems is well established (Sinha et al., 1999). Information on winter maize with intercrop like radish, spinach and carrot (*Daucus carota*) and their nutrient uptake is very meagre. Hence, present study were undertaken to generate information on best suited winter maize-based intercropping system under Central plain zone of Uttar Pradesh, regarding the yield and nutrient uptake.

An experiment was carried out during the winter (*rabi*)

season of 2010-2011 at the Agriculture Research Farm, Institute of Agricultural Sciences, Banaras Hindu University, Varanasi. The soil is alluvial with sandy loam texture and deep. The pH range is neutral to slightly alkaline in reaction (PH 7.6), well drained and moderately fertile being medium in organic carbon (0.52%), low in available nitrogen (145.0 kg ha<sup>-1</sup>), medium in available phosphorus (15.5 kg ha<sup>-1</sup>) and available potassium (162.5 kg ha<sup>-1</sup>). There were 11 treatments combination comprising T<sub>1</sub> - sole maize 'Bio-seed-9544' (normal), T<sub>2</sub> - sole maize (paired), T<sub>3</sub> - sole radish 'Hill queen', T<sub>4</sub> - sole spanish 'Local', T<sub>5</sub> - sole carrot 'E.N.', T<sub>6</sub> - maize normal + radish, T<sub>7</sub> - maize paired + radish, T<sub>8</sub> - maize normal + spanish, T<sub>9</sub> - maize paired + spanish, T<sub>10</sub> - maize normal + carrot, and T<sub>11</sub> - maize paired + carrot, tried in randomized block design with four replications. The experimental crops were sown on 14 December 2010 and harvested at different dates i.e. maize on 16 May 2011, radish and carrot uprooted on 13 February 2011 and 09 April 2011, respectively, 2 cuttings of spinach taken from first fortnight of February to end of February during the experimentation year. Maize was sown at two different spacing 75 cm space in between rows and paired row planting 100/50 cm space in between rows and 20 cm plant to plant spacing in both of planting method. Among the intercrops (radish, spinach and carrot) row to row

spacing was 25 cm. Recommended package of practices was followed to raise the healthy crops. Maize was fertilized with 150 kg N, 90 kg P and 90 kg K, while 50 kg N, 100 kg P and 50 kg K in radish, 120 kg N, 60 kg P and 60 kg K in carrot and 35 kg N, 50 kg P and 50 kg K ha<sup>-1</sup> in spinach. In intercropping, the crops received the fertilizers on the basis of proportionate area under each crop. Full recommended doses of P and K along with one-third N to maize, 50% N to radish, spinach and carrot was applied as basal to all the crops in sole as well as intercropping system. Remaining two-third N to winter maize was top-dressed in 2 equal splits at knee high and teaselling stage. Rest 50% N was applied after first irrigation to all the intercrops. Fertilizer requirement of all the crops was met through urea, single super phosphate (SSP) and murate of potash (MOP). For the computation of maize grain equivalents and economics per 100 kg market price of ₹ 800, ₹ 80, ₹ 500, ₹ 500 and ₹ 1000 for maize, maize stover, radish, spinach and carrot, respectively were used. Soil samples after harvest of crops were collected for analysing the organic carbon and available nitrogen, phosphorus and potassium contents as per the standard analytical method. The Initial available nutrients found in experimental field soil were 145.0, 15.5 and 162.5 (NPK kg ha<sup>-1</sup>).

There was significant reduction in shelling % and harvest index of winter maize in intercropping system which was higher in normal planting maize (sole) (Table 1). Cobs plot<sup>-1</sup> was maximum in maize (paired) + spinach (43.75) and maize (paired) + carrot (43.75) intercropping systems. Straw yield was recorded maximum in maize (normal) + spinach (10.14 t ha<sup>-1</sup>), followed by maize (paired) + carrot (8.40 t ha<sup>-1</sup>) and lowest in maize (paired) + radish (7.31 t ha<sup>-1</sup>). Reduced yield attributes of maize when intercropped with radish, spinach and carrot might be due to less nutrient availability to maize plant and competitions for resources like light and moisture, and also due to antagonistic effect of allelo-chemicals by radish, spinach, carrot and maize plant roots. These results are in accordance with the result of Jha *et al.* (2002) and this was also supported by Jha *et al.* (2000) which reported higher yield attributes of winter maize under sole cropping. Shelling % was found to be increased in paired sole planting of maize (100/50 cm) over normal sole maize

(75 cm x 20 cm). Significantly better yield parameters under sole maize system (normal and paired) and reduced values of those attributes under intercropping systems (maize + radish, maize + spinach, and maize + carrot planted in normal and paired pattern), also affected the grain yield of maize under both the conditions. Sole production system of maize produced greater grain yield, while comparatively low grain yield was obtained under intercropping systems. Karim *et al.* (1989) opined with the finding of present study.

Land equivalent ratio (LER) of all the intercropping combination at both the planting methods (normal and paired planting) were greater than unity indicating higher land-use efficiency of intercropping over the respective sole cropping (Table 1). The highest LER recorded from maize + spinach (1.59) resulted in 59% more land-use efficiency. Sodani *et al.* (1993) also found that intercropping systems gave higher maize equivalent yield and land equivalent ratio than their respective sole crops.

Benefit: cost ratio (B:C) and net return recorded higher in intercropping system than their respective sole crops, except sole carrot (Table 1). Intercropping of carrot with winter maize registered higher B:C ratio (4.97), followed by maize + radish (3.62 and 3.86) and minimum in maize + spinach intercropping system, which was better than the sole systems of maize. Similar results were reported by Singh *et al.* (1993), who showed that maize + lentil fetched maximum monetary returns. Maize + carrot intercropping system fetched highest net return (₹ 184.8 x 10<sup>3</sup> ha<sup>-1</sup>), which were higher over the sole cropping of maize.

Intercropping system influenced the uptake of NPK nutrients in plant and found to be reduced in those treatments which showed poor growth as well as lower dry matter accumulation (Table 2). Sole maize removed comparatively higher amount of N (126.89 kg ha<sup>-1</sup>), P (25.15 kg ha<sup>-1</sup>), and K (280.36 kg ha<sup>-1</sup>), which might be attributed to its inherent exhaustive nature and ability to produce higher total biomass production resulting in higher absorption of nutrient. The total K uptake was quite high in comparison to uptake of N and P in all the treatments. Total N, P and K uptake of the system was

Table 1. Effect of intercropping on yield and land equivalent ratio (LER) of winter maize-based cropping systems

| Treatment                            | Number of cobs plot <sup>-1</sup> | Shelling % | Grain yield (q ha <sup>-1</sup> ) |           | Straw yield (q ha <sup>-1</sup> ) | Harvest index (%) | Maize equivalent yield (t ha <sup>-1</sup> ) | LER   |           |       | Net return (x 10 <sup>3</sup> ₹ ha <sup>-1</sup> ) | B:C ratio |
|--------------------------------------|-----------------------------------|------------|-----------------------------------|-----------|-----------------------------------|-------------------|----------------------------------------------|-------|-----------|-------|----------------------------------------------------|-----------|
|                                      |                                   |            | Maize                             | Intercrop |                                   |                   |                                              | Maize | Intercrop | Total |                                                    |           |
| T <sub>1</sub> : Maize (N) sole      | 43.50                             | 84.44      | 67.37                             |           | 74.46                             | 47.37             | 6.74                                         |       |           |       | 34.6                                               | 1.37      |
| T <sub>2</sub> : Maize (P) sole      | 43.50                             | 84.01      | 69.99                             |           | 82.60                             | 45.90             | 7.00                                         |       |           |       | 37.4                                               | 1.48      |
| T <sub>3</sub> : Radish (sole)       |                                   |            |                                   | 275.79    |                                   |                   | 17.23                                        |       |           |       | 109.2                                              | 3.80      |
| T <sub>4</sub> : Spinach (sole)      |                                   |            |                                   | 124.35    |                                   |                   | 7.77                                         |       |           |       | 43.9                                               | 2.41      |
| T <sub>5</sub> : Carrot (sole)       |                                   |            |                                   | 297.70    |                                   |                   | 37.21                                        |       |           |       | 247.8                                              | 4.97      |
| T <sub>6</sub> : Maize (N) + radish  | 43.75                             | 83.65      | 51.20                             | 168.53    | 67.49                             | 42.95             | 15.65                                        | 0.76  | 0.61      | 1.37  | 90.3                                               | 2.24      |
| T <sub>7</sub> : Maize (P) + radish  | 43.25                             | 83.67      | 56.58                             | 172.71    | 73.13                             | 43.50             | 16.43                                        | 0.81  | 0.63      | 1.44  | 97.0                                               | 2.41      |
| T <sub>8</sub> : Maize (N) + spinach | 43.50                             | 83.24      | 65.41                             | 73.45     | 101.39                            | 39.40             | 11.13                                        | 0.97  | 0.59      | 1.56  | 51.1                                               | 1.11      |
| T <sub>9</sub> : Maize (P) + spinach | 43.75                             | 83.95      | 67.26                             | 78.36     | 98.85                             | 40.48             | 11.62                                        | 0.96  | 0.63      | 1.59  | 54.8                                               | 1.19      |
| T <sub>10</sub> : Maize (N) + carrot | 43.25                             | 83.50      | 57.94                             | 168.53    | 79.65                             | 42.09             | 26.86                                        | 0.86  | 0.57      | 1.43  | 173.3                                              | 3.62      |
| T <sub>11</sub> : Maize (P) + carrot | 43.75                             | 84.00      | 63.21                             | 175.40    | 83.96                             | 43.11             | 28.25                                        | 0.91  | 0.59      | 1.51  | 184.8                                              | 3.86      |
| SEm±                                 | 0.364                             | 0.41       |                                   |           | 4.746                             | 1.813             | 0.82                                         |       |           |       |                                                    |           |
| CD (P=0.05)                          | NS                                | 1.23       |                                   |           | 13.959                            | 5.333             | 2.40                                         |       |           |       |                                                    |           |

N = Normal sowing; P = Paired sowing; B:C ratio = Benefit: cost ratio

Table 2. Effect of intercropping on N, P and K uptake (kg ha<sup>-1</sup>), protein content and protein yield (kg ha<sup>-1</sup>) of winter maize and soil organic carbon status

| Treatment                            | N uptake by grain | N uptake by stover | Total N uptake | P uptake by grain | P uptake by stover | Total P uptake | K uptake by grain | K uptake by stover | Total K uptake | Protein (%) | Protein yield by grain | Organic carbon (%) | Organic matter (%) |
|--------------------------------------|-------------------|--------------------|----------------|-------------------|--------------------|----------------|-------------------|--------------------|----------------|-------------|------------------------|--------------------|--------------------|
| T <sub>1</sub> : Maize (N) sole      | 94.95             | 31.95              | 126.89         | 16.18             | 7.66               | 23.84          | 34.38             | 209.05             | 243.43         | 8.59        | 593.41                 | 0.24               | 0.44               |
| T <sub>2</sub> : Maize (P) sole      | 84.18             | 37.41              | 121.59         | 16.46             | 8.70               | 25.15          | 36.23             | 244.12             | 280.36         | 7.50        | 526.11                 | 0.24               | 0.43               |
| T <sub>6</sub> : Maize (N) + radish  | 45.95             | 22.26              | 68.21          | 10.33             | 6.43               | 16.76          | 20.69             | 171.12             | 191.81         | 5.47        | 287.21                 | 0.30               | 0.53               |
| T <sub>7</sub> : Maize (P) + radish  | 57.64             | 31.25              | 88.89          | 12.00             | 7.71               | 19.71          | 24.71             | 201.82             | 226.53         | 6.25        | 360.24                 | 0.28               | 0.45               |
| T <sub>8</sub> : Maize (N) + spinach | 70.47             | 57.71              | 128.18         | 15.06             | 13.31              | 28.38          | 33.55             | 316.23             | 349.78         | 6.72        | 440.44                 | 0.26               | 0.43               |
| T <sub>9</sub> : Maize (P) + spinach | 78.36             | 62.13              | 140.49         | 15.55             | 13.13              | 28.68          | 34.49             | 305.57             | 340.06         | 7.19        | 489.72                 | 0.26               | 0.44               |
| T <sub>10</sub> : Maize (N) + carrot | 55.49             | 33.90              | 89.40          | 12.09             | 9.20               | 21.29          | 24.77             | 229.66             | 254.43         | 5.94        | 346.84                 | 0.29               | 0.50               |
| T <sub>11</sub> : Maize (P) + carrot | 70.85             | 39.17              | 110.03         | 13.64             | 9.82               | 23.45          | 29.83             | 243.08             | 272.92         | 6.88        | 442.83                 | 0.26               | 0.44               |
| SEm±                                 | 10.34             | 6.48               | 13.52          | 1.29              | 0.95               | 1.73           | 3.41              | 19.75              | 20.74          | 0.64        | 64.61                  | 0.01               | 0.02               |
| CD (P=0.05)                          | 30.40             | 19.06              | 39.76          | 3.79              | 2.80               | 5.10           | 10.03             | 58.08              | 60.99          | 1.88        | 190.01                 | 0.04               | 0.07               |

N = Normal sowing; P = Paired sowing

relatively higher than sole cropping, irrespective of the normal and paired row planting of maize. In intercropping system, normal maize with radish had lowest grain yield and paired maize with spinach had maximum grain yield. Hence, normal maize with radish exhibited lowest nutrient uptake and paired maize with spinach showed maximum nutrient uptake than the sole crop of maize. All the physiological and edaphological conditions related to growth and development and moreover, the maturity was conductive and congenial during growing time. Because of this vital reason maize plants showed greater uptake of N, P and K in the grain and stover and similarly the protein yield by grain. Similar results were also obtained by Kanakeri (1991), where the uptake of N, P and K by maize was reduced significantly due to intercropping as against sole cropping.

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## SHORT COMMUNICATION

# Production potential and quality of rice (*Oryza sativa*) varieties as influenced by date of transplanting in Southern Telangana

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## ABSTRACT

A field experiment was undertaken at Hyderabad, Telangana during *kharif* 2010 to study the effect of date of transplanting and varieties on production potential and quality of rice (*Oryza sativa* L.). Early planting (23<sup>rd</sup> July) recorded significantly higher yield and yield attributing characters, viz. panicles (338.3 no. m<sup>-2</sup>), spikelets panicle<sup>-1</sup> (176.7), test weight (18.7 g) and grain yield (6.23 t ha<sup>-1</sup>), while sterility % (14.2) was lowest under 5<sup>th</sup> August planting. Variety MTU 1010 registered higher grain yield (5.98 t ha<sup>-1</sup>) and yield attributing characters, viz. panicle (341.3 no. m<sup>-2</sup>), lowest sterility % (13.4), test weight (28.8 g) and harvest index (0.48) over other varieties, viz. RNR 2354, RNR 2458 and JGL 384. On the other hand, the drastic changes in rice quality before and after cooking were also noted with planting date and varieties.

**Key words:** Dates of transplanting, *Oryza sativa*, Quality, Production potential, Rice, Varieties

Rice (*Oryza sativa* L.) is specifically important to India because it is a choice crop of the millions of poor, small and marginal farmers not only for income but also for household food security. Paradigm shifts in planning research programmes and new approaches will need to be followed to enhance rice production and productivity in the decades ahead, to meet not only the domestic demand but also to have an exportable surplus. One of the normal scientific approaches, is to increase the yield of this crop through manipulation of agronomic practices. The optimal date of transplanting of any field crop depends on the environmental conditions required for good crop growth and development. The early seeding produces higher grain yield while delayed sowing generally decreases yield (Pandey *et al.* 2007). The importance of continuing to develop new rice varieties to guarantee India's food security and support the region's economic development needs no special emphasis. A significant interaction was found between transplanting date and variety on growth traits and grain yield (Vange and Obe, 2006). Hence, the present study was undertaken to find out the effect of date of transplanting and varieties on production potential and quality of rice under Southern Telangana Zone of Andhra Pradesh.

The experiment was conducted during *kharif* 2010 in sandy clay loam soil at the Agriculture Research Institute,

Acharya N.G. Ranga Agricultural University, Rajendranagar, Hyderabad, Telangana. The site is geographically situated at 17° 19'N Latitude, 78° 23' E Longitude and at an altitude of 542.3 m above mean sea level. The soil of experimental site was slightly alkaline in reaction (pH 7.8) medium in organic carbon (0.75%), low in available nitrogen (255 kg ha<sup>-1</sup>), medium in P<sub>2</sub>O<sub>5</sub> (49 kg ha<sup>-1</sup>) and K<sub>2</sub>O (218 kg ha<sup>-1</sup>) with electrical conductivity of 1.46 dS m<sup>-1</sup>. The treatments consisted of four dates of transplanting, viz. D<sub>1</sub> (23 July), D<sub>2</sub> (5 August), D<sub>3</sub> (20 August) and D<sub>4</sub> (5 September) and four varieties, viz. V<sub>1</sub> (MTU 1010), V<sub>2</sub> (RNR 2354), V<sub>3</sub> (RNR 2458) and V<sub>4</sub> (JGL 384) laid out in randomized block design with three replications. Nursery of four varieties was raised under field conditions according to suit the different dates of transplanting. Transplanting was done as per treatments with 30 days old seedlings raised in nursery plots @ 2 to 3 seedling hill<sup>-1</sup>, adopting a spacing of 15 cm × 15 cm. The rice varieties received fertilizer dose of 120 kg N, 60 kg P<sub>2</sub>O<sub>5</sub> and 40 kg K<sub>2</sub>O ha<sup>-1</sup>, respectively. An observation on yield attributing and yield characters was recorded. Quality characters such as preference for aroma, appearance and taste were assessed by score method as suggested by Juliano *et al.* (1965). The cooked samples were randomly tasted by nine judges and score was awarded as per the score evaluation chart given in Table 1.

Early transplanting i.e., 23<sup>rd</sup> July recorded significantly higher yield contributing characters, viz. panicles (no. m<sup>-2</sup>), spikelets panicle<sup>-1</sup> and test weight than other dates of transplanting (Table 2). Grain yield and harvest index were also highest under 23<sup>rd</sup> July transplanting which was significantly

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Table 1. Score evaluation chart

| Eating quality score | Aroma                   |
|----------------------|-------------------------|
| 1                    | No scent at all         |
| 2                    | Slightly weak scent     |
| 3                    | Moderately weak scent   |
| 4                    | Weak scent              |
| 5                    | Very weak scent         |
| 6                    | Moderately strong scent |
| 7                    | Strong scent            |
| 8                    | Very strong scent       |
| 9                    | Mostly strong scent     |

superior to other date; however straw yield was recorded higher under 20 August transplanting. The significantly linear decline in grain yield with every 15 days delay in planting was recorded from 23<sup>rd</sup> July to 5<sup>th</sup> September. The reduction in grain yield was recorded to the tune of 9.21, 19.05 and 27.80%, respectively under 5 August, 20 August and 5 September transplanting compared with 23<sup>rd</sup> July of planting. These results are in consonance with the findings of Yadav and Tripathi (2008) who also observed more number of yield contributing characters and higher yield in July planting than in delayed plantcrop. Regards to varieties, MTU 1010 was recorded significantly higher yield attributing parameters, viz. panicles (341.3 m<sup>-2</sup>), lower sterility % (13.4), test weight (24.8 g) and grain yield (5.98 t ha<sup>-1</sup>) and harvest index (0.48) but spikelets panicle<sup>-1</sup> (168.7) was found maximum in JGL 384 (Table 2). Whereas, straw yield was obtained higher in RNR 2458 which was comparable to MTU 1010 and significantly superior over rest of varieties. Finally the earlier transplanted crop on 23 July and variety MTU 1010 benefited from appropriate temperature

regimes, better sunshine, ample rainfall relative humidity that resulted into a more vigorous and extensive root system leading to increased vegetative growth means more efficient sink formation and greater sink size, ultimately reflected in higher grain yield of rice in these treatments were also observed by Murthy *et al.* (2007); Yadav and Tripathi (2008); Jena *et al.* (2010).

Quality parameters before cooking i.e. length and breadth of rice grain was not markedly influenced by dates of transplanting (Table 3). Higher length (5.8 mm), breadth (1.9 mm) and more hardness (6.3 kg m<sup>-2</sup>) of rice grain was noted under 5 August date of transplanting but % sound grains was found higher with 5 September planted crop with having smooth nature of breakage. However, variability in after cooking quality parameters was also observed with different date of transplanting. Quality parameters, viz. grain elongation ratio, colour, size, shape, texture, taste, flavour and overall appearance was observed better under 5 August transplanting (Table 4). These results are in close conformity with findings of Rao *et al.* (1996). Varieties showed marked effect on quality parameters of rice grain before (Table 3) and after cooking (Table 4). MTU 1010 registered higher length, breadth, hardness and % of sound grains with having smooth nature of breakage in before cooking and this variety follow similar trends in after cooking quality parameters, viz. grain elongation, colour, size, shape, texture, taste, flavour and overall appearance of grain over other varieties. Hardness of grain (4.6 kg m<sup>-2</sup>) before cooking was found lower in variety RNR 2458.

Table 2. Effect of planting dates and varieties on yield attributes and yield of rice

| Treatment                    | Panicles<br>(no. m <sup>-2</sup> ) | Spikelets<br>panicle <sup>-1</sup> | Sterility<br>% | Test weight<br>(g) | Grain yield<br>(t ha <sup>-1</sup> ) | Straw yield<br>(t ha <sup>-1</sup> ) | Harvest index<br>(%) |
|------------------------------|------------------------------------|------------------------------------|----------------|--------------------|--------------------------------------|--------------------------------------|----------------------|
| <i>Date of transplanting</i> |                                    |                                    |                |                    |                                      |                                      |                      |
| D <sub>1</sub> : July 23     | 338.3                              | 176.7                              | 15.7           | 18.7               | 6.23                                 | 5.75                                 | 0.52                 |
| D <sub>2</sub> : August 5    | 320.2                              | 164.5                              | 14.2           | 18.3               | 5.65                                 | 6.09                                 | 0.48                 |
| D <sub>3</sub> : August 20   | 296.7                              | 140.8                              | 20.4           | 18.5               | 5.04                                 | 6.64                                 | 0.43                 |
| D <sub>4</sub> : September 5 | 305.3                              | 123.1                              | 20.5           | 16.7               | 4.50                                 | 6.32                                 | 0.42                 |
| CD (P=0.05)                  | 17.4                               | 5.22                               | 1.05           | 0.68               | 0.25                                 | 0.16                                 | 0.02                 |
| <i>Varieties</i>             |                                    |                                    |                |                    |                                      |                                      |                      |
| V <sub>1</sub> : MTU 1010    | 341.3                              | 123.3                              | 13.4           | 24.8               | 5.98                                 | 6.46                                 | 0.48                 |
| V <sub>2</sub> : RNR 2354    | 317.8                              | 146.8                              | 15.4           | 16.1               | 5.08                                 | 5.97                                 | 0.46                 |
| V <sub>3</sub> : RNR 2458    | 297.5                              | 166.2                              | 17.7           | 16.7               | 5.52                                 | 6.56                                 | 0.46                 |
| V <sub>4</sub> : JGL 384     | 303.3                              | 168.7                              | 24.3           | 14.6               | 4.85                                 | 5.76                                 | 0.45                 |
| CD (P=0.05)                  | 17.4                               | 5.22                               | 1.05           | 0.68               | 0.25                                 | 0.16                                 | 0.02                 |

Table 3. Effect of planting dates and varieties on before cooking quality parameters of rice

| Treatment                    | Length (mm) | Breadth (mm) | Hardness (kg m <sup>-2</sup> ) | % sound grains | Nature of breakage |
|------------------------------|-------------|--------------|--------------------------------|----------------|--------------------|
| <i>Date of transplanting</i> |             |              |                                |                |                    |
| D <sub>1</sub> : July 23     | 5.6         | 1.9          | 5.1                            | 64.1           | Smooth             |
| D <sub>2</sub> : August 5    | 5.8         | 1.9          | 6.3                            | 72.5           | Smooth             |
| D <sub>3</sub> : August 20   | 5.6         | 1.8          | 5.7                            | 71.6           | Smooth             |
| D <sub>4</sub> : September 5 | 5.6         | 1.9          | 5.8                            | 75.4           | Smooth             |
| <i>Varieties</i>             |             |              |                                |                |                    |
| V <sub>1</sub> : MTU 1010    | 6.0         | 2.0          | 6.9                            | 75.0           | Smooth             |
| V <sub>2</sub> : RNR 2354    | 5.8         | 1.8          | 5.9                            | 74.0           | Smooth             |
| V <sub>3</sub> : RNR 2458    | 5.5         | 1.9          | 4.6                            | 71.2           | Smooth             |
| V <sub>4</sub> : JGL 384     | 5.2         | 1.8          | 5.4                            | 63.3           | Smooth             |

Table 4. Effect of planting dates and varieties on quality parameters after cooking of rice

| Treatment                    | Length<br>(mm) | Breadth<br>(mm) | Grain<br>elongation<br>ratio (mm) | Colour | Size | Shape | Texture | Taste | Flavour | Overall<br>appearance |
|------------------------------|----------------|-----------------|-----------------------------------|--------|------|-------|---------|-------|---------|-----------------------|
| <i>Date of transplanting</i> |                |                 |                                   |        |      |       |         |       |         |                       |
| D <sub>1</sub> : July 23     | 8.5            | 3.2             | 2.7                               | 5.5    | 6.3  | 6.3   | 6.0     | 5.8   | 5.8     | 5.8                   |
| D <sub>2</sub> : August 5    | 8.4            | 2.7             | 3.1                               | 7.5    | 7.8  | 8.0   | 7.5     | 7.0   | 6.8     | 7.3                   |
| D <sub>3</sub> : August 20   | 7.9            | 2.8             | 2.9                               | 7.5    | 7.5  | 7.5   | 7.3     | 7.0   | 7.5     | 7.0                   |
| D <sub>4</sub> : September 5 | 8.0            | 2.7             | 3.0                               | 3.8    | 4.5  | 4.3   | 5.0     | 4.8   | 4.5     | 5.0                   |
| <i>Varieties</i>             |                |                 |                                   |        |      |       |         |       |         |                       |
| V <sub>1</sub> : MTU 1010    | 8.9            | 3.0             | 3.0                               | 6.3    | 7.3  | 7.0   | 6.8     | 6.8   | 6.5     | 6.8                   |
| V <sub>2</sub> : RNR 2354    | 8.3            | 2.8             | 3.0                               | 6.8    | 6.8  | 6.3   | 6.3     | 6.3   | 6.5     | 6.5                   |
| V <sub>3</sub> : RNR 2458    | 7.8            | 2.8             | 2.8                               | 6.0    | 6.3  | 6.8   | 6.5     | 5.8   | 6.0     | 6.3                   |
| V <sub>4</sub> : JGL 384     | 7.8            | 2.7             | 3.0                               | 5.3    | 5.8  | 6.0   | 6.3     | 5.8   | 5.5     | 5.5                   |

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SHORT COMMUNICATION

## Sustainable production of maize (*Zea mays*)–wheat (*Triticum aestivum*) cropping system with agronomic management

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### ABSTRACT

The field experiment was carried out in twotehsils namely Salumber and Sarada of Udaipur district situated in Humid Southern Plain Zone of Rajasthan (IVb) on 36 farmers' fields from *kharif* 2007 to *rabi* 2009-10. Application of complete recommended package of practices i.e. improved varieties seeds, seed treatment with *Azotobacter*, recommended spacing and fertilizer doses, irrigations at critical stages, plant protection measures and weed management, resulted in significantly higher maize-grain equivalent yield ( $10.5 \text{ t ha}^{-1}$ ), net returns ( $\text{₹ } 85.7 \times 10^3 \text{ ha}^{-1}$ ), benefit: cost ratio (4.09) and maximum production efficiency ( $44.9 \text{ kg ha}^{-1}\text{day}^{-1}$ ) of maize-wheat cropping system over rest of the treatments. Hence, farmers should adopt complete recommended package of practices in maize-wheat cropping system for realizing sustainable production. However, resource-poor and marginal farmers who cannot afford to apply the full package should adopt at least one most critical yield limiting factor.

**Key words:** Agronomic management, Economics, Maize–wheat cropping system, Production efficiency, Productivity

Agronomic management practices are the most important non-monetary inputs and play very important role for obtaining potential yields and higher net returns from any crop or crop sequence. Jain *et al.* (2008) and Kumar and Singh (2010) also studied the role of agronomic management practices in maximizing productivity of maize (*Zea mays* L.)–wheat [*Triticum aestivum* (L.) emend. Fiori & Paol.] and rice (*Oryza sativa* L.)–wheat cropping systems, respectively. Maize-wheat is the most popular staple food-grain cropping system of Humid Southern Plain Zone of Rajasthan but farmers are unable to get maximum returns from this cropping system. A diagnostic survey of 180 farmers conducted in 2007 of Humid Southern Plain Zone of Rajasthan (Salumber and Sarada tehsils of Udaipur district) revealed that non-availability of good quality seeds of improved/hybrid varieties, inadequate or imbalanced fertilization, poor weed management, attack of insect-pests and diseases, no or less use of pesticides and bio-fertilizers for seed treatment, improper plant population, lack of technical knowledge, etc. are some of reasons responsible for low productivity. Information on agronomic management practices on individual crops is available, while for cropping system, very limited available information is available. Therefore, the present investigation was undertaken to find out the effect of

different agronomic management practices on production potential, profitability and production efficiency over farmers' practice in maize-wheat cropping system.

An on-farm experiment was carried out in twotehsils namely Salumber and Sarada of Udaipur district situated in Humid Southern Plain Zone of Rajasthan (IVb) continuously for 3 years from *kharif* 2007 to *rabi* 2009-10. The soils of the experimental sites were sandy clay loam, having pH 8.5, low to medium available nitrogen, medium available phosphorus and high available potassium status. The experiment consists of four treatments (T<sub>1</sub>: Farmers' practice, T<sub>2</sub>: Farmers' practice+improved varieties seeds of maize 'PM 5' and wheat 'Raj 4037', T<sub>3</sub>: Complete recommended package of practices (Improved varieties seeds, seed treatment with *Azotobacter*, recommended spacing and fertilizer doses, irrigations at critical stages, plant protection measures and weed management), and T<sub>4</sub>: Farmers' practice+improved variety seeds of maize 'PM 5' and wheat 'Raj 4037'+recommended dose of fertilizers), were applied to both maize and wheat crops at the same site during a year. The experiment was evaluated in randomized block design considering every farmer (12) as replication. Maize was sown during second fortnight of June to first fortnight of July and wheat was sown during third week of November to first week of December. Net plot size under each treatment was 100 m<sup>2</sup>. Farmers' practice include use of own local seeds, sowing by broadcasting method in maize and line sowing in wheat,

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imbalance use of fertilizers i.e. high/low nitrogen, less or no phosphorus and potassium, inadequate plant protection measures and weed management. In recommended package of practices, sowing was done at 60 cm x 25 cm spacing in maize and 20-22.5 cm in wheat. Recommended doses of fertilizers for maize and wheat were 90:40:30 and 120:40:30 kg NPK ha<sup>-1</sup>, respectively. Urea, single super phosphate and muriate of potash were used to supply nitrogen, phosphorus and potassium, respectively in both the crops. Seeds were also treated with *Azotobacter*. Weeds were controlled by pre-emergence spray of atrazine @ 0.5 kg ha<sup>-1</sup> followed by manual hand weeding at 25 days after sowing in maize and by post-emergence spray of 2,4-D ester @ 0.5 kg ha<sup>-1</sup> in wheat under treatment use of recommended package of practices (T<sub>3</sub>). The crops were evaluated in terms of grain and stover/straw yields, net returns and benefit: cost ratio. Maize-grain equivalent yield (MGEY) was also calculated by converting the wheat grain yield data to make grain equivalent on present market price basis. The production efficiency (PE) was calculated by using the formula (Nanda *et al.*, 2007):

$$PE \text{ (kg ha}^{-1}\text{day}^{-1}\text{)} = \frac{\text{MGEY (kg ha}^{-1}\text{)}}{\text{Duration of the cropping system (days)}}$$

Maximum grain (3.0 t ha<sup>-1</sup>) and stover yields (4.8 t ha<sup>-1</sup>) of maize were recorded with the treatment having complete recommended package of practices (T<sub>3</sub>) and these yields were significantly superior to rest of the treatments (Table 1). The improvement in grain yield with treatment T<sub>3</sub> was 87.5, 25.0 and 7.1% over farmers' practice (T<sub>1</sub>), farmers' practice+improved variety (T<sub>2</sub>) and farmers' practice+improved variety+recommended doses of fertilizers (T<sub>4</sub>), respectively. The respective increase in stover yield was 92.0, 37.1 and 9.1%. Similarly, the highest grain (5.2 t ha<sup>-1</sup>) and straw yields (7.8 t ha<sup>-1</sup>) of wheat were recorded with the treatment having complete recommended package of practices (T<sub>3</sub>) and these yields were also significantly superior to rest of the treatments. The improvement in grain yield with treatment T<sub>3</sub> was 67.7, 26.8 and 13.0% over farmers' practice (T<sub>1</sub>), farmers' practice+improved variety (T<sub>2</sub>) and farmers' practice+improved variety+recommended doses of fertilizers (T<sub>4</sub>), respectively. The respective increase in straw yield was 66, 23.8 and 9.9%. Treatment having improved varieties and recommended doses of fertilizers in both the crops along with farmers' practice (T<sub>4</sub>)

also gave significantly higher grain and stover/straw yields over the farmers' practice (T<sub>1</sub>) and farmers' practice along with improved varieties (T<sub>2</sub>). Treatment having improved varieties of both the crops along with farmers' practice (T<sub>2</sub>) also gave significantly higher grain and stover/straw yields over the farmers' practice (T<sub>1</sub>).

When treatments were compared for maize-grain equivalent yield of maize-wheat system as a whole, it was found that treatment T<sub>3</sub> (complete recommended package of practices i.e. improved varieties seeds, seed treatment with *Azotobacter*, recommended spacing and fertilizer doses, irrigations at critical stages, plant protection measures and weed management) had higher maize-grain equivalent yield (10.5 t ha<sup>-1</sup>) and it was significantly higher than rest of the treatments, followed by T<sub>4</sub> (Farmers' practice+improved varieties seeds of maize 'PM 5' and wheat 'Raj 4037'+recommended dose of fertilizers) (Table 1). Jain *et al.* (2008) also reported that complete recommended package of practices resulted in significantly higher maize-grain equivalent yield (10.2 t ha<sup>-1</sup>) compared with maize-grain equivalent yield (6.9 t ha<sup>-1</sup>) obtained under farmers' practice. Farmers' practice (T<sub>1</sub>) produced the lowest maize-grain equivalent yield (5.5 t ha<sup>-1</sup>).

Economic analysis of treatments showed that significantly higher net returns (₹ 85.7 × 10<sup>3</sup> ha<sup>-1</sup>) and benefit: cost ratio (4.09) were accrued with the application of complete recommended package of practices in maize-wheat cropping system (T<sub>3</sub>) over rest of the treatments (Table 1). Patil *et al.* (2007) also reported similar findings in maize-wheat cropping system. The highest total system incremental net returns (₹ 50.7 × 10<sup>3</sup> ha<sup>-1</sup>) were also recorded with recommended packages of practices over farmers' practice. The treatment T<sub>4</sub> also had significantly higher net returns (₹ 76.4 × 10<sup>3</sup> ha<sup>-1</sup>) and benefit: cost ratio (3.90) over farmers' practice (₹ 35 × 10<sup>3</sup> ha<sup>-1</sup> and 2.52) and farmers' practice+improved varieties (₹ 65.1 × 10<sup>3</sup> ha<sup>-1</sup> and 3.59).

Production efficiency was also maximum (44.9 kg ha<sup>-1</sup> day<sup>-1</sup>) in the treatment having complete recommended package of practices (T<sub>3</sub>) and it was higher by 21.5, 9.1 and 4.4 kg ha<sup>-1</sup> day<sup>-1</sup> over farmers' practice (T<sub>1</sub>), farmers' practice+improved varieties (T<sub>2</sub>) and farmers' practice+improved varieties+recommended doses of fertilizers (T<sub>4</sub>), respectively. Treatment having improved varieties and recommended doses of fertilizers

Table 1. Influence of various agronomic management practices on yield and economics of maize-wheat cropping system (mean of 3 years)

| Treatment                                                         | Grain yield<br>(t ha <sup>-1</sup> ) |       | Stover / straw yield<br>(t ha <sup>-1</sup> ) |       | MGEY<br>(t ha <sup>-1</sup> ) | Net returns<br>(x 10 <sup>3</sup> ₹ ha <sup>-1</sup> ) | Incremental net<br>returns over T <sub>1</sub><br>(x 10 <sup>3</sup> ₹ ha <sup>-1</sup> ) | BCR  | PE<br>(kg ha <sup>-1</sup> day <sup>-1</sup> ) |
|-------------------------------------------------------------------|--------------------------------------|-------|-----------------------------------------------|-------|-------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------|------|------------------------------------------------|
|                                                                   | Maize                                | Wheat | Maize                                         | Wheat |                               |                                                        |                                                                                           |      |                                                |
| Farmers' practice (T <sub>1</sub> )                               | 1.6                                  | 3.1   | 2.5                                           | 4.7   | 5.5                           | 35.0                                                   |                                                                                           | 2.52 | 23.4                                           |
| Farmers' practice + improved<br>varieties (T <sub>2</sub> )       | 2.4                                  | 4.1   | 3.5                                           | 6.3   | 8.4                           | 65.1                                                   | 30.1                                                                                      | 3.59 | 35.8                                           |
| Complete recommended<br>package of practices (T <sub>3</sub> )    | 3.0                                  | 5.2   | 4.8                                           | 7.8   | 10.5                          | 85.7                                                   | 50.7                                                                                      | 4.09 | 44.9                                           |
| Farmers' practice + Improved<br>varieties + RDF (T <sub>4</sub> ) | 2.8                                  | 4.6   | 4.4                                           | 7.1   | 9.5                           | 76.4                                                   | 41.4                                                                                      | 3.90 | 40.5                                           |
| CD (P=0.05)                                                       | 0.05                                 | 0.12  | 0.07                                          | 0.16  | 0.16                          | 1.60                                                   |                                                                                           | 0.06 |                                                |

MGEY = Maize-grain equivalent yield; NR = Net returns; BCR = Benefit: cost ratio; PE = Production efficiency; RDF = Recommended dose of fertilizers

in both the crops along with farmers' practice ( $T_4$ ) also gave higher production efficiency over the farmers' practice ( $T_1$ ) and farmers' practice along with improved varieties ( $T_2$ ). Treatment having improved varieties of both the crops along with farmers' practice ( $T_2$ ) also registered higher production efficiency over the farmers' practice ( $T_1$ ).

On the basis of three years' experimentation, it could be concluded that farmers should adopt complete recommended package of practices for sustainable production of maize-wheat cropping system in Humid Southern Plain Zone of Rajasthan (IVb).

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## SHORT COMMUNICATION

# Response of rice (*Oryza sativa*) to integrated nutrient management under temperate condition of Kashmir

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## ABSTRACT

A field experiment was carried out at Kashmir during *kharif* season 2012 to assess the response of rice to integrated nutrient management. Grain yield ( $8.06 \text{ t ha}^{-1}$ ), straw yield ( $10.40 \text{ t ha}^{-1}$ ) and growth parameters, viz. plant height, number of tillers  $\text{hill}^{-1}$  and leaf area index were recorded higher in treatment  $T_4$  (recommended dose of fertilizers-RDF + poultry manure @  $2 \text{ t ha}^{-1}$ ) compared to other treatments. The yield attributes like number of panicles  $\text{m}^2$ , number of grains panicle $^{-1}$ , panicle weight and harvest index was also observed significantly higher in treatment  $T_4$ . The economic analysis showed that treatment  $T_{12}$  (75% RDF + inoculation of *Azospirillum*) showed maximum net return ( $\text{₹ } 88.72 \times 10^3 \text{ ha}^{-1}$ ) and benefit: cost ratio (3.79), followed by  $T_6$  (RDF + *Azospirillum*) and  $T_5$  (RDF + sheep manure @  $2 \text{ t ha}^{-1}$ ).

**Key words:** Growth, Integrated nutrient management, Poultry manure, Rice, Yield

Rice (*Oryza sativa* L.) is the premier food crop of India; therefore national food security system of India largely depends on productivity and quality of rice ecosystem. Among the factors responsible for low productivity of rice, inadequate fertilizer use and emergence of multiple-nutrient deficiencies due to poor recycling of organic resources and unbalanced use of fertilizers are common. The soils although being rich in nutrients but only small proportion of it becomes available to plants especially under temperate agro climates. Further the organic sources unlike inorganic ones have the substantial residual effect on succeeding crops (Shivakumar and Ahlawat, 2008). Kashmir division of J & K state is endowed with organic sources particularly FYM, poultry manure, sheep manure, vermicompost, crop residues and bio-fertilizers. Since the organic alone cannot supply the whole nutrient demands of the crop in view of the very low nutrient availability as well as their restricted availability on large scale, the use of chemical fertilizers would continue to play their main role in the enhancement of soil productivity to encouraging level till organic reserve of the soil is raised to a very high level (Laxminarayana, 2006). The present study was, therefore, designed to study the response of rice to integrated nutrient management under temperate conditions of Kashmir.

A field experiment was conducted at the Research Farm, Division of Agronomy, Sher-e-Kashmir University of Agricultural Sciences and Technology of Kashmir, Shalimar,

Srinagar (Jammu and Kashmir), India during *kharif* season of 2012 on silty clay loam soil, high in organic carbon (0.78%), medium in available nitrogen ( $314.24 \text{ kg ha}^{-1}$ ), high in available phosphorus ( $37.43 \text{ kg ha}^{-1}$ ) and low in available potassium ( $159.22 \text{ kg ha}^{-1}$ ) with neutral pH (6.8). The experimental site is located at  $34^{\circ}08' \text{ N}$  latitude and  $74^{\circ}83' \text{ E}$  longitude at a height of 1587 m above mean sea level. The experiment comprised of 13 treatments, viz.  $T_0$  (control),  $T_1$  (recommended dose of fertilizer-RDF + farmyard manure-FYM @  $10 \text{ t ha}^{-1}$ ),  $T_2$  (RDF + crop residue @  $15 \text{ t ha}^{-1}$ ),  $T_3$  (RDF + vermicompost @  $2 \text{ t ha}^{-1}$ ),  $T_4$  (RDF + poultry manure @  $2 \text{ t ha}^{-1}$ ),  $T_5$  (RDF + sheep manure @  $2 \text{ t ha}^{-1}$ ),  $T_6$  (RDF + *Azospirillum* inoculation @ 5 packs 200 g each),  $T_7$  (75% RDF + FYM @  $10 \text{ t ha}^{-1}$ ),  $T_8$  (75% RDF + crop residue @  $15 \text{ t ha}^{-1}$ ),  $T_9$  (75% RDF + vermicompost @  $2 \text{ t ha}^{-1}$ ),  $T_{10}$  (75% RDF + poultry manure @  $2 \text{ t ha}^{-1}$ ),  $T_{11}$  (75% RDF + sheep manure @  $2 \text{ t ha}^{-1}$ ), and  $T_{12}$  (75% RDF + *Azospirillum* inoculation @ 5 packs 200 g each) were laid out in a randomized block design with 3 replications on gross plot size of  $5.4 \text{ m} \times 2.1 \text{ m}$ . The different sources of organic manures were analyzed for nutrient content as shown in Table 1. After land preparation, the organic manures, viz. FYM, crop residue and vermicompost were applied. Bio-fertilizers were given before transplanting by dipping the roots of seedlings for 10-15 minutes [using trench of  $4 \text{ m} \times 4 \text{ m}$  with 10 cm depth filled with slurry of 1 kg of *Azospirillum* (5 pkts) + 1000 g peat based inoculation]. Recommended dose of nitrogen, phosphorus and potassium fertilizer (120:60:30) in the form of urea, diammonium phosphate

and muriate of potash, respectively were applied to plots as per treatments just before transplanting rice seedlings. Full recommended dose of  $P_2O_5$ ,  $K_2O$  and half recommended dose of N were applied as basal and remaining half of the N was applied in two equal splits at tillering and panicle initiation stages. Thirty five days old seedling of the variety 'Jhelum' uprooted from the nursery beds were transplanted on the same day (9<sup>th</sup> June) in the experimental field. Three to four robust seedlings per hill were transplanted at spacing of 15 cm x 15 cm by line planting. Pre emergence application of butachlor 5% G @ 1.5 kg a.i. ha<sup>-1</sup> was applied 3 days after transplanting in uniform layer and water was impounded in the field for three days. All the agronomic practices were carried out uniformly to raise the crop. The crop was harvested on 27 September, 2012. The total rainfall during the crop season was 252.7 mm. Mean maximum and minimum temperatures varied from and 21.57-33.71°C and 8.95-19.64°C, respectively. The data collected was subjected to analysis of variance technique as described by Cochran and Cox (1958).

Growth attributes, viz. plant height, number of tillers m<sup>-2</sup> and leaf area index (LAI) were significantly influenced by different treatments (Table 1). Treatment T<sub>4</sub> (RDF + poultry manure @ 2 t ha<sup>-1</sup>) recorded highest plant height at harvest, followed by T<sub>10</sub> (75% RDF + poultry manure @ 2 t ha<sup>-1</sup>) and T<sub>3</sub> (RDF + vermicompost @ 2 t ha<sup>-1</sup>) as compared to other treatments. The might be due to the fact that there was greater availability of available plant nutrients through poultry manure (Shivakumar and Ahlawat, 2008), besides it also had a solubilizing effect on fixed forms of nutrients in soil (Garg and Bahla, 2008). Higher number of tillers m<sup>-2</sup> was recorded in treatment T<sub>4</sub> (RDF + poultry manure @ 2 t ha<sup>-1</sup>) which was at par with other treatments. The application of poultry manure might have promoted nitrogen supply which is essential for vegetative growth. These results collaborate with the findings of Hasannuzaman *et al.* (2010). Treatment T<sub>4</sub> (RDF + poultry manure @ 2 t ha<sup>-1</sup>) recorded significantly higher values of LAI, followed by T<sub>10</sub> (75% RDF + poultry manure @ 2 t ha<sup>-1</sup>) as compared to rest of the treatments. There was a marked

significant difference in LAI between control and rest of the INM treatments. This might be because of the fact that concurrent increase in plant height leading to an increase in leaf area and LAI. Hasannuzaman *et al.* (2010) also reported increased LAI by poultry manure along with RDF.

Application of RDF + poultry manure @ 2 t ha<sup>-1</sup> (T<sub>4</sub>) recorded the highest values of yield attributes, viz. number of panicles m<sup>-2</sup> (343.5), number of grains panicle<sup>-1</sup> (123.3), panicle weight (3.57 g) and 1000-grain weight (26.4 g), which was significantly superior over other treatments (Table 1). The number of panicles is generally associated with tiller production capacity of the crop which in turn is dependent on balanced nutrition (Uddin *et al.*, 2002). However, balanced nutrient supply and capacity for translocation of assimilates from source to sink determines the production of effective tillers. Poultry manure supplies both macro and micro nutrients which are essential for plant growth, besides, encourages microbial population and improves soil health thereby, affects yield contributing characters. These results corroborate the findings of Garg and Bahla (2008). The highest grain yield (8.06 t ha<sup>-1</sup>), biological yield (18.46 t ha<sup>-1</sup>) and straw yield (10.40 t ha<sup>-1</sup>) were recorded in treatment T<sub>4</sub> (RDF + poultry manure @ 2 t ha<sup>-1</sup>) (Table 1), which was significantly superior over other treatments. Further, lowest grain yield (4.26 t ha<sup>-1</sup>), biological yield (10.17 t ha<sup>-1</sup>) and straw yield (5.91 t ha<sup>-1</sup>) and harvest index (41.84%) were recorded in treatment T<sub>0</sub> (control). Treatment T<sub>4</sub> also recorded significantly higher harvest index (43.66%) and lowest harvesting index was obtained with the control. This might be due to the fact that poultry manure offered better nutritional quality and favourable balance of nutrients when supplemented with NPK which provided the maximum yield. Rakshit *et al.* (2008) also observed similar findings. Moreover, application of organic manures adds and exports the fixed nutrients of soil in available form and regulates its supply to the crop through mineralization and prevents them from leaching and volatilization losses leading to increased yields (Singh, 1996).

Table 1. Effect of integrated nutrient management on growth, yield and yield attributes of rice

| Treatment                                                          | Plant height (cm) | Number of tillers m <sup>-2</sup> | leaf area index (LAI) | Panicles m <sup>-2</sup> | Grains panicle <sup>-1</sup> | 1000-grain weight (g) | Panicle weight (g) | Grain yield (t ha <sup>-1</sup> ) | Straw yield (t ha <sup>-1</sup> ) | Harvest index (%) |
|--------------------------------------------------------------------|-------------------|-----------------------------------|-----------------------|--------------------------|------------------------------|-----------------------|--------------------|-----------------------------------|-----------------------------------|-------------------|
| T <sub>0</sub> ; Control                                           | 111.3             | 315.7                             | 103.6                 | 308.4                    | 73.5                         | 23.1                  | 2.24               | 4.26c                             | 5.91f                             | 41.84ef           |
| T <sub>1</sub> ; RDF + FYM (10 t ha <sup>-1</sup> )                | 141.6             | 337.7                             | 123.8                 | 330.0                    | 110.6                        | 24.2                  | 3.45               | 7.45ab                            | 10.02b                            | 42.59bc           |
| T <sub>2</sub> ; RDF + crop residue (15 t ha <sup>-1</sup> )       | 136.5             | 338.8                             | 122.7                 | 332.1                    | 108.7                        | 24.6                  | 3.50               | 7.30b                             | 9.96b                             | 42.00def          |
| T <sub>3</sub> ; RDF + vermicompost (2 t ha <sup>-1</sup> )        | 142.0             | 344.2                             | 124.0                 | 333.9                    | 120.0                        | 24.9                  | 3.15               | 7.51ab                            | 10.02b                            | 42.47bcd          |
| T <sub>4</sub> ; RDF + poultry manure (2 t ha <sup>-1</sup> )      | 145.8             | 347.9                             | 125.8                 | 343.5                    | 123.3                        | 26.4                  | 3.57               | 8.06a                             | 10.40a                            | 43.66a            |
| T <sub>5</sub> ; RDF + sheep manure (2 t ha <sup>-1</sup> )        | 137.1             | 345.2                             | 118.5                 | 341.5                    | 110.7                        | 25.1                  | 3.04               | 7.44ab                            | 9.78cd                            | 43.20a            |
| T <sub>6</sub> ; RDF + <i>Azospirillum</i>                         | 132.5             | 341.6                             | 115.9                 | 337.2                    | 114.4                        | 24.0                  | 3.10               | 7.14b                             | 9.80cd                            | 42.14cdef         |
| T <sub>7</sub> ; 75% RDF+ FYM (10 t ha <sup>-1</sup> )             | 133.4             | 335.2                             | 116.2                 | 331.9                    | 115.8                        | 25.1                  | 3.36               | 7.22b                             | 9.71de                            | 42.30bcdef        |
| T <sub>8</sub> ; 75% RDF + crop residue (15 t ha <sup>-1</sup> )   | 132.5             | 338.8                             | 119.3                 | 335.5                    | 114.7                        | 24.7                  | 3.08               | 7.29b                             | 9.91bc                            | 42.38bcde         |
| T <sub>9</sub> ; 75% RDF + vermicompost (2 t ha <sup>-1</sup> )    | 138.2             | 337.6                             | 119.9                 | 333.9                    | 111.6                        | 24.4                  | 3.40               | 7.29b                             | 9.80cd                            | 42.70b            |
| T <sub>10</sub> ; 75% RDF + poultry manure (2 t ha <sup>-1</sup> ) | 142.8             | 346.2                             | 124.0                 | 342.6                    | 117.0                        | 25.5                  | 3.47               | 7.58a                             | 10.22a                            | 42.18bcdef        |
| T <sub>11</sub> ; 75% RDF + sheep manure (2 t ha <sup>-1</sup> )   | 135.0             | 341.2                             | 118.2                 | 337.6                    | 114.6                        | 24.5                  | 3.09               | 7.12b                             | 9.61e                             | 42.55bc           |
| T <sub>12</sub> ; 75% RDF + <i>Azospirillum</i>                    | 135.5             | 342.9                             | 117.9                 | 336.2                    | 110.2                        | 24.7                  | 2.90               | 7.10b                             | 9.96b                             | 41.89ef           |
| SE±                                                                | 2.66              | 4.80                              | 1.90                  | 0.33                     | 3.40                         | 0.50                  | 0.38               | 0.16                              | 0.06                              | 0.15              |
| CD (P=0.05)                                                        | 8.00              | 14.10                             | 5.70                  | 1.10                     | 10.20                        | 1.50                  | 1.14               | 0.48                              | 0.18                              | 0.46              |

\*Symbols denote comparison of means through DMRT (0.05)

Table 2. Relative economics of rice as affected by different INM treatments

| Treatment                                                          | Cost of cultivation<br>(x 10 <sup>3</sup> ₹ ha <sup>-1</sup> ) | Gross return<br>(x 10 <sup>3</sup> ₹ ha <sup>-1</sup> ) | Net return<br>(x 10 <sup>3</sup> ₹ ha <sup>-1</sup> ) | Benefit:<br>cost ratio |
|--------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------|------------------------|
| T <sub>0</sub> ; Control                                           | 18.64                                                          | 66.79                                                   | 48.15                                                 | 2.58                   |
| T <sub>1</sub> ; RDF + FYM (10 t ha <sup>-1</sup> )                | 44.70                                                          | 115.52                                                  | 70.81                                                 | 1.58                   |
| T <sub>2</sub> ; RDF + crop residue (15 t ha <sup>-1</sup> )       | 39.70                                                          | 114.30                                                  | 74.60                                                 | 1.88                   |
| T <sub>3</sub> ; RDF + vermicompost (2 t ha <sup>-1</sup> )        | 54.70                                                          | 115.79                                                  | 61.09                                                 | 1.12                   |
| T <sub>4</sub> ; RDF + poultry manure (2 t ha <sup>-1</sup> )      | 44.70                                                          | 123.37                                                  | 78.67                                                 | 1.76                   |
| T <sub>5</sub> ; RDF + sheep manure (2 t ha <sup>-1</sup> )        | 34.70                                                          | 114.35                                                  | 80.05                                                 | 2.31                   |
| T <sub>6</sub> ; RDF + <i>Azospirillum</i>                         | 24.90                                                          | 111.53                                                  | 86.63                                                 | 3.48                   |
| T <sub>7</sub> ; 75% RDF+ FYM (10 t ha <sup>-1</sup> )             | 43.18                                                          | 112.16                                                  | 68.97                                                 | 1.60                   |
| T <sub>8</sub> ; 75% RDF + crop residue (15 t ha <sup>-1</sup> )   | 38.18                                                          | 113.41                                                  | 75.22                                                 | 1.97                   |
| T <sub>9</sub> ; 75% RDF + vermicompost (2 t ha <sup>-1</sup> )    | 53.18                                                          | 113.21                                                  | 60.03                                                 | 1.13                   |
| T <sub>10</sub> ; 75% RDF + poultry manure (2 t ha <sup>-1</sup> ) | 43.18                                                          | 117.52                                                  | 74.33                                                 | 1.72                   |
| T <sub>11</sub> ; 75% RDF + sheep manure (2 t ha <sup>-1</sup> )   | 33.18                                                          | 110.61                                                  | 77.43                                                 | 2.33                   |
| T <sub>12</sub> ; 75% RDF + <i>Azospirillum</i>                    | 23.38                                                          | 112.10                                                  | 88.72                                                 | 3.79                   |

**Cost of inputs and selling price of the produce:** DAP = ₹ 2400 q<sup>-1</sup>; Tractorization = ₹ 4000 ha<sup>-1</sup>; Crop residue = ₹ 100 q<sup>-1</sup>; Sheep manure = ₹ 500 q<sup>-1</sup>; Urea = ₹ 540 q<sup>-1</sup>; Poultry manure = ₹ 10 kg<sup>-1</sup>; Azospirillum = ₹ 100 kg<sup>-1</sup>; Labour = ₹ 125 day<sup>-1</sup>; ZnSO<sub>4</sub> = ₹ 67 kg<sup>-1</sup>; MOP = ₹ 1650 q<sup>-1</sup>; Vermicompost = ₹ 15 kg<sup>-1</sup>; Butachlor = ₹ 28 kg<sup>-1</sup>; FYM = ₹ 200 q<sup>-1</sup>; Grain = ₹ 1100 q<sup>-1</sup>; Straw = ₹ 35 shief<sup>-1</sup>; Seed = ₹ 22 kg<sup>-1</sup>

The highest net return (₹ 88.72 x 10<sup>3</sup> ha<sup>-1</sup>) and benefit: cost ratio (3.79) was registered with treatment T<sub>12</sub> (75% RDF + inoculation of *Azospirillum*), followed by T<sub>6</sub> (RDF + inoculation of *Azospirillum*) with net return of ₹ 86.63 × 10<sup>3</sup> ha<sup>-1</sup> and B:C ratio of 3.48 (Table 2). This might be due to the fact that the increased yield in treatment T<sub>4</sub> (RDF + poultry manure @ 2 t ha<sup>-1</sup>) can not compensate the high cost of poultry manure and at the same time the lower cost invested for bio-fertilizer was incurred in treatment T<sub>12</sub>. In addition to this the cost incurred on the manure was too high to show least profit made by the treatments. Further, the least benefit: cost ratio in treatment T<sub>3</sub> (RDF + vermicompost @ 2 t ha<sup>-1</sup>) was because of the fact that increased yield shown by the addition of vermicompost can not nullify its incurred cost. Similar results were reported by Hussain *et al.* (2012).

Thus it is concluded that the application of poultry manure established its superiority over other organic manures in combination with RDF in terms of crop establishment, growth characters, yield attributing characters and finally yield under Kashmir conditions. However, sustainable yields and soil health (nutrient reserves) can be improved outstandingly by use of integrated nutrient management practices in rice.

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## SHORT COMMUNICATION

# Yield and nutrient uptake of transplanted rice (*Oryza sativa*) with different moisture regimes and integrated nutrient supply

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## ABSTRACT

A field experiment was conducted during *kharif* 2010 at Faizabad (Uttar Pradesh) to assess the effects of moisture regime and integrated nutrient supply on yield, yield attributes and nutrients uptake of transplanted rice (*Oryza sativa* L.). Irrigation with 7 cm water at 1 day after disappearance of ponded water (DADPW) was superior to 3 and 5 DADPW in respect of yield, yield attributes and nutrient uptake. The highest grain yield (5.44 t ha<sup>-1</sup>) and straw yield (7.68 t ha<sup>-1</sup>), was recorded with 1 DADPW. Application of recommended dose of NPK (120:60:40 kg ha<sup>-1</sup>) applied through inorganic fertilizers was at par with green manuring + 75% NPK through inorganic fertilizers in respect of yield attributes, yield and total nutrient uptake. The maximum grain yield (5.21 t ha<sup>-1</sup>) and straw yield (7.45 t ha<sup>-1</sup>) was obtained with 100% NPK (120:60:40 kg ha<sup>-1</sup>), which was at par with green manuring + 75% NPK applied through inorganic fertilizers.

**Key words:** Biocompost, Moisture regime, Nutrient uptake, Water use efficiency, Yield

Rice (*Oryza sativa* L.) is an important food grain crop grown extensively in tropical and sub-tropical regions of the world. In India, rice plays a vital role in our national food security and a means of livelihood for millions of rural household. Yield decline under intensive rice-wheat cropping system in the Indo-Gangetic plain region is associated with the imbalanced application of NPK and emerging deficiency of micronutrients. Chemical fertilizer indeed boots up crop production initially; however, it causes several problems related to soil health and grains quality. Integration of chemical fertilizers with organic manures has been found to be quite promising not only in maintaining higher productivity but also in providing greater stability in crop production. It is well known fact that water management is one of the major factor responsible for achieving better harvest in crop production. In general farmers use to keep rice field submerged throughout the growth period on the basis of assumption in their mind that higher grain yield of rice can be achieved only by doing this practice. However, now it has been proved that intermittent drainage increases the growth as well as grain yield of rice (Dwivedi, 2008). Thus, the judicious use of available irrigation water and application of integrated

nutrient supply in respect to available soil moisture may play an important role in minimizing the present large gap between yield achieved and yield achievable.

The experiment was conducted during the *kharif* season of 2010 at Agronomy Research Farm, Narendra Deva University of Agriculture and Technology, Faizabad (Uttar Pradesh) located at 26.47°N latitude, 82.12°E longitude and at altitude of 113 msl from Indo-Gangetic region of eastern U.P. The soil samples were collected randomly from 10 places of the experimental field with the help of soil auger and analyzed it. The soil of experimental field was silt loam in texture having pH 7.8, EC 0.28 dSm<sup>-1</sup>, organic carbon 0.39% and medium in nitrogen (189.2 kg ha<sup>-1</sup>), low in phosphorus (13.15 kg ha<sup>-1</sup>), and high in potassium (255.73 kg ha<sup>-1</sup>). The experiment comprised three moisture regime in main plots (7cm irrigation 1, 3 and 5 days after disappearance of ponded water-DADPW) with four integrated nutrient management as sub-plots [S<sub>1</sub>: 100% NPK (120:60:40 kg ha<sup>-1</sup>) through inorganic fertilizer, S<sub>2</sub>: 25% N through FYM + 75% NPK through inorganic fertilizers, S<sub>3</sub>: 25% N through bio-compost + 75% NPK through inorganic fertilizers, and S<sub>4</sub>: green manuring + 75% NPK through inorganic fertilizers]. The experiment was laid out in split plot design with four replications.

Irrigation treatments based on DADPW was started just after transplanting with 7 cm depth of water in each irrigation as per treatment. Twenty-five days old age seedling was transplanted in the field. Rice 'Sarjoo-52' was planting at 20 cm

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x 10 cm spacing. The data were recorded randomly from five places in each plot, while for nutrient uptake as, the N P K content in grain and straw determined separately and multiplied by their respectively yields. The data recorded in respect to different observations were analyzed as per standard statistical procedure.

Various moisture regimes and integrated nutrient management were affected significantly on the yield attributing characters of the crop (Table 1). Among moisture regimes, the highest number of effective shoots running meter<sup>-1</sup> (121.54), length of panicles (22), number of grains panicles<sup>-1</sup> (180.14) and weight of grains panicles<sup>-1</sup> (4.34 g) were recorded with application of 7 cm irrigation 1 DADPW, which was significantly superior over the 7 cm irrigation 3 and 5 DADPW. This might be due to favourable vegetable growth and development as they received adequate and sufficient moisture at proper amount and critical stages during entire period of growth. As the results of which the plant height, number of shoots, leaf area index and dry matter accumulation were recorded which contributed to highest yield attributes through increased photosynthetic activity of leaves. Besides, translocation of photosynthesis from source to sink might have also increased adequate moisture conditions through higher uptake of nutrients which led to better yield attributes. Lowest yield attributes were recorded under 7 cm irrigation 5 DADPW as growing plants suffered due to moisture stress; hence plants were unable to extract more water and nutrients from deep in soil under moisture deficit conditions which ultimately led to poor growth and yield attributes. Similar results were reported by Luikham and Anal (2008).

In case of integrated nutrient management, the highest number of effective shoots running meter<sup>-1</sup> (118.94), length of panicles (21.14), number of grains panicles<sup>-1</sup> (174.33), and weight of grains panicles<sup>-1</sup> (4.11g) were recorded with S<sub>1</sub> treatment [100% NPK (120:60:40 kg ha<sup>-1</sup>)], which was at par with S<sub>4</sub> treatment (green manuring + 75% NPK applied through inorganic fertilizers) and was found significantly superior over

the rest treatment. This was due to fact that better vegetative growth and development of plant received adequate nutrients during entire period of growth. The plant height, number of shoots running meter<sup>-1</sup>, leaf area index, and dry matter accumulation was higher under sufficient availability of nutrients. Hence, it contributed higher value of yield attributing characters through increased photosynthetic activity of leaves and translocations of assimilates. Similar results were recorded by Zaidi *et al.* (2006).

The grain and straw yield of rice was influenced significantly due to the effect of moisture regimes and integrated nutrient supply (Table 1). Among the moisture regimes, the highest grain (5.45 t ha<sup>-1</sup>) and straw yield (7.68 t ha<sup>-1</sup>) was recorded under 7 cm irrigation 1 DADPW, which was significantly superior over the 7 cm irrigation 3 and 5 DADPW. This might be due to adequate moisture availability which contributed to increased dry matter accumulation. Better vegetative growth coupled with higher yield attributes resulted in higher grain and straw yield. Lowest grain (4.76 t ha<sup>-1</sup>) and straw yield (6.70 t ha<sup>-1</sup>) was recorded in 7 cm irrigation 5 DADPW which might be owing to water scarcity during both vegetative and reproductive phase of growth and water stress during critical stages hampered the source and sink ratio to the large extent which reduced yield attributing characters hence resulted poor grain and straw yield. The similar results were reported by Zaidi *et al.* (2006).

In case of various treatments of integrated nutrient supply, the maximum grain yield (5.21 t ha<sup>-1</sup>) and straw yield (7.45 t ha<sup>-1</sup>) was obtained under S<sub>1</sub> treatment [100% NPK (120:60:40 kg ha<sup>-1</sup>)], which was at par with S<sub>4</sub> treatment (green manuring + 75% NPK applied through inorganic fertilizers). The minimum grain yield (4.75 t ha<sup>-1</sup>) and straw yield (6.64 t ha<sup>-1</sup>) was obtained with the application of 25% N through FYM + 75% NPK through inorganic fertilizers. The higher yield might be due to more number of effective panicles, number of grains panicle<sup>-1</sup>, length of panicle and test weight. Similar findings were reported by Tripathi *et al.* (2007).

Table 1. Yield and yield attributes of rice as influenced by moisture regime and integrated nutrient supply system

| Treatment                     | Number of effective shoots running meter <sup>-1</sup> | Length of panicle (cm) | Number of grains panicles <sup>-1</sup> | Weight of grains panicles <sup>-1</sup> | Test weight (g) | Grain yield (t ha <sup>-1</sup> ) | Straw yield (t ha <sup>-1</sup> ) | Harvest index (%) |
|-------------------------------|--------------------------------------------------------|------------------------|-----------------------------------------|-----------------------------------------|-----------------|-----------------------------------|-----------------------------------|-------------------|
| <i>Moisture regime</i>        |                                                        |                        |                                         |                                         |                 |                                   |                                   |                   |
| I <sub>1</sub>                | 121.54                                                 | 22.00                  | 180.14                                  | 4.34                                    | 24.44           | 5.45                              | 7.68                              | 41.51             |
| I <sub>2</sub>                | 113.82                                                 | 20.01                  | 168.18                                  | 3.84                                    | 23.71           | 4.85                              | 6.85                              | 41.48             |
| I <sub>3</sub>                | 101.39                                                 | 18.64                  | 160.67                                  | 3.18                                    | 23.11           | 4.68                              | 6.70                              | 41.11             |
| SEm±                          | 2.26                                                   | 0.41                   | 3.41                                    | 0.08                                    | 0.48            | 0.10                              | 0.15                              | 0.20              |
| CD(P=0.05)                    | 7.24                                                   | 1.30                   | 10.91                                   | 0.24                                    | NS              | 0.32                              | 0.47                              | NS                |
| <i>Nutrient supply system</i> |                                                        |                        |                                         |                                         |                 |                                   |                                   |                   |
| S <sub>1</sub>                | 118.94                                                 | 21.14                  | 174.33                                  | 4.11                                    | 24.35           | 5.21                              | 7.45                              | 41.12             |
| S <sub>2</sub>                | 102.17                                                 | 19.05                  | 163.17                                  | 3.31                                    | 23.07           | 4.75                              | 6.64                              | 41.65             |
| S <sub>3</sub>                | 109.50                                                 | 19.72                  | 167.48                                  | 3.64                                    | 23.65           | 4.92                              | 6.89                              | 41.66             |
| S <sub>4</sub>                | 118.38                                                 | 20.95                  | 173.68                                  | 4.08                                    | 23.94           | 5.13                              | 7.39                              | 41.03             |
| SEm±                          | 2.42                                                   | 0.47                   | 3.93                                    | 0.09                                    | 0.55            | 0.12                              | 0.17                              | 0.18              |
| CD(P=0.05)                    | 7.61                                                   | 1.36                   | 11.76                                   | 0.25                                    | NS              | 0.34                              | 0.49                              | NS                |

I<sub>1</sub>, I<sub>2</sub> and I<sub>3</sub> = 7 cm irrigation 1, 3 and 5 days after disappearance of ponded water-DADPW, respectively; S<sub>1</sub> = 100% NPK (120:60:40 kg ha<sup>-1</sup>) through inorganic fertilizer; S<sub>2</sub> = 25% N through FYM + 75% NPK through inorganic fertilizers; S<sub>3</sub> = 25% N through bio-compost + 75% NPK through inorganic fertilizers; S<sub>4</sub> = green manuring + 75% NPK through inorganic fertilizers

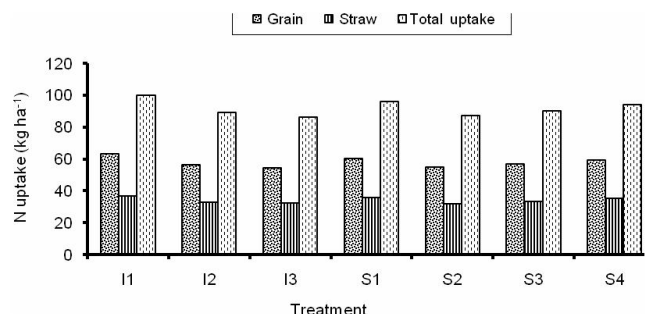


Fig. 1. Influence of moisture regime and integrated nutrient supply system on N uptake in rice crop

I<sub>1</sub>, I<sub>2</sub> and I<sub>3</sub> = 7 cm irrigation 1, 3 and 5 days after disappearance of ponded water-DADPW, respectively; S<sub>1</sub> = 100% NPK (120:60:40 kg ha<sup>-1</sup>) through inorganic fertilizer; S<sub>2</sub> = 25% N through FYM + 75% NPK through inorganic fertilizers; S<sub>3</sub> = 25% N through bio-compost + 75% NPK through inorganic fertilizers; S<sub>4</sub> = green manuring + 75% NPK through inorganic fertilizers

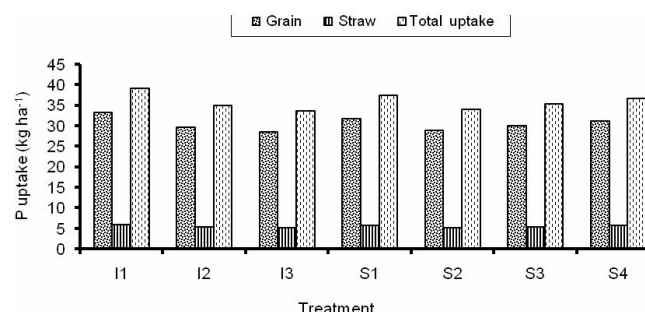


Fig. 2. Influence of moisture regime and integrated nutrient supply system on P uptake in rice crop

The maximum N, P and K uptake (100.02, 39.12 and 89.41 kg ha<sup>-1</sup>, respectively) was recorded under 7 cm irrigation 1 DADPW, which was found significantly superior over 7 cm irrigation 3 and 5 DADPW (Fig. 1, 2 and 3). The minimum total N, P and K uptake was recorded under 7 cm irrigation 5 DADPW. The similar results were reported by Luikham and Anal (2008). The maximum total N, P and K uptake of 96.17, 42.18, 86.14 kg ha<sup>-1</sup>, respectively, was recorded with S<sub>1</sub> treatment [100% NPK (120:60:40 kg ha<sup>-1</sup>)], which was at par with S<sub>4</sub> (green manuring + 75% NPK applied through inorganic fertilizers). Minimum NPK

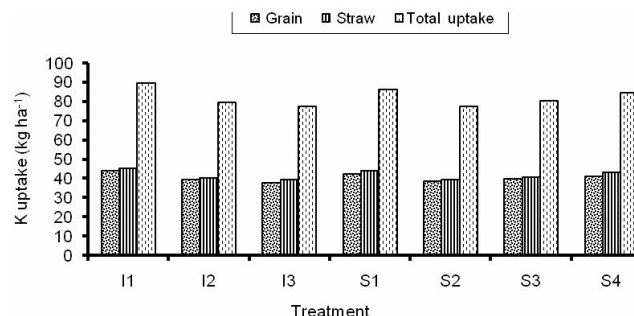


Fig. 3. Influence of moisture regime and integrated nutrient supply system on K uptake (kg ha<sup>-1</sup>) in rice crop

uptake was recorded with S<sub>2</sub> treatment (25% N through FYM + 75% NPK through inorganic fertilizers). The present results are in close proximity with the findings of Pandey *et al.* (2007) and Tripathi *et al.* (2007).

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## SHORT COMMUNICATION

# Effect of boron application on seed yield and protein content of mungbean (*Vigna radiata* L.)

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## ABSTRACT

The experiment was conducted at Kanpur, Uttar Pradesh during *kharif* 2012 to evaluate the effect of boron on yield and protein content of mungbean (*Vigna radiata* L.) variety 'K 851'. The highest yield attributes, viz. pods plant<sup>-1</sup>, seeds pod<sup>-1</sup> and 1000-seed weight was recorded in boron 0.2% foliar spray, followed by application of boron 10 kg ha<sup>-1</sup> at flowering, while the lowest was in the control. This treatment also registered the highest yields plant<sup>-1</sup> than other treatments. The highest protein content (25.23%) was recorded with 0.2% foliar spray of boron, which was at par with boron 0.1% foliar spray (25.03%).

**Key words:** Foliar spray, Mungbean, Protein content, Seed soaking, *Vigna radiata*

Mungbean [(*Vigna radiata* (L.) Wilczek)] is an important crop in India and serves as a major source of dietary protein for majority of people. The nutritive value of mungbean lies in its high and easily digestible protein and contains approximately 25-28% protein, 1.0 % oil, 3.5-4.5% fiber and vitamins on dry weight basis. Boron influenced the absorption of NPK and its deficiency changed the equilibrium in optimization of these macronutrients. Field studies revealed that deficiency of boron cause considerable reduction in nodulation growth and yield of mungbean (Howeler *et al.*, 1978). Keeping importance of boron in view, an experiment was undertaken with an objective to evaluate the effect of doses of boron fertilization on growth, yield and protein content of mungbean.

An experiment was conducted during *kharif* 2012 at Department of Crop Physiology of C.S. Azad Agriculture University and Technology, Kanpur. The soil of the experimental field was sandy loam in texture with 7.5 pH and had 0.52% organic carbon, 0.056% total nitrogen, 0.127% total phosphorus and 0.580% total potassium. Eight treatments of boron were assigned in randomized block design with three replications (Table 1). Mungbean 'K 851' was sown on 24 July 2011. All the recommended agronomic package of practices was followed for good crop husbandry. Threshing was done manually and weight of seed obtained from each plant was recorded separately by weighing balance. Ten plants were taken from each plot to measure the yield attributes. To obtain protein content in mungbean seed, N content was multiplied with the factor 6.25. The data were analyzed through a statistical computer software Minitab.

The number of pods plant<sup>-1</sup> did not differ statistically among the treatments but higher values were recorded under application of 5 kg boron ha<sup>-1</sup> at flowering (21.6) and the lowest with the control (Table 1). The effect of boron level on number of seeds pod<sup>-1</sup> was significant. The highest number of seeds pod<sup>-1</sup> (12.44) was recorded with boron 0.2% foliar spray, closely followed by 0.1% foliar spray (11.88) and application of boron 10 kg ha<sup>-1</sup> at flowering (11.55), while the lowest was in the control. Rerkasen and Jamjod (1997) also recorded higher number of seeds pod<sup>-1</sup> in mungbean with the application of borax. The highest 1000-seed weight was recorded in boron 0.2% foliar spray treatment (67.20 g), closely followed by treatment with boron 10 kg ha<sup>-1</sup> (65.80 g) at flowering, while lowest was in the control. This might be due to that boron application affects cell division, carbohydrate metabolism, sugar and starch formation, which increases size and weight of grain (Kalyani *et al.*, 1993).

Seed and straw yield plant<sup>-1</sup> of mungbean affected significantly by different doses and methods of boron application (Table 1). It is clear from the data that application of boron increased seed yield significantly over the control. Boron applied through foliar spray was found effective in increasing yields and its attributes. Treatment T<sub>8</sub> (boron 0.2% foliar spray) recorded higher seed yield plant<sup>-1</sup>, closely followed by treatment with application of boron 10 kg ha<sup>-1</sup> at flowering. This may be due to that boron makes the stigma receptive and sticky and making pollen grain fertile enhancing the pollination. Thus increased fruit setting reduces sterility of flower resulting in increased number of grains pod<sup>-1</sup>. The application of boron

Table 1. Effect of boron on yield attributes, yield and protein content of mungbean

| Treatment                                                         | Number of pods plant <sup>-1</sup> | Number of seeds pod <sup>-1</sup> | Seed yield plant <sup>-1</sup><br>(g) | Straw yield plant <sup>-1</sup><br>(g) | Biological yield plant <sup>-1</sup><br>(g) | 1000- seed weight<br>(g) | Harvest index<br>(%) | Protein content |
|-------------------------------------------------------------------|------------------------------------|-----------------------------------|---------------------------------------|----------------------------------------|---------------------------------------------|--------------------------|----------------------|-----------------|
| T <sub>1</sub> : Control                                          | 17.4                               | 9.66                              | 2.00                                  | 4.06                                   | 6.06                                        | 60.10                    | 33.00                | 23.60           |
| T <sub>2</sub> : Boron 0.2% as seed soaking                       | 18.2                               | 10.21                             | 2.03                                  | 4.16                                   | 6.19                                        | 60.40                    | 33.79                | 24.75           |
| T <sub>3</sub> : Boron 0.4% as seed soaking                       | 18.0                               | 10.33                             | 2.20                                  | 4.00                                   | 6.20                                        | 61.40                    | 35.48                | 24.60           |
| T <sub>4</sub> : Boron 5 kg ha <sup>-1</sup> as soil application  | 19.2                               | 10.77                             | 2.41                                  | 4.50                                   | 6.91                                        | 61.60                    | 34.88                | 24.75           |
| T <sub>4</sub> : Boron 10 kg ha <sup>-1</sup> as soil application | 21.4                               | 10.99                             | 2.50                                  | 5.08                                   | 7.58                                        | 61.70                    | 32.98                | 24.71           |
| T <sub>5</sub> : Boron 5 kg ha <sup>-1</sup> at flowering         | 21.6                               | 11.32                             | 2.66                                  | 5.00                                   | 7.66                                        | 63.80                    | 34.22                | 24.92           |
| T <sub>6</sub> : Boron 10 kg ha <sup>-1</sup> at flowering        | 21.4                               | 11.55                             | 2.83                                  | 4.80                                   | 7.63                                        | 65.80                    | 37.09                | 24.87           |
| T <sub>7</sub> : Boron 0.1% as foliar spray                       | 21.0                               | 11.88                             | 2.73                                  | 6.55                                   | 9.63                                        | 64.90                    | 31.98                | 25.03           |
| T <sub>8</sub> : Boron 0.2% as foliar spray                       | 21.0                               | 12.44                             | 3.08                                  | 6.80                                   | 10.53                                       | 67.20                    | 35.42                | 25.23           |
| SEd±                                                              | 2.04                               | 0.42                              | 0.17                                  | 0.29                                   | 0.47                                        | 1.24                     | 0.86                 | 0.16            |
| CD (P=0.05)                                                       | NS                                 | 0.90                              | 0.35                                  | 0.62                                   | 1.01                                        | 2.63                     | 1.83                 | 0.34            |

Table 2. Correlation coefficient between yield parameter and yield of mungbean

| Yield parameter                          | Pods plant <sup>-1</sup> | Seed pod <sup>-1</sup> | Seed yield plant <sup>-1</sup><br>(g) | Straw yield plant <sup>-1</sup><br>(g) | Biological yield plant <sup>-1</sup><br>(g) | 1000-seed weight<br>(g) | Harvest index<br>(%) | Protein content |
|------------------------------------------|--------------------------|------------------------|---------------------------------------|----------------------------------------|---------------------------------------------|-------------------------|----------------------|-----------------|
| Number of pods plant <sup>-1</sup>       | 1.000                    | 0.8410**               | 0.8650**                              | 0.6800                                 | 0.7057                                      | 0.7579*                 | 0.1036               | 0.7064          |
| Number of seeds pod <sup>-1</sup>        |                          | 1.000                  | 0.9760**                              | 0.9034**                               | 0.9416**                                    | 0.9495**                | 0.2089               | 0.8376**        |
| Seed yield plant <sup>-1</sup> (g)       |                          |                        | 1.000                                 | 0.8335*                                | 0.8883**                                    | 0.9639**                | 0.3239               | 0.7565*         |
| Straw yield plant <sup>-1</sup> (g)      |                          |                        |                                       | 1.000                                  | 0.9960**                                    | 0.8167**                | 0.1585               | 0.6587          |
| Biological yield plant <sup>-1</sup> (g) |                          |                        |                                       |                                        | 1.000                                       | 0.8760*                 | -0.0332              | 0.6916          |
| 1000-seed weight (g)                     |                          |                        |                                       |                                        |                                             | 1.000                   | 0.3817               | 0.7150*         |
| Harvest index (%)                        |                          |                        |                                       |                                        |                                             |                         | 1.000                | 0.2736          |
| Protein content                          |                          |                        |                                       |                                        |                                             |                         |                      | 1.000           |

\*, \*\*Significant at P=0.05 and P=0.01 level, respectively

exerted significant effect on straw and biological yield plant<sup>-1</sup>. Boron at 0.2% and 0.1% through foliar spray recorded significantly higher straw yield over the control and most of the boron treatments and the lowest with the control. Maximum harvest index was recorded under application of boron 10 kg ha<sup>-1</sup> at flowering. There was significant effect of boron on 1000-seed weight (Table 1). The highest protein content was registered with boron 0.2% foliar spray treatment, which is at par with 0.1% concentration of boron through foliar spray. This might be due to that boron plays an important role in protein synthesis in plants. Similar results were also reported by Patra and Bhattacharya (2009), who observed positive effect of boron on total protein and amino acid content in mungbean.

Number of pods plant<sup>-1</sup>, number of seed pod<sup>-1</sup>, straw yield plant<sup>-1</sup>, biological yield plant<sup>-1</sup>, 1000-seed weight and protein content have positive significant correlation with seed yield plant<sup>-1</sup> have to give due consideration to these characters during selection for higher yield (Table 2). Number of seeds pod<sup>-1</sup> has significant positive correlation with number of pods plant<sup>-1</sup>, straw yield plant<sup>-1</sup>, biological yield plant<sup>-1</sup>, 1000-seed weight

and protein content. One thousand seed weight have significant positive correlation with number of pods plant<sup>-1</sup>, straw yield plant<sup>-1</sup> and protein content. Protein content has significant positive correlation with number of pods plant<sup>-1</sup> and 1000-seed weight for the selection of high protein content mungbean varieties.

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## SHORT COMMUNICATION

# Screening of potato (*Solanum tuberosum*) genotypes for morphology and yield attributes in mid-hill rainfed condition of Uttarakhand

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## ABSTRACT

Eight new potato (*Solanum tuberosum* L.) hybrids including two checks (Kufri Giriraj and Kufri Jyoti) were evaluated at Ranichauri, Tehri Garhwal, Uttarakhand during *kharif* 2008. Most of the traits differed significantly except shoot numbers hill<sup>-1</sup>, number of compound leaves plant<sup>-1</sup>, shoot weight, 'A' and 'B' grade tubers and specific gravity. Among hybrids, SM/91-1515 gave highest yield and most tolerant to late blight, even at the late stage of the crop. Thus SM/91-1515 hybrid was found to be suitable for Garhwal Himalayas of Uttarakhand.

**Key words:** Genotype, Morphology, Potato, Screening, Yield

Potato (*Solanum tuberosum* L.) is grown all over the world as a staple food next to rice and wheat. It is used as vegetable, stock feed and in industries for manufacturing starch, beverages and other processed products. Currently, it occupies a prominent place in our cropping systems followed in different agro-climatic situations. In Uttarakhand state, this crop, being off season in nature, has a special status. Though, the average yield of potato in the state is low, but potential exists to increase it through genetical and agronomical manipulations. Little attention has been paid so far on the development of suitable early maturing, high yielding varieties for the North-West Himalaya, having disease resistance, particularly to the late blight, which has been observed to be a menace and responsible for low productivity. Presently, a large number of high yielding disease resistant varieties of potato are available for the plain regions of the country but their number for the rainfed hill region is only four. Kufri Jyoti is the first well known cultivar of potato that was recommended for cultivation in Uttarakhand hills way back in 1968. In its initial years of commercialization, it had not only produced high yields but also had resistance to late blight. With passage of time, this variety became susceptible to various diseases. After a long gap, three more varieties, viz. Kufri Giriraj, Kufri Shailja and Kufri Himalini were released for cultivation in the year 1998, 2004 and 2005, respectively. However, Kufri Giriraj and Kufri Shailja could not survive longer

in farmers' fields due to one or other reasons. Considering the above problems in this important off season vegetable crop, a study was conducted to evaluate some of the new potato hybrids developed by Central Potato Research Institute (CPRI), Shimla.

The present investigation was carried out at research block of Department of Vegetable Science, G.B. Pant University of Agriculture and Technology, Hill Campus, Ranichauri, Uttarakhand, India during *kharif* 2008. The research material comprised of six hybrids namely, SM/98-239, SM/95-43, SM/96-127, SM/87-185, KS/96-725 and SM/91-1515 with two checks, viz. Kufri Giriraj and Kufri Jyoti. The Planting material (seed tubers) procured from CPRI Shimla (Himachal Pradesh), India. The experiment was conducted in randomized complete block design with three replications. Disease free, medium sized (2.5 to 5 cm in diameter) and well-sprouted seed tubers were selected. These were treated with Mancozeb (0.2%) for 1 hour before planting. The treated tubers were planted in furrows and covered with soil by making ridges in the last week of February 2008. A basal application of 60 kg N (as urea), 100 kg P<sub>2</sub>O<sub>5</sub> (as diammonium phosphate), 100 kg K<sub>2</sub>O (as muriate of potash) along with 20 tonnes of FYM per hectare was given in different plots. Top dressing of remaining 50% nitrogen (60 kg ha<sup>-1</sup> as urea) was done after 60 days of planting. Life saving irrigation was given as and when needed and harvesting was done in the last week of August 2008.

Uniform cultural practices were adopted for all genotypes. The present investigation was focused on morphological, yield and yield contributing characters, viz. plant emergence, plant

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height, number of shoots hill<sup>-1</sup>, number of compound leaves plant<sup>-1</sup>, leaflet area (cm<sup>2</sup>), shoot fresh weight (g plant<sup>-1</sup>), average number of tubers, average weight of tuber (g), tuber yield (t ha<sup>-1</sup>), tuber grades, specific gravity (g cm<sup>-3</sup>) and tuber dry matter (%).

**Tuber grades:** The grading was done on the basis of tuber diameter at the center. The various grades of tuber are: A grade: > 7.5 cm diameter, B grade: 5.0-7.5 cm diameter and C grade: < 5 cm diameter. The per cent of different grades of tubers was worked out as per following formula:

$$\% \text{ grade of tuber} = \frac{\text{Number of tubers in a grade}}{\text{Total number of tubers}} \times 100$$

**Specific gravity:** Selected sample units (five kg each) were first weighed in air and then the same unit was re-weighed suspended in water. The specific gravity of tuber was calculated as per method followed by Birhman *et al.* (1988).

Specific gravity = Weight in air / (Weight in air - Weight in water)

**Tuber dry matter content:** Dry matter content was determined by oven drying finely chopped tuber pieces first at 80°C for six hours and then at 65°C till constant weight.

The late blight reaction was recorded on Malcalmsom's 1-9 scale basis, where, 1 is highly susceptible and 9 is highly resistant.

It is apparent from Table 1 that potato hybrids significantly differed in per cent plant emergence at 45 days after planting (DAP). Amongst 8 hybrids, KS/96-725 showed highest emergence of 71.26%, followed by Kufri Jyoti (71.05%), whereas, it was minimum in Kufri Giriraj (53.56%). Differences for this character were not visible in KS/96-725, Kufri Jyoti, SM/96-127 and SM/91-1515. Similar variation in plant emergence in different potato hybrids/ varieties was also reported by Mishra *et al.* (2005). The explanation given by these workers for lesser per cent of emergence was mainly the rottage of mother tubers of some of the hybrids in the field. The height of tagged plants at both stages of observations was found significantly different in various hybrids/ varieties (Table 1). Maximum plant height was measured in hybrid SM/91-1515 (91.46 cm at 100 DAP), although it was minimum in KS/96-725. On an average, all the hybrids/ varieties were superior in respect to plant height than that of check Kufri Giriraj. This kind of variability in potato genotypes has also been reported

by Nanedkar and Sharma (1998) and they attributed it due to the inheriting characters of the hybrids. The maximum shoot production was recorded at 100 DAP in hybrid SM/91-1515 (6.33), it was lowest (4.00) in SM/98-239 (Table 1). The differences for shoot number at 100 days of planting were statistically non-significant. Variations in respect to number of compound leaves at 100 DAP among the hybrids were also observed insignificant. It was found highest (70.26) in hybrid SM/91-1515 and lowest (66.46) in SM/87-185.

Leaflet size of all the hybrids increased continuously till 100 days of planting (Table 1). At this stage, hybrid SM/96-127 reported largest size (32.04 cm<sup>2</sup>) of the leaflet, which was smallest (17.48 cm<sup>2</sup>) in the Kufri Giriraj. Contrary to these findings, significance variations among the hybrids for these characters were also reported by various workers (Raghav and Singh, 2004; Kamal *et al.*, 2006; Sharma and Lavatre, 2006). They were of the view that these characters in different genotypes are much influenced by genetic composition as well as environmental factors. Significant differences for shoot fresh weight at the time of dehauling (110 DAP) were not visualized in the hybrids (Table 1).

Amongst the hybrids, SM/91-1515 produced maximum number of tubers in a plant (10.20) but this number was not significantly better to that of produced by the hybrids SM/95-43, SM/96-127 and prevalent variety Kufri Jyoti (Table 2). The minimum numbers of tubers (6) were counted in another check variety Kufri Giriraj. Difference in tuber number in potato hybrids believed to be due to differences in genetic makeup of the cultures as reported by Kamal *et al.* (2006). It is evident from the results (Table 2) that hybrids varied significantly in their tuber weight too. In present investigation character like tuber number plant<sup>-1</sup>, SM/91-1515 was also produced largest sized tubers (113 g) but these were not statistically superior over the tubers of commercialized variety Kufri Jyoti (93 g). However, this increase was observed to the tune of 21.5%. Another hybrid which ranked second in average weight of tuber was SM/87-185 (111 g) and smallest tubers of 73 and 63 g were recorded in hybrids KS/96-725 and Kufri Giriraj, respectively.

It is generally believed that the yield of potato (Table 2) is largely governed by two factors i.e. tuber number and average weight of tuber. Similar observations were also recorded in this study. The hybrid SM/91-1515, which had more number of tubers

Table 1. Growth parameters of different potato genotypes

| Hybrid      | Emergence (%) | Average plant height (cm) | Number of shoots hill <sup>-1</sup> | Number of compound leaves plant <sup>-1</sup> | Average leaflet area (cm <sup>2</sup> ) | Average shoot fresh weight (g plant <sup>-1</sup> ) |
|-------------|---------------|---------------------------|-------------------------------------|-----------------------------------------------|-----------------------------------------|-----------------------------------------------------|
| SM/98-239   | 81.33 (9.01)* | 68.80                     | 4.00                                | 68.06                                         | 29.80                                   | 677.3                                               |
| SM/95-43    | 78.33(8.85)   | 71.66                     | 4.86                                | 69.73                                         | 22.28                                   | 588.0                                               |
| SM/96-127   | 88.00 (9.38)  | 65.80                     | 4.86                                | 66.80                                         | 32.04                                   | 625.3                                               |
| SM/87-185   | 77.33 (8.79)  | 72.53                     | 4.93                                | 66.46                                         | 22.96                                   | 509.3                                               |
| KS/96-725   | 89.33 (9.45)  | 61.80                     | 5.06                                | 67.40                                         | 18.03                                   | 280.0                                               |
| SM/91-1515  | 86.66 (9.30)  | 91.46                     | 6.33                                | 70.26                                         | 18.92                                   | 978.0                                               |
| K. Giriraj  | 64.66 (8.04)  | 61.13                     | 5.26                                | 69.06                                         | 17.48                                   | 585.3                                               |
| K. Jyoti    | 89.33 (9.45)  | 70.66                     | 5.53                                | 67.20                                         | 24.09                                   | 808.0                                               |
| CD (P=0.05) | 6.23          | 7.25                      | NS                                  | NS                                            | 4.83                                    | NS                                                  |

\*Figures in parentheses represent square root transformed values

Table 2. Tuber and yield parameters of different potato genotypes

| Hybrid         | Average tuber number plant <sup>-1</sup> | Average weight of tuber (g) | Yield (t ha <sup>-1</sup> ) | Tuber grade       |                   |                   | Specific gravity | Tuber dry matter (%) |
|----------------|------------------------------------------|-----------------------------|-----------------------------|-------------------|-------------------|-------------------|------------------|----------------------|
|                |                                          |                             |                             | A                 | B                 | C                 |                  |                      |
| SM/98-239      | 8.20                                     | 109.67                      | 33.05                       | 55.28<br>(48.09)* | 30.82<br>(33.63)* | 13.88<br>(21.69)* | 1.042            | 18.32                |
| SM/95-43       | 9.26                                     | 94.00                       | 36.02                       | 47.19<br>(43.38)  | 31.84<br>(34.22)  | 20.95<br>(26.88)  | 1.032            | 18.48                |
| SM/96-127      | 8.93                                     | 96.67                       | 33.56                       | 60.07<br>(51.20)  | 25.52<br>(29.89)  | 14.39<br>(21.86)  | 1.027            | 19.65                |
| SM/87-185      | 7.40                                     | 111.00                      | 34.02                       | 55.46<br>(48.15)  | 23.59<br>(29.00)  | 17.09<br>(24.41)  | 1.054            | 21.91                |
| KS/96-725      | 8.43                                     | 73.00                       | 27.90                       | 41.30<br>(39.75)  | 36.00<br>(36.74)  | 21.94<br>(27.80)  | 1.303            | 19.56                |
| SM/91-1515     | 10.20                                    | 113.00                      | 36.71                       | 53.31<br>(46.89)  | 27.25<br>(31.39)  | 20.65<br>(26.80)  | 0.985            | 18.28                |
| K. Giriraj (c) | 6.00                                     | 63.67                       | 22.07                       | 28.30<br>(31.76)  | 30.51<br>(33.33)  | 41.17<br>(39.85)  | 0.993            | 15.04                |
| K. Jyoti (c)   | 8.86                                     | 93.00                       | 31.34                       | 45.85<br>(42.60)  | 29.48<br>(32.78)  | 24.65<br>(29.75)  | 1.016            | 18.28                |
| CD (P=0.05)    | 1.93                                     | 21.99                       | 6.24                        | NS                | NS                | 8.17              | NS               | 2.63                 |

\*Figures in parentheses represent arcsine transformed values

plant<sup>-1</sup> along with larger size of tubers was also had highest tuber yield (36.71 t ha<sup>-1</sup>). The yield of best hybrid (SM/91-1515) was not only 17.13% more than the yield of check variety Kufri Jyoti but also had given about 66.56% higher yield than another cultivated variety Kufri Giriraj. However statistically, difference in the yield of SM/91-1515 and other new hybrids, viz. SM/95-43, SM/89-185 and SM/96-127 was not significant. More translocation of photosynthates from the side of production to side of utilization and as well as the genetically character of rapid bulking and faster tuberization seen to be the reason for high yield in new hybrids. This variability in tuber yield is in conformity with the earlier findings given by Kamal *et al.* (2006). The other reasons looked for higher yield in SM/91-1515 hybrid were higher bulking rate and moderate resistance for late blight diseases.

It is revealed from Table 2 that the hybrids did not significantly differ for 'A' (above 7.5 cm in diameter) and 'B' (5.0-7.5 cm in diameter) grade tubers; however, significant differences were seen for 'C' grade tubers (<5.0 cm). Maximum 'C' grade tubers were obtained in released varieties Kufri Jyoti and Kufri Giriraj; however, this number was minimum in hybrid SM/98-239. Most hybrids, in general, produced highest number of 'A' grade tubers. Specific gravity did not differed significantly among the tested hybrids/varieties. Highest specific gravity of 1.303 was recorded in KS/96-725, whereas it was lowest in SM/91-1515.

However, tuber dry matter content was observed significantly for all genotypes. Hybrid SM/87-185 showed the maximum tuber dry matter content (21.91%); therefore, this

variety may be suggested for processing, while it was minimum in Kufri Giriraj (15.04%). Singh *et al.* (2008) found significantly different dry matter content in potato varieties. Hybrid SM/91-1515 and SM/96-127 proved to be the most blight tolerant genotypes even in the late stage of the crop.

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## SHORT COMMUNICATION

# Yield and yield attributes of brinjal (*Solanum melongena*) germplasms of North East India

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## ABSTRACT

An investigation was carried out during *rabi* season of 2009-10 and 2010-11 at the Horticulture Experimental Farm of the Assam Agricultural University, Jorhat (Assam), India to assess the yield and other characteristics including reaction to bacterial wilt of the available germplasms of brinjal. The highest fruit yield of 2.5 kg plant<sup>-1</sup> was observed in GB 09-16-02, followed by GB 09-02-02, GB 09-05 and GB 09-12 and JB 09-03 with fruit yield of 2.3, 2.2, 2.1 and 2.1 kg plant<sup>-1</sup>, respectively. GB 09-16-02 exhibited highest fruit number plant<sup>-1</sup> (30) and plant height (33 cm). GB 09-02-02 was the best for fruit length (33 cm) and early flowering (117 days). GB 09-05 exhibited best performance for fruit diameter (12 cm) and single fruit weight (300 g). The long genotypes, viz. GB 09-16-02, GB 09-02-02 and GB 09-12 and round brinjals, viz. GB 09-05 and JB 09-03 were found to be resistant to bacterial wilt also. Hence, these genotypes may be promoted to develop as new varieties or may be utilized in the crossing programme as parents.

**Key words:** Bacterial wilt, Brinjal, Germplasm, *Per se* performance, Yield

Brinjal is one of the most common, highly productive and popular vegetable crop widely cultivated in India. It was originated in India including North Eastern Region (Doijode, 2001; Tsao and Lo, 2006). This part of the country has accumulated wide range of variability in this crop and many local and exotic germplasm are available in this region. In AVRDC, a centre for world vegetable germplasm, Rashid *et al.* (1999) studied 270 germplasm for resistance to various biotic stresses including bacterial wilt. But the work on the available germplasms in North East India is limited. Therefore, it is very essential to study the germplasms for desirable characteristics in order to identify promising genotypes for commercial cultivation and use in the future breeding programmes.

The experiment was conducted at the Experimental Farm of the Department of Horticulture, Assam Agricultural University, Jorhat (Assam) during 2009-10 and 2010-11 with sixty-five germplasm of brinjal including PPL, JC-1 and Arka Nidhi as three National check varieties. The trial was laid out in randomized block design with three replications. The seedlings were transplanted at a spacing of 60cm×45cm. All the recommended cultural practices were followed. The observations were recorded on five randomly selected plants per replication for each genotype for most of the desirable characters like plant height, number of fruits plant<sup>-1</sup>, fruit length, fruit yield, duration to 50% flowering, single fruit weight (g) and days to 1<sup>st</sup> harvest. The plants were also observed for bacterial wilt incidence under wilt sick plot by counting the

number of plants wilted out of the total number of plants expressed in percentage. The grading of the genotypes was done following the scale of Mew and Ho (1976) modified by Hussain *et al.* 2005 as: resistant (R)- 0 to 20%, moderately resistant (MR)- 20 to 40%, moderately susceptible (MS)- 40 to 60% and susceptible (S)- above 60%.

Great diversity among the genotypes was seen for characters like fruit shape, colour, yield, fruit length and fruits plant<sup>-1</sup> (Table 1). Shape of the fruits varied from long to round or oblong. Different fruit colours like purple, white, pink etc were observed. The height of the plants of the diverse germplasms ranged from 34 cm in JB 10-13 to 100 cm in genotype GB 09-16-02 (Table 2). Fruits plant<sup>-1</sup> ranged from 3 to 30. The highest number of fruits plant<sup>-1</sup> was recorded in GB 09-16-02 (30 fruits) and the lowest was recorded in GB 09-07-02 and JB 09-01 (3 fruits each). Out of 14 genotypes showing high number of fruits plant<sup>-1</sup>, GB 09-02-02, SM6-6, GB 09-02-01, JB 10-13, JB 10-15, JB 10-16 and JB 10-22 produced 13 to 16 fruits plant<sup>-1</sup>. The highest single fruit weight was recorded in GB 09-05 (300 g) and lowest was observed in BB-54 (35 g). All total 9 genotypes showed high single fruit weight. Among the 65 entries, fruit length ranged from 12 cm in GB 09-05 to 33 cm in GB 09-02-02. Diameter of fruits ranged from 3 cm to 12 cm, the highest was recorded in GB 09-05 and the lowest in GB 09-02-02. The range of duration of 50% flowering was from 117 days in GB 09-02-02 to 136 days in GB 09-10-14. Results further revealed that genotypes such as 09/BRBW RES-01, 09/BRBW

Table 1. Diversity of 65 brinjal germplasms for different quantitative characters

| Characters                           | High | Medium | Low |
|--------------------------------------|------|--------|-----|
| Plant height (cm)                    | 22   | 20     | 23  |
| Number of fruits plant <sup>-1</sup> | 14   | 21     | 30  |
| Fruit length (cm)                    | 9    | 20     | 36  |
| Fruit diameter (cm)                  | 12   | 15     | 38  |
| Fruit yield plant <sup>-1</sup>      | 25   | 20     | 20  |
| Duration to 50% flowering            | 32   | 15     | 18  |
| Single fruit weight (g)              | 25   | 18     | 22  |
| Days to 1 <sup>st</sup> harvest      | 22   | 27     | 16  |

another study, Bora *et al.*, 2011 reported ‘Utsav’ as a very good brinjal variety resistant to bacterial wilt disease.

From the study, long genotypes of brinjal, viz. GB 09-16-02, GB 09-02-02 and GB 09-12 were the best for yield, duration, tolerance to diseases; whereas the round genotypes, viz. GB 09-05, JB 09-03 and GB 09-11-01 were best for similar characters. These genotypes may be promoted for development as new variety or may be utilized as parents in the hybridization programmes.

Table 2. Mean performance of promising brinjal germplasms (mean of 2 years)

| Genotype      | Fruit yield plant <sup>-1</sup> (kg) | Fruits plant <sup>-1</sup> | Fruit length (cm) | Fruit diameter (cm) | Single fruit weight (g) | Plant height (cm) | Days to 50% flowering | Days to 1 <sup>st</sup> harvest | Reaction to bacterial wilt |
|---------------|--------------------------------------|----------------------------|-------------------|---------------------|-------------------------|-------------------|-----------------------|---------------------------------|----------------------------|
| GB 09-16-02   | 2.5 I                                | 30 I                       | 32                | 4                   | 80                      | 100 I             | 128                   | 144                             | R                          |
| GB 09-02-02   | 2.3 II                               | 13                         | 33 I              | 3                   | 225                     | 68                | 117 I                 | 130                             | R                          |
| GB 09-12      | 2.1 IV                               | 11 IV                      | 28                | 7                   | 180                     | 56                | 120 III               | 140                             | R                          |
| GB 09-05      | 2.2 III                              | 10                         | 12                | 12 I                | 300 I                   | 54                | 125                   | 146                             | R                          |
| GB 09-02-01   | 1.2                                  | 8                          | 21                | 4                   | 200                     | 64                | 130                   | 129                             |                            |
| GB 09-05-03   | 0.5                                  | 6                          | 30                | 4                   | 100                     | 71                | 121                   | 135                             |                            |
| JB 09-03      | 2.1 IV                               | 5                          | 14                | 8 IV                | 250 II                  | 60                | 134                   | 150                             | R                          |
| GB 09-11-01   | 1.9                                  | 5                          | 14                | 10                  | 150                     | 45                | 130                   | 145                             | MR                         |
| GB 09-10-14   | 1.2                                  | 4                          | 12                | 5                   | 150                     | 52                | 136                   | 146                             | R                          |
| JB 10-13      | 1.5                                  | 14                         | 15                | 3                   | 113                     | 34                | 132                   | 150                             | MR                         |
| JB 10-14      | 1.2                                  | 11                         | 11                | 6                   | 163                     | 36                | 128                   | 155                             | MR                         |
| JC-1©         | 0.6                                  | 8                          | 16                | 5                   | 195                     | 85                | 130                   | 150                             | R                          |
| JC-2©         | 0.8                                  | 5                          | 14                | 9                   | 250                     | 66                | 118                   | 140                             | MR                         |
| SM 6-6©       | 1.2                                  | 13                         | 15                | 7                   | 58                      | 43                | 135                   | 150                             | MR                         |
| PPL©          | 0.6                                  | 8                          | 15                | 5                   | 150                     | 69                | 130                   | 150                             | S                          |
| ArkaNidhi©    | 0.9                                  | 7                          | 14                | 4                   | 40                      | 54                | 120                   | 140                             | R                          |
| CD (P = 0.05) | 0.8                                  | 5.0                        | 6.0               | 4.0                 | 51.0                    | 10.0              | 7.0                   | 6.5                             |                            |
| CV (%)        | 8.5                                  | 6.3                        | 5.3               | 3.6                 | 8.3                     | 6.3               | 2.8                   | 3.5                             |                            |

R = Resistant; MR = Moderately resistant; MS= Moderately susceptible; S = Susceptible

RES-04, GB 09-02-02, GB 09-12, GB 09-02-01, GB 09-05-03, JB 10-18, JB 10-19 and JB 10-22 were found to be early maturing. The highest fruit yield plant<sup>-1</sup> was recorded in GB 09-16-02 (2.5 kg) and the lowest was recorded in JB 09-01 (0.3 kg) which also had the least number of fruits plant<sup>-1</sup>. The long genotypes, viz. GB 09-16-02, GB 09-02-02, GB 09-12 and round genotypes such as GB 09-05, JB 09-03 and GB 09-11-01 were found to be good yielders.

Bacterial wilt disease in brinjal is a serious problem in Assam and other North Eastern region of India. In the present study, the genotypes, viz. BB-54, Utsav, Singnath, Bholanath, Arka Nidhi, GB 09-02-02, GB 09-16-02, GB 09-05, GB 09-12, JB 09-13, GB 09-05-03 and GB 09-10-14 were identified as resistant; whereas GB 09-11-01, 10/BRBW RES-4, RCMBL-3, Khoruah-3, JB 10-13, JB 10-15, JB 10-18 and SM 6-6 as moderately resistant against bacterial wilt under naturally wilt sick plot (Table 2). In

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## SHORT COMMUNICATION

# Impact of weather on phenology of guava in Uttar Pradesh: A cursory analysis

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## ABSTRACT

A study was conducted to analyze the reasons for failure of guava crop during rainy season of 2010 through roving survey of principal guava growing belt around Lucknow and Kanpur in Uttar Pradesh. The study indicated that in order to mitigate the adverse impacts of dry weather, irrigation of guava orchards at 7-10 days intervals during March – April to keep the temperatures around (38.1°C) and ensure relative humidity of 85% appears to be a practical approach for ensuring a reasonable rainy season crop as against total failure. Intercropping of shallow rooted short duration vegetables in guava orchards, besides providing additional income and employment opportunities, helps in modifying microclimate in guava orchards insulating from total crop failure and improving sustainable livelihood options.

**Key words:** Guava, Microclimate manipulation, Temporal relations, Weather dynamics

Guava, often called as poor man's apple, is an important fruit crop of the Indo-Gangetic plains and is considered a hardy fruit tree besides being rich in nutrients (Morton, 1987). It has a tendency to produce profusely under good management practices two to three crops annually offering sustained livelihood options throughout the year. Uttar Pradesh is an important guava growing state and best quality guavas are grown here. Major guava producing districts are Allahabad, Kaushambi, Kanpur, Farrukhabad, Lucknow, Badaun, Aligarh and Shahjahanpur. Uttar Pradesh is the highest guava producing state with an annual production of about 412.6 thousand metric tonnes and productivity is 16.5 t ha<sup>-1</sup> which is more than national productivity of guava (12.0 t ha<sup>-1</sup>). Guava is now an export commodity and 289.3 MT of guavas were exported from India to different countries (NHB, 2011).

Although rainy season crop offers large volume of relatively poor quality fruits as compared to winter crop, it is commercially important owing to ease of production naturally and reasonably good demand of fruits in markets. Commercial production of rainy season guava is thus one of the livelihood options for guava growers across the country in spite of relatively low prices. In the present scenario of climate change and especially during 2010 fruiting season, deviation from the normal crop phenology leading to impacts on production in guava producing areas of Uttar Pradesh was discernible. During rainy season of 2010 very little fruiting of guava in Lucknow, Kanpur and nearby areas was observed. In this background, a study was conducted to understand the reasons for poor productivity through roving survey of principal guava growing belts around Lucknow and Kanpur. Interactions were made

with guava growers of five villages (Shivdeenpurva, Jaggupurva, Kallupurva, Peeperkhura and Gurripurva) Kanpur and Unnao districts and a sample size of sixty five farmers was considered in the survey.

All the guava growers were given a pre-designed questionnaire to give their responses with regard to different aspects of their cultural management practices and yield performance. After analysis of feedback received from farmers, it was brought out that there was normal flowering in rainy season crop during March, 2010 but flowers and fruit sets dropped due to high temperature and hot winds that prevailed in the months of March and April, 2010. Eighty per cent farmers responded that even after providing 1-2 irrigations, the crop could not be saved. Only 20 per cent farmers around Kanpur were successful in getting satisfactory yield of rainy season crop that raised intercrops of short duration shallow rooted vegetables, viz. okra, bitter melon and chillies. They provided regular irrigation to intercrops at weekly intervals during March – April 2010 and in this way the main crop was also benefited in minimizing flowers and fruit drop. Through five irrigations, about 4.0 m<sup>3</sup> water was given per acre to reduce the heat load. Flowering during rainy season crop conditions occurs during March – April and fruits are harvested from July to September under Lucknow. The weather parameters such as maximum and minimum temperature and rainfall with the crop phenology indicated that under normal conditions in Lucknow, as against the mean maximum and minimum temperatures in March of 32.2°C and 14.2°C, 34.41°C and 15.87°C, respectively (Fig. 1a and 1b), prevailed during 2010 an increase of 6.86% and 11.76% over the normal. Similarly in April 2010, maximum and minimum

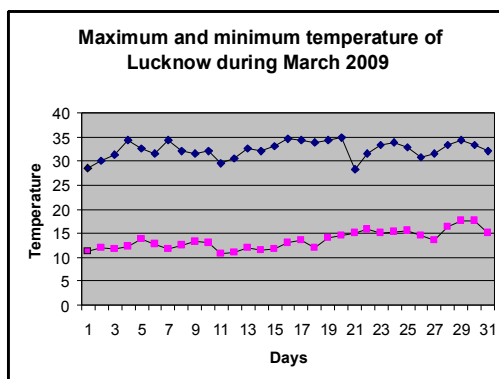


Fig. 1(a)

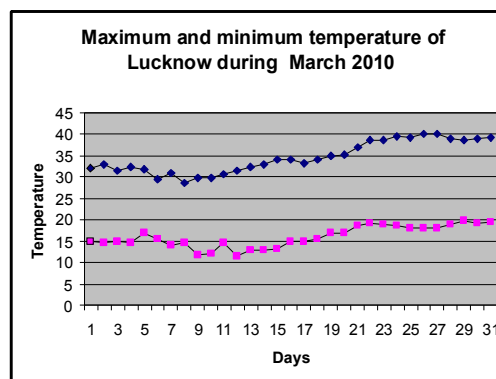


Fig. 1(b)

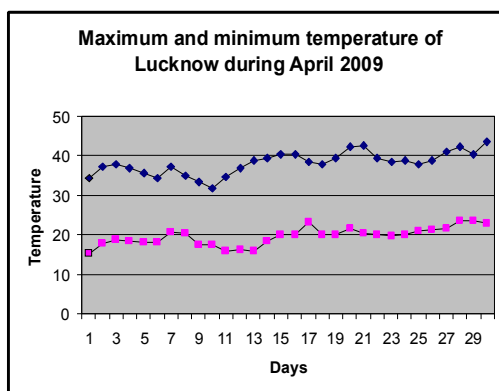


Fig 2(a)

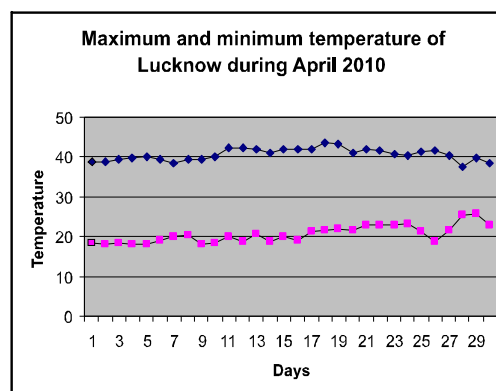


Fig 2(b)

temperatures were 40.53°C and 20.63°C which were also quite high (Fig. 2a and 2b). During fourth week of March 2010, maximum temperature levels rose to 39.5 to 40°C continuously for five days and it reached up to 43.5°C on 18<sup>th</sup> and 19<sup>th</sup> April, 2010. Relative humidity was low (69.1%) during the first fortnight of April. On 9<sup>th</sup> April 2010, it was the lowest (43%). Prevalence of high temperatures, low relative humidity (RH) and hot winds during the critical periods of flowering and fruit set adversely impacted the rainy season crop prospects of 2010 due to dropping of flowers and fruit lets. Climatic factors namely temperature, relative humidity and wind have been considered to have a profound influence on fruit drop in fruits.

Some of the earlier reports described high temperature is the main cause of immature fruit drop in fruits (Chandler, 1925; Powell, 1930 and Kinnison and Finch, 1934). Climatic conditions have been variously associated with fruit drop in many fruit crops. It is felt that during abnormally high temperature, the rate of transpiration increased from the leaves and young fruits and consequently sufficient quantity of moisture was withdrawn from the fruits which were unable to withstand the stress and were shed easily (Randhava and Chaddha, 1994). It was also reported that temperature associated with continuous high winds often upset the normal moisture which may not keep pace with increased transpiration resulting in conditions leading to incipient wilting which may in its turn trigger off premature abscission of fruits. Wagle (1928) reported that application of irrigation during flowering season to trees made a very substantial difference in the yield of fruits. Rainfall has been considered to be main limiting resource for crop growth

(Barron *et al.*, 2003). Guava, being a shallow rooted crop with its feeder roots distribution only up to 20 - 25 cm depth (<http://www.fruitipedia.com/guava.htm>). Dry spells during flowering and fruit growth period could cause flower and fruit drop. There was no rainfall during March and April 2010 when compared with average rainfall of 7.7 mm and 4.9 mm during respective months of the previous year.

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## SHORT COMMUNICATION

# Growth, yield and quality of phalsa (*Grewia subinaequalis*) with pruning intensity and nitrogen levels in sodic soil

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## ABSTRACT

The experiment was conducted during two consecutive years of 2007-08 and 2008-09 with four levels of pruning ( $P_0$ : control,  $P_1$ : 30 cm,  $P_2$ : 60 cm, and  $P_3$ : 90 cm above the ground level) and four levels of nitrogen ( $N_0$ : control,  $N_1$ : 40 g,  $N_2$ : 80 g and  $N_3$ : 120 g plant<sup>-1</sup>) in phalsa (*Grewia subinaequalis* D.C.) under sodic soil conditions of Lucknow (Uttar Pradesh). Mean data revealed that maximum shoot length (2.09 m) was recorded with treatment combination  $P_0N_2$ , while maximum plant spread (2.01 m), number of canes bush<sup>-1</sup> (18.47), number of fruits bush<sup>-1</sup> (1478) and yield bush<sup>-1</sup> (0.825 kg) were recorded with  $P_3N_2$  treatment. However, the highest average values of 100-fruit weight (61.72 g) in  $P_1N_2$ , juice (54.30%) in  $P_1N_1$ , TSS (17.7 °B) in  $P_3N_3$ , acidity per cent (4.42) in  $P_2N_1$  and ascorbic acid (38.65 mg 100<sup>-1</sup> g) in  $P_2N_2$  treatment were recorded. Overall, treatment combination of  $P_3N_2$  (90 cm pruning + 80 g N bush<sup>-1</sup>) performed better.

**Key words:** Growth, Nitrogen levels, Phalsa, Pruning intensity, Quality, Sodic soil, Yield

Phalsa (*Grewia subinaequalis* D.C.) is an indigenous perennial bush, hardy minor fruit and can be grown successfully under rainfed conditions on marginal, sub marginal and saline soils through minimum use of inputs (Singh *et al.*, 1986). It is popular fruit of subtropical and tropical regions (Singh and Singh, 2003) and commercially cultivated in Punjab, Haryana, Uttar Pradesh and Andhra Pradesh (Kandolia and Bhua, 1996). Phalsabears small roundish pinkish fruits called berry, is used as table purpose and it has the medicinal properties also (Awasthi, 1997). It is a rich source of vitamins A and C and comes in market in the month of June when other fresh fruits are not available (Sharma *et al.*, 2008). Phalsa bears on current year's growth and proper pruning provides a long productive life to the plant, improves yield and fruit quality, besides regulating the growth and vigour of the plant. Nitrogen governs plant growth by virtue of being a major constituent of chlorophyll, protein, amino acids and photosynthetic activity (Rathore *et al.*, 2008). However, very little information is available on the effect of pruning intensity and nitrogen fertilization in phalsa under sodic soil conditions.

The experiment was conducted under sodic soil conditions during two consecutive years of 2007-08 and 2008-09 at the Horticulture Research Farm of the Department of Applied Plant Science (Horticulture), Babasaheb Bhimrao Ambedkar University, Lucknow, India. It was conducted on 3-

year old phalsa plants of local cultivar and laid out with two factors randomized block design. Treatments comprised four levels of pruning, viz. control ( $P_0$ ), 30 cm ( $P_1$ ), 60 cm ( $P_2$ ), and 90 cm ( $P_3$ ) above ground level and four levels of nitrogen, viz. control ( $N_0$ ), 40 g ( $N_1$ ), 80g ( $N_2$ ), and 120 g ( $N_3$ ) plant<sup>-1</sup>. The pruning was done after commencing the dormancy in the month of December. Nitrogen was applied in the form of urea. The data recorded was analyzed to test the level of significance.

Growth parameters, viz. shoot length, plant spread and number of canes bush<sup>-1</sup> was affected significantly by severity of pruning and levels of nitrogen during both the years of study (Table 1). Maximum shoot length (1.87 and 1.80 m), plant spread (1.70 and 1.92 m) and number of canes bush<sup>-1</sup> (13.90 and 18.50), respectively, was recorded with  $P_3$  (90 cm pruning) during both the years. Application of 80 g N bush<sup>-1</sup> recorded maximum shoot length (1.79 and 1.70 m), plant spread (1.73 and 1.90 m, respectively) and number of canes bush<sup>-1</sup> (13.55 and 16.72, respectively) during both the years. Due to interaction effect of pruning intensity x nitrogen, maximum shoot height was recorded (1.94 and 2.24 m) in  $P_0N_2$ , followed by  $P_3N_2$  (2.12 and 1.95 m) during both the year of experiment, respectively. While, maximum plant spread (1.83 and 2.20 m) and number of canes bush<sup>-1</sup> (16.40 and 20.54) was observed in  $P_3N_2$  during the both years, respectively. Plant height, spread and number of canes bush<sup>-1</sup> decreased with increase in the severity of pruning. Increase in number of canes might have been due to optimum availability of plant nutrition. Similar findings were obtained by Singh *et al.* (2004) who reported minimum height of new

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Table 1. Effect of pruning intensity and nitrogen levels on growth, yield and quality of phalsa

| Treatment                     | Shoots length<br>(m) | Plant spread<br>(m) | Number of canes<br>bush <sup>-1</sup> | Number of fruits<br>bush <sup>-1</sup> | 100-fruit weight<br>(g) | Yield bush <sup>-1</sup><br>(kg) | Juice<br>(%) | TSS<br>(°B) | Acidity<br>(%) | Ascorbic acid<br>(mg 100 <sup>-1</sup> g fresh<br>fruit weight) |       |       |       |       |       |       |      |      |       |       |
|-------------------------------|----------------------|---------------------|---------------------------------------|----------------------------------------|-------------------------|----------------------------------|--------------|-------------|----------------|-----------------------------------------------------------------|-------|-------|-------|-------|-------|-------|------|------|-------|-------|
|                               | 2007-08              | 2008-09             | 2007-08                               | 2008-09                                | 2007-08                 | 2008-09                          | 2007-08      | 2008-09     | 2007-08        | 2008-09                                                         |       |       |       |       |       |       |      |      |       |       |
| Pruning intensity             |                      |                     |                                       |                                        |                         |                                  |              |             |                |                                                                 |       |       |       |       |       |       |      |      |       |       |
| P <sub>0</sub>                | 1.77                 | 1.74                | 1.67                                  | 1.66                                   | 8.53                    | 8.74                             | 1162         | 1096        | 52.75          | 53.87                                                           | 0.604 | 0.586 | 42.32 | 41.50 | 14.63 | 14.46 | 3.94 | 3.41 | 30.17 | 32.05 |
| P <sub>1</sub>                | 1.33                 | 1.31                | 1.27                                  | 1.28                                   | 11.90                   | 13.81                            | 1054         | 1190        | 57.90          | 53.30                                                           | 0.615 | 0.636 | 53.22 | 52.87 | 15.06 | 15.08 | 3.81 | 3.79 | 33.20 | 31.96 |
| P <sub>2</sub>                | 1.41                 | 1.52                | 1.44                                  | 1.78                                   | 13.27                   | 16.57                            | 1215         | 1206        | 54.57          | 54.00                                                           | 0.669 | 0.649 | 52.02 | 52.27 | 15.40 | 15.66 | 3.72 | 3.89 | 36.80 | 35.53 |
| P <sub>3</sub>                | 1.87                 | 1.80                | 1.70                                  | 1.92                                   | 13.90                   | 18.50                            | 1342         | 1276        | 54.34          | 55.35                                                           | 0.730 | 0.708 | 47.80 | 48.50 | 16.23 | 16.27 | 3.48 | 3.57 | 36.96 | 35.25 |
| CD (P=0.05)                   | 0.05                 | 0.07                | 0.06                                  | 0.07                                   | 0.53                    | 0.57                             | 44           | 49          | 1.47           | 1.48                                                            | 0.016 | 0.028 | 0.69  | 0.75  | 0.50  | 0.58  | 0.07 | 0.04 | 0.07  | 0.15  |
| Nitrogen level                |                      |                     |                                       |                                        |                         |                                  |              |             |                |                                                                 |       |       |       |       |       |       |      |      |       |       |
| N <sub>0</sub>                | 1.51                 | 1.37                | 1.43                                  | 1.37                                   | 10.60                   | 13.02                            | 1017         | 1057        | 53.30          | 53.62                                                           | 0.542 | 0.565 | 45.57 | 47.80 | 14.87 | 14.75 | 3.16 | 3.51 | 30.91 | 33.13 |
| N <sub>1</sub>                | 1.56                 | 1.63                | 1.36                                  | 1.71                                   | 11.92                   | 14.38                            | 1193         | 1166        | 58.28          | 55.27                                                           | 0.695 | 0.646 | 47.95 | 48.15 | 14.16 | 14.28 | 4.19 | 3.81 | 35.10 | 32.68 |
| N <sub>2</sub>                | 1.79                 | 1.70                | 1.73                                  | 1.90                                   | 13.55                   | 16.72                            | 1345         | 1276        | 59.51          | 57.57                                                           | 0.795 | 0.734 | 51.42 | 50.22 | 15.72 | 15.67 | 3.63 | 3.55 | 36.57 | 35.47 |
| N <sub>3</sub>                | 1.51                 | 1.67                | 1.55                                  | 1.65                                   | 11.54                   | 13.50                            | 1219         | 1268        | 48.48          | 50.07                                                           | 0.585 | 0.635 | 50.42 | 49.57 | 16.57 | 16.77 | 3.99 | 3.79 | 33.50 | 33.50 |
| CD (P=0.05)                   | 0.05                 | 0.07                | 0.06                                  | 0.07                                   | 0.53                    | 0.57                             | 44           | 49          | 1.47           | 1.48                                                            | 0.016 | 0.028 | 0.69  | 0.75  | 0.50  | 0.58  | 0.07 | 0.04 | 0.07  | 0.15  |
| Interaction                   |                      |                     |                                       |                                        |                         |                                  |              |             |                |                                                                 |       |       |       |       |       |       |      |      |       |       |
| P <sub>0</sub> N <sub>0</sub> | 1.80                 | 1.24                | 1.55                                  | 1.13                                   | 8.20                    | 4.84                             | 956          | 961         | 52.40          | 55.70                                                           | 0.501 | 0.535 | 38.70 | 39.50 | 14.25 | 14.75 | 3.20 | 3.60 | 26.10 | 31.50 |
| P <sub>0</sub> N <sub>1</sub> | 1.72                 | 1.81                | 1.64                                  | 1.70                                   | 7.48                    | 8.95                             | 1009         | 927         | 57.50          | 55.00                                                           | 0.580 | 0.510 | 41.20 | 42.00 | 13.80 | 13.60 | 4.65 | 3.25 | 33.00 | 30.33 |
| P <sub>0</sub> N <sub>2</sub> | 1.94                 | 2.24                | 1.82                                  | 2.00                                   | 10.00                   | 10.80                            | 1208         | 1142        | 58.40          | 56.50                                                           | 0.705 | 0.645 | 43.90 | 43.50 | 15.10 | 14.50 | 4.14 | 3.30 | 31.60 | 34.00 |
| P <sub>0</sub> N <sub>3</sub> | 1.62                 | 1.68                | 1.67                                  | 1.84                                   | 8.45                    | 10.39                            | 1475         | 1354        | 42.70          | 48.30                                                           | 0.630 | 0.654 | 45.50 | 41.00 | 15.40 | 15.00 | 3.80 | 3.50 | 30.00 | 32.40 |
| P <sub>1</sub> N <sub>0</sub> | 1.22                 | 1.40                | 1.20                                  | 1.00                                   | 10.20                   | 13.00                            | 935          | 1109        | 55.60          | 52.30                                                           | 0.520 | 0.580 | 51.70 | 52.50 | 14.40 | 14.80 | 2.90 | 3.60 | 28.00 | 32.00 |
| P <sub>1</sub> N <sub>1</sub> | 1.45                 | 1.30                | 1.00                                  | 1.42                                   | 12.40                   | 13.60                            | 1119         | 1183        | 59.40          | 54.50                                                           | 0.665 | 0.645 | 54.20 | 54.60 | 14.15 | 13.85 | 4.15 | 3.55 | 35.50 | 29.50 |
| P <sub>1</sub> N <sub>2</sub> | 1.51                 | 1.12                | 1.48                                  | 1.52                                   | 11.50                   | 17.50                            | 1177         | 1346        | 66.25          | 57.20                                                           | 0.780 | 0.770 | 55.60 | 50.60 | 15.80 | 15.20 | 4.40 | 3.80 | 35.40 | 33.80 |
| P <sub>1</sub> N <sub>3</sub> | 1.14                 | 1.42                | 1.43                                  | 1.20                                   | 13.50                   | 11.14                            | 984          | 1122        | 50.40          | 49.20                                                           | 0.496 | 0.552 | 51.40 | 53.80 | 15.90 | 16.50 | 3.80 | 4.20 | 33.90 | 32.50 |
| P <sub>2</sub> N <sub>0</sub> | 1.32                 | 1.24                | 1.24                                  | 1.66                                   | 11.20                   | 16.44                            | 966          | 983         | 50.60          | 54.40                                                           | 0.489 | 0.535 | 50.40 | 51.50 | 15.10 | 14.85 | 3.65 | 3.60 | 32.80 | 32.85 |
| P <sub>2</sub> N <sub>1</sub> | 1.28                 | 1.52                | 1.35                                  | 1.89                                   | 13.60                   | 16.80                            | 1302         | 1235        | 56.20          | 52.50                                                           | 0.732 | 0.648 | 52.60 | 53.50 | 13.50 | 14.20 | 3.85 | 5.00 | 37.30 | 36.20 |
| P <sub>2</sub> N <sub>2</sub> | 1.62                 | 1.50                | 1.81                                  | 1.91                                   | 16.30                   | 18.04                            | 1420         | 1235        | 59.50          | 58.70                                                           | 0.845 | 0.725 | 53.80 | 51.80 | 15.40 | 16.00 | 3.00 | 3.50 | 39.80 | 37.50 |
| P <sub>2</sub> N <sub>3</sub> | 1.45                 | 1.84                | 1.38                                  | 1.66                                   | 12.00                   | 15.00                            | 1173         | 1369        | 52.00          | 50.40                                                           | 0.610 | 0.690 | 51.30 | 52.30 | 17.60 | 17.60 | 4.40 | 3.45 | 37.30 | 35.60 |
| P <sub>3</sub> N <sub>0</sub> | 1.72                 | 1.60                | 1.76                                  | 1.72                                   | 12.80                   | 17.80                            | 1208         | 1176        | 54.62          | 52.00                                                           | 0.660 | 0.610 | 41.50 | 45.30 | 15.76 | 14.60 | 2.90 | 3.25 | 36.75 | 36.20 |
| P <sub>3</sub> N <sub>1</sub> | 1.80                 | 1.92                | 1.48                                  | 1.84                                   | 14.20                   | 18.20                            | 1340         | 1318        | 60.00          | 59.10                                                           | 0.804 | 0.780 | 43.80 | 42.50 | 15.20 | 15.50 | 4.10 | 3.45 | 34.80 | 34.70 |
| P <sub>3</sub> N <sub>2</sub> | 2.12                 | 1.95                | 1.83                                  | 2.20                                   | 16.40                   | 20.54                            | 1576         | 1380        | 53.95          | 57.90                                                           | 0.850 | 0.800 | 52.40 | 55.00 | 16.60 | 17.00 | 3.00 | 3.60 | 39.50 | 36.60 |
| P <sub>3</sub> N <sub>3</sub> | 1.84                 | 1.74                | 1.73                                  | 1.93                                   | 12.22                   | 17.46                            | 1242         | 1229        | 48.80          | 52.40                                                           | 0.606 | 0.644 | 53.50 | 51.20 | 17.40 | 18.00 | 3.95 | 4.00 | 32.80 | 33.50 |
| CD (P=0.05)                   | 0.10                 | 0.13                | 0.12                                  | 0.14                                   | 1.06                    | 1.15                             | 87           | 99          | 2.95           | 2.96                                                            | 0.033 | 0.056 | 1.38  | 1.50  | 1.00  | 1.15  | 0.14 | 0.07 | 0.14  | 0.31  |

growth and canes bush<sup>-1</sup> when pruned severely, whereas maximum was recorded in control and light pruned plants. The results of Singh (2000) for plant spread corroborates the present findings.

A significant response on yield parameters, viz. number of fruits bush<sup>-1</sup> and fruit yield bush<sup>-1</sup> with the different levels of pruning and nitrogen were recorded during both the year of study (Table 1). However for 100 fruit weight, a non-significant variation was observed due to single effect of pruning and nitrogen but recorded significant with interaction of both. In different pruning severity, highest number of fruits bush<sup>-1</sup> (1342 and 1276) and yield bush<sup>-1</sup> (0.730 and 0.708 kg) recorded with P<sub>3</sub> (90 cm pruning) during first and second year, respectively; whereas, maximum average 100 fruit weight (57.9 and 53.30 g) was recorded in severe pruning (P<sub>1</sub>). Highest number of fruits bush<sup>-1</sup> (1345 and 1276), 100 fruit weight (59.51 and 57.57 g) and yield bush<sup>-1</sup> (0.795 and 0.734 kg) were recorded with the application of 80 g N bush<sup>-1</sup> during the both year of experimentation, respectively. In combination of pruning with nitrogen, highest number of fruits bush<sup>-1</sup> (1576 and 1381) and highest yield bush<sup>-1</sup> (0.85 and 0.80 kg) was noticed in P<sub>3</sub>N<sub>2</sub> during both the years, respectively; however, maximum 100 fruit weight (66.25 and 57.20 g) was noticed with P<sub>1</sub>N<sub>2</sub>. The 100 fruit weight might be increased due to the adequate amount of nitrogen and heavy pruning of the plants which results decreased number of fruits with increased fruit weight. Higher yield might be due to the accommodation of more branches and higher number of fruits in light pruning which directly attributed to increased fruit yield. These results corroborate with the findings of Kandolia and Bhuva (1996) for 100 fruit weight and yield bush<sup>-1</sup>, while results of Singh (2000) for number of fruits is in favor of present finding.

Significantly higher juice % was recorded in severe pruned plants (P<sub>1</sub>). Application of 80 g N bush<sup>-1</sup> (N<sub>2</sub>) recorded maximum juice %, while, lowest in control (N<sub>0</sub>) (Table 1). Under combined effect of pruning intensity and nitrogen, higher juice % was recorded under P<sub>1</sub>N<sub>1</sub>, which might be due to bigger size and higher weight of fruit in severe pruned plants. However, lowest juice % was recorded under control treatments (P<sub>0</sub>N<sub>0</sub>). Higher values of total soluble solids (TSS) was recorded under light pruned plants (P<sub>3</sub>) and was found decreasing with increasing severity of pruning, while lowest TSS was recorded in control treatment (P<sub>0</sub>) (Table 1). Higher TSS was recorded under higher dose of nitrogen i.e. 120 g bush<sup>-1</sup> (N<sub>3</sub>) and was lowest with 40 g N bush<sup>-1</sup>. Interaction effect of P and N also influenced significantly in respect of TSS, which was highest under P<sub>3</sub>N<sub>3</sub> while, was lowest in P<sub>0</sub>N<sub>1</sub> during both the years of experimentation. It may be due to smaller fruit contains higher TSS obtained from lightly pruned bushes than the heavily

pruned ones. There was no any linear impact on acidity with pruning severity and nitrogen level. However, the highest % of acidity was recorded under the control (P<sub>0</sub>) during first year, while it was higher in P<sub>2</sub> treatment during second year. In combined effect of pruning and nitrogen, on an average, higher acidity was noticed with P<sub>2</sub>N<sub>1</sub>. The highest average ascorbic acid recorded under in medium pruning-P<sub>2</sub> and N<sub>2</sub> treatment, while it was lowest under the control (N<sub>0</sub> and no pruning). Further under interaction effect, P<sub>2</sub>N<sub>2</sub> recorded highest values of average ascorbic acid, followed by P<sub>3</sub>N<sub>2</sub> and it was lowest under P<sub>1</sub>N<sub>1</sub> treatment. Higher ascorbic acid might be due to bigger and better quality of fruit in medium pruning. The findings in respect of ascorbic acid are also in consonant with the findings of Singh *et al.* (2006).

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## SHORT COMMUNICATION

# Bio-efficacy of tolfeipyrad against diamond back moth (*Plutella xylostella* Linn.) infesting cabbage

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## ABSTRACT

Three dosages of the tolfeipyrad 15% EC, viz. 100, 125 and 150 g along with two treated checks [cartap hydrochloride 50% SP @ 500 g ai ha<sup>-1</sup> and chlorphenapyr (Lepido) 10% SC @ 75 g ai ha<sup>-1</sup>] were evaluated against diamond back moth (DBM) *Plutella xylostella* Linn. in cabbage (*Brassica oleraceae* var. *capitata* Linn.) during rabi 2009-10 and 2010-11. All the treatments were significantly superior over untreated check. Among the treatments, maximum reduction in larval population of DBM after 3<sup>rd</sup>, 7<sup>th</sup> and 14<sup>th</sup> days after spray ranged from 25.01 to 96.39% in tolfeipyrad @ 150 and 125 g ai ha<sup>-1</sup>. Minimum larval population of DBM with maximum marketable yield was recorded in tolfeipyrad @ 150 g ai ha<sup>-1</sup>, followed by cartap hydrochloride and tolfeipyrad @ 125 g ai ha<sup>-1</sup>.

**Key words:** Cartap hydrochloride, Chlorphenapyr, Diamond back moth, *Plutella xylostella*, Tolfenpyrad

Cabbage (*Brassica oleraceae* var. *capitata* Linn.) is a major vegetable crop in India. It occupies 3.69 lac hectares of area with production of 79.49 million tonnes during 2010-2011 (Anonymous, 2011). Over the years it has been cultivated more intensively, which has resulted in high pest infestation. Diamond back moth (*Plutella xylostella*), head caterpillar (*Crociodolomia binotalis*), web worm (*Hellula undalis*), cabbage butterflies (*Pieris brassicae* and *P. rapae*), aphids (*Lipaphis erysimi* and *Brevicoryne brassicae*) and flea beetle (*Phyllotreta brassicae*) are common insect pests on vegetable brassicas in India. Among them, diamondback moth (DBM) is the most serious pest of cabbage and responsible for low productivity of this crop in India. The yield loss caused by DBM varies from 31 to 100% (Lingappa *et al.*, 2004). The extensive and indiscriminate use of pesticides led to the development of the resistance to most of the frequent used chemicals causing control failures of DBM.

Tolfenpyrad is a broad spectrum foliar contact insecticide used in wide range of crops against several economically important insect pests of order hemiptera, thysanoptera, lepidoptera, coleoptera, diptera, orthoptera, eriophid and tarsonemid mites. It has activity on all developmental stages of target insect. It works by inhibiting the cellular respiration in mitochondria, as a result, the compound causes rapid cessation of feeding and death of pests usually occurs in 24-48 hours. As a novel pyrazole insecticide and miticide, it acts mainly through the inhibition mitochondrial electron transport system. By binding to the electron transport complex inhibitor in

mitochondrial metabolic system, it could block the electron transportation and inhibit the energy supply and storage process of the insect. And so on, tolfeipyrad is termed as mitochondrial electron transport complex inhibitor (METI). It has a broad insecticidal spectrum and contact activity, especially to DBM, thrips (*Thysanoptera*), and leafhopper (*Empoasca pirusuga Matumura*). Keeping this in view, the present investigation was undertaken to evaluate the insecticidal property of the compound tolfeipyrad against DBM in cabbage.

A field experiment was conducted during rabi 2009-10 and 2010-11 at Horticulture Farm of Rajasthan College of Agriculture, Udaipur in randomized block design with four replication, each in 7 m x 5 m plots keeping spacing of 45 cm row to row and 30 cm plant to plant. The transplanting of cabbage variety 'Snowball' was done in the field. All the agronomic practices recommended in the state package and practices were followed to grow the healthy crop. Three dosages of the test compound tolfeipyrad 15 % EC, viz. 100, 125 and 150 g ai ha<sup>-1</sup> (T<sub>1</sub>, T<sub>2</sub>, T<sub>3</sub>), cartap hydrochloride 50% SP @ 500 g ai ha<sup>-1</sup> (T<sub>4</sub>) and chlorfenapyr (Lepido) 10% SC @ 75 g ai ha<sup>-1</sup> (T<sub>5</sub>) were used for the insecticidal treatments. The pre and post treatment population of DBM was recorded at 24 hours before spray and 3, 7 and 14 days after spraying on outer three leaves of randomly selected five plants of each plot in each replication through visual counting during early hours of the day. Similarly, impact of treatments on major bio agents especially on coccinellids, chrysopids, preying mantids and spiders was also recorded in each replication. Efficacy of

various treatments in controlling the pest was analysed using analysis of variance. The population data thus obtained was converted into per cent reduction in population using the method employed by Henderson and Tilton (1955) as under :

$$\text{Per cent reduction in population} = 100 \left\{ 1 - \frac{T_a \times C_b}{T_b \times C_a} \right\}$$

Where;  $T_a$  = Number of insects after treatment

$T_b$  = Number of insects before treatment

$C_a$  = Number of insects in untreated check after treatment

$C_b$  = Number of insects in untreated check before treatment

It is apparent from the Table 1 that pre-treatment larval population of diamond back moth ranged from 3.25 to 4.09 larvae plant<sup>-1</sup> in opened leaves in different treatments, whereas significant reduction in DBM population was observed after spray. Lowest population, 0.73 larvae plant<sup>-1</sup> in opened leaves was observed after 3<sup>rd</sup> day of first spray in  $T_3$ , followed by 0.84, 1.69, 1.83, 1.98 and 4.76 larvae plant<sup>-1</sup> in  $T_2$ ,  $T_5$ ,  $T_1$ ,  $T_4$  and  $T_6$ , respectively. Similarly after 7<sup>th</sup> days of first spray,  $T_3$  and  $T_2$  was proven very effective with 0.35 and 0.47 larvae plant<sup>-1</sup> compared to 0.54, 0.57 and 1.79 larvae plant<sup>-1</sup> in  $T_5$ ,  $T_4$  and  $T_1$ , respectively. The impact of treatments after 14<sup>th</sup> days of 1<sup>st</sup> spray showed the similar results wherein lowest population 0.20 larvae plant<sup>-1</sup> was observed in  $T_3$ , followed by 0.45, 0.69, 0.73 and 1.83 larvae plant<sup>-1</sup> in opened leaves in treatment  $T_2$ ,  $T_4$ ,  $T_5$  and  $T_1$ , respectively. Larval population of DBM untreated control ranged from 3.73 to 4.76 larvae plant<sup>-1</sup> in opened leaves during first spray observations.

The pre-treatment population before second spray clearly shows the persistent toxicity, wherein population ranged from 0.41 to 4.83 larvae plant<sup>-1</sup> in the opened leaves (Table 1). The larval population of DBM after 3<sup>rd</sup> day of second spray resulted in lowest population, 0.25 plant<sup>-1</sup> in  $T_2$ , followed by 0.31, 0.49,

0.51 and 1.12 in  $T_3$ ,  $T_4$ ,  $T_5$  and  $T_1$ , respectively compared to 4.87 in untreated control. The populations were drastically reduced after 7<sup>th</sup> day of 2<sup>nd</sup> spray with minimum 0.15 larvae opened leaf<sup>-1</sup> in  $T_3$ , followed by 0.19, 0.35, 0.37 and 0.98 plant<sup>-1</sup> in opened leaves in  $T_2$ ,  $T_4$ ,  $T_5$  and  $T_1$ , respectively, whereas 4.91 larvae were recorded in untreated control. The resultant population after 14 days of second spray ranged from 0.04 to 1.03 in test treatments and 4.33 larvae plant<sup>-1</sup> in opened leaves in untreated control. The application of tolfeprpyrad 15% EC @ 150 g ai ha<sup>-1</sup> ( $T_3$ ) was found highly effective with lowest population, 0.04 larvae plant<sup>-1</sup>, followed by 0.10, 0.22, 0.33 and 1.03 in  $T_2$ ,  $T_5$ ,  $T_4$  and  $T_1$ , respectively.

The data showed the similar trend wherein significant maximum population, 6.8 to 8.9 larvae plant<sup>-1</sup> in the opened leaves, was recorded in the control (Table 2). Lowest population, 0.70 larvae plant<sup>-1</sup> in opened leaves was observed after 3<sup>rd</sup> day of spray in  $T_3$ , followed by 1.3, 1.6 and 1.7 in  $T_2$ ,  $T_4$ ,  $T_1$  and  $T_5$ , respectively. Similarly after 7<sup>th</sup> days of first spray,  $T_3$ ,  $T_2$  and  $T_4$  was proven very effective with 0.3 and 0.7 larvae plant<sup>-1</sup> compared to 1.2 and 1.3 in  $T_1$  and  $T_5$ , respectively. The impact of treatments after 14<sup>th</sup> days of 1<sup>st</sup> spray showed the similar results wherein lowest population 0.40 larvae plant<sup>-1</sup> was observed in  $T_3$ , followed by 0.7, 0.69, 0.9, 1.2 and 1.3 larvae plant<sup>-1</sup> in opened leaves in  $T_2$ ,  $T_4$ ,  $T_5$  and  $T_1$ , respectively. It is visible from the Table 2 that the larval population of DBM in untreated control ranged from 6.8 to 8.4 larvae plant<sup>-1</sup> in opened leaves.

The pre-treatment population before second spray clearly shows the persistent toxicity of treatments wherein population ranged from 0.4 to 8.2 larvae plant<sup>-1</sup> in the opened leaves. The larval population of DBM after 3<sup>rd</sup> day of second spray resulted in lowest population, 0.20 plant<sup>-1</sup> in  $T_3$ , followed by 0.5, 0.7, 0.9 and 1.1 in  $T_2$ ,  $T_4$ ,  $T_5$  and  $T_1$ , respectively compared to 7.9 in untreated control. The populations were drastically reduced after 7<sup>th</sup> day of 2<sup>nd</sup> spray with minimum 0.09 larvae opened larvae plant<sup>-1</sup> in  $T_3$ , followed by 0.31, 0.49, 0.69 and 0.95 plant<sup>-1</sup> in opened leaves in  $T_2$ ,  $T_4$ ,  $T_5$  and  $T_1$ , respectively; whereas 6.52

Table 1. Bio-efficacy of tolfeprpyrad 15% EC against diamond back moth on cabbage during 2009-10

| Treatment                                           | Dose<br>(g ai<br>ha <sup>-1</sup> ) | I spray        |                |                  |                |                  |                |                  |                | II spray       |                  |                |                  |                |                  | Yield<br>(t ha <sup>-1</sup> ) |
|-----------------------------------------------------|-------------------------------------|----------------|----------------|------------------|----------------|------------------|----------------|------------------|----------------|----------------|------------------|----------------|------------------|----------------|------------------|--------------------------------|
|                                                     |                                     | PT             | 3<br>DAT       | % Red-<br>uction | 7<br>DAT       | % Red-<br>uction | 14<br>DAT      | % Red-<br>uction | PT             | 3<br>DAT       | % Red-<br>uction | 7<br>DAT       | % Red-<br>uction | 14<br>DAT      | % Red-<br>uction |                                |
| T <sub>1</sub> ; Tolfenpyrad<br>15% EC              | 100                                 | 3.25<br>(1.79) | 1.83<br>(1.34) | 50.11            | 1.79<br>(1.33) | 56.84            | 1.83<br>(1.34) | 51.72            | 1.82<br>(1.34) | 1.12<br>(1.06) | 66.46<br>(0.99)  | 0.98<br>(0.99) | 47.03            | 1.03<br>(1.01) | 36.87            | 18.01                          |
| T <sub>2</sub> ;Tolfenpyrad<br>15% EC               | 125                                 | 3.65<br>(1.90) | 0.84<br>(0.92) | 79.61            | 0.47<br>(0.68) | 89.90            | 0.45<br>(0.66) | 89.43            | 0.49<br>(0.68) | 0.25<br>(0.49) | 49.40<br>(0.43)  | 0.19<br>(0.43) | 61.85            | 0.10<br>(0.30) | 77.23            | 19.26                          |
| T <sub>3</sub> ;Tolfenpyrad<br>15% EC               | 150                                 | 4.09<br>(2.00) | 0.73<br>(0.85) | 84.18            | 0.35<br>(0.59) | 93.30            | 0.20<br>(0.44) | 95.80            | 0.41<br>(0.62) | 0.31<br>(0.54) | 25.01<br>(0.39)  | 0.15<br>(0.39) | 64.01            | 0.04<br>(0.19) | 89.11            | 19.84                          |
| T <sub>4</sub> ; Cartap<br>hydrochlori<br>de 50% SP | 22.5                                | 3.98<br>(1.96) | 1.98<br>(1.40) | 55.92            | 0.57<br>(0.75) | 88.78            | 0.69<br>(0.77) | 85.13            | 0.95<br>(0.92) | 0.49<br>(0.69) | 48.84<br>(0.59)  | 0.35<br>(0.59) | 63.76            | 0.33<br>(0.56) | 61.25            | 18.27                          |
| T <sub>5</sub> ;Chlorfenapyr<br>10% SC              | 20                                  | 4.01<br>(2.00) | 1.69<br>(1.29) | 62.66            | 0.54<br>(0.73) | 89.45            | 0.73<br>(0.85) | 84.39            | 0.98<br>(0.91) | 0.51<br>(0.70) | 48.38<br>(0.60)  | 0.37<br>(0.60) | 62.86            | 0.28<br>(0.53) | 68.13            | 18.36                          |
| T <sub>6</sub> ;Control                             |                                     | 3.73<br>(1.93) | 4.21<br>(2.05) |                  | 4.76<br>(2.18) |                  | 4.35<br>(2.08) |                  | 4.83<br>(2.20) | 4.87<br>(2.21) |                  | 4.91<br>(2.21) |                  | 4.33<br>(2.08) |                  | 15.16                          |
| CD (P=0.05)                                         |                                     | 0.76           | 0.63           |                  | 0.56           |                  | 0.55           |                  | 0.58           | 0.53           |                  | 0.51           |                  | 0.48           |                  | 1.41                           |

Value in the parentheses represents the square root transformed value; DAT = Days after transplanting; PT = Pre-treatment; g ai = gram active ingredient

Table 2. Bio-efficacy of tolfenpyrad 15% EC against diamond back moth on cabbage during 2010-11

| Treatment                                          | Dose<br>(g<br>aiha <sup>-1</sup> ) | I spray       |               |                     |               |                  |               |                  |               | II spray      |                  |                |                  |                |                  | Yield<br>(t ha <sup>-1</sup> ) |
|----------------------------------------------------|------------------------------------|---------------|---------------|---------------------|---------------|------------------|---------------|------------------|---------------|---------------|------------------|----------------|------------------|----------------|------------------|--------------------------------|
|                                                    |                                    | PT            | 3<br>DAT      | %<br>Red-<br>uction | 7<br>DAT      | % Red-<br>uction | 14<br>DAT     | % Red-<br>uction | PT            | 3<br>DAT      | % Red-<br>uction | 7<br>DAT       | % Red-<br>uction | 14<br>DAT      | % Red-<br>uction |                                |
| T <sub>1</sub> ; Tolfenpyrad<br>15% EC             | 100                                | 5.6<br>(2.37) | 1.6<br>(1.25) | 73.74               | 1.3<br>(1.14) | 79.76            | 1.2<br>(1.10) | 80.30            | 1.3<br>(1.14) | 1.1<br>(1.05) | 17.64            | 0.95<br>(0.97) | 15.32            | 1.01<br>(1.00) | 30.98            | 18.94                          |
| T <sub>2</sub> ; Tolfenpyrad<br>15% EC             | 125                                | 7.4<br>(2.72) | 1.3<br>(1.14) | 83.85               | 0.7<br>(0.83) | 91.75            | 0.7<br>(0.83) | 91.30            | 0.7<br>(0.83) | 0.5<br>(0.71) | 30.47            | 0.33<br>(0.56) | 45.37            | 0.23<br>(0.48) | 44.60            | 21.53                          |
| T <sub>3</sub> ; Tolfenpyrad<br>15% EC             | 150                                | 5.1<br>(2.25) | 0.7<br>(0.83) | 87.38               | 0.3<br>(0.55) | 94.87            | 0.2<br>(0.55) | 96.39            | 0.4<br>(0.63) | 0.2<br>(0.44) | 51.33            | 0.09<br>(0.3)  | 73.92            | 0.05<br>(0.22) | 78.92            | 24.54                          |
| T <sub>4</sub> ; Cartap<br>hydrochloride<br>50% SP | 22.5                               | 6.0<br>(2.45) | 1.6<br>(1.26) | 75.49               | 0.9<br>(0.95) | 86.92            | 0.7<br>(0.83) | 89.28            | 0.9<br>(0.95) | 0.7<br>(0.83) | 24.30            | 0.49<br>(0.7)  | 36.91            | 0.52<br>(0.72) | 2.59             | 19.00                          |
| T <sub>5</sub> ; Chlorfenapyr 10%<br>SC            | 20                                 | 6.8<br>(2.61) | 1.7<br>(1.30) | 77.02               | 1.1<br>(1.05) | 85.90            | 1.3<br>(1.14) | 82.43            | 1.2<br>(1.09) | 0.9<br>(0.95) | 27.0             | 0.69<br>(0.83) | 33.37            | 0.93<br>(0.96) | 30.82            | 18.73                          |
| T <sub>6</sub> ; Control                           |                                    | 6.8<br>(2.61) | 7.4<br>(2.90) |                     | 7.8<br>(2.79) |                  | 7.4<br>(2.85) |                  | 7.3<br>(2.86) | 7.5<br>(2.63) |                  | 6.3<br>(2.55)  |                  | 4.33<br>(2.08) |                  | 12.37                          |
| CD (P=0.05)                                        |                                    | 0.82          | 0.61          |                     | 0.63          |                  | 0.52          |                  | 0.49          | 0.51          |                  | 0.57           |                  | 0.53           |                  | 1.53                           |

Value in the parentheses represents the square root transformed value; DAT = Days after transplanting; PT = Pre-treatment; g ai = gram active ingredient

larvae plant<sup>-1</sup> were recorded in untreated control. The resultant population after 14 days of second spray ranged from 0.04 to 1.03 in treatment and 4.33 larvae plant<sup>-1</sup> in opened leaves in untreated control. The application of T<sub>3</sub> was found highly effective with lowest population, 0.05 larvae plant<sup>-1</sup>, followed by 0.23, 0.52, 0.93 and 1.01 in T<sub>2</sub>, T<sub>4</sub>, T<sub>5</sub> and T<sub>1</sub>, respectively.

Amongst all the tested doses of tolfenpyrad 15% EC (100, 125 and 150 g ai ha<sup>-1</sup>), tolfenpyrad @ 150 g ha<sup>-1</sup> (T<sub>3</sub>) has recorded on an average the highest yield of 19.84 and 24.54 t ha<sup>-1</sup> compared to the other treatments and standard check during first and second year, respectively (Table 1 and 2). The next maximum yield, 19.26 and 21.53 t ha<sup>-1</sup>, was obtained in T<sub>2</sub>. The doses of cartap hydrochloride 50% SP @ 22.5 g ai ha<sup>-1</sup> and chlorfenapyr 10% SC @ 20 g ai ha<sup>-1</sup> were at par with tolfenpyrad 15% EC @ 100 g ai ha<sup>-1</sup> during both the years. The untreated control recorded the lowest yield (15.16 and 12.37 t ha<sup>-1</sup> during first and second year, respectively) compared to all other treatments.

The results of present investigations against DBM are in agreement with the earlier observations made by Shindo *et al.* (2008) and Okazaki *et al.* (2010), who reported tolfenpyrad as most effective insecticide against garden pea leaf minor (*Chromatomyia horticola* Goureaux) on turnip and greenhouse jassids (*Trialeurodes vaporariorum* Westwood) on tomato and strawberry. Saini *et al.* (2010), observed tolfenpyrad as significantly superior to imidacloprid 17.8 L and thiomethoxam 24 WG in suppressing the population of jassids on cotton. The superiority of cartap hydrochloride against DBM as revealed in the present studies is in close agreement with the studies of Kumar *et al.* (2007), who reported that cartap hydrochloride

(0.05%) caused 58.8 to 60.1% reduction in larval population within 14 days of first and second spray.

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## SHORT COMMUNICATION

# Population dynamics of Hemipteran insects on pigeonpea (*Cajanus cajan*) and its correlation with abiotic factors

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## ABSTRACT

A field experiment was carried out during rainy season of 2012 at Jabalpur, Madhya Pradesh to determine the population and correlation with abiotic factors of Hemipteran insects on pigeonpea (*Cajanus cajan* L.). Population of different Hemipteran insects was observed from 36<sup>th</sup> standard week (first week of September 2012) to 5<sup>th</sup> standard week (last week of January to first week of February 2013). The abiotic factors also affected the population built up of the insect pests. Jassid (*Empoasca fabae* Harris) population exhibited significant correlation with maximum and minimum temperatures and evaporation, whereas stink bug population was correlated with wind speed. However, there was negative correlation between evaporation and pod bug nymph and adult population.

**Key words:** Abiotic, Hemipteran, Population, Standard week

In India, pigeonpea (*Cajanus cajan* L.) is mostly cultivated during rainy season (*khariif*) as sole or intercrop for green vegetable and 'dhal' purpose. Among the various constraints limiting pigeonpea production, insect pests are the major ones. More than 200 insect species have been reported feeding on pigeonpea at various stages of its growth in India (Lateef and Read, 1990). Theiaswi *et al.* (2008) recorded a total of 22 species of insect pests feeding or damaging field bean (*Lablab purpureus* L.) during the crop period of which majority were sucking pests followed by borers and defoliators. Hence, the objective of this research was to determine the population dynamics of hemipteran insect on pigeonpea.

A field experiment was carried out on population dynamics of hemipteran insects and their correlation with abiotic factors on pigeonpea variety 'JA-4' in the Experimental Field of Department of Entomology, Livestock Farm, Adhartal, Jawaharlal Nehru Krishi Vishwavidyalaya, Jabalpur (Madhya Pradesh) during rainy season 2012. The experiment was laid down in 5 m x 12.6 m plot size with 21 rows of 5.0 meter row length at 60 cm x 20 cm row x plant spacing. Crop was grown under rainfed condition and kept unprotected from plant protection measures. Regular observations were recorded starting immediately after germination and continued up to harvest. Observation on different pod sucking insect pests was recorded on 25 tagged plants, twice in a standard week (SW) on the pigeonpea crop. The counts of pod sucking insects

(nymphs and adults) i.e. jassid were recorded on 6 leaves 2 each from upper, middle and lower plant canopy, Aphids were recorded on 10 cm twig per plant, Green stink bug were recorded per 5 plants and pod bug were recorded on 25 pods. Simultaneously a corresponding weekly record of meteorological data, viz. minimum and maximum temperature, morning and evening relative humidity, morning and evening vapour pressure, total rainfall per week, number of rainy days per week, wind speed, sunshine hours and evaporation were collected (Table 1). The influence of the different meteorological parameters on insect population was studied by graphical super imposition technique.

The jassids (*E. fabae*) was first recorded during the first week of September i.e. on 36<sup>th</sup> SW (3.9.2012 to 9.9.2012) and the activity of the test continued to third SW of January (15.1.2013 to 21.1.2013) on pigeonpea. The number of jassid was worked out as weekly average per leaf (Table 2). The peak population of the pest was observed during 42<sup>nd</sup> SW (third week of October) and 48<sup>th</sup> SW (last week of November – first week of December). During this period maximum and minimum temperature ranged from 28.4 to 32.5 and 11.5 to 18.6°C, respectively; whereas morning and evening relative humidity ranged from 79 to 83 and 32 to 33%, respectively. Further wind speed, sunshine hours, morning and evening vapour pressure and evaporation ranged from 2.1 to 2.4 km hr<sup>-1</sup>, 8.4 to 9.4 hrs, 9.7 to 14.2 mm, 9.1 to 11.3 mm and 2.3 to 3.4 mm, respectively.

Correlation between various abiotic factors and jassid population exhibited significant positive impact of maximum

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and minimum temperature and evaporation on pest population (Table 4). The present findings are in accordance with the findings of Reddy *et al.* (2001) and Kaushik *et al.* (2008). They also reported significantly positive influence of minimum temperature on jassid population. The present findings contradict the findings of Kumar and Nath (2003). They reported that maximum temperature of the preceding week showed negative influence on jassid population. Results further revealed that both morning and evening relative humidity showed negative impact on jassid population, but were not-significant. On the contrary, Reddy *et al.* (2001) and Kaushik *et al.* (2008) reported positive impact on the pest population. The findings on evaporation and sunshine showed positive correlation on the jassid population. It contradicts the findings of Kumar and Nath (2003), they reported negative influence of evaporation and sunshine of the preceding week on pest population. The present findings are in conformity with the findings of Kumar *et al.* (2010). They also reported that morning and evening relative humidity, wind speed, rainfall and rainy days had negative impact on pest population.

The aphid, *A. craccivora* was first recorded during first week of September i.e. on 3<sup>rd</sup> September (36<sup>th</sup> SW). The activity of the pest continued from 4<sup>th</sup> September to third week of January. The peak population of the pest were observed during 46<sup>th</sup> SW (third week of November) and 51<sup>st</sup> SW (third week of December) (Table 3). During this period maximum and minimum temperature ranged from 10.0 to 28.6°C to and 7.1 to 25.3°C, respectively, whereas morning and evening relative humidity ranged from 28 to 85% and 29 to 88%, respectively. Further wind speed, sunshine, morning and evening vapour pressure and evaporation ranged from 2.3 to 2.7 km hr<sup>-1</sup>, 8.4 to 8.6 hrs, 7.3 to 10.0 mm, 6.7 to 7.8 mm and 2.4 to 3.0 mm, respectively. There was no rainfall received during this period (Table 1). Correlations

between various abiotic factors and aphid population showed significant negative impact of evening relative humidity and wind speed on pest population (Table 4).

Table 2. Incidence of jassid, *Empoasca fabae* Harris on different plant canopy of pigeonpea at Jabalpur

| Standard Week | Mean jassid (adult and nymph) population per leaf in different plant canopy |                |            |      |
|---------------|-----------------------------------------------------------------------------|----------------|------------|------|
|               | Upper leaf                                                                  | Middle leaf    | Lower leaf | Mean |
| 36            | 0.60                                                                        | 0.63           | 0.44       | 0.56 |
| 37            | 0.58                                                                        | 0.55           | 0.42       | 0.52 |
| 38            | 0.49                                                                        | 0.72           | 0.44       | 0.55 |
| 39            | 0.53                                                                        | 0.81           | 0.53       | 0.62 |
| 40            | 0.63                                                                        | 0.92           | 0.55       | 0.70 |
| 41            | 0.69                                                                        | 0.94           | 0.54       | 0.72 |
| 42            | 0.63                                                                        | 1.00           | 0.61       | 0.75 |
| 43            | 0.49                                                                        | 0.86           | 0.52       | 0.62 |
| 44            | 0.45                                                                        | 0.85           | 0.48       | 0.59 |
| 45            | 0.49                                                                        | 0.82           | 0.48       | 0.60 |
| 46            | 0.51                                                                        | 0.78           | 0.51       | 0.60 |
| 47            | 0.46                                                                        | 0.76           | 0.48       | 0.57 |
| 48            | 0.55                                                                        | 0.78           | 0.51       | 0.61 |
| 48            | 0.48                                                                        | 0.70           | 0.48       | 0.55 |
| 50            | 0.49                                                                        | 0.84           | 0.51       | 0.61 |
| 51            | 0.37                                                                        | 0.71           | 0.29       | 0.46 |
| 52            | 0.36                                                                        | 0.50           | 0.27       | 0.38 |
| 01            | 0.39                                                                        | 0.48           | 0.29       | 0.39 |
| 02            | 0.48                                                                        | 0.70           | 0.50       | 0.56 |
| 03            | 0.49                                                                        | 0.78           | 0.48       | 0.58 |
| Mean          | 0.50                                                                        | 0.75           | 0.46       | 0.58 |
| SD            | 0.08                                                                        | 0.13           | 0.08       |      |
| SE±           | 0.50± 0.01                                                                  | 0.75± 0.02     | 0.46± 0.01 |      |
| Variance      | 0.008                                                                       | 0.01           | 0.008      |      |
| T cal.        | Upper leaf vs                                                               | 9.69*          | 2.89*      |      |
| T cal.        | -                                                                           | Middle leaf vs | 2.89*      |      |

NS= Non significant; \*Significant at P = 0.05; T cal = T calculated; SD = Standard deviation; SE = Standard Error; SW = Standard weeks; vs = versus

Table 1. Weekly meteorological data of experimental field during rainy season 2012

| Standard week | Date       |            | Temperature (°C) |      | Relative humidity (%) |         | Wind speed (km hr <sup>-1</sup> ) | Sunshine (hrs) | Rainfall (mm) | Rainy days (No.) | Vapour pressure (mm) |         | Evaporation (mm) |
|---------------|------------|------------|------------------|------|-----------------------|---------|-----------------------------------|----------------|---------------|------------------|----------------------|---------|------------------|
|               | From       | To         | Max              | Min  | Morning               | Evening |                                   |                |               |                  | Morning              | Evening |                  |
| 36            | 03/09/2012 | 09/09/2012 | 30.8             | 23.9 | 93                    | 79      | 5.4                               | 2.8            | 52.2          | 3                | 23.8                 | 23.4    | 3.0              |
| 37            | 10/09/2012 | 16/09/2012 | 30.7             | 23.4 | 91                    | 79      | 5.1                               | 2.8            | 87.4          | 2                | 22.1                 | 22.6    | 3.1              |
| 38            | 17/09/2012 | 23/09/2012 | 30.8             | 23.1 | 89                    | 69      | 3.9                               | 4.6            | 11.0          | 1                | 20.9                 | 20.7    | 2.6              |
| 39            | 24/09/2012 | 30/09/2012 | 31.9             | 22.6 | 88                    | 51      | 3.1                               | 8.7            | 0.0           | 0                | 20.0                 | 16.3    | 3.6              |
| 40            | 01/10/2012 | 07/10/2012 | 33.4             | 21.9 | 91                    | 44      | 3.3                               | 7.3            | 2.3           | 0                | 20.1                 | 16.2    | 3.4              |
| 41            | 08/10/2012 | 14/10/2012 | 32.4             | 18.5 | 83                    | 35      | 2.9                               | 9.0            | 0.0           | 0                | 14.8                 | 12.9    | 4.0              |
| 42            | 15/10/2012 | 21/10/2012 | 32.5             | 18.6 | 79                    | 32      | 2.4                               | 9.4            | 0.0           | 0                | 14.2                 | 11.3    | 3.4              |
| 43            | 22/10/2012 | 28/10/2012 | 31.6             | 15.2 | 82                    | 35      | 1.9                               | 8.4            | 0.0           | 0                | 13.7                 | 11.8    | 3.2              |
| 44            | 29/10/2012 | 04/11/2012 | 27.9             | 15.6 | 91                    | 51      | 3.6                               | 6.7            | 0.0           | 0                | 13.5                 | 12.9    | 2.7              |
| 45            | 05/11/2012 | 11/11/2012 | 28.2             | 14.6 | 88                    | 30      | 3.0                               | 7.7            | 0.0           | 0                | 11.7                 | 9.5     | 2.8              |
| 46            | 12/11/2012 | 18/11/2012 | 28.6             | 10.0 | 85                    | 28      | 2.7                               | 8.4            | 0.0           | 0                | 10.0                 | 7.8     | 3.0              |
| 47            | 19/11/2012 | 25/11/2012 | 27.9             | 10.6 | 88                    | 29      | 2.2                               | 7.6            | 0.0           | 0                | 9.3                  | 8.2     | 2.4              |
| 48            | 26/11/2012 | 02/12/2012 | 28.4             | 11.5 | 83                    | 33      | 2.1                               | 8.4            | 0.0           | 0                | 9.7                  | 9.1     | 2.3              |
| 49            | 03/12/2012 | 09/12/2012 | 28.7             | 10.6 | 85                    | 26      | 2.1                               | 9.0            | 0.0           | 0                | 9.4                  | 7.8     | 2.4              |
| 50            | 10/12/2012 | 16/12/2012 | 29.0             | 14.0 | 92                    | 41      | 2.2                               | 7.2            | 3.2           | 1                | 12.3                 | 11.8    | 2.7              |
| 51            | 17/12/2012 | 23/12/2012 | 25.3             | 7.1  | 88                    | 29      | 2.3                               | 8.6            | 0.0           | 0                | 7.3                  | 6.7     | 2.4              |
| 52            | 24/12/2012 | 31/12/2012 | 23.8             | 5.0  | 87                    | 42      | 2.2                               | 9.1            | 0.0           | 0                | 6.8                  | 7.4     | 2.3              |
| 1             | 01/01/2013 | 07/01/2013 | 23.3             | 7.2  | 87                    | 32      | 3.6                               | 5.9            | 0             | 0                | 7.7                  | 8.7     | 2.2              |
| 2             | 08/01/2013 | 14/01/2013 | 23               | 5.2  | 84                    | 36      | 2.6                               | 9.2            | 0             | 0                | 6.7                  | 7.3     | 2.6              |
| 3             | 15/01/2013 | 21/01/2013 | 26.7             | 10.1 | 86                    | 36      | 4                                 | 8.1            | 0             | 0                | 9                    | 9.1     | 3.1              |
| 4             | 22/01/2013 | 28/01/2013 | 21.4             | 5    | 81                    | 36      | 2.9                               | 7.6            | 0             | 0                | 6.6                  | 7       | 2.5              |
| 5             | 29/01/2013 | 04/02/2013 | 24.6             | 7.4  | 87                    | 42      | 2.3                               | 7.4            | 0             | 0                | 7.9                  | 8.8     | 2.5              |

Table 3. Incidence of hemipteran insects infesting pigeonpea

| SW | Mean population of Hemipterans (Nymph + Adult) |                              |                     |
|----|------------------------------------------------|------------------------------|---------------------|
|    | Aphids per 10 cm twig                          | Green stink bug per 5 plants | Pod bug per 25 pods |
| 36 | 8.10                                           |                              |                     |
| 37 | 7.30                                           |                              |                     |
| 38 | 7.80                                           |                              |                     |
| 39 | 8.70                                           |                              |                     |
| 40 | 10.40                                          |                              |                     |
| 41 | 10.40                                          |                              |                     |
| 42 | 10.90                                          |                              |                     |
| 43 | 11.70                                          |                              |                     |
| 44 | 15.40                                          |                              | 6.88                |
| 45 | 21.80                                          |                              | 6.94                |
| 46 | 22.60                                          | 25.34                        | 6.88                |
| 47 | 20.40                                          | 30.08                        | 7.12                |
| 48 | 17.40                                          | 30.06                        | 7.28                |
| 49 | 15.40                                          | 25.94                        | 7.00                |
| 50 | 11.00                                          | 30.04                        | 7.32                |
| 51 | 12.00                                          | 30.18                        | 7.62                |
| 52 | 10.40                                          | 30.26                        | 7.22                |
| 1  | 7.30                                           | 25.76                        | 7.16                |
| 2  | 5.60                                           | 25.72                        | 6.76                |
| 3  | 3.20                                           | 25.63                        | 6.68                |

SW = Standard weeks

The green stink bug, *N. viridula* was first recorded during the second week of November i.e. on 12<sup>th</sup> November (46<sup>th</sup> SW). The activity of the pest continued from 12<sup>th</sup> November to third week of January. The peak population of the pest were observed during 47<sup>th</sup> SW (third-fourth week of November) and 52<sup>nd</sup> SW (last week of December) (Table 3). During this period maximum and minimum temperature ranged from 23.8 to 27.9°C and 5.0 to 10.6°C, respectively; whereas morning and evening relative humidity ranged from 87 to 88% and 29 and 42%, respectively. Further wind speed, sunshine, morning and evening vapour pressure and evaporation ranged from 2.2 to 2.2 km hr<sup>-1</sup>, 7.6 to 9.1 hrs, 6.8 to 9.3 mm, 7.0 to 8.2 mm and 2.3 to 2.4 mm, respectively. Correlations between various abiotic factors and stink bug population exhibited significant influence of wind speed on pest population (Table 4).

The pod bug, *C. gibbosa* was first recorded during the last week of October i.e. on 29<sup>th</sup> October (44<sup>th</sup> SW). The activity of the pest continued from 29<sup>th</sup> October to third week of January. The peak population of the pest was observed during 48<sup>th</sup> SW (last week of November) and 51<sup>st</sup> SW (third week of December) (Table 3). During this period maximum and minimum temperature ranged from 25.3 to 28.4°C and 7.1 to 11.5°C, respectively, whereas morning and evening relative humidity ranged from 83 to 88% and 33 to 39%, respectively. Further wind speed, sunshine, morning and evening vapour pressure and evaporation ranged from 2.1 to 2.3 km hr<sup>-1</sup>, 8.4 to 8.6 hrs, 7.3 to 9.7 mm, 6.7 to 9.1 mm and 2.3 to 2.4 mm, respectively. Correlations between various abiotic factors and pod bug nymph and adult population exhibited that evaporation had significant negative impact on pest population (Table 4).

In the present study two peaks were recorded during 48<sup>th</sup> and 51<sup>st</sup> SW, whereas Mishra and Dash (2001) reported two

Table 4. Correlation coefficient (r) of abiotic factors on hemipteran insects (nymph + adult) on pigeonpea

| Weather factors                    | Jassid   | Aphids   | Green sting bug | Pod bug  |
|------------------------------------|----------|----------|-----------------|----------|
|                                    | r        | r        | r               | r        |
| Max. temperature °C                | 0.77**   | 0.04 NS  | 0.15 NS         | -0.04 NS |
| Min. temperature °C                | 0.52*    | -0.18 NS | 0.15 NS         | -0.14 NS |
| Morning RH (%)                     | -0.30 NS | -0.10 NS | 0.45 NS         | 0.25 NS  |
| Evening RH (%)                     | -0.09 NS | -0.45 *  | 0.31 NS         | -0.13 NS |
| Wind speed ( km hr <sup>-1</sup> ) | -0.11 NS | -0.44 *  | -0.63 *         | -0.57 *  |
| Sunshine ( hrs)                    | 0.24 NS  | 0.26 NS  | 0.04 NS         | -0.03 NS |
| Rainfall (mm)                      | -0.15 NS | -0.28 NS | 0.32 NS         | 0.29 NS  |
| Rainy days (Nos.)                  | -0.11 NS | -0.30 NS | 0.32 NS         | 0.29 NS  |
| Morn. Vapour pressure (mm)         | 0.37 NS  | -0.26 NS | 0.12 NS         | -0.17 NS |
| Even. Vapour pressure (mm)         | 0.10 NS  | -0.38 NS | 0.16 NS         | -0.15 NS |
| Evaporation (mm)                   | 0.71**   | -0.21 NS | -0.45 NS        | -0.63 *  |

NS = Non significant ; \*Significant at P = 0.05 ; \*\*Significant at P = 0.01

peaks but during 50<sup>th</sup> and 4<sup>th</sup> SW, respectively. Further evening relative humidity had negative impact on bug population. The present results confirm the findings of Mishra and Dash (2001) and Kaushik *et al.* (2008), they also reported negative impact of relative humidity on the pest population. Maximum and minimum temperature showed negative effect on pest population but were non-significant. On the contrary, Kaushik *et al.* (2008) reported maximum and minimum temperature to exhibit positive impact on the pest population.

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## SHORT COMMUNICATION

# Screening of mungbean (*Vigna radiata*) genotypes against major insects

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## ABSTRACT

An experiment was conducted at Banaras Hindu University, Varanasi, Uttar Pradesh during July to October 2012 for screening of 30 genotypes of mungbean [*Vigna radiata* (L.) Wilczek] against white fly (*Bemisia tabaci*), jassid (*Empoasca kerri*), flower thrips (*Caliothrips indicus*) and pod borer (*Maruca vitrata*). The minimum population of white fly was recorded on genotype TMB-36, followed by RMG-1004 and maximum in BM-2003-2 and HUM-12. The minimum population of jassid was recorded on KM-2293, followed by TMB-36 and Pusa-1271, while maximum in BM-4, followed by ML-1628 and BM-2003-2. In case of thrips, minimum infestation was found in ML-1628, followed by Pusa-1171, ML-1464 and maximum in BPMR-145, followed by HUM-12 and Pusa-0672. According to pod damage, minimum damage was recorded in RVSm-11-9, followed by IPM-306-6, IPM 2K-15-4, MH-805, BM-4, DGGS-4 and BM-2002-2, and maximum in Pusa-1171, followed by HUM-12, BM-2003-2, Unnati, KM-2293 and ML-1628. Based on yield obtained at the time of harvesting, highest yield was recorded in AKM-4, followed by KM-2293, AKM-09-2, IPM-3066 and ML-1628. These genotypes were found resistant/tolerant against infestation of major insect pests of mungbean. The lowest yield was recorded with AKM-8802, followed NDMR-10-35, BM-4, ML-1464 and IPM 2K-15-4.

**Key words:** Genotypes, Insect pests, Mungbean, Screening

Mungbean [*Vigna radiata* (L.) Wilczek], (synonyms, *Phaseolus aurius* Roxb., *Phaseolus radiatus* L.) is the third most important pulse crop of India after chickpea and pigeonpea. India alone accounts for 65% of its world acreage and 54% of the production. It is grown in about 3.50 million hectares (Mha) in the country mainly in Rajasthan, Maharashtra, Andhra Pradesh, Karnataka, Orissa and Bihar. A phenomenal increase in area, production and productivity has occurred since 1964-65. The area has increased from 1.99 Mha in 1964-65 to 3.54 Mha in 2010-11. The production has increased from 0.60 million tonnes (Mt) to 1.81 Mt during the same period, ([www.aicrpmullarp.res.in](http://www.aicrpmullarp.res.in)). In India nearly 60 species of insect pest have been recorded from mungbean but only some cause economic damage and are more common over large areas. The major insect pests infesting the crop are whitefly, jassid, thrips, pod sucking bug, pod borers, stem fly and aphid which are major constraints in production of mungbean. For managing these pests chemical insecticides are used indiscriminately which cause various adverse effects on environment as well as on animal and human health including beneficial organisms. For obtaining higher yield of a good quality, it is necessary to adopt insect pest management practices which are economically feasible, environmentally sound and sociologically acceptable.

The present investigation was conducted at the field of Agricultural Research Farm, Institute of Agricultural Sciences,

Banaras Hindu University, Varanasi, Uttar Pradesh during July to December 2012 in randomized block design with 30 genotypes of mungbean, viz. BPMR-145, BM-2002-1, NDMR-10-35, MH-805, DGGS-4, Pusa-1271, Pant M-09-11, MH-2-15, Pusa-0672, BM-2003-2, BM-4, DGG-1, KM-2293, AKM-09-2, RVSm-11-9, Pusa-1171, TBM-36, AKM-4, IPM-02-14, IPM-02-3, AKM-8802, IPM-306-6, ML-1907, ML-1628, ML-818, RMG-1004, IPM 2K-15-4, Unnati, ML-1464 and HUM-12. The row to row and plant to plant distance was maintained as 30 cm and 10 cm, respectively. The unit plot size was 4 m x 0.60 m. For recording the infestation of insect pests, five plants were selected from each genotype and each unit of plot and the mature and immature stages of major insect pests present on them were counted at 7 days interval, starting from 25<sup>th</sup> August 2012. The insect pests recorded were: white fly (*Bemisia tabaci*), jassid (*Empoasca kerri*), flower thrips (*Caliothrips indicus*) and pod borer (*Maruca vitrata*). The number of insects counted from all the three replications for all the genotypes were averaged separately for each genotype on standard week basis. The damage caused by the pod borer observed on the basis of per cent pod and grain damage in the samples collected from all the replications of 30 genotypes.

Under the experiment 30 genotypes of mungbean were screened against white fly, jassid, flower thrips and pod borer and data were presented in Table 1 showed that maximum white

fly population was recorded in BM-2003-2 (1.74 whitefly split<sup>-1</sup> cage), followed by HUM-12 (1.71 whitefly split<sup>-1</sup> cage) and minimum in TMB-36 (0.84 whitefly split<sup>-1</sup> cage). Kumar *et al.* (2006) reported that the population of the whiteflies on various test entries fluctuated between 1.84 and 6.25 per plant compared to 6.30 whiteflies recorded on variety T 44.

On the basis of insect count on genotypes it was found that BM-4 had maximum population of insect (1.79 jassid split<sup>-1</sup> cage), followed by ML-1628 (1.65 jassid split<sup>-1</sup> cage) and minimum population was recorded in KM-2293 (1.0 jassid split<sup>-1</sup> cage). Anonymous (2011) screened 18 mungbean genotypes of ML series against jassid and reported that minimum infestation was found in ML 1799 and ML 1774 (1.66 jassid cage<sup>-1</sup>) and maximum in ML 1958, ML 1919, ML 1907 (6.66 jassid cage<sup>-1</sup>).

Overall maximum thrips population was found in genotype BPMR-145 (1.50 thrips 5 flower<sup>-1</sup>), followed by HUM-12 (1.27 thrips 5 flower<sup>-1</sup>) and minimum in ML-1628 (0.69 thrips 5 flower<sup>-1</sup>). Anonymous (2010) conducted advance stage screening of five entries and reported minimum population on COGG 912 (4.55 thrips 10 flower<sup>-1</sup>) and maximum in PKVAKM-4 (8.22 thrips 10 flower<sup>-1</sup>). Anonymous (2011) conducted advance stage screening of two entries i.e. COGG 912 and HUM-12. The entries COGG 912 (0.77 thrips 10 flower<sup>-1</sup>) was found superior

than HUM 12 (3.04 thrips 10 flower<sup>-1</sup>) in showing less susceptibility to thrips damage.

According to damage caused by insects discussed above, 30 genotypes were screened further for infestation of pod borers. Per cent pod damage and seed damage were considered for this purpose (Table 2). The maximum pod damage was recorded in Pusa-1171 (21.67%), followed by HUM-12 (20.0%), BM-2003-2 (19.33%) and Unnati (19.30%) and minimum damage were recorded in RVSm-11-9 (9.0%). Mandal (2005) reported cultivars ML 5, ML 408 and RMG 266 were resistant (less than 5% pod damage), cultivars ML 131, ML 505, RMG 275, Pusa 8971 and Pusa 8972 were moderately resistant (5.1-10% pod damage); PDM 219, RMG 175, RMG 202, Pusa 8974, Pusa Baisakhi and K851 were moderately susceptible (10.1-15% pod damage); PDM 216, ML 537, PDM 86-199 and WBM 202 were susceptible (>15% pod damage).

On the basis of grain yield obtained from different genotypes, this was showed that genotype AKM-4 produced maximum grain yield (8.77 q ha<sup>-1</sup>) and genotype AKM-8802 produced minimum grain yield (4.0 q ha<sup>-1</sup>) (Table 2). Genotypes KM-2293 (8.12 q ha<sup>-1</sup>), AKM-09-2 (7.78 q ha<sup>-1</sup>), IPM-3066 (7.78 q ha<sup>-1</sup>), and ML-1628 (7.77 q ha<sup>-1</sup>), RMG-1004 (7.74 q ha<sup>-1</sup>) were showed comparatively higher yield. The genotypes, viz. NDMR-10-35 (4.07 q ha<sup>-1</sup>), BM-4 (4.10 q ha<sup>-1</sup>), ML-1464 (4.25 q

Table 1. Population of major insects in mungbean genotypes

| Sl. No.     | Genotypes    | Whitefly cage <sup>-1</sup> |                    | Jassids cage <sup>-1</sup> |                    | Thrips 5 flowers <sup>-1</sup> |                    |
|-------------|--------------|-----------------------------|--------------------|----------------------------|--------------------|--------------------------------|--------------------|
|             |              | Mean                        | Transformed values | Mean                       | Transformed values | Mean                           | Transformed values |
| 1           | BPMR-145     | 0.89                        | <b>1.18</b>        | 1.39                       | <b>1.37</b>        | 1.50                           | <b>1.41</b>        |
| 2           | BM-2002-1    | 1.20                        | <b>1.30</b>        | 1.44                       | <b>1.39</b>        | 1.19                           | <b>1.30</b>        |
| 3           | NDMR-10-35   | 1.00                        | <b>1.23</b>        | 1.33                       | <b>1.35</b>        | 1.22                           | <b>1.31</b>        |
| 4           | MH-805       | 1.12                        | <b>1.27</b>        | 1.35                       | <b>1.36</b>        | 1.07                           | <b>1.25</b>        |
| 5           | DGGS-4       | 1.07                        | <b>1.25</b>        | 1.25                       | <b>1.32</b>        | 1.16                           | <b>1.29</b>        |
| 6           | PUSA-1271    | 1.11                        | <b>1.27</b>        | 1.17                       | <b>1.29</b>        | 1.11                           | <b>1.27</b>        |
| 7           | Pant M-09-11 | 1.16                        | <b>1.29</b>        | 1.41                       | <b>1.38</b>        | 0.99                           | <b>1.22</b>        |
| 8           | MH-2-15      | 1.37                        | <b>1.37</b>        | 1.29                       | <b>1.34</b>        | 1.15                           | <b>1.28</b>        |
| 9           | Pusa-0672    | 1.19                        | <b>1.30</b>        | 1.30                       | <b>1.34</b>        | 1.26                           | <b>1.33</b>        |
| 10          | BM-2003-2    | 1.74                        | <b>1.50</b>        | 1.65                       | <b>1.47</b>        | 1.14                           | <b>1.28</b>        |
| 11          | BM-4         | 1.64                        | <b>1.46</b>        | 1.79                       | <b>1.51</b>        | 1.17                           | <b>1.29</b>        |
| 12          | DGG-1        | 1.20                        | <b>1.30</b>        | 1.49                       | <b>1.41</b>        | 0.89                           | <b>1.18</b>        |
| 13          | KM-2293      | 1.22                        | <b>1.31</b>        | 1.00                       | <b>1.22</b>        | 0.98                           | <b>1.22</b>        |
| 14          | AKM-09-2     | 1.15                        | <b>1.29</b>        | 1.56                       | <b>1.43</b>        | 1.20                           | <b>1.30</b>        |
| 15          | RVSm-11-9    | 1.25                        | <b>1.32</b>        | 1.29                       | <b>1.34</b>        | 1.01                           | <b>1.23</b>        |
| 16          | Pusa-1171    | 1.28                        | <b>1.34</b>        | 1.22                       | <b>1.31</b>        | 0.72                           | <b>1.11</b>        |
| 17          | TMB-36       | 0.84                        | <b>1.16</b>        | 1.12                       | <b>1.27</b>        | 1.12                           | <b>1.27</b>        |
| 18          | AKM-4        | 1.29                        | <b>1.34</b>        | 1.39                       | <b>1.38</b>        | 1.16                           | <b>1.29</b>        |
| 19          | IPM-02-14    | 1.17                        | <b>1.29</b>        | 1.22                       | <b>1.31</b>        | 1.19                           | <b>1.30</b>        |
| 20          | IPM-02-3     | 1.38                        | <b>1.37</b>        | 1.28                       | <b>1.34</b>        | 1.08                           | <b>1.26</b>        |
| 21          | AKM-8802     | 1.66                        | <b>1.47</b>        | 1.54                       | <b>1.43</b>        | 1.24                           | <b>1.32</b>        |
| 22          | IPM-306-6    | 1.07                        | <b>1.25</b>        | 1.45                       | <b>1.39</b>        | 0.91                           | <b>1.19</b>        |
| 23          | ML-1907      | 0.97                        | <b>1.21</b>        | 1.27                       | <b>1.33</b>        | 0.80                           | <b>1.14</b>        |
| 24          | ML-1628      | 1.07                        | <b>1.25</b>        | 1.69                       | <b>1.48</b>        | 0.69                           | <b>1.09</b>        |
| 25          | ML-818       | 1.28                        | <b>1.33</b>        | 1.45                       | <b>1.40</b>        | 1.21                           | <b>1.31</b>        |
| 26          | RMG-1004     | 0.86                        | <b>1.17</b>        | 1.19                       | <b>1.30</b>        | 0.81                           | <b>1.15</b>        |
| 27          | IPM 2K-15-4  | 1.62                        | <b>1.46</b>        | 1.26                       | <b>1.33</b>        | 1.19                           | <b>1.30</b>        |
| 28          | Unnati       | 1.29                        | <b>1.34</b>        | 1.28                       | <b>1.33</b>        | 1.24                           | <b>1.32</b>        |
| 29          | ML-1464      | 1.41                        | <b>1.38</b>        | 1.43                       | <b>1.39</b>        | 0.78                           | <b>1.13</b>        |
| 30          | HUM-12       | 1.71                        | <b>1.49</b>        | 1.62                       | <b>1.45</b>        | 1.27                           | <b>1.33</b>        |
| CD (P=0.05) |              |                             | <b>0.16</b>        |                            | <b>NS</b>          |                                | <b>0.12</b>        |

Note: Bold values are  $\sqrt{x+0.5}$  transformed values

Table 2. Damage and grain yield of mungbean genotypes

| Sl. No.     | Genotypes              | Pod damage (%) |                    | Seed damage (%) |                    | Grain yield<br>(q ha <sup>-1</sup> ) |
|-------------|------------------------|----------------|--------------------|-----------------|--------------------|--------------------------------------|
|             |                        | Mean           | Transformed values | Mean            | Transformed values |                                      |
| 1           | BPMR-145               | 14.33          | <b>22.25</b>       | 11.00           | <b>19.37</b>       | 5.41                                 |
| 2           | BM-2002-1              | 12.00          | <b>20.27</b>       | 15.67           | <b>23.32</b>       | 4.07                                 |
| 3           | NDMR-10-35             | 14.67          | <b>22.52</b>       | 12.50           | <b>20.70</b>       | 6.97                                 |
| 4           | MH-805                 | 10.33          | <b>18.75</b>       | 8.67            | <b>17.12</b>       | 7.20                                 |
| 5           | DGGS-4                 | 11.33          | <b>19.67</b>       | 11.67           | <b>19.97</b>       | 5.52                                 |
| 6           | PUSA-1271              | 12.33          | <b>20.56</b>       | 10.33           | <b>18.75</b>       | 7.50                                 |
| 7           | Pant M-09-11           | 15.33          | <b>23.05</b>       | 10.20           | <b>18.63</b>       | 6.95                                 |
| 8           | MH-2-15                | 17.33          | <b>24.60</b>       | 12.83           | <b>20.99</b>       | 7.43                                 |
| 9           | Pusa-0672              | 12.67          | <b>20.85</b>       | 10.50           | <b>18.91</b>       | 6.73                                 |
| 10          | BM-2003-2              | 19.33          | <b>26.08</b>       | 9.33            | <b>17.79</b>       | 4.86                                 |
| 11          | BM-4                   | 11.17          | <b>19.52</b>       | 16.00           | <b>23.58</b>       | 4.10                                 |
| 12          | DGG-1                  | 16.67          | <b>24.09</b>       | 13.33           | <b>21.42</b>       | 7.73                                 |
| 13          | KM-2293                | 18.33          | <b>25.35</b>       | 9.33            | <b>17.79</b>       | 8.12                                 |
| 14          | AKM-09-2               | 13.00          | <b>21.13</b>       | 12.67           | <b>20.85</b>       | 7.78                                 |
| 15          | RVS <sub>m</sub> -11-9 | 9.00           | <b>17.46</b>       | 10.33           | <b>18.75</b>       | 6.79                                 |
| 16          | Pusa-1171              | 21.67          | <b>27.74</b>       | 11.00           | <b>19.37</b>       | 8.07                                 |
| 17          | TMB-36                 | 17.33          | <b>24.60</b>       | 10.67           | <b>19.06</b>       | 7.63                                 |
| 18          | AKM-4                  | 17.67          | <b>24.85</b>       | 12.50           | <b>20.70</b>       | 8.77                                 |
| 19          | IPM-02-14              | 18.00          | <b>25.10</b>       | 11.33           | <b>19.67</b>       | 7.12                                 |
| 20          | IPM-02-3               | 15.33          | <b>23.05</b>       | 5.00            | <b>12.92</b>       | 7.60                                 |
| 21          | AKM-8802               | 14.67          | <b>22.52</b>       | 14.00           | <b>21.97</b>       | 4.00                                 |
| 22          | IPM-306-6              | 9.33           | <b>17.79</b>       | 6.67            | <b>14.96</b>       | 7.78                                 |
| 23          | ML-1907                | 13.00          | <b>21.13</b>       | 7.67            | <b>16.07</b>       | 7.25                                 |
| 24          | ML-1628                | 18.33          | <b>25.35</b>       | 9.17            | <b>17.62</b>       | 7.77                                 |
| 25          | ML-818                 | 14.00          | <b>21.97</b>       | 9.50            | <b>17.95</b>       | 6.69                                 |
| 26          | RMG-1004               | 15.00          | <b>22.79</b>       | 9.00            | <b>17.46</b>       | 7.74                                 |
| 27          | IPM 2K-15-4            | 9.67           | <b>18.11</b>       | 8.67            | <b>17.12</b>       | 4.94                                 |
| 28          | Unnati                 | 19.00          | <b>25.84</b>       | 9.67            | <b>18.11</b>       | 8.30                                 |
| 29          | ML-1464                | 14.00          | <b>21.97</b>       | 8.17            | <b>16.61</b>       | 4.25                                 |
| 30          | HUM-12                 | 20.00          | <b>26.57</b>       | 11.50           | <b>19.82</b>       | 6.17                                 |
| CD (P=0.05) |                        |                | <b>5.19</b>        |                 | <b>3.96</b>        | 2.42                                 |

Note: Bold values are Arc sin transformed values

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ha<sup>-1</sup>) and IPM2K-15-4 (4.94 q ha<sup>-1</sup>) showed comparatively lower yield.

On the basis of experimental findings, it may be concluded that genotypes TMB-36 and RMG-1004 showed resistance against white fly, while KM-2293 and TMB-36 against jassid and ML-1628 and Pusa-1171 showed resistance against flower thrips. According to damage caused by pod borer, genotypes IPM-306-6 and RVS<sub>m</sub>-11-9 showed resistance against pod borer damage. Genotype AKM-4 produced maximum yield.

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## SHORT COMMUNICATION

# Efficacy and economics of new insecticides for management of aphid (*Lipaphis erysimi*) in Indian mustard

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## ABSTRACT

An experiment was conducted during 2008-09 and 2009-10 for determining the efficacy and economics of newer insecticides for the management of mustard aphid (*Lipaphis erysimi* Kalténbach) on Indian mustard (*Brassica juncea* (L.) Czern. & Coss). Lower aphid intensity of 7.2 and 8.2 aphids plant<sup>-1</sup> were recorded with imidacloprid 17.8 SL treatment, which produced highest seed yield of 2487 and 2377 kg ha<sup>-1</sup> during first and second year, respectively. Application of dimethoate 30 EC was second best treatment with 8.7 and 9.7 aphids plant<sup>-1</sup> providing seed yield of 2377 and 2353 kg ha<sup>-1</sup> in the respective years. Maximum net return of ₹ 30425 ha<sup>-1</sup> and cost: benefit ratio of 1:15.7 was gained from the application of imidacloprid 17.8 SL, followed by INMR of ₹ 28714 ha<sup>-1</sup> and cost: benefit ratio of 1:14.8 from dimethoate 30 EC treated mustard crop. The ranking of insecticides for the management of aphid was imidacloprid 17.8 SL < acetamiprid 20 SP < dimethoate 30 EC < thiomethoxam 25 WG < clothianidin 50 WDG < fipronil 5 SL < spinosad 45 SC.

**Key words:** Economics, Efficacy, Insecticides, Mustard aphid

Indian mustard (*Brassica juncea* (L.) Czern. & Coss) is an important oilseed crop, which is a valuable source of edible oil. Average productivity of rapeseed-mustard in India is only 1014 kg ha<sup>-1</sup>, which is far below than 1883 kg ha<sup>-1</sup> yield at global level. Mustard aphid (*Lipaphis erysimi* Kalt.) have been reported as a major constraint responsible for this low yield level, which causes average yield losses ranging 27.3-94.5% in Indian mustard due to aphid in Uttar Pradesh (Singh *et al.*, 2000; Malik *et al.*, 2003). The infestation of aphid in rapeseed-mustard causes losses in seed yield along with the oil content in seeds. However, a number of chemicals has been recommended for the management of this pest by several workers, but their indiscriminate use for suppressing the pest population increases the environmental hazards (Kumar *et al.*, 2010). Use of insecticides for the management of insect-pests in different crops is an integral part of integrated pest management, which should be eco-friendly, economically viable and socially acceptable. A number of newer chemicals have been registered in different groups for their better efficacy against different insect-pests. Thus, it is imperative to find out a selective molecule for the cost effective management of mustard aphid. Therefore, efforts were made to determine the efficacy and economics of some newer insecticides for the management of mustard aphid on Indian mustard in central Uttar Pradesh.

The experiment on efficacy and economics of newer insecticides for the management of mustard aphid on Indian mustard, were carried out in the Insectary of the Department of

Entomology of the C.S. Azad University of Agriculture and Technology, Kanpur during 2008-09 and 2009-10. Indian mustard variety 'Maya' was sown in 3.15 m x 2.75 m plot size with 45 cm x 10 cm spacing in first week of November during both the years and the crop was raised under recommended agronomic practices except insect-pest management. Two fortnightly sprays of imidacloprid 17.8% SL @ 0.0045%, acetamiprid 20% SP @ 0.004%, thiomethoxam 25 WG @ 0.005%, fipronil 5% SL @ 0.01%, dimethoate 30 EC @ 0.03%, spinosad 45% SC @ 0.015%, clothianidin 50% WDG @ 0.005% were applied for their evaluation against aphid. First application of insecticides was done at the economic threshold level of 15.6 aphids plant<sup>-1</sup>, which is reported by Singh *et al.* (2000) for central Uttar Pradesh. Mean aphid intensity was calculated after each spray and finally over all mean intensity was also calculated based on weekly observations. Seed production of the crop was recorded at harvest. Thus, the data recorded on aphid intensity as well as yield were analyzed for their critical differences. The economic analysis for the application of insecticides was also done for recommending the effective and economic treatment.

All the treatments were found significantly effective in reducing the aphid population over control during both the years. After first spraying, imidacloprid proved to be the best treatment showing significantly minimum mean aphid intensity of 10.7 aphids plant<sup>-1</sup>, which was on par with 11.5, 11.9, 12.0, 12.3 and 12.7 aphids plant<sup>-1</sup> observed on dimethoate, clothianidin, acetamiprid, thiomethoxam and fipronil sprayed crop during

2008-09, respectively (Table 1). However, the application of spinosad in mustard showed significantly higher aphid intensity in comparison to the imidacloprid sprayed crop. Similar trend of efficacy of different insecticides was recorded after their second application and reduction in aphid intensity was also noticed in comparison to previous spray. Like-wise first application, significantly lowest aphid intensity of 3.8 aphids plant<sup>-1</sup> was recorded in imidacloprid treatment showing statistical parity with 4.9, 5.8, 5.9 and 6.0 aphids plant<sup>-1</sup> counted in acetamiprid, dimethoate, thiomethoxam and clothianidin treated crop, respectively. Aphid intensity was found to increase after second spray (298.5 aphids plant<sup>-1</sup>) in comparison to the first application (162.3 aphids plant<sup>-1</sup>) in unsprayed crop. The overall impact of insecticide application on mustard aphid during 2008-09 revealed that minimum intensity of this pest being 7.2 aphids plant<sup>-1</sup> was observed on imidacloprid treated mustard. However, the intensity of aphids in acetamiprid, dimethoate, clothianidin and thiomethoxam treatment was 8.4, 8.7, 9.0 and 9.1 aphids plant<sup>-1</sup>, respectively. Significantly highest overall aphid intensity being 230 aphids plant<sup>-1</sup> was recorded in untreated crop during 2008-09. The performance of imidacloprid against mustard aphid continued as a best treatment during 2009-10, which had 11.7 and 4.7 aphids plant<sup>-1</sup> after first and second spray, respectively. However, the aphid intensity noticed in acetamiprid, dimethoate, thiomethoxam and clothianidin showed on par aphid intensity to those of imidacloprid sprays after both the applications. Significantly lowest over all aphid intensity of 8.2 aphids plant<sup>-1</sup> was observed on imidacloprid sprayed mustard, which showed statistical superiority over 8.1 and 9.1 aphids plant<sup>-1</sup> recorded on the crop sprayed with fipronil and spinosad, respectively. Thus, it can be concluded that two fortnightly applications of imidacloprid 17.8 SL (0.0045%) provided significantly lowest aphid intensity in comparison to the other treatments during both the years. The rank of efficacy of the

tested insecticides against mustard aphid was imidacloprid < acetamiprid < dimethoate < thiomethoxam < clothianidin < fipronil < spinosad. These results are in conformity with those of Akhauri and Singh (2009) and Singh *et al.* (2011), who reported effective control of mustard aphid by imidacloprid.

Significantly highest seed production of 2487 and 2377 kg ha<sup>-1</sup> was recorded from the mustard crop sprayed with imidacloprid 17.8 SL during 2008-09 and 2009-10, respectively (Table 1). Statistically at par seed yields were harvested from all the treatments except spinosad in first year. Application of spinosad yielded 1700 kg ha<sup>-1</sup> mustard. However, two sprays of imidacloprid (2377 kg ha<sup>-1</sup>), dimethoate (2353 kg ha<sup>-1</sup>), acetamiprid (2263 kg ha<sup>-1</sup>), clothianidin (2235 kg ha<sup>-1</sup>) and thiomethoxam (2187 kg ha<sup>-1</sup>) produced at par yields during second year, while fipronil and spinosad produced statistically poor production of 2063 and 1580 kg ha<sup>-1</sup>, respectively. It can be inferred that two fortnightly sprays of imidacloprid 17.8 SL were found superior for the management of mustard aphid, which, thereby, produced significantly highest seed yield followed by dimethoate, acetamiprid, clothianidin and thiomethoxam. These results are in conformity with the findings of Akhauri and Singh (2009). The maximum seed yield was obtained from application of imidacloprid, followed by dimethoate.

Economic analysis for the application of different insecticides for the management of mustard aphid on mustard was computed on the basis of mean seed yield (Table 2). The cost of increased seed yield of mustard over control ranged between ₹ 11804 to 32370 based on the increased yield and the whole sale market price of mustard @ ₹ 2600. Cost of different treatment varied between ₹ 1900-4040 ha<sup>-1</sup>. Maximum incremental net monetary return ₹ 30425 ha<sup>-1</sup> was realized from the imidacloprid 17.8 SL treated crop, whereas the minimum INMR of ₹ 8480 was obtained from the spinosad sprayed crop.

Table 1. Effect of newer insecticide on the intensity of aphid and seed yield of Indian mustard

| Treatment            | Dose                                 | Mean aphid intensity (No. plant <sup>-1</sup> ) |                  |                  |                  |                  |                  | Seed yield (kg ha <sup>-1</sup> ) |         |
|----------------------|--------------------------------------|-------------------------------------------------|------------------|------------------|------------------|------------------|------------------|-----------------------------------|---------|
|                      |                                      | 2008-09                                         |                  |                  | 2009-10          |                  |                  | 2008-09                           | 2009-10 |
|                      |                                      | I spray                                         | II spray         | Mean             | I Spray          | II Spray         | Mean             |                                   |         |
| Imidacloprid 17.8 SL | 0.0045%<br>(0.2 ml l <sup>-1</sup> ) | 10.7<br>(3.28)                                  | 3.8<br>(1.95)    | 7.2<br>(2.70)    | 11.7<br>(3.43)   | 4.7<br>(2.19)    | 8.2<br>(2.88)    | 2487                              | 2377    |
| Acetamiprid 20 SP    | 0.004%<br>(0.2 g l <sup>-1</sup> )   | 12.0<br>(3.47)                                  | 4.9<br>(2.22)    | 8.4<br>(2.91)    | 13.0<br>(3.61)   | 5.9<br>(2.44)    | 9.4<br>(3.08)    | 2357                              | 2263    |
| Thiomethoxam 25 WG   | 0.005%<br>(0.2 g l <sup>-1</sup> )   | 12.3<br>(3.52)                                  | 5.9<br>(2.43)    | 9.1<br>(3.02)    | 13.3<br>(3.66)   | 6.9<br>(2.63)    | 10.1<br>(3.18)   | 2327                              | 2187    |
| Fipronil 5 SL        | 0.01%<br>(2 ml l <sup>-1</sup> )     | 12.7<br>(3.57)                                  | 7.1<br>(2.67)    | 9.9<br>(3.16)    | 13.7<br>(3.71)   | 8.1<br>(2.85)    | 10.9<br>(3.31)   | 2237                              | 2063    |
| Dimethoate 30 EC     | 0.03%<br>(1 ml l <sup>-1</sup> )     | 11.5<br>(3.40)                                  | 5.8<br>(2.42)    | 8.7<br>(2.96)    | 12.6<br>(3.55)   | 6.8<br>(2.62)    | 9.7<br>(3.12)    | 2377                              | 2353    |
| Spinosad 45 SC       | 0.015%<br>(0.2 ml l <sup>-1</sup> )  | 13.9<br>(3.73)                                  | 8.1<br>(2.86)    | 11.0<br>(3.32)   | 15.6<br>(3.96)   | 9.1<br>(3.03)    | 12.0<br>(3.47)   | 1700                              | 1580    |
| Clothianidin 50 WDG  | 0.005%<br>(0.2 g l <sup>-1</sup> )   | 11.9<br>(3.45)                                  | 6.0<br>(2.46)    | 9.0<br>(3.00)    | 12.8<br>(3.59)   | 7.0<br>(2.66)    | 9.9<br>(3.16)    | 2368                              | 2235    |
| Untreated            |                                      | 162.3<br>(12.74)                                | 298.5<br>(17.28) | 230.0<br>(15.18) | 160.0<br>(12.65) | 295.4<br>(17.19) | 227.7<br>(15.09) | 1220                              | 1153    |
| SE (d) ±             |                                      | 0.14                                            | 0.29             | 0.19             | 0.11             | 0.24             | 0.16             | 145                               | 134     |
| CD (P=0.05)          |                                      | 0.32                                            | 0.62             | 0.42             | 0.23             | 0.52             | 0.36             | 315                               | 287     |

Figures in parentheses are  $\sqrt{x} + 0.5$  transformed values

Table 2. Economic analysis of insecticide application for the management aphid in Indian mustard (mean of two years)

| Treatment            | Dose                              | Quantity of insecticide (ml or g ha <sup>-1</sup> ) | Cost of insecticide (₹ ha <sup>-1</sup> ) | Labour cost + sprayer charge (₹ ha <sup>-1</sup> ) | Cost of treatment (₹ ha <sup>-1</sup> ) | Seed yield (kg ha <sup>-1</sup> ) | Seed yield increased over control (kg ha <sup>-1</sup> ) | Cost of increased seed yield | Incremental net monetary return (₹ ha <sup>-1</sup> ) | Cost: benefit ratio |
|----------------------|-----------------------------------|-----------------------------------------------------|-------------------------------------------|----------------------------------------------------|-----------------------------------------|-----------------------------------|----------------------------------------------------------|------------------------------|-------------------------------------------------------|---------------------|
| Imidacloprid 17.8 SL | 0.0045% (0.2 ml l <sup>-1</sup> ) | 252.8 ml                                            | 404                                       | 1540                                               | 1944                                    | 2431                              | 1245                                                     | 32370                        | 30425                                                 | 1:15.7              |
| Acetamiprid 20 SP    | 0.004% (0.2 g l <sup>-1</sup> )   | 200 ml                                              | 360                                       | 1540                                               | 1900                                    | 2310                              | 1124                                                     | 29224                        | 27324                                                 | 1:14.4              |
| Thiomethoxam 25 WG   | 0.005% (0.2 g l <sup>-1</sup> )   | 200 g                                               | 500                                       | 1540                                               | 2040                                    | 2256                              | 1070                                                     | 27820                        | 25780                                                 | 1:12.6              |
| Fipronil 5 SL        | 0.01% (2 ml l <sup>-1</sup> )     | 200 ml                                              | 2500                                      | 1540                                               | 4040                                    | 2150                              | 964                                                      | 25064                        | 21024                                                 | 1:5.2               |
| Dimethoate 30 EC     | 0.03% (1 ml l <sup>-1</sup> )     | 1000 ml                                             | 400                                       | 1540                                               | 1940                                    | 2365                              | 1179                                                     | 30654                        | 28714                                                 | 1:14.8              |
| Spinosad 45 SC       | 0.015% (0.2 ml l <sup>-1</sup> )  | 200 ml                                              | 1784                                      | 1540                                               | 3324                                    | 1640                              | 454                                                      | 11804                        | 8480                                                  | 1:2.6               |
| Clothianidin 50 WDG  | 0.005% (0.2 g l <sup>-1</sup> )   | 200 g                                               | 1333                                      | 1540                                               | 2873                                    | 2301                              | 1115                                                     | 28990                        | 26117                                                 | 1:9.0               |
| Untreated            |                                   |                                                     |                                           |                                                    |                                         | 1186                              |                                                          |                              |                                                       |                     |

**Price of insecticides:** Imidacloprid ₹ 160 per 100 ml, Acetamiprid ₹ 1800 kg<sup>-1</sup>, Thiamethoxam ₹ 25 per 5 g, Fipronil ₹ 125 per 100 ml, Dimethoate ₹ 40 per 100 ml, Spinosad ₹ 900 per 100 g and Clothianidin ₹ 40 per 6 g; Wage of labour = ₹ 120 day<sup>-1</sup>; Charge of sprayer = ₹ 25 day<sup>-1</sup>; Sale price of mustard = ₹ 26 kg<sup>-1</sup>; IBC = Incremental cost: benefit ratio

Cost: benefit of treatment was calculated to be 15.7 in the application of imidacloprid 17.8 SL, which ranged between 2.6 to 14.8 in remaining treatments. It can be summarized that maximum net return) of ₹ 30425 ha<sup>-1</sup> and cost : benefit ratio 1:15.7 was gained from the application of imidacloprid 17.8 SL followed by ₹ 28714 ha<sup>-1</sup> and cost: benefit ratio 1:14.8 from dimethoate treated mustard crop.

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## SHORT COMMUNICATION

# Growth and metabolism of sesame (*Sesamum indicum* L.) plants in relation to lead toxicity

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## ABSTRACT

Overall growth was found to be adversely affected at increasing doses of lead (Pb) content in sesame (*Sesamum indicum* L.) plants, commonly called Til. This plant showed mild chlorosis at early stages and severe chlorosis after a long period of exposure. Necrosis and small brown spots were observed in petiolar inter- and intra-veinal regions of the leaves. Increasing doses of Pb cause significant reduction in chlorophyll content while only higher doses caused reduction in protein content of plants.

**Key words:** Catalase, Chlorophyll, Lead, Peroxidase, Protein, Sesame, Sugar

By and large excess doses of all the heavy metals had toxic effects in living organisms. Lead is quite toxic metal which causes damage to brain and kidney in both human beings and animals. Common sources of lead include earthenware, glazed surface and pigments of older paints. Lead is particularly harmful to unborn and young children. Normally lead affects most of the organs but particularly it affects the central nervous system. The seeds of sesame (*Sesamum indicum* L.) are used in making various sweet preparations in India. Sesame oil has both medicinal and religious values. Its oil is also used in preparation of paints, perfumed oils and soaps. Heavy metals including lead used to cause contamination of agricultural soil due to sewage sludge application. Waste lands can be created by different industrial and mining activities such land masses can also be polluted by number of heavy metals including Pb, which in turn may adversely affect pH value and mobility of some essential nutrients. Keeping all these aspects in view, this study was carried out to investigate the effect of different doses of Pb on the growth and metabolism of til plants.

An experiment was conducted in til plants in sand culture under controlled glass house condition. Controlled plants were provided only basal nutrient solution whereas test plants were treated with different doses of lead which were superimposed on basal nutrient solution, prepared using the method of Hewitt (1966). Micro and macro nutrient solutions were prepared by A.R. grade chemical reagents. Graded levels of Pb were prepared from its salt, viz. lead acetate. This heavy metal was supplied in the doses of 0.25, 0.5, 1.0 and 2.0 mM. The basal nutrient solution along with various treatments of lead was supplied regularly for proper growth of plants. Around 50 ml

each of basal nutrient solution and lead treatments were supplied to each pot regularly. The observations were taken daily for any abnormal symptoms such as chlorosis, necrosis, browning of leaf tissues and any other typical toxic symptom resulted due to effect of lead. Plants were harvested after a full season (~3 months). After harvesting plants were taken out from pots and thoroughly washed with distilled water and after blotting the moisture fresh and dry weights of plants were taken. For dry weight plants were cut into pieces and kept into oven at 70°C for three days for drying purpose and after that dry weight of each plant was measured. Both fresh and dry weights were taken by using electronic balance. Length of the stem was taken in each pot from the upper surface of the sand till the apex of shoot. In the fresh leaf extract, toxic effects of Pb on metabolic parameters such as content of chlorophyll, sugar, protein and activities of antioxidative enzymes, viz. catalase and peroxidase were measured. Fresh leaf tissue was ground in ice chilled pestle mortar. One gram of leaf tissue was extracted in 10 ml of glass distilled water. The homogenate was filtered through two fold muslin cloth with the help of buchner funnel and stored at freezing temperature in a refrigerator. Leaf extract was used for estimation of various metabolic parameters as per methods: (i) Total chlorophyll content was measured by the method of Petering (1940), (ii) Protein content was estimated by the method of Lowry *et al.* (1951), (iii) Total sugars was estimated by the method of Dubias *et al.* (1953), (iv) Catalase activity was assayed by the method of Bisht (1972) a modified method of Euler and Josephson (1927), and (v) Peroxidase (POD) enzyme was estimated by the modified method of Luck (1963).

Table 1. Effect of different doses of lead on stem length and biomass of sesame plants

| Treatment     | Stem length (cm) | Fresh weight (g) | Dry weight (g) |
|---------------|------------------|------------------|----------------|
| Control       | 67.50±4.50       | 90.00±3.00       | 11.20±0.90     |
| 0.25 m M Pb   | 59.25±3.40       | 65.00±2.00       | 8.80±0.90      |
| 0.50 m M Pb   | 53.95±2.70       | 49.00±2.00       | 4.30±0.60      |
| 1.0 m M Pb    | 48.65±4.90       | 29.00±4.00       | 3.30±0.40      |
| CD (P = 0.05) | 18.62            | 16.17            | 1.50           |

Table 2. Effects of different doses of lead on certain metabolic parameters of sesame plants

| Treatment     | mg Chlorophyll g <sup>-1</sup> FW | µg Protein g <sup>-1</sup> FW | mg Total Sugar g <sup>-1</sup> FW | Catalase activity µmoles <sup>-1</sup> H <sub>2</sub> O <sub>2</sub> split mg <sup>-100</sup> FW | Peroxidase activity (ΔOD mg <sup>-100</sup> FW) |
|---------------|-----------------------------------|-------------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Control       | 1.07±0.019                        | 129.51±3.56                   | 0.406±0.817                       | 105.0±5.00                                                                                       | 16.50±0.50                                      |
| 0.25 mM Pb    | 0.62±0.017                        | 169.52±5.54                   | 0.975±0.100                       | 47.50±2.50                                                                                       | 22.00±2.00                                      |
| 0.50 mM Pb    | 0.52±0.007                        | 99.81±2.73                    | 0.725±0.025                       | 52.5±2.50                                                                                        | 65.00±4.00                                      |
| 1.00 mM Pb    | 0.429±0.011                       | 83.57±9.90                    | 0.806±0.018                       | 95.00±5.00                                                                                       | 53.00±2.00                                      |
| CD (P = 0.05) | 0.1                               | 20.338                        | 0.321                             | 21.75                                                                                            | 15.76                                           |

Growth of plants was adversely affected by lead toxicity in sesame plants. Mild chlorosis was noticed at early stages but excessive chlorosis of whole leaves was noticed after a long period of exposure. Necrosis and small brown spots were observed in petioler, inter and intra-veinal regions of the leaves. The length and thickness of the stem and number of branches were found to be reduced in lead treated plants as compared to controlled plants (Table 1). At later stages of growth bending of leaves as well as basal leaf shedding were also observed.

Reduction in fresh and dry weight of plants was observed at increasing level of lead. Lead was found to induce chlorosis in younger leaves followed by older leaves with much severity. This in turn inhibited overall plant growth also. Several workers have reported adverse effects of heavy metals on chlorophyll and Fe content in plants (Singh *et al.*, 2005 and 2006). The results of present study are in conformity with earlier reports. Reduced plant growth might be attributed to the adverse effect of Pb. Tandon *et al.* (2000) have reported reduced root and shoot growth in germinating seeds of gram at increasing level of Pb.

A significant reduction in total chlorophyll content was observed at all the doses of lead as compared to the controlled plants. Reduction in chlorophyll content might be due to the disturbance in Fe metabolism. Total sugar content increased at 0.25 mM lead, which subsequently reduced at higher doses of Pb.

The protein content increased at 0.25 mM lead application and decreased subsequently at higher doses (Table 2). The reduction in protein content at higher doses of Pb might be due to the interference of this metal in transcription of DNA and translocation at the time of gene expression.

The activity of anti-oxidative iron enzymes, viz. catalase and peroxidase showed different pattern of results. On one hand catalase activity was inhibited at all doses of lead

Application, whereas peroxidase activity was stimulated at all the doses of lead. The catalase activity was suppressed due to supply of lead to the plant at higher level. Heavy metal including Pb pollution induces plants to produce more peroxidase and thus enhances POD activity.

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## SHORT COMMUNICATION

# Biochemical responses of wheat (*Triticum aestivum*) varieties under salinity stress

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## ABSTRACT

An experiment with six concentrations of saline water (0, 200, 400, 600, 800 and 1000 ppm) was conducted in pots during January-April 2012 to study their impacts on wheat (*Triticum aestivum* L. emend. Fiori & Paol.). Saline water considerably reduced the plant growth and influenced several biochemical parameters. Saline water did not inhibit total chlorophyll, carbohydrate and protein content in leaves. Activities of the antioxidant enzymes like CAT (Catalase), GR (Glutathione reductase), SOD (Superoxide dismutase) and lipid peroxidation, however, were affected with increasing salinity.

**Key words:** Catalase, Glutathione reductase, Lipid peroxidation, Saline water, Superoxide dismutase

High concentrations of salts cause ionic, osmotic and associated secondary stresses to plants. Plant responses to these stresses are complex but can be grouped into three general categories: homeostasis, detoxification and growth control. Polyamines are involved in the growth control of the cells subjected to salt stress. Salt stress causes inhibition of growth and development, reduction in photosynthesis, respiration, and protein synthesis and disturbs nucleic acid metabolism. Decrease in uptake of  $K^+$ ,  $Mg^{2+}$ ,  $Ca^{2+}$  and thereby decrease in growth at higher sodium concentration has also been reported. Biotic stresses also lead to oxidative stress through an increase in reactive oxygen species (ROS), such as superoxide ( $O_2^-$ ), hydrogen peroxide ( $H_2O_2$ ) and hydroxyl radicals ( $OH^\cdot$ ). Some authors have reported carbohydrate accumulation in various plants under salinity conditions. On the other hand, some scientists observed that at low and moderate salinity levels, sugars and consequently the total carbohydrates are decreased. Soluble protein is generally decreased in response to salinity (Parida *et al.*, 2002). It has also been reported that high salt concentration increase soluble proteins (Shaddad *et al.*, 2006). Proline accumulation under stress conditions may be due to induction of proline biosynthesis enzymes, reduction in the rate of proline oxidation conversion to glutamate, decreased utilization of proline in protein synthesis and enhanced protein turnover (Claussen, 2005).

The experiment was carried out during the period January to April 2012 in the Department of Biochemistry and Biochemical engineering, Sam Higginbottom Institute of Agriculture, Technology and Sciences, Allahabad (Uttar Pradesh), India.

The materials and methods used in the present study were standard routine analytical methods. Wheat variety K 9533 and PBW 502 were used for study. Seeds were first washed under running tap water for five minutes, then surface sterilized by treatment with 1% sodium hypochlorite followed by four distilled water rinses. The surface sterilized seeds were sown in pots and allowed to germinate. Pots were irrigated with normal water till germination of the seeds. Thereafter, pots were irrigated with saline water ( $NaCl+CaCl_2$ ) at the time interval of 3 days till completion of life cycle and during rest days pots were treated with normal water. Seedlings were planted in plastic and mud pots in completely natural environment. Plants were treated with different concentrations of saline water (mentioned previously) for 30 days. Roots and leaves from treatments were taken for various biochemical analyses. Chlorophyll, protein and carbohydrate contents were estimated by methods given by Arnon *et al.* (1949), Lowry *et al.* (1951) and Hedge and Hofreiter (1962), respectively. For SOD, CAT and GR extraction, leaf samples (0.5 g) were homogenised in ice cold 0.1 M phosphate buffer (pH 7.5) containing 0.5 mM EDTA with pre-chilled pestle and mortar. Each homogenate was transferred to centrifuge tubes and was centrifuged at 4°C for 30 minutes and supernatant was used for enzyme estimation. CAT activity was measured according to Aebi (1984). SOD activity was estimated by recording the decrease in absorbance of superoxide nitroblue tetrazolium complex by the enzyme (Sengupta *et al.*, 1993). GR activity was assayed by recording the increase in absorbance in the presence of oxidized glutathione (GSSG) and 5,5-dithiois-2-nitrobenzoic acid (DTNB) (Sairam *et al.*, 2002). Malondialdehyde (MDA) was measured by colorimetric method

(Stewart and Bewly, 1980).

The maximum chlorophyll content was recorded in wheat variety K 9533 in  $T_1$  ( $11.8 \text{ mg g}^{-1}$ ) and minimum in treatment  $T_6$  ( $6.72 \text{ mg g}^{-1}$ ), whereas in wheat variety PBW502, maximum chlorophyll was recorded in  $T_1$  ( $12.5 \text{ mg g}^{-1}$ ) and minimum in  $T_6$  ( $9.95 \text{ mg g}^{-1}$ ) (Table 1). This decrease in chlorophyll may be associated directly with light harvesting component in photosynthesis. Salt stress causes a significant reduction in biosynthesis of chlorophyll and carotenoids. Chlorophyll a decreased more than Chlorophyll b, which resulted in a lower ratio of Chlorophyll a/b. The results of present study are in accordance with Ashraf (1989). The decreased accumulation of chlorophyll was due to reduced biosynthesis of chlorophyll molecules and enhanced degradation of pigments as reported by Delfine *et al.* (1999). The present study revealed higher concentration of protein in both the varieties on increasing salinity (Table 1). This could be due to reactive oxygen species (ROS), which may lead to unspecific oxidation of proteins and membrane lipids. Consequently, tissues injured by oxidative stress generally contain increased concentrations of carboxylated proteins, as has already been reported by Ames *et al.* (2004). PBW 502 wheat variety showed highest soluble carbohydrate content ( $18.59 \pm 0.89 \text{ mg kg}^{-1}$ ) than variety K 9533 ( $15.5 \pm 0.046 \text{ mg kg}^{-1}$ ) at 1000 ppm saline (Table 1). These results are in agreement with the findings of Dash and Panda (2001). Similarly, Hamid *et al.* (2011) also reported similar results when these tests were performed on other wheat varieties.

Catalase activity in K 9533 variety showed a sharp increase in  $T_2$  ( $464.91 \pm 36.32 \text{ unit mg}^{-1}$ ), which further declined and again showed an elevation in  $T_6$  ( $466.66 \pm 39.38 \text{ unit mg}^{-1}$ ). On the other hand, PBW 502 variety in  $T_2$  showed  $30.53 \text{ unit mg}^{-1}$  catalase activity, which further declined to  $5.76 \text{ unit mg}^{-1}$  in  $T_6$  (Table 2). Such photo-inactivation of catalase and photosystem II represent specific and widely occurring early symptoms of incipient photo damage indicating stress conditions. The repair

of activity of denatured enzyme is not sufficient to cope up the photo inactivation. During prolonged exposures to NaCl, chlorophyll is bleached in light and rate of its photo degradation increases in proportion as catalase level declines. The results suggest that the enhanced susceptibility of leaf tissues to photo oxidative damage is widely observed in stressed plants, which is related to early loss of catalase as described by Peter *et al.* (2006).

The PBW 502 wheat variety showed an increasing trend for SOD to  $7.49 \pm 0.43 \text{ unit mg}^{-1}$  in  $T_6$ , whereas the K 9533 variety showed maximum SOD of  $18.15 \pm 3.5 \text{ unit mg}^{-1}$  in  $T_2$ , which declined subsequently till  $T_5$  but increased marginally in  $T_6$  (Table 2). Many workers have suggested that the salt tolerance character is related to increased SOD activity in salt-tolerant cultivars (Noreen *et al.*, 2009). Decline in SOD activity at second and third growth stages could be related to proline accumulation (Koca *et al.*, 2007). When SOD activity was high, ROS especially superoxide radical scavenging was done properly and thus damage to membranes and oxidative stress decreased, leading to increase of tolerance to oxidative stress. Salt stress increased superoxide level in cells. If this radical is not scavenged by SOD, it causes disturbance (Mittler, 2000).

Wheat variety K 9533 recorded a declining trend in glutathione reductase activity from  $0.392 \pm 0.062 \text{ unit ml}^{-1} \text{ mg}$  in  $T_1$  to  $0.070 \pm 0.008 \text{ unit ml}^{-1} \text{ mg}$  in  $T_6$ , whereas PBW 502 showed a slight increase from  $0.120 \pm 0.024 \text{ unit ml}^{-1} \text{ mg}$  in  $T_1$  to  $0.160 \pm 0.032 \text{ unit ml}^{-1} \text{ mg}$  in  $T_2$  and further declined to  $0.04 \pm 0.026$  in  $T_6$ . Glutathione reductase activates glutathione-ascorbate cycle and converts GSSG to GSH as reported by Asada (2006). In addition, glutathione reductase regulates GSH/GSSG ratio and supplies GSH for GPX and DHAR, which converts  $\text{H}_2\text{O}_2$  to  $\text{H}_2\text{O}$  and reduced oxidized ascorbate, respectively. Although glutathione reductase acquires the reduction power from NADPH,  $\text{H}^+$ , it dissipates the power and in turn, increases  $\text{NADP}^+/\text{NADPH}$ ,  $\text{H}^+$  ratio. In roots of *Chrysanthemum*

Table 1. Effect of saline water concentrations on chlorophyll, protein and carbohydrates of wheat

| Saline water concentration | Total chlorophyll ( $\text{mg g}^{-1}$ ) |                  | Total protein ( $\text{mg g}^{-1}$ ) |                 | Total carbohydrate ( $\text{mg kg}^{-1}$ ) |                  |
|----------------------------|------------------------------------------|------------------|--------------------------------------|-----------------|--------------------------------------------|------------------|
|                            | K 9533                                   | PBW 502          | K 9533                               | PBW 502         | K 9533                                     | PBW 502          |
| $T_1$ : 0 ppm              | $11.80 \pm 1.16$                         | $12.50 \pm 1.32$ | $3.0 \pm 0.0089$                     | $6.8 \pm 0.068$ | $6.43 \pm 0.10$                            | $7.6 \pm 0.016$  |
| $T_2$ : 200 ppm            | $9.85 \pm 0.12$                          | $12.95 \pm 1.38$ | $3.8 \pm 0.0150$                     | $7.2 \pm 0.16$  | $7.8 \pm 0.046$                            | $9.56 \pm 0.02$  |
| $T_3$ : 400 ppm            | $8.20 \pm 4.96$                          | $11.40 \pm 1.09$ | $5.4 \pm 0.0160$                     | $8.8 \pm 0.83$  | $9.5 \pm 0.03$                             | $10.8 \pm 0.38$  |
| $T_4$ : 600 ppm            | $7.70 \pm 1.41$                          | $10.90 \pm 0.89$ | $6.8 \pm 0.060$                      | $9.8 \pm 0.83$  | $11.33 \pm 0.014$                          | $12.7 \pm 0.44$  |
| $T_5$ : 800 ppm            | $7.10 \pm 0.40$                          | $10.45 \pm 0.63$ | $10.0 \pm 0.160$                     | $9.8 \pm 0.83$  | $12.8 \pm 0.106$                           | $15.96 \pm 0.16$ |
| $T_6$ : 1000 ppm           | $6.72 \pm 0.58$                          | $9.95 \pm 0.16$  | $11.8 \pm 1.2$                       | $10.4 \pm 1.57$ | $15.5 \pm 0.046$                           | $18.59 \pm 0.89$ |
| CD ( $P=0.05$ )            | 1.66                                     | 1.13             | 1.20                                 | 11.11           | 0.33                                       | 0.39             |

The values are mean  $\pm$  SD ( $n=3$ )

Table 2. Effect of saline water concentrations on enzyme activities of wheat

| Saline water concentration | Specific activity of catalase ( $\text{unit min}^{-1} \mu\text{l}^{-1} \text{ mg}^{-1}$ ) |                  | Specific activity of super oxide dismutase ( $\text{unit mg}^{-1}$ ) |                 | Specific activity of glutathione reductase ( $\text{unit ml}^{-1} \text{ mg}^{-1}$ ) |                   | Leaf malondialdehyde (MDA) level ( $\text{nMol ml}^{-1}$ ) |                   |
|----------------------------|-------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------|-------------------|
|                            | K 9533                                                                                    | PBW 502          | K 9533                                                               | PBW 502         | K 9533                                                                               | PBW 502           | K 9533                                                     | PBW 502           |
| $T_1$ : 0 ppm              | $90.25 \pm 2.5$                                                                           | $3.81 \pm 0.58$  | $2.03 \pm 0.09$                                                      | $0.60 \pm 0.09$ | $0.392 \pm 0.068$                                                                    | $0.120 \pm 0.024$ | $32.33 \pm 4.21$                                           | $14.15 \pm 0.11$  |
| $T_2$ : 200 ppm            | $464.91 \pm 36.32$                                                                        | $30.53 \pm 1.24$ | $18.15 \pm 3.5$                                                      | $3.05 \pm 0.08$ | $0.303 \pm 0.045$                                                                    | $0.160 \pm 0.032$ | $42.33 \pm 4.21$                                           | $20.32 \pm 0.15$  |
| $T_3$ : 400 ppm            | $281.47 \pm 12.2$                                                                         | $20.45 \pm 0.86$ | $8.0 \pm 0.66$                                                       | $4.20 \pm 0.06$ | $0.210 \pm 0.029$                                                                    | $0.150 \pm 0.086$ | $53.33 \pm 6.21$                                           | $30.81 \pm 0.17$  |
| $T_4$ : 600 ppm            | $197.54 \pm 6.25$                                                                         | $12.58 \pm 0.22$ | $7.30 \pm 0.14$                                                      | $4.61 \pm 0.08$ | $0.160 \pm 0.015$                                                                    | $0.098 \pm 0.046$ | $75.0 \pm 6.0$                                             | $37.43 \pm 0.04$  |
| $T_5$ : 800 ppm            | $192.06 \pm 5.63$                                                                         | $8.6 \pm 0.20$   | $6.11 \pm 0.44$                                                      | $5.69 \pm 0.15$ | $0.098 \pm 0.005$                                                                    | $0.090 \pm 0.046$ | $121.66 \pm 5.55$                                          | $42.52 \pm 0.008$ |
| $T_6$ : 1000 ppm           | $466.66 \pm 39.38$                                                                        | $5.76 \pm 0.59$  | $7.79 \pm 0.13$                                                      | $7.49 \pm 0.43$ | $0.070 \pm 0.008$                                                                    | $0.04 \pm 0.026$  | $141.66 \pm 5.55$                                          | $50.13 \pm 0.26$  |
| CD ( $P=0.05$ )            | 6.84                                                                                      | 0.84             | 4.82                                                                 | 0.34            | 0.02                                                                                 | 0.00              | 1.29                                                       | 0.48              |

The values are mean  $\pm$  SD ( $n=3$ )

*morifolium*, with the increase in NaCl concentration there was an increase in the glutathione reductase activity (Hossain *et al.*, 2004).

Results further revealed that leaf malondialdehyde (MDA) levels in variety K 9533 recorded a marked increase from  $32.33 \pm 4.21$  nmol ml<sup>-1</sup> in T<sub>1</sub> to  $141.66 \pm 5.55$  nmol ml<sup>-1</sup> in T<sub>6</sub> and also PBW 502 variety showed an increase from  $14.15 \pm 0.11$  nmol ml<sup>-1</sup> (T<sub>1</sub>) and  $50.13 \pm 0.25$  nmol ml<sup>-1</sup> (T<sub>2</sub>) (Table 2). MDA is regarded as the marker for evaluation of lipid peroxidation or damage to plasma membrane and organelle membranes that increase with environmental stresses. MDA content is the product of lipid peroxidation and is a prominent indicator of membrane impairment in plants exposed to saline regimes (Katsuhara *et al.*, 2005). Lipid peroxidation is linked to the activity of antioxidant enzyme, e.g. with the increase of SOD, CAT, GR etc, the oxidative stress tolerance is enhanced and the MDA is also increased.

On the basis of the above results it can be concluded that wheat variety PBW 502 was found to be more salt stress resistant as compared to K 9533.

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## SHORT COMMUNICATION

# Effect of agricultural advisory and trainings on knowledge and attitude of dairy farmers in Nalgonda, Andhra Pradesh

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## ABSTRACT

**Adiagnostic study was conducted in Nalgonda district of Andhra Pradesh with a sample size of 675 dairy farmers belonging to 27 villages. There was a 180.7% increase in knowledge of the dairy farmers after exposure to training programme. There was a 96.7% change in the attitude of farmers who underwent trainings on scientific dairy farming practices. Combination of different teaching methods (TV + charts and posters + lecture method + exposure visit) was effective in imparting knowledge to the dairy farmers. The trainings formulated based on the needs of the farmers revealed a positive indication of adoption of scientific dairy farming practices.**

**Key words:** Attitude, Breeding practices, Feeding practices, Fodders, Knowledge, Milking

Dairy enterprise is considered a 'treasure' of the Indian economy, particularly for rural systems. It provides nutrition, draft animal power, organic manure, supplementary employment, cash income and a 'cushion' for 'drought proofing' in India (Patel, 1993; Paroda, 1998). The sector involves millions of resource-poor farmers, for whom animal ownership ensures critical livelihood, sustainable farming, and economic stability. Dairying in the recent decades has been considered a vital component in the diversification of Indian agriculture, where crop farming is beset with stagnating growth and low absorption of unskilled agricultural labourers. In order to alleviate the problem of unemployment/under-employment, diversification of crop production into non-crop enterprises like dairy farming is of vital importance (Pandey, 2000; Alagh, 2002).

Training the unskilled/ignorant farmers will be a critical input, which not only hastens the agricultural and livestock production but also generating the much needed employment and income. Improper development is mainly due to non-adaption of scientific farming practices, lack of knowledge and stringent. It may be achieved by conducting trainings for dairy farmers to impart knowledge on the latest dairy farming technologies (Harikrishna *et al.*, 2012). Training is usually an integral part of human resource development, which helps in enhancing the knowledge level and changing the attitude of the trainees (Kumar *et al.*, 2008). It is obvious that farmers are trying their best for increasing production levels by using the traditional methods and their experiences. But due to lack of

scientific knowledge and favourable environment towards the recommended practices, they cannot use available resources in proper and scientific manner (Harikrishna *et al.*, 2012). Keeping these things in view, the present diagnostic investigation was undertaken to study the effect of District Agricultural Advisory and Transfer of Technology (DAATT) Centre trainings on knowledge and attitude of dairy farmers about scientific dairy farming practices in Nalgonda district of Andhra Pradesh State in India.

The study was conducted for DAATT centre in Nalgonda under Acharya N.G Ranga Agricultural University, Hyderabad. As per the DAATT centre approved technical programme, 27 training programmes in scientific dairying were conducted for dairy farmers of 27 different villages, viz. Alair, Kangal, Mothkur, Turkapally, Bhongir, Vangapally, Miryalguda, Pochampally, Mall, Suryapet, Nidamanur, Thammadpally, Munugode, Gurrampode, Kondrapolu, Halia, Palivela, Ramapuram, Chintapally, Singaram, Charlapally, Marrigudem, S. lingotam, Appajipet, Kethapally, Deverkonda and Nakrekal etc. The farmers who had participated in these training programmes of two days duration were considered as respondents. Thus, there were 675 respondents as a sample size for this study. The data were collected to measure the change in knowledge (Kumar *et al.*, 2008). The change in knowledge score worked out by administering knowledge test before and after the trainings. Attitude of the respondents towards trainings was measured with the help of modified scale developed by Sankhala and Chand (1998). Change in attitude towards training programmes

provided was measured and mean difference was calculated. The collected data were tabulated and analyzed.

The main objective of providing trainings was to acquaint the respondents with adequate knowledge and inculcate positive attitude towards the scientific dairying and to improve the productivity of bovines. The effect of training programmes on scientific dairy farming practices was measured for knowledge and attitude.

The analysis on extent of change in knowledge presented in Table 1 revealed that the mean knowledge score on dairy management practices after imparting the training was 48.46 compared to the mean knowledge score 22.26 before training. The results revealed improvement and manifested in the form of higher mean knowledge score (117.70) after the training programme. It is in accordance with the findings of Harikrishna *et al.* (2012). Similarly, the mean knowledge score regarding breeding and feeding practices, cultivation of fodders and their conservation, health care, milking methods, clean milk production and clean milk products technologies after imparting training was relatively high. The overall gain in knowledge was 180.7%, which appears to be a very good outcome of trainings. The results were in agreement with the findings of Kumar *et al.* (2008) and Harikrishna *et al.* (2012). The present study indicates that, the training programmes of DAATT centre, Nalgonda concerning scientific dairy farming had significantly enhanced knowledge of trainees for better adaption.

It is a well-known fact that individual attitude has a major influence on his/her behavior in respect to a definite objective. People's attitude towards a training programme will fundamentally determine the nature and extent of their participation, which is the basis for success. This signifies the enormity of identifying the participant's attitude towards the training programmes. The DAATT centre organizes training programmes to the farmers who are the ultimate users and adopt the recommended practices. As the farmers are expected to have a positive attitude toward training programmes, they should endow with feedback in order to make changes in the

suggested practices for their betterment.

The mean score of attitude of respondents towards training programmes conducted by DAATT centre after training was 47.44, while before training, it was 24.12 (Table 2). The difference in the means of attitude scores before and after training programmes was 23.3. The per cent change in attitude score recorded in the present study was 96.7. It shows that the training programmes conducted on various areas of scientific dairy farming practices were successful in changing the attitude of the respondents in an encouraging direction. The results of the present study were in agreement with the findings of Kumar *et al.* (2008) and Harikrishna *et al.* (2012). The possible rationale for higher mean attitude score was that the farmers possessed more knowledge on scientific dairy farming practices by trainings of DAATT centre, resulting in more positive attitude towards training programmes.

TV, charts and posters, lecture method and exposure visit were used in trainings in a combination. It was evident with the Table 3 that the combination of different teaching methods (88.6%) was effective in imparting knowledge to the farmers, because these teaching methods targeted the audio and visuals senses. The other effective teaching methods were television (82.2%), lecture method (32.3%) and charts and posters (28.2%) in that order. These results are corroborating the reports of Harikrishna *et al.* (2012). In fact the above results helped to arrive at a conclusion that combination of various teaching methods makes trainings more effective than using single method. The results recorded in the current study were also consistent with the findings of Thammi Raju and Deepa (2011).

Table 3. Perception of trainees on the teaching methods (multiple responses)

| S.No. | Teaching method                                                        | Good  | Excellent |
|-------|------------------------------------------------------------------------|-------|-----------|
| 1.    | Television                                                             | 27.06 | 82.16     |
| 2.    | Charts and Posters                                                     | 17.62 | 28.24     |
| 3.    | Lecture method                                                         | 21.48 | 32.26     |
| 4.    | Combination<br>(TV + charts and posters +<br>lecture + exposure visit) | 26.24 | 88.62     |

Table 1. Effect of training programme on change in knowledge of dairy farmers

| S.No. | Topics covered in training programme | Stage of training (knowledge score) |                |                 | Per cent change |
|-------|--------------------------------------|-------------------------------------|----------------|-----------------|-----------------|
|       |                                      | Before training                     | After training | Mean difference |                 |
| 1.    | Management practices                 | 22.26                               | 48.46          | 26.20           | 117.7           |
| 2.    | Breeding practices                   | 18.34                               | 49.82          | 31.48           | 171.7           |
| 3.    | Feeding practices                    | 24.02                               | 54.64          | 30.62           | 127.5           |
| 4.    | Fodder crops cultivation             | 23.22                               | 56.36          | 33.14           | 142.7           |
| 5.    | Conservation of fodders              | 09.18                               | 46.82          | 37.64           | 410.0           |
| 6.    | Diseases and their control           | 22.16                               | 52.20          | 30.04           | 135.6           |
| 7.    | Clean milk production technologies   | 20.24                               | 64.42          | 44.18           | 218.3           |
| 8.    | Milking methods                      | 12.66                               | 38.24          | 25.58           | 202.1           |
| 9.    | Clean milk products technologies     | 17.12                               | 34.30          | 17.18           | 100.4           |
|       | Overall score                        | 18.80                               | 49.47          | 30.67           | 180.7           |

Table 2. Effect of training programme on change in attitude of dairy farmers

| Particular         | Stage of training (attitude score) |                |                 | Per cent change |
|--------------------|------------------------------------|----------------|-----------------|-----------------|
|                    | Before training                    | After training | Mean difference |                 |
| Change in attitude | 24.12                              | 47.44          | 23.32           | 96.7            |

Based on the results obtained in the present study, conducting training programmes frequently for dairy farmers on scientific dairy farming practices would help in improvement in their knowledge and attitude levels for competently enhancing the acceptance and dissemination of dairy farming practices.

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## NEW EXECUTIVE COUNCIL OF THE SAP (2014-2016)

| Office/Position          | Members of Executive Council                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>President</b>         | <ul style="list-style-type: none"><li>• <b>Dr. A.K. Singh</b>, ADG (Extension), ICAR, New Delhi</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Vice Presidents</b>   | <ul style="list-style-type: none"><li>• <b>Dr. P.S. Kendurkar</b>, Ex-Prof. &amp; Head, Dept. of Agricultural Biochemistry, C.S. Azad University of Agric. &amp; Tech., Kanpur</li><li>• <b>Dr. Sanjeev Gupta</b>, Principal Scientist, IIPR., Kanpur</li><li>• <b>Dr. Sunil Kumar</b>, Head, Crop Production Division, IGFR, Jhansi</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>General Secretary</b> | <ul style="list-style-type: none"><li>• <b>Dr. A.K. Tripathi</b>, C.S. Azad University of Agric. &amp; Tech., Kanpur</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Joint Secretaries</b> | <ul style="list-style-type: none"><li>• <b>Dr. Poonam Singh</b>, Prof. &amp; Head, Dept. of Seed Science &amp; Technology, C.S. Azad University of Agric. &amp; Tech., Kanpur</li><li>• <b>Dr. S.N. Singh</b>, Principal Scientist (Agronomy), Indian Institute of Sugarcane Research, Lucknow</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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## The Society of Agricultural Professionals C.S. Azad University of Agriculture and Technology, Kanpur-208 002

### SAP Awards-2014

The Society of Agricultural Professionals invites applications/nominations for “**SAP Awards-2014**” to honour eminent scientists/agricultural professionals in following categories. This invitation circular is issued as per decision taken in the Executive Council meeting of the SAP held on August 11, 2009 at CSAUAT, Kanpur and thereafter approval of General Body Meeting of the society held on March 27, 2010 at CSAUAT, Kanpur.

**SAP Gold Medal:** The Society Gold Medal is given to the outstanding scientists of Indian origin in recognition of their professional leadership, outstanding contributions and service to The Society of Agricultural Professionals. The number of SAP gold medalist shall not exceed three in the award year.

**SAP Recognition Award:** This award will be given for extra-ordinary research contributions in the field of Crop Improvement, Crop Protection, and Natural Resource Management including Crop Production and Social Sciences. The selection will be based on research contribution as evident by development of products/varieties/patents/technologies and publications in high impact journals. The award is made to honour distinguished scientists who are founder/life members of the Society and above 45 years of age. The number of Recognition Awards shall not exceed three in the award year.

**SAP Young Scientist Award:** This award is instituted by the Society to promote qualitative efforts in terms of education/research/extension in the field of agriculture and allied sciences and to provide encouragement to brilliant young scientists for their original work (age below 45 years). The age relaxation for this award shall be maximum of two years for founder members only. Not more than three scientists, who are founder/life member of the society, would be awarded in the award year.

#### NOTE:

1. Eligible scientists are required to send their curriculum vitae (three copies) along with full documentary evidences (separate application for each award) by Registered Post in the prescribed proforma to “**General Secretary, The Society of Agricultural Professionals, Department of Agronomy, C.S. Azad University of Agriculture and Technology, Kanpur-208 002 (UP)**”. The complete application on prescribed proforma including a separate one page summary (not exceeding 500 words) of your resume, one passport size photograph and soft copy on C.D. must reach to the General Secretary, SAP latest by **September 30, 2014**.
2. The age of various awards will be computed as on 1st January of 2014.
3. Those who are already recipient of these awards need not apply.

The decision of Executive Council of the SAP regarding conferment of awards shall be final.

(A.K. Tripathi)  
General Secretary, SAP  
sap.kanpur@gmail.com



**The Society of Agricultural Professionals  
C.S. Azad University of Agriculture and Technology, Kanpur-208 002**

**NOMINATION FORM**

**SAP Awards-2014**

1. Name of the nominee:
2. Name of the SAP award applied for:
3. Date and place of birth (attach age proof certificate):
4. Address (with Telephone No., Email and Fax)

Office: \_\_\_\_\_

Residence: \_\_\_\_\_

Passport size  
recent photo

5. Academic credentials (beginning with Bachelor's Degree):

| Qualification | Name of the University/Institution | Year | Class/ Division/<br>Distinction, if any |
|---------------|------------------------------------|------|-----------------------------------------|
|               |                                    |      |                                         |
|               |                                    |      |                                         |

6. Employment Records:

| Post held | Organization | Period |    |
|-----------|--------------|--------|----|
|           |              | From   | To |
|           |              |        |    |

7. Fellowships/ Awards/Honours/ Recognitions (if any):
8. Significant contribution of the nominee in scientific field (within 200 words):
9. List of major publications of the nominee (if required separate sheets may be used):
10. Membership status to SAP: Founder / Life / Annual / 5-year / 10-year
11. Significant contribution of the nominee for the SAP (within 100 words):
12. International Exposure, if any:
13. Services rendered to Professional Bodies:

Place: \_\_\_\_\_

Date: \_\_\_\_\_

Signature of the Applicant

**CERTIFICATE BY THE HEAD OF THE INSTITUTION**

This is to certify that the above information given by the applicant is correct. I recommend him/her for consideration for the **"SAP Awards-2014"** (*Name of award applied for*) of The Society of Agricultural Professionals, C.S. Azad University of Agriculture and Technology, Kanpur-208 002, Uttar Pradesh.

Place:

Signature of the Head of the Institution

Date:

Name and Designation



## GUIDELINES AND INSTRUCTIONS FOR CONTRIBUTORS

**Current Advances in Agricultural Sciences-An International Journal** is published by The Society of Agricultural Professionals, Department of Agronomy, Chandra Shekhar Azad University of Agriculture & Technology, Kanpur 208 002, India half yearly in June and December every year. It publishes **original researches** in all areas of Agricultural Sciences including Agricultural Engineering and Home Science. **Review articles** from relevant fields reviewing research status, research lacunae and lines of future research are also published. Articles submitted for publication should be exclusive for this journal and must not be submitted elsewhere. **All the authors must be the member of the Society** and shall be solely responsible for the factual accuracy of their contribution. Language of publication is English (British).

### SUBMISSION AND PREPARATION OF MANUSCRIPT:

Authors are requested to submit the manuscript through Email only ([sapjournal.kanpur@gmail.com](mailto:sapjournal.kanpur@gmail.com)) and **two hard copies** in a format that is compatible with MS Word 2000 or later version. The manuscript should be submitted to **The General Secretary, The Society of Agricultural Professionals, Department of Agronomy, Chandra Shekhar Azad University of Agriculture & Technology, Kanpur 208 002 (Uttar Pradesh), India**. Manuscripts should follow the Journal style (see the latest issue).

Each full-length article should not exceed **6 printed pages**, i.e. **3,000 words**, including figures, tables and list of references. Short communications should be restricted to about **2 printed pages**, i.e. **1,000 words**. When preparing the articles, use only **Times New Roman (12 point font)** typed in double space throughout on one side of bond paper (A-4 size). No editing or material changes at the proof stage will be allowed. **Short communications** will have only title, author's name, address with Email of corresponding author, **ABSTRACT, key words, INTRODUCTION** followed by text, **CONCLUSIONS** and **REFERENCES**.

**Full length articles** should be suitably divided into following main headings typed in capital letters in the centre of the page: **ABSTRACT, Key words, INTRODUCTION, MATERIALS AND METHODS, RESULTS AND DISCUSSION, CONCLUSIONS, ACKNOWLEDGEMENTS, and REFERENCES**. Tables and Figures should be given separately at the end and the headings **INTRODUCTION** and **CONCLUSIONS** need not be mentioned in the text in both full and short communications.

- 1 **Title:** It should be short (not exceeding 50 letters), specific and informative typed in small bold letters with first letter capital and Latin names in italics. The **Byline** should contain, in addition to the names and initials of authors (in capital letters), the address of the institution where research was carried out (in italics). Change of address of corresponding author should be given as a footnote. Also mention the **Email ID of the corresponding author**.
- 2 **ABSTRACT:** A brief abstract (not exceeding 200 words), of the principal information and the most important conclusions should be typed after address in one paragraph and indented on both sides.
- 3 **Key words:** Key words (not exceeding 8 words) should be given at the end of the abstract in alphabetical order, separated by commas, with first letter of each key word in capital.
- 4 **INTRODUCTION:** It should be brief and to the point, limited to the statement of the problem and objective of the work. The literature reviewed should be pertinent to the problem under study.
- 5 **MATERIALS AND METHODS:** It should comprise all relevant details of the methodology including experimental design, treatment details and statistical methods used. For well established methods of analysis, citation of reference will be appropriate. All statistical comparisons among treatments may be made at  $P = 0.05$  level of probability.
- 6 **RESULTS AND DISCUSSION:** Results may be reported and discussed together to avoid repetition. The information should be divided into suitable sub-headings, typed in italics and bold. Undue and vague claims and stretched explanations may be avoided. Conclusions of results (2-3 sentences) may be given as a last paragraph of this section.
- 7 **TABLES, FIGURES AND PHOTOGRAPHS:** Tables and figures should be limited to the necessary minimum, typed on separate sheets with appropriate titles. All weights and measurements must be in SI or metric units. Data presented in the figures should not be duplicated in Tables and vice versa. It is required to submit reproducible artwork. It is essential that figures are submitted as high-resolution scans. Photographs must be on glossy paper. Colour photographs will be printed on request and on payment of Rs. 5000/- or US \$ 200 as processing charges.
- 8 **ACKNOWLEDGEMENT:** It should mention only assistance received in terms of financial support from any source for this study. If not necessary, it can be omitted.
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 ICAR, 2007. Annual Report, 2007-08, pp. 34-35. Indian Council of Agricultural Research, New Delhi.  
 Mishra JS and Bhan VM, 1997. Effect of cultivar and weed control on weed growth and yield of pea. *Indian Journal of Agronomy* 42: 316-319.  
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 Reddy TY and Reddi GHS, 2006. Principles of Agronomy, pp. 527. Kalyani Publishers, B1-1292, Rajendra Nagar, Ludhiana.

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